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THE MERCHANT ADVENTURER'S GUIDEBOOK

BY
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PROLOGUE

....it occurs to me that there is a proper balance between not asking enough of oneself and asking or expecting too much.

.... Not easy to find the balance, for if one does not have wild dreams of achievement, there is no spur....

One must think like a hero to behave like a merely decent human being.

- Sartre, May. Journal of a Solitude

This book is about being a merely decent investment professional. Each section offers some perspective, and if you are open to it, some insight on our way of investing. There are many ways to successfully invest – so the guidebook isn't some sort of truth, or a general-purpose guide to investing in any fashion. It is specific to us. All the ways to invest successfully are specific to the investor, and consistent over time.

Learning any specific skill means developing certain habits of mind. This guidebook describes the habits of mind essential to our investment process. Each of the three main parts of the book are describing the same habits from different perspectives:

1. Notes & Letters: Descriptions of events as they happened and in describing events, they also describe how to look at the events and the habits of mind or principles involved.
2. Principles & Processes: Short pieces discussing specific habits of mind or analytical frameworks intended as anchors for formal policies and procedures - anchoring corporate processes in certain habits of mind.
3. Readings: Excerpts from different writers offering additional perspective or support on why we adopted different habits of mind.

The habits of mind described in the following pages are tools for “seeing” the present from a shared perspective and operating with shared processes. It may be that all we need to achieve excellence isn't to have the “correct” perspective, but just that we have a shared perspective and shared processes.

Mastering the perspectives in the guidebook is not easy. It will require heroic efforts to develop the habits of mind described and apply them daily in our work. And at the end of that heroic effort, we will only be doing decent or acceptable work. ***To paraphrase Sartre, one must think like a hero to do the work of a merely decent investment analyst.***



PART 1

NOTES

&

LETTERS

JANUARY 2022

VALUE INVESTING, CHANGE & GORILLA GAMES

Summary: The punchline is that the only opportunities we worth researching are what I am calling "value gorillas" - these are large economic opportunities which are obvious problems with non-obvious / complex / hard solutions (gorilla scale payoff to the winner) but which are also invisible or ignored like the invisible gorilla in the famous attention experiment. This "invisible" property tends to go along with low valuation or the expectations that the "gorilla" problem won't or can't be solved.

NUCLEAR AS AN INSTANCE OF A GENERAL CASE

The long-term opportunity in nuclear energy is obvious and has been for decades. The regulatory issues and risks with nuclear have also been obvious. And this is a key part of what's interesting and what makes for a large investment opportunity. Big opportunities are NEVER esoteric or niche - they tend to be obvious, but hard and risky and complex in some ways. Big opportunities end up being NEW COMBINATORIAL SOLUTIONS TO OLD KNOWN PROBLEMS.

New combinations of technologies or techniques to solve old problems may include small pieces which are new -- but usually the advantage comes from assembling technologies and techniques which are known to work in a new way. It's not an invention, but a new combination of old inventions. That's why the new combinatorial solution can scale as a business -- whereas a true invention will take time to learn about and learn how to operate. This "combinatorial" idea is my own way of stating what folks from Joe Henrich the anthropologist states in his *Secrets of Our Success* to the process which science writer Amir Aczel describes in the Riddle of the Compass. The key insight one gets when you realize you are looking for scalable combinations of reliable tech, is that it means you can see these things coming in the potential unlocked by existing technologies.

Thus, we have a model for understanding change by studying choke points or limiting steps in the value chains of different industries and asking experts or researching what potential existing technologies could be applied to those choke points. Then we also look for people and businesses working on playing with or implementing those technologies, or who have intellectual capital in those areas which could become more valuable.

Nuclear energy is simply an instance of this general case. It sits at a point in the energy value chain. Energy enables all industrial activity and frees people from labor allowing them to create and allocate their time to other activities, and the cost of energy is a key constraint in all economies so a technological implementation which could shift the cost of energy down is important.

Other instances of this general case include many instances of automation and virtualization, where the advanced manufacturing model using virtual design, a heterodox combination of constantly advancing manufacturing technologies including 3D printing as well other automated tools and robotics is "eating" legacy manufacturing. This is the business Jabil and Flex are in. Like the situation with GE & Westinghouse, these were also companies which had a large opportunity which was and still is emerging, but which traded at low valuations (single digit PE) on their existing/legacy businesses.

My original interest in crypto-currencies came from the simple fact that the charges from Visa and Mastercard or the "vig" on transactions in the US has gone up over time as technology has improved, whereas in other parts of the world like China - it has gone down. This is a critical choke point that one can see with Visa/Mastercard as one of the largest expense items for folks like Home Depot behind lumber and labor. And you could see people like Walmart and others working on finding ways around it. Ultimately, with the payment systems - I think what is coming is that governments will offer different versions of CBDC's which function as public payment rails. The

CBDC becomes a publicly available open payments back-end which will open up competition in the payments space. I know this is an explicit objective of the EU and Israel and China - and I believe it is part of the plan everywhere. Not every choke point will have a private solution - but that's part of what makes it interesting. In the case of payments and CBDC's -- these new networks will enable innovation on top of them. That's what we will watch for. Any bank with modern digital infrastructure will be able to plug into these systems - so the emergence of CBDC's will be an important event for banking generally. We will finally see who has invested in their infrastructure and who hasn't -- and whoever hasn't will be quickly left behind. (Note: crypto is not a value gorilla - it is a gorilla large opportunity, but it is priced as though crypto has or will solve the problem - it's discounting success and not "invisible".)

Iraq is arguably also an instance of this where the new combinations are driven by the younger generation - so its demographics but specifically growing up digital. The key lessons to learn from Iraq involve how developing economies onboard new consumers using technology. Our hypothesis on the role of technology enabling emerging world consumers draws from Mark Miller's 2003 Digital Path essay. Miller proposed that the internet could provide digital institutional infrastructure as an alternative to government and the traditional model. Miller's Digital Path is an alternative to Acemoglu's Narrow Path or DeSoto's Other Path. The models of how we onboard new consumers through slowly evolving government institutions which are fundamentally rent seeking themselves won't work in the coming years if 3 billion consumers arrive from the developing world. We need to watch this. The Digital Path is the only way, but technology is so abundant that the pervasive shifts between competition, conflict and cooperation between technology and governments is easy to forget.

Nuclear is one example of a general set of opportunities which involve:

1. The unlocking of a large amount of value through applying a combination of multiple old known technology to a choke point in a key value chain (Gorilla large, inspired by the old Geoffrey Moore book - <https://www.harpercollins.com/products/the-gorilla-game-revised-edition-geoffrey-a-moore>).
2. Which is obvious, although not necessarily believable (Gorilla in the picture but did you notice it - http://theinvisiblegorilla.com/gorilla_experiment.html).

We can then look for companies which are working on these opportunities leveraging existing businesses and clients, and whose businesses trade as though the opportunities will not be realized (invisible gorilla / value stock / free real options).

ALLOCATING OUR TIME: ONLY ONE GAME FOR US - VALUE GORILLA'S

Any framework is only as good as the use to which we put it. In the case of this Value Gorilla Framework, the use is in targeting and sorting ideas as well as in checking the potential of current holdings.

John has discussed how the world is rapidly changing and when he is talking about that, where my mind always goes is to "what does that mean for what I do as an analyst?" As an analyst the challenges are at least two-fold: (1) it's easy to get distracted by short-term events, which you don't want to ignore because they may create price volatility which is an opportunity to buy or sell at a better price - but which aren't worth more than a few percent at the margin long-term; (2) and then it is easy to get overwhelmed and fall back into looking at things that are smaller opportunities but feel easier to understand - and investing is not like baseball - in investing the home runs are more important than just getting on base.

As analysts -- in a world characterized by rapid change it becomes essential to focus on the large opportunities. The pace of change means that you can ignore everything else. Previously, the way I had

articulated this idea was by saying that returns don't mean revert -- so this is another way of saying the old value investing model of buying cheap assets and waiting for mean reversion is dead. We can argue about whether or not this model is really dead (it probably isn't) -- but the key idea is that there are better return opportunities in finding the companies creating new value than there is in reversion to mean valuations. The creation of new value is a much bigger pie. So that is what we need to focus on.

But we do not need to invest in opportunities which the market is already pricing as if they are a certainty. We want the invisible gorillas. And this is where the real work is. The overall opportunities will be obvious - like the proverbial dollar bill on the street, such that the instinctive response will be "duh" or "but if it were going to happen it would have" - modern versions of if men were meant to fly they would have wings sort of thing. So the work will be in understanding businesses which are working to solve these known problems.

This is where the idea of combinatorial innovation should be useful to us as analysts -- because we aren't looking for people really working on the bleeding edge, but on scaling known technologies and applying them in a new process. The new combinations may not be easy to understand, and they may be complex -- but the ideas won't be new or esoteric and the applications will be global. This makes it a gorilla. The fact that the market is discounting a low probability of success makes it invisible.

These are what we need to be focused on in a time of rapid change.

THE BASIC STORY OR THESIS FOR EACH INVESTMENT IS WHY ITS A VALUE GORILLA

The framework also works for how we communicate our ideas to each other and out investors.

Our elevator pitch of each investment should be why each holding is a value gorilla. What is the economic opportunity in front of it and what are they doing to exploit this opportunity? Why is it likely to succeed while the market is pricing the stock as if they will not succeed? That's it.

Massmart is a value gorilla because of the scope of the retail opportunity over time in Africa, and the capabilities and ambitions of Walmart. It's invisible now because of the poor performance of the business in the recent past, but new management and a fresh commitment from Walmart after the parent exited many other markets makes us think there is a will and a plan to succeed over time which isn't in the price of the stock. Over the next 30 years, consumer growth in Africa should dwarf consumer growth in the rest of the world as the continent catches up economically and keeps growing in population long after populations in the rest of the world have stabilized or begun to fall.

Brookfield is a value gorilla because they have a disciplined global investment platform focused on refreshing the infrastructure assets of the developed world and building the new capacity that will be needed in the developing world. Brookfield's investment capabilities are well known - but the magnitude of the long-term opportunity and the potential for extraordinary profits created by low-cost capital / low rates -- we believe this is what the market is underestimating. Brookfield could compound at a 20% rate for a long-time, such that it is own sort of long-term compounding gorilla. What is invisible in this case is the compound, or the old exponential vs linear case.

CONCLUSION

Overall, I believe this simple framework offers us a unique and clear heuristic for allocating our time (managing ourselves as analysts) as well as prioritizing and sorting opportunities and communicating the thesis behind each idea within the team and with investors.

JULY 2022

WHAT JUST HAPPENED

“If you hold a cat by the tail, you will learn things you cannot learn any other way”.

- Mark Twain

We launched the Merchant Adventure Fund in May of 2018 as an unconstrained global value oriented public equity portfolio. Since then the world has surprised almost everyone with negative interest rates, negative oil prices, a global pandemic, the invention of new vaccines in record time, a land war in Europe supported by China, unprecedented developed world central bank and fiscal stimulus, followed now by rapid inflation, financial market enthusiasm for anything virtual and ethereal arguably leading to overinvestment only to be complemented with underinvestment in anything which could be touched – including oil, which after having a negative price two years ago now appears to be in short supply. To cite a partial list. The market has always been a harsh taskmaster. We were holding a cat by the tail and were learning things one couldn't learn any other way.

Our mandate was to get out and learn about publicly traded companies around the world, looking for quality management teams and business franchises with significant growth opportunities and trading at low valuations. We did this for a few years. Then the world closed. We kept the same focus but worked virtually for the next two years. In March 2022, a team member left the group. We took the opportunity to reflect on what we had learned through the pandemic, what had worked and what hadn't. The condensed list is below.

What worked:

1. Constructing a long-term perspective based on domain expertise and financial analysis.
2. Teaming up or gang-tackling the analysis of a company and industry.
3. Being able to act on high conviction but non-consensus beliefs.

What didn't:

1. Working Remotely... didn't work.
2. Consensus Decision Making wasted a lot of energy but never achieved agreement.
3. Everything, everywhere, all at once – too many different businesses in too many places.

All organizations are learning and evolving. By spending the last four years looking at “everything, everywhere all at once” we accelerated the learning curve for the team and have been able to define a focus and processes that we couldn't have come to any other way. Amar Bhidé described this sort of trial-and-error process of building norms in a professional service firm in his classic [1995 paper on the evolution of McKinsey](#). But I think Mark Twain captured the gist of it. There are some things you can't learn unless you hold a cat by the tail. Going forward, we need to do less of what didn't work and more of what worked.

This boils down to two principles:

1. **Focus** on where we have an edge or can build an edge as an organization and in our investment process.
2. **Imagine** how human agency¹ and technological capabilities create future economic potential.

¹ One can think of human agency as labor productivity and technological capabilities as capital productivity. The role of imagination is to project forward in time paths for labor and capital productivity changes, which we all know are the key drivers of per capita income growth or increasing wealth in economies.

WHERE WE GO FROM HERE

“...millennials and Gen-Z share values on a global scale.... For no previous generation could we so confidently pinpoint these or other common traits as we can for billions of young people today. The great divide in the world is therefore not East versus West or North versus South but young versus old.” -- [Parag Khanna, Move](#).

We invest in human progress² based on our understanding of human agency, technological capabilities, and economic potential. Businesses uncover the economic potential and are the vehicles we invest in. But businesses are driven by motivated individuals (human agency) and they exploit scientific advances (technological capabilities). This is a very simple and timeless model, but the implementation is complex and evolving.

We know that the future hasn't happened yet and that the most accurate way to model things that haven't happened yet is in terms of probability trees. Many probabilities don't fit normal distributions but are multi-modal - and sequences of events are path dependent. The natural way for us to manage this complexity is to build stories or scenarios based on what we know – to imagine probable futures. The value of this analysis is often not necessarily in predicting what path the future will take but in analytically preparing for multiple paths and having a list of events to watch for, which would represent important forks in the probability tree.

We must try to imagine the future - building scenarios based on (a) country demographics informed by (b) cultural anthropology and imagining major potential technological advances like (c) the digital path of development for emerging consumers or (d) nuclear fusion or (e) cellular rejuvenation.

The literature on scenarios goes in a number of different directions, but our general guides are [Superforecasting](#) by Tetlock and [Using Scenarios](#) by Chermak – both of which take a more function approach. ***Our approach to scenarios and forecasting is described further in the Scenario Process in Part 2.***

ENDING WITH A NEW BEGINNING

“All beginnings are hard.” -- Talmud

Business and investment is an [adventure](#) – there are neither guarantees nor certainty, only the adventure. The first iteration of the fund challenged investors to hold diverse global risks in a single portfolio and trust our expertise on that diverse set of holdings. While the thesis behind each of the positions may have been valid, what we learned was that it was impossible to (1) establish expertise on a diverse set of businesses and risks, and (2) maintain a unified team perspective while staying abreast of [“everything everywhere all at once.”](#)

During the first iteration of the fund, we added team members focused on different geographies and sectors, added junior analysts and interns as well as domain experts in software and biotech. We added knowledge and expertise as the team grew, but we never reconciled divergent perspectives of experienced analysts. The largest gap in perspectives always surfaced in what the key drivers of change were -- rather than any disagreement about current facts. Based on the problem of unifying analyst perspectives – we need a unified set of key drivers of change.

The farm team concept also failed for similar reasons. In the process of taking small farm team positions which could be advocated by a single analyst, we ended up with holdings based on divergent perspectives which couldn't be reconciled. Small “farm” holdings ended up being a distraction and creating conflicts more than they

² Jason Crawford's Roots of Progress offers useful perspective on human progress, which is fundamentally a philosophical justification for human freedom and scientific inquiry as drivers of economic growth and improved standards of living: <https://rootsofprogress.org/>

created learning or opportunities. The hypothesis behind having a farm team was as a learning mechanism – but this didn't pan out. There are better ways to build learning into the culture.

Our best work involved teams of a (1) junior analyst who brought a younger perspective and could also help with blocking and tackling aspects of the analysis, (2) a domain expert with decades of knowledge and experience in the field the subject company was focused on, (3) a senior analyst to integrate the domain expertise with the young optimism and energy and a financial model and overall investment thesis. Building cross functional teams and working more in these teams is one of the best ways to drive organizational performance, unify perspectives, build culture and encourage learning.

Our experiences investing in Japan, Russia and China also taught us a diverse set of lessons which boiled down to one thing: societies whose populations aren't growing will face difficulties³. It is obvious – young people are the future. If societies have low birth rates and emigration, then those societies are failing and shrinking. Our experience investing in what looked like low valuation assets that just tended to get cheaper and cheaper in shrinking states led us to step back and consider what to do differently.

Remembering that investing is about the future and that it works best when we are investing in human progress, we chose to focus our attention on countries or markets where:

1. ***Large enough domestic market to have some scale (40 million or more in population).***
2. ***Not shrinking (population hasn't declined over the last 5 years).***
3. ***Connected to the global economy (at least one [mega city](#)).***

We decided to focus on these demographically well positioned countries – but it is important to note that we are choosing an area to focus on rather than creating negative screens or limits saying we won't invest in other places. Rather than constructing limits, we are quantifying and enumerating preferences or advantages.

Additionally, to remain a research-driven organization we needed to accept that administrative costs limit where we invest. *Country funds are the appropriate way for investors to put money to work in attractive geographies which have limited capital markets and/or administrative and capital constraints – part of focusing the fund is to focus on open capital markets with lower administrative costs and complexity.*

The new beginning for the fund is the simple process of putting the lessons of our first four years to work:

1. ***Focusing on areas of structural change in the global economy.***
 2. ***Focusing on economically and financially functional geographies.***
 3. ***Focusing on fewer companies.***
 4. ***Focusing on building knowledge of these opportunities in teams.***
- ... Focus, focus, focus.***

³ "I and the public know
What all schoolchildren learn,
Those to whom evil is done
Do evil in return."

- W.H. Auden, September 1st, 1939

WHEN ON THE BATTLEFIELD

“I often say that when you can measure what you are speaking about, and express it in numbers, you know something about it: but when you cannot measure it, when you cannot express it in numbers, your knowledge is of a meagre and unsatisfactory kind...”

-- William Thomson (Lord Kelvin)

Gerald Loeb famously wrote about the battle for investment survival during the great depression. Most companies are losing investments over time, not winners. The market tends to appreciate because the winners create more wealth than the losers destroy over time. To survive and prosper on the investment battlefield one needs to identify these winners – but the market is perverse, and profits come from the gap in the expectations in the purchase price of an asset versus future expectations. Profits over time come from rising expectations. So, our task is identifying the winners who will drive the markets in the future, but at a time when they aren’t expected to win.

Our task in the battle for investment survival is quantifying the gap between the expectations in current company prices and our expectations for what a company can achieve given the economic potential, technological capabilities, and human talent it has and is targeting. While quantifying expectations in a current stock price is a pedestrian task, imagining the future paths of a business and putting numbers to it is the furthest thing from pedestrian and involves a high degree of creativity, persistence, and humility.

Since investing is all about the future – the starting point for our analysis is to construct and maintain a perspective on where the world is going in the long-term. This starts with the economic geography of the world, and then adds an analysis of key drivers of change in the world. The maps and models and frameworks are all about being able to express in nodes and edges (graphs and numbers) the current state and economic context of companies. Measuring their capabilities and potential. We build our map, invest based on it. ***It’s very simple at a conceptual level – the devil is in the details of how we collect the information, process it as a team and implement what we learn in the portfolio.***

THE LOOP: Economic Potential, Technological Capabilities & Human Agency

With any framework, its value is in its use⁴. The framework’s value isn’t in being “true” – but in providing a shared language and terms for the implementation and coordination of a research team for the purpose of building a global equity portfolio.

Operationally, the research and portfolio management process is a battlefield decision-making loop which starts with identifying where we will fight for profits (areas of economic potential – products, industries and geographies), who and what are exceptional in those areas (technological capabilities and human agency – companies and managements), then observing and calibrating (doing the work to gather the information, test assumptions, learn and adapt). The devil is in the details of each step – and in each section below we discuss what work products are involved in each step, with an eye toward the tasks ahead of us as we build our organizational capabilities.

⁴ All references to “value” in an economic context is “value in use” or instrumental value. Although investors use the term intrinsic value all over the place from a strict logical and definitional perspective to talk about objects having intrinsic value is like saying a chair thinks because it was designed. It’s logical gobbledygook. We seek to avoid logical nonsense.

ECONOMIC POTENTIAL

“Long-term trends can be predicted, although their specifics cannot.” -- Kevin Kelly

We break long-term trends and structural changes into three categories or domains: biological, physical, and digital. Since we are investing based on where the world is going, we think of diversification in the portfolio in terms of forces driving change in these three domains of the economy. These different domains each map to multiple sectors in the economy as shown in the table below.

		Biological	Physical	Digital
		Biotech & Health Technology	Advanced Materials & Manufacturing	Distributed compute and networks
Communications			+	+
Consumer Discretionary		+	+	+
Consumer Staples		+	+	
Energy		+	+	+
Financials				+
Health Care		+	+	+
Industrials			+	+
Materials		+	+	
Real Estate		+	+	+
Technology		+	+	+
Utilities			+	+

Our perspectives on change in the biological, physical, and digital worlds will be captured in a series of scenarios following the process described in Part 2.

In assessing economic potential, in addition to broad economic trends, we need to pay attention to changes in human geography – which is basically where people are today versus where they are expected to be tomorrow. Human geography in this context is best expressed, as Ai Weiwei put it, as [human flow](#). While this is complicated, if we take a few basic criteria – the world’s human geography comes into a clear focus.

Only seven countries have open capital markets, at least [one mega city](#), a domestic population above 40 million, and a population which is growing over the last five years:

Brazil, Indonesia, Mexico, South Africa, Turkey, United Kingdom, United States.

In terms of economic potential, our focus is on businesses positioned to benefit from innovation in those growing geographies which are also open to foreign capital.

The countries below meet all the criteria above, except they have certain capital market or government restrictions that cause us to exclude them from our open markets list:

Argentina, Bangladesh, Democratic Republic of Congo, Egypt, Ethiopia, India, Iran, Iraq, Kenya, Nigeria, Pakistan, Philippines, Vietnam.

Note: A discussion of the country framework is included in the Principles and Process section on the Composition of Economic Growth.

TECHNOLOGICAL CAPABILITY & HUMAN AGENCY

“The future is already here, it’s just not evenly distributed.” -- William Gibson

Businesses make money by mobilizing technological capabilities and human agency – and in the process of doing this they take something that was hard and make it easier, lowering transaction costs and driving human progress.

Businesses don’t do well with fundamental discovery and long-term research. What businesses do well is scale and breadth – to commodify & distribute innovations and discoveries. Businesses are the mechanism for distributing the future. And in distributing the future, they often create wealth.

The purpose of our company research process is to identify exceptional businesses which are characterized by some combination of unique and leading technical capabilities and people – distributing the future and driving human progress. Our company research framework is outlined below.

1. Business Franchise

How does the business take one dollar and turn it into two?

What are the technical skills and advantages the company has?

Nuts and bolts of the financial model and scaling.

2. Management Quality / Culture

Leadership & culture are key to re-investment of capital & organizational resilience.

There are no exceptional results without exceptional people doing exceptional work.

3. Valuation:

Gap between business potential and expectations discounted in current price.

Napkin math captured in versions of a Gordon Growth Model:

$Value = Profits / (Target\ Return - Growth\ Rate)$

$Value\ (Price / Sales) = Normal\ Net\ Margin / (Target\ Return - Sales\ Growth\ Rate)$

$Value\ (Price / Book) = Normal\ ROE / (Target\ Return - Book\ Growth\ Rate)$

Perspectives on business franchise, management quality, and valuation combine to form our investment thesis on each company in the portfolio. Technical capabilities influence the business franchise. Management quality and corporate culture are all about human agency – motivating, engaging, growing, compounding and retaining human intellectual capital. Valuation is the step of identifying the gap between our expectation and the expectations in the current price, to maximize the probability of profiting from rising expectations (or multiple expansion). ***This information will all be captured in the Investment Thesis Notes for each company in Sentieo.***

A NOTE ON VALUATION

We are not trying to create complex models for the valuation thesis, but to imagine a business at scale operating in a steady state. The purpose of the valuation framework is to get a target price to sales or price to book multiple.

From a nuts-and-bolts perspective, our target return is 15% and the maximum long-term sales or book growth rate for the purpose of deriving a target valuation multiple is 10%, thus, for any subscale business with long-term growth potential the target multiple to sales = Normal Net Margin / 0.05; or target multiple to book value = Normal ROE / 0.05. Different businesses are best characterized as being driven either by invested capital producing a return (book value multiple, ROE & book growth framework) or by cranking up sales (sales multiple, net margin, sales growth framework). Part of knowing a business is knowing which is a better characterization.

Once we really know a business, figuring out the target multiple and our target price becomes the easy part. This is an important characteristic of a useful valuation discipline – it focuses one on understanding businesses and offers a quick and easy way to compare opportunities and doesn't get bogged down in technical details of modelling. Implementations of technically precise models all get precision at the price of understanding the way a business is evolving and being able to compare that precise model to other companies. This is like Heisenberg's uncertainty principle applied to investments: you can know exactly what a business is now or where it's going, but not both. Investing is about the future, not the past – so we want to know where the businesses are going.

OBSERVATION & CALIBRATION

Scenario planning and investing are both focused on possible and probable futures, so it makes sense that Peter Schwartz's description of scenario planning also applies to investing:

“... planning involves imagining different possibilities and then how we can get from here to there. And so it requires a combination of two things: imagination and analytical realism.”

Most investment operations do analytical realism very well but building a culture that supports imagining an uncertain future and positing scenarios which will certainly be wrong in many ways – well, very few firms do that well. ***Our goal is to build a culture which both supports a disciplined analytical process (realism) as well as cultivating a long-term view drawn from imagined paths from here to the future.***

All company observations are tested against the investment thesis. The investment thesis includes a set of factors to watch for or observe to confirm or refute the investment thesis⁵. Calibration is the modification of the investment thesis and the decisions to buy and sell based on the thesis or changes in the thesis. All that matters is exactly what we buy and sell, when we do it and where the expectations are in the price at the time of purchase or sale. All the steps, frameworks, principles, and processes leading to this point are like the gears that deliver force from the engine (the economy) to the wheels (the portfolio). Buys and sells are where the rubber meets the road.

The best description of what it is to be an observer is captured in Peter Drucker's autobiography [The Adventures of a Bystander](#). As observers, we ask questions and listen to the response. Observing is a mix of the art of being curious and asking the right questions and being an active listener – hearing and seeing what is there, and not just watching for what one expects.

Our portfolio construction guidelines describe when we buy or sell shares, and which companies we consider buying or selling shares in. The portfolio construction guidelines are also subject to observation, calibration, and change.

The changes in the portfolio construction guidelines from the first iteration of the fund to now include:

1. Private company investments as described in Part 2,
2. Focusing on companies with smaller capitalizations, limited to less than USD \$25 billion,
3. Diversifying the portfolio across drivers of economic change rather than economic sectors,
4. Focusing on fewer companies (30-40 holdings) and eliminating the farm team of small holdings, and
5. Owning larger positions of each of the fewer in the portfolio.

The purpose of all these adjustments were to incorporate what we learned from “holding the market” by the tail. The portfolio construction guidelines are listed below.

Rules:

- Minimum 70% Public Equities / Maximum 10% Cash,
- Room in between for select Private Company investments, or Convertible debt,
- Maximum Market Capitalization of \$25 Billion.

Diversification:

⁵ Has to be falsifiable (Popper) and has to watch out for confirmation bias (Cathcart).

- *Portfolio diversification is viewed in terms of the different themes or areas of structural economic change. The portfolio is invested across this set of structural changes which the manager view as significant but does not seek to be broadly economically diversified in any traditional sense.*
- *The portfolio is concentrated, holding 20 to 40 positions.*

Position Sizing – Buy/Sell Discipline:

- *Research process constantly calibrates, questions, and refines each company investment thesis.*
- *The investment thesis focuses on business franchise, management quality and low valuation.*
- *If we are confident of 1 of the 3 factors, the position is 1%, 2 of 3 is 2%, 3 of 3 is 3%.*
- *Weights over 3% reflect confidence in all 3 factors plus something exceptional.*
- *When our assessment changes significantly the position size is adjusted.*

CLOSING: COMING BACK TOGETHER AS A TEAM

We do our best work as a team. The greatest cost of the covid epidemic for us as an organization is that “sheltering in place” robbed us of the experience of being in the “foxhole” together during the epidemic. We didn’t come together as a team because we just weren’t together very much.

At this point, the fund has largely transitioned to this new approach. We have 36 holdings in the portfolio excluding the Euphrates Iraq Fund, down from a pandemic high of 85 names at one point. All 36 of those holdings are in the way of biotech or health technology, advanced materials and manufacturing or distributed compute and networking. None of the companies are over \$25 billion in market cap. The Euphrates Fund interest will be redeemed effective 9/30 with the cash returned to the fund around mid-October.

Operationally, the key steps between now and mid-October are to implement the new process and re-introduce the fund. But organizationally, we need to make sure we come together as a team and that requires that members of the portfolio management team be in the same place working together as much as possible, and that we adhere to a disciplined investment meeting schedule for the fund.

Specifically, we will review the portfolio each Thursday afternoon and make any changes to the guidelines in the portfolio in the weekly meeting. Outside of the weekly meeting, portfolio guidelines will not be changed.

Trading toward portfolio guidelines may happen throughout the week based on written instructions/recommendations/targets from the portfolio management team and included in the notes from the weekly team meeting. This process will be detailed in an Observable or a Sentio notebook (tbd) and the portfolio guidelines spreadsheet and notes will be kept in a format linked to the process notebook.

All of these lessons, principles and processes could be perfectly correct – but none of that will matter without our agency, without us coming together as a team with shared principles and a common set of tools and processes so that our efforts magnify each other rather than competing with each other. This magnification is difficult to achieve – but at its root I think [Ed Catmull from Pixar captured it best](#) in a practice he called “plus-ing”⁶ which can be summarized by saying that you have to say both what you like and don’t like about something if you are going to say anything at all. If all you have to say is no, then you don’t have anything to say. And if all you have to say is “yes” – well you might not have been paying attention. A quality dialog is driven by detailed observations – good and bad both, and then listening and adaptation or calibration.

⁶ Since Catmull’s plus-ing comes from improvisation which I think is exemplified by [Freestyle Love Supreme](#), we will just call it freestyling.

Excerpts from past Fund Letters

JUNE 2018

*And this was one of the wonders of the new company: **the adventurers knew so little about the world.** They believed they had some geographical knowledge; they had their investors' money; they had confidence. And they took such a big step, feeling as they went the shape of a world known only from maps and the pronouncements of the cosmographical authorities. Within a single reflex were contained both formidable ambition and startling ignorance.*

In a calculating, pragmatic and essentially conservative world, it was a bold and imaginative leap into the unknown. And it would have huge consequences for England. The ancestry of what later became Britain's mature mercantile empire needs to be traced back to this root – a single, ambitious and (from one perspective at least) hare-brained project that grew up in and around London in the early 1550s.

Alford, Stephen. London's Triumph: Merchants, Adventurers, and Money in Shakespeare's City

CLEAR COMPASS

The Merchant Adventure Fund is designed to get in the way of the three areas listed below, and other, “misunderstood and mispriced” phenomena. The first point focuses on corporations as the relevant entities to understand and invest in, avoiding the circular and cyclical asset allocation madness many go through. The second point asks whether the way we live in the West reflects a culture or whether the way we live is just how humans live when they have a certain level of health care and material wealth. [The Gapminder Foundation](#) would call this living on income Level 4, and assert that our income levels define us more than other factors. On the third point, the network connects and coordinates the agents in the economy.

The three key systematic factors are:

Creation of wealth through corporations. I am ceaselessly amazed that investors don't get this. Around the world, the corporation is the vehicle for wealth creation. But it gets the same respect Churchill gave democracy as the least bad form of government.

Expansion of Basic Health Care, including maternal and infant care. The average number of births per woman fell from about 5.6 in 1966 (close to where it had been for all human history) to a level just over 2 where it sits today. The median family in the world today is 4 people everywhere but in parts of Africa. First time ever. The flip side of births is life expectancy. As life expectancy increased, births fell. And falling births are a precondition for women's rights and equality movements. The significance of the global public health efforts which catalyzed this change cannot be overstated.

Exponential expansion of networks. We thought the late 1990's internet was big – but that was just wiring a few hundred million fixed machines together. What's happening now involves over a billion smart phones, data centers full of servers, a network upgrade to 5G and new generations of autonomous devices. We haven't even started yet. This connectivity is more like the development of a new sensory function than anything else, like eyes during the pre-Cambrian explosion.

In these three points, we have a powerful model of how a single dominant and stable household structure for consumption and a single dominant and stable wealth creation structure have both emerged. This sort of “standard consumption and production model” hasn't ever existed before on a global scale – and it only emerged in the last few decades enabled by advances in global health. With the network connectivity, we have a substrate that connects and coordinates these consumption and production entities. This is what our society has evolved into, and once we discern that reality – we can start to ask where this all leads and how to profit and prosper in a future which will be fundamentally new in many ways.

SEPTEMBER 2018

A good opening sentence, like a good investment theme isn't a summary – it's the entrance. The Merchant Adventure Fund itself is built on a few core themes. They are our assumptions, or compass. Many market participants don't agree with our assumptions, so it is worth explaining and exploring them. The first assumption we make is that:

The best way to profit over time is through the ownership of publicly traded corporations that are in the way of economic growth.

This simple sentence is an entrance to its own story. The story starts with the creation of corporations as a way for people to coordinate and combine labor and capital to buy resources for a dollar and sell products for two or more. Few people focus on the corporation in this way – as the opening sentence of the story of global economic growth.

Ownership, or “title” to assets or corporations is the instrument used to mobilize capital. In Elizabethan England, the role of ownership wasn't only critical to the colonies which were created by companies of merchant adventurers – it was also critical to the publication of the First Folio and Shakespeare's immortality. As Emma Smith documents in *The Making of Shakespeare's First Folio*, the key to the publication of the folio in the years after Shakespeare's death was the fact that the Kings Men held title to the plays and the aging players, Heminge and Condell, hatched a monetization strategy. Understanding the importance of ownership leads one to look for clean title and adopt a long-term perspective.

.... A key part of this is the role of **price**, or *the expectations discounted in the price* at which the public markets trade ownership interests in businesses. Can we become part owners for a price that reflects lower expectations for the business than it is likely to achieve? In other words, is it a low valuation company with good long-term prospects? The expectations in the price of a stock are like height of the hurdles in a race, and the company is the runner. We like to invest in businesses with the hurdling ability of Edwin Moses, but where the market's price sets the hurdles at the height of a matchbox.

... Our goal in building the business is to be small but fierce. To punch above our weight. We won't accomplish this by being smarter or dressed better than others. We will do it by working more efficiently and focusing on a different set of key factors and drivers. Our assumptions are different and the differentiator. As any logician knows, different assumptions necessarily lead to different conclusions.

... We are focused on what will be next. Specifically, we believe that the falling cost of sequencing and manipulating genetic information may be crossing a point after which it will rapidly change the way we all live. Owning a “bundle of biotechs” isn't a normal position for a value portfolio to carefully research, build and maintain. But investment value isn't based on past performance or past business results. Investment value is driven by the difference between the expectations in the market price and the future performance of a company.

... All values are values in use, and businesses continually change and evolve. Our focus is on understanding businesses as they are now and as they will be. Effort spent explaining what was, while trying to manage a portfolio based on what will create value is a misallocation of scarce intellectual capital and a “bad” anchor – and a disadvantage. We pay close attention to the anchors we set for ourselves and others through our activities. When we look at the task before us – stubbornly avoiding accidental anchors in the analysis of companies and economies is a key advantage we have.

DECEMBER 2018

“It is a world of change in which we live, and a world of uncertainty. We live only by knowing something about the future; while the problems of life, or of conduct at least, arise from the fact that we know so little. This is as true of business as of other spheres of activity. The essence of the situation is action according to opinion, of greater or less foundation and value, neither entire ignorance nor complete and perfect information, but partial knowledge.”

Knight, Frank H.. Risk, Uncertainty, and Profit

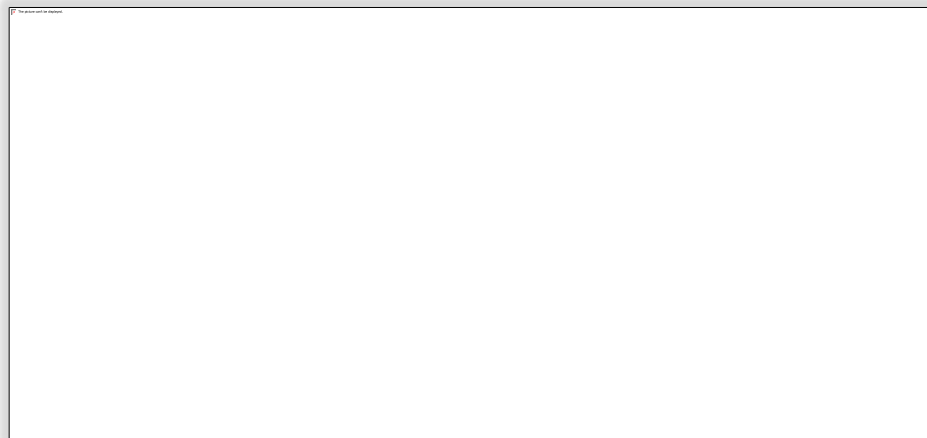
UNCERTAINTY & PROFIT

Given the recent market volatility and chatter about being late in the cycle or early in a trade war, we thought it might be useful to go back to the basics, which means going back to Frank Knight’s classic book Risk, Uncertainty & Profit. Knight asserts that profit in organizations comes from confronting uncertainty with partial knowledge.

We follow Knight’s advice and look for profit in areas of uncertainty – in the elements which evade any reliable quantification and where we must rely on partial information, opinion, and conjecture. We believe our fascination with uncertainty gives us a key competitive advantage as investors. Most analysts and investors are either trained to avoid structural uncertainty, or they find uncertainty too hard to deal with because of organizational pressures to be “right.”

When wading into areas of economic uncertainty, we will make mistakes. The only question is how wrong we will be. Will we make big mistakes or small mistakes? Being wrong bothers us, but it bothers us less than missing big opportunities for profit. We try to make a lot of “small” mistakes in the pursuit of “big” opportunities. Profit comes from uncertainty, and confronting uncertainty comes with a high error rate.

But what does this mean? How does one confront an uncertain future and manage through it? This is best captured by a simple picture from a research report written in the time before spreadsheets. The drawing shows the investors as stick figures stuck in the present: faces up against a wall, trying to look into the future. The lone investor at the bottom has a different perspective, separate from the others and focused on future profits. That is what we are trying to do (below).



from a 1980’s Research Report by John Gunn

MARCH 2019

Models create structures within which we can think logically.... But the logic comes at a cost... All models are wrong.... They omit details.

Page, Scott E.. The Model Thinker: What You Need to Know to Make Data Work for You (p. 6). Basic Books. Kindle Edition.

F. Scott Fitzgerald noted “the test of a first-rate intelligence is the ability to hold two opposed ideas in mind at the same time and still retain the ability to function.” The two opposed ideas we grapple with are captured in Page’s quote above (which is a reprise of George Box’s original observation): models give us the structure to think but, are also wrong because they simplify.

This battle between simplicity and complexity has been famously mapped onto hedgehogs and foxes: hedgehogs see & seek simplicity; foxes see & seek complexity. Jim Collins popularized the metaphor in Good to Great and Phil Tetlock also utilized it in his book Superforecasting. The tension between seeing the layers of complexity like a fox, and the need to use simplified models to analyze and communicate information is also the fundamental challenge of investment research.

We use accounting to summarize sets of transactions, but in building the accounts in financial statements – information is omitted. Some of these omissions are filled in by financial footnotes – but much of the work of investment research is going from the simplified accounting model in the financial statements back to the complexity of the underlying business, without ever being able to see all the complexity because of a host of constraints. A first-rate investment research process has to pass the same test Fitzgerald poses for a “first rate intelligence” – holding opposing ideas in the mind at the same time and maintaining the ability to buy, sell or wait.

Some have suggested to us that the idea of being a value investor and a biotech investor at the same time is a specific example of opposed ideas, which we hold at the same time. We would like to take a little time now to explain the reasoned basis for this position as well as the basic theses behind the different biotech holdings in the fund.

To begin with, we look for a basic set of factors which genuinely aren’t unique: strong balance sheets, quality management and consequential target markets. But then we add our own perspective. In this context as well as others, our own perspective is often about being patient when others aren’t – when we are dealing with a business where the basic factors of quality management, cash and consequential targets markets all check out.

For instance, if a quality management team has been working on a specific set of problems or pathways with limited success for decades and the market has become disillusioned such that the stock trades at a discount to the amount of capital which has been invested in research over time – well that interests us. We take a used car approach to buying what we believe are quality management teams when their stocks are trading at a discount to what they have spent in research. We are always attempting to buy research in process at a discount.

Each of the standard factors which we look for serves a practical purpose:

strong balance sheet - having cash on the balance sheet to buy time and allow for the complexity and challenges that are part of any genuine research process;

quality management - the resilience of a management team to keep pushing for the goal or intelligently pivot;

consequential target markets - the relevance and economic importance of the treatment or technology if the company does achieve its goals.

JUNE 2019

THE HUSTLE

Someone's always buying, someone's always selling, and the hustle is about trying to be in the middle of that whole thing. None of it was legal. Nobody knew where anything came from. The guy who got us Nikes, did he really have a "staff discount?" You don't know. You don't ask. It's just, "Hey, look what I found" and "Cool, how much do you want?"

Trevor Noah, Born a Crime.

Trevor Noah captured the insanity of bad governance and its' consequences in his autobiography Born a Crime. Bad or inadequate governance forces people to get what they need to survive through what we call “informal” means. Noah called it “the hustle” – and that captures it almost perfectly. Everyone is just trying to hustle to stay alive or get what they can to pull themselves up just a little bit. The important distinction we need to make from Noah's description of “the hustle” is that in the way we think of it, “hustling” includes both legal and illegal activities in the informal economy.

As long as you are caught in *the hustle*, you can't get far because as Hernando de Soto noted -- you don't own what you own. *The hustle* isn't about Confucian or Protestant values – you don't get into or out of *the hustle* because of your beliefs. Beliefs have nothing to do with it. *The hustle* is the natural human state in the absence of functioning sets of rules and institutions.

The study of how humanity escapes *the hustle* is often referred to as global development. One of the best resources on global development is the Gapminder Foundation (gapminder.org). Gapminder breaks the world into [four income levels](#) and shows pictures on their Dollar Street website (dollarstreet.org) of what life looks like in different countries at the different income levels. Each “figure” in the graphic below represents one billion people, and the graphic illustrates roughly how many people in the world today live at different income levels.



Level 1 is “extreme poverty” – people living on less than \$2 per day. In the early 1970's, one out of two people in the world lived in extreme poverty. Half of the global population of humans was malnourished and miserable. Now that number is down to roughly one in ten.

The shift of the global population out of extreme poverty, beginning to escape *the hustle*, is extraordinary. Forty percent of the population of the world is moving up to higher income levels and hopefully on their way to joining the global consumer economy as a result. In the graphic, the upwardly mobile forty percent is represented by the 3 billion people currently in Income Level 2 – or people living on between 2 to 8 USD per day. To offer a more complete picture of global development, Gapminder has created their “map of the world” – shown below with 2015 data:



The map offers another useful view of the data. The details of development in each country are messy and confusing – but the near universal fact is that most of the world is on the march toward longer life spans and greater wealth. Whenever one hears talk of low interest rates and secular stagnation, it is useful to remember the facts about global development and that 40% of the global population is on the way to the global consumer party but haven’t arrived yet. Those 3 billion or so people will be arriving in the next two decades. Fewer than 3 billion people are in the consumer economy today (those in Level 3 and Level 4), so the number of people at this “party” will more than double in the next twenty years. Those 3 billion people coming to the consumer party have all already been born. This isn’t theoretical.

So how will these people escape *the hustle* and be able to build material wealth? The specifics will differ for each person and in each country – but building wealth will generally require accounts of some sort. Extending the financial system by banking the unbanked (or under-banked) continues to be one of the largest business opportunities in the world.

To repeat ourselves (which we do often) we think the recent market gyrations confirm the enduring wisdom of Josef de La Vega’s from 1688: “*Whoever wishes to win in this game must have patience and money, since the values are so little constant and rumors so little founded on truth.*” What would have meaning for us would be a shift or shuffling in valuation gaps in the market. Dramatic moves in the market in the last few quarters have not been accompanied by any significant “shuffling” of the deck or shifts in valuations. It has been as if a dealer threw a deck of cards into the air and they landed back on the table in the same order they were in before, or as if the dealer kept dealing the same cards unshuffled. The game will change if the market (the dealer) starts shuffling the deck.

CLOSING

In their the June 2019 letter, the managers of the Euphrates Iraq Fund observed that:

Our recent trip made it plain that everyone from the Iraqi entrepreneur to the C-suite executive in downtown Baghdad to the average guy taking Uber to work are driven not by the headlines of war and misery that distort our understanding of their country, but by the actual conditions in the world in which they live. Those conditions are far from ideal, but they are also much improved from the conditions that prevailed five years ago. That very real improvement is responsible for the growing sense of optimism we saw in Baghdad.

When one gets out into the world and looks – people everywhere are working to improve their lives and build something for their families. In Iraq, Turkey, as well as the US and the UK and in the banks and most other holdings in the fund -- we get to invest in the success of people and businesses at multiples of earnings which are lower than the sales multiples of most of the supposedly safe investments in US technology, payments, media or consumer stocks.

We believe that in time, the investments at low multiples in businesses or geographies currently perceived as high risk will be safer than all the money many feel safe investing at high multiples in business franchises and places that they believe are safe. Time will tell. But as de la Vega observed, the only certainty is that values will be little constant and the rumors little founded on truth. Given these facts of the market, we suggest anchor yourself against the inconstancy of the market by getting out in the world and looking around -- the billions of people hustling to build material wealth are hard to miss, and amazing to see at work.

SEPTEMBER 2019

*Shakespeare was a master of ellipsis, of leaving things out...*Bloom, Harold. Cleopatra: I Am Fire and Air

In How to Read and Why, Harold Bloom exhorts us to “read deeply, not to believe, not to accept, not to contradict, but to learn.” Reading closely or “deeply” is much like listening -- it is about hearing both what is said and what is not said, as well as how it is said. In other of his books, Bloom repeatedly highlights one Shakespeare’s skills as knowing what to leave out or unsaid – the “master of ellipsis” was how he would express it.

Many will wonder what in the world a literary critic has to do with investing and business analysis – but I would point out two facts (1) much of business is just drama acted off-stage, and (2) most of a business or security analyst’s time is spent reading or interviewing professionals about their businesses. As business analysts, we are business critics – the domain is business, but large parts of the skillset involved in performing criticism are the same as what’s used in literature, drama or any other field. Much of what we do as investment analysts is reading, so the question of how to read well and actively is essential. Just as one must interpret Shakespeare – one must interpret and be able to fill in the ellipses and gaps left in the narratives of CEO’s annual letters or business reports in the Financial Times or Wall Street Journal. Our job is to read and listen with a critical eye and ear -- to mind the gaps in what is said as well as the gaps in our own understanding.

The point of criticism in any domain is not just to criticize – but to see clearly. In The Anatomy of Influence, Bloom cites one of his mentors teaching him that the heart of criticism is appreciation. The opposite of appreciation and criticism is mediocrity. The mediocre eye sees what it is told to see and the mediocre ear hears what it is told to hear. The mediocre reader catches the headline and whatever points reinforce his or her existing beliefs and learns nothing. In investing, systematic effects or factors and mediocrity are synonymous. Excellence and learning are idiosyncratic. Our fund portfolio is idiosyncratic as well. Time will tell if the fund achieves excellence, but that is our unrelenting goal. Although he has passed away, Bloom left clear thoughts on mediocrity:

... we certainly owe mediocrity nothing, whatever collectivity it purports to advance or at least represent. (How to Read and Why).

Collective forms of mediocrity dominate the current business environment, from index funds to negative interest rates. We believe that one should be careful of these institutionalized forms of mediocrity. Blindly relying on them for your future wealth and investment returns might not work as well as the crowds suggest.

A COMMENT ON NEGATIVE RATES & OWNER OPERATORS

It is important to understand what negative rates mean. The market clearing interest rate is the return on capital in an economy for activity at different time horizons. Negative interest rates mean “investing” capital at negative returns -- at a loss. Negative rates tell us that a lot of businesses in an economy should be out of business, and that the default, bankruptcy or capital recycling mechanisms in an economy are broken. In a business context, negative rates are a quantitative indicator of economies which have acquiesced to mediocrity.

We continue to believe that one wants to own the equity of companies that produce a positive return on capital. Specifically, within the universe of companies which are building economic value, we like owning shares in businesses where management is also a large shareholder. We refer to these businesses as owner-operators.

We do not have a precise quantitative definition for owner operators. We apply a threshold test for the overlap of ownership of the shares and operational control or influence. The test resembles Supreme Court Justice Potter's famous "I know it when I see it" test for obscenity. As of the end of the 3rd quarter, approximately 38% of the fund portfolio was invested in 25 different companies we categorize as owner-operators. A quick scan of the list's sectors and countries of the companies shows that when we discuss owner-operators we aren't talking about any specific industry or country. What makes owner-operators special isn't the specific business or its location, but the higher probability of alignment of management interests and efforts with our own long-term goals as shareholders.

As stated before: our reason for investing in a business, or our *investment hypothesis*, is made up of four general parts: **business franchise, growth opportunities, management and valuation**. We call it a hypothesis instead of a thesis because we are always learning and never entirely correct. In investing, one makes educated guesses and takes profits or losses. The investment hypothesis is a structure for making educated guesses about the future value of a business.

CLOSING

If one takes Mr. Bloom's advice on criticism and appreciation to heart, and rejects the rolling tide of mediocrity – a close reading of current events offers some clues of where the world may be going and how we might profit:

1. WeWork may be a sign that multiple expansion in one part of the markets, driven by a single large investor, is spent. If WeWork signals the end of the willingness of investors to 'suspend disbelief' and the use of mathematics in valuing large private companies then this could be the beginning of a return to sanity in private markets, where valuations have been above public market valuations for quite a while now.
2. Approximately 45% of the S&P 500 trades above 4x price-to-sales (P/S) at the end of September. Since the S&P 500 is made up of companies that are already large, the idea that well over a third of these large companies are going to be able to maintain profitability and growth at high levels perpetually into the future is very, very, very improbable. Large, quickly growing, highly profitable companies tend to attract competition, government scrutiny or both.
1. After the most recent ECB rate cut, the "knives" come out for Mr. Draghi before he has even left the building. The open criticism of the failure of negative rates and continued bond purchases to provide economic stimulus is refreshing. The ECB needs a simple lesson in epidemiology – no virus spreads well if it kills its host too quickly. The virus of monetary stimulus uses the banks as the host, but negative rates kill the banks quickly, so the negative rates won't ever work. \
2. China has shifted from pumping capital out into the world, to needing external capital to fund its economy. China needs outside investors -- but it still wants to be able to play by its own unique rules. Note that if China and India start putting in bids for significant amounts of capital to build their countries, that would put upward pressure on interest rates. In a world where three billion consumers are coming, interest rates don't stay at the zero lower bound.
3. The attack on Abqaiq may be a sign that power is shifting in the Middle East in a way that few understand and that the market isn't discounting yet. Saudi Aramco's spring 2019 debt prospectus showed its largest oil field, Ghawar, is already well into its decline, and recent headlines have begun to take note that US Shale production may soon be in decline as well.

A close reading of events leads to an appreciation of what's happening but not discounted in current market prices. As we look for investments, we like opportunities where we can clearly articulate the business or economic opportunity which isn't in the current stock price -- especially when we are effectively partnering with company management because they are also large shareholders.

In closing, no letter can include everything. We are open to questions about the inevitable ellipses – both intentional and unintentional. **We are students of ellipsis, not its masters.**

DECEMBER 2019

The investment business is one of committing your clients' capital to quality growing companies and leaving it there until the company changes or gets grossly overvalued. You ought to have to disclose whether you're in that business or the screwing-around business.

Robert Kirby, from Wall Street People by Charlie Ellis

VALUE, GROWTH & PATIENCE

Effective investment management over time requires four key elements: (1) a **point of view** or the set of assumptions and beliefs to help cut through the noise, (2) **valuation** – buy ownership at a price that's less than your estimate of the value of the business; (3) **growth** – buy ownership of businesses whose value and total addressable market are both increasing over time; (4) **patience** – when you get a fair price for a growing business, let time and the mathematics of compounding work in your favor. We have discussed our point of view, valuation, and growth in previous letters. These first three are not simple, but also aren't that difficult to articulate. However, patience – well, that is hard to explain. When one is patient with an investment it is akin to “standing still like a hummingbird” ([Henry Miller](#)). Patience isn't passive – it's an active process.

Robert Kirby is a somewhat surprising advocate of patience. One of his favorite activities, outside of his efforts building Capital Guardian Trust Co., was racing cars. But if you understand how having the patience to let wealth compound in a growing business is an active decision – one where you go back and regularly check on the business, its progress and management, well then it isn't hard to see why Kirby advocated what he also described at times as a passive approach to active management. The patient investor who selects growing companies, purchased at attractive prices, also has the benefit of lower taxes and transaction costs. Patience, properly applied, gives you an edge in the race to produce returns in the long run.

Ben Graham said that “in the short run, the market is a voting machine but in the long run, it is a weighing machine.” As value investors we tend to buy when the market votes are against a company with a solid underlying business franchise and management team or culture. With many of the larger positions in the fund, we are now in a position of waiting for the voting booths to close and the scales to get rolled out. Many investors will talk about how they do this – but we believe it is more useful to show a few concrete examples from the fund where the voting has gone against us. Graham and others have many fine quotes, but we believe the real information about an investor is in their transaction ledger. The information isn't quoted sentiments, but the quoted prices and share counts where the trades were made.

...As a reminder: our reason for investing in a business, or our **investment hypothesis**, is made up of four general parts: **business franchise, growth opportunities, management and valuation**. The investment hypothesis is why we are invested in different businesses. But, deciding to invest in a company is just the beginning. Other difficult decisions follow the decision to buy an initial position: how much to own and when to increase, reduce or eliminate a position, and what new facts we learn over time. The way an investment develops over time is best expressed as a story.

MARCH 2020

The great virtue of a free-market system is that it does not care what color people are; it does not care what their religion is; it only cares whether they can produce something you want to buy. It is the most effective system we have discovered to enable people who hate one another to deal with one another and help one another. -- Friedman, Milton. Why Government Is the Problem (Essays in Public Policy)

WHAT'S GOING ON: SELECTION, ADAPTATION & PROGRESS

The world is confronting a highly contagious and rapidly spreading virus that is on course to become another of several endemic seasonal diseases which include the common cold (also known as human endemic coronavirus) and the flu. This new virus is much more dangerous than the flu or the common cold, although we do not know how much more dangerous. We have only been observing it for a few months.

In the face of this unknown threat, most of the economies of the world have limited contact between people to slow the spread of the virus. The limitations on human contact also limit or shift economic activity. From an investment research perspective, our focus is not on the economic damage, but how people and companies are adapting to the new environment. ***As investors, we seek to steer around the economic impairments while focusing on what has changed in the long-term potential and opportunities for different businesses. We ask ourselves “What is a business likely to look like three to five years from now?”***

We are optimists and believe in people and the ability of individuals everywhere to adapt and innovate. Our long-term optimism is anchored in the economic power of human agency and free markets to enable and coordinate the creation of wealth. This isn't an idealistic position – but comes from the brutal experience of investing in markets around the world for decades and realizing that Dr. Friedman is right – free markets are *“the most effective system we have to enable people who hate one another to deal with one another and help one another.”* But this virus has led to the expansion of the state and limits on individual freedom and agency everywhere. We believe that state control and interventions limiting human agency destroy wealth in the long-term. The state does not build wealth over time, it provides a foundation of safety and stability for individuals and companies to build wealth. We expect that three to five years from now the world will have devised multiple approaches to manage the health risk from the coronavirus. ***The risk which concerns us in the long-term comes from the expanded role of the state and the emerging barriers to the free flow of commerce, capital and people.***

In response to the crisis, the Merchant Adventure team has been working remotely since March 6th. We hold two team meetings a day: one in the morning just after the US markets open and one in the afternoon after the US market closes. Because we set up our systems in the cloud and online to accommodate frequent travel, we were able to quickly shift to working from home. In the short-term, we have been surprised by how easily we shifted to virtual meetings. In the longer-term, working remotely results in a loss of efficiency in communication and workflows – for example, when we are making difficult decisions necessary to navigate a crisis, it is better to be talking face to face.

SELECTION: VALUATION SPREADS BLOW OUT

One core element of how we select securities is valuation. Valuation is the ritual of comparing the current market prices with accounting measures of business performance. While this is described as a “style” of investing, we do ask ourselves what the people who don't relate prices to measures of business performance are doing? I think we all know that without reference to valuations one is not investing but speculating. Speculation can be wildly successful and profitable. Until, at last – it isn't.

We have discussed the valuation spreads within the S&P 500 Index (“SP50”) and the MSCI All Country World Index (“ACWI”) before, but its important to update those numbers. In the first quarter of 2020, stocks in the S&P 500 which traded at over 4x sales declined approximately 9.9% while stocks in the S&P 500 which traded at less than 1x sales declined approximately 36.0%. ***For the 5 quarters leading up to March 31, 2020 the stocks in the S&P 500 which traded at over 4x sales appreciated approximately 25.0% while the below 1x sales stocks declined 24.7%.***⁷ In the last 15 months, the largest valuation gap in the history of the S&P 500 has opened up. It shows no sign of closing. As a result, we have two markets, not one – the glamour stocks are up 25% and the value stocks are down 25% since December of 2018. We are invested largely on the wrong side of this valuation gap.

We believe that accounting, cash flows and earnings matter: valuation matters. When the markets decide earnings matter and focuses on them, the valuation gap will close.

ADAPTATION: MANAGEMENT

No one knows exactly how dangerous this virus will be over time, or how we will manage it. Even though our knowledge of the virus is limited by the fact that it is new, we do get to see how everyone reacts. How each company is managed in a crisis. ***This is a real time test of management quality, vigor and ability to adapt to rapidly changing economic conditions.***

None of the adaptations which leading portfolio companies made in response to the novel virus were actions which we anticipated when we purchased the securities, but the adaptations were enabled by qualities and capabilities which were part of our criteria for selecting securities. Management matters because management’s job is to identify and pursue new opportunities which emerge. While we do not know what changes will come out of the current crisis – we do know that the businesses with vigorous leadership who are able to adapt and help their customers adapt will win. Rather than spending our time predicting the course of the virus, we are looking to fill as much of the portfolio as possible with ownership interests of companies led by management teams who are working to find ways to prosper in the face of the extreme and unanticipated challenges presented by the current crisis.

CLOSING

We will sometimes deliver dramatically different short-term investment results than the market indices because our portfolio is a collection of idiosyncratic businesses, not systematic factors. Our goal is to beat the market over time by owning shares of growth businesses when they are priced as if they will never grow again. This is traditionally referred to as “value investing” but with the caveat that because of the way today’s economies are driven by innovation and creative destruction, we are loath to invest in declining businesses at any price.

Five years from now, we expect the virus will be behind us and scientists will have invented new tools or have repurposed old ones to address Covid-19 and the subsequent viruses which will come after it. Although the virus will be behind us, we are likely to be living in a world with an expanded state presence in the economy and our lives. Just as Milton Friedman had his way of expressing what’s unique about markets he also captured the real problem with the expansion of the state: *“the tragedy is that because government is doing so many things it ought not to be doing, it performs the functions it ought to be performing badly.”* The current crisis we are living through is a compound tragedy – the tragedy of both a deadly new disease and the tragedy of world which was caught unprepared for it. That is what has happened. Going forward, the question is how we will all adapt.

⁷ Calculated using Bloomberg’s Portfolio Analysis application and the holdings of the iShares ETF’s for the S&P 500 (ticker: IVV) and the ACWI (ticker: ACWI).

JUNE 2020

COVID-19: WHY WE NEED TO BE RESILIENT & ADAPT

Responses to the pandemic have varied from totalitarian to laissez faire to tragicomic. The current judgement of financial markets is that the worst of the turmoil is past. Responses from both monetary (central banks) and fiscal (legislative or executive) authorities have been swift and of unprecedented magnitude. The risk to markets at this point is that the government stimulus is in the past, but the financial reckoning with the losses from economic shutdowns is still ahead of us.

Economic shocks (like what we are experiencing now) are disorienting, but they are not mysterious. They can be a playground for both innovation and opportunism as Bernard-Henri Levy reminds us in [The Virus in the Age of Madness](#). But, then Stuart Kauffman reminds us that chaos and instability have a structure and logic of their own in [At Home in the Universe](#). At the end of the day, instability in human affairs is itself often what is temporary and unstable.

We would suggest now is the time for investors to closely review their own investment options or opportunity costs. We will return to more individual company discussions in future letters, but this quarter seems a useful time to step back and take a broader perspective.

“Our intuitions about the requirements for order have, I contend, been wrong for millennia. We do not need careful construction; we do not require crafting. We require only that extremely complex webs of interacting elements are sparsely coupled.” -- Stuart Kauffman, [At Home in the Universe](#)

Kauffman observed that densely connected networks are unstable and networks with points of sparse connection tend toward stability. This simple idea links network structure to stability and applies to the networks of transactions which make up companies, industries, and economies. For instance, network stability is one reason that systems tend to consolidate into a few leading companies or political parties, and the chaos and complexity get encapsulated within the organizations⁸.

Disruptive technologies have torn down, or are tearing down, many legacy economic networks. As Mark Andreessen famously noted, “software is eating the world.” We are only beginning to understand the characteristics of the new networks we have built in their place. At the time of his departure from Instagram in late 2018, [Kevin Systrom told Kara Swisher](#) “*Social media is in a pre-Newtonian moment, where we all understand that it works, but not how it works. There are certain rules that govern it and we have to make it our priority to understand the rules, or we cannot control it.*”

The defenestration of traditional media by social media is a useful example of how technological change is more nuanced than just “more for less.” While the traditional media had a myriad of problems, it was sparsely connected and stable; so stable that it had a role in the formal power structure as the fourth estate. In contrast, we are now struggling through the consequences of “densely connected” social media networks playing a dominant role during a crisis. Effective response to the novel coronavirus requires coordinated, collective action. Instead of trusted stable information channels to coordinate responses to the novel coronavirus, our technological progress brings us cacophony and chaos:

... a collective panic, aggravated by news channels and social networks that, by reporting to us, day after day, the numbers of those who had died, were still in critical condition, or had recovered, herded us into a parallel universe in which nothing else, anywhere, was news—and, in so doing, drove us quite literally mad. After all, that is how Chinese water torture works—the sound of the drop, repeated

⁸ This echoes Coase in an interesting way, because complex densely networked interactions tend to also be high-cost interactions, so it isn't clear that Kauffman isn't just describing the general fundamentals which underly the specific emergent phenomena Coase observed in where organizations are formed in an economy.

endlessly, becomes a fearsome dragon. How would we react if the traffic safety authorities decided to install giant loudspeakers at one-mile intervals to inform us of the day's highway fatalities?

Levy, Bernard-Henri. The Virus in the Age of Madness.

The qualitative experience of a merciless assault of noise that Levy captures as the “*sound of the drop, repeated endlessly*” -- a description of being tuned in to a dense network of connections, robs us of the ability to consider and calculate. Substantive debates of trade-offs are ignored, while we seek out news of the latest calamity, the “*sound of the drop, repeated endlessly.*” It is precisely at these critical moments in crises that we must weigh the opportunity costs of the alternatives available to us, and not become distracted.

The economic shocks from COVID-19 and our responses to it will fill history books. But our task is to manage wealth through the crisis, not to evaluate it after the fact. As we look out at a world dominated by governments printing money, bonds without yields and private investments trading at high prices for high fees, we continue to believe that publicly traded equities provide the best alternative to protect and build real wealth. In the process of looking for the best individual companies to protect and build wealth, we also need to make sure that we are aware of our public equity investment alternatives. The decision of what not to buy is just as important as the decision of what to buy.

CLOSING

Markets have gone from expecting a broad economic collapse in March to pricing in a technology led resurgence in July. In this environment, we have focused on our opportunity costs, assessing tradeoffs between what we own and what we could own. This opportunity cost analysis is analogous to a broader analysis and planning exercise which leaders in different organizations across economic sectors and geographies are either working on or will be working on to prepare for 2021.

A world which flits from one certainty to its opposite may be tinged by madness as Bernard-Henri Levy reminds us, but as Stuart Kauffman carefully explains, our intuitions about order are generally too pessimistic; stability does not require much. The world naturally finds order despite our fears.

SEPTEMBER 2020

A businessman isn't a mathematician. Remember that. Never become hypnotized by the beauty of numbers. A businessman is someone who buys at ten and is happy to get out at twelve. The other kind of man buys at ten, sees it rise to eighteen and does nothing. He is waiting for it to get to twenty. The beauty of numbers. When it drops to ten again he waits for it to get back to eighteen. When it drops to two he waits for it to get back to ten. Well, it gets back there. But he has wasted a quarter of his life. And all he's got out of his money is a little mathematical excitement.

V. S. Naipaul, [A Bend in the River](#), p.24

The abundance of numbers in the financial markets makes it easy for *businessmen to mistake themselves for mathematicians and become hypnotized by the beauty of numbers*, as Naipaul puts it. One should not ignore the numbers, but the numbers must connect with a narrative or story. The numbers alone do not mean anything. The story that can be told with the numbers, that is everything.

MEAN REVERSION IS DEAD, VALUATION ISN'T

The economy evolves and changes. We all know this – but chasing down what economic changes mean in a way that can inform investment decisions in the present is always a matter of educated guesswork. The structure of the global economy before 1989 was very different than it is now. In 1989, the Berlin Wall fell and the Eastern Block began to open up to the West, and China continued opening up while clamping down at the same time. The end of the military confrontation that defined the post WWII period would lead to the commercialization of a host of technologies, including the internet.

Before 1989, public companies enjoyed a privileged level of permanence within the economy. By undertaking the significant investment required to provide a standard of transparency and governance, they could raise public capital. Furthermore, if they were dealt some blow, their scale and status granted them access to funding that would allow time to work in their favor. This literal margin of safety allowed for the process of mean reversion in returns on capital to be probable. Unless mortally wounded, most companies could recover and reward shareholders for their patience. Mean reversion of returns on capital was a key economic phenomenon that generated returns for value investors.

But American capitalism was also transforming in a process described by John Weir Close as [A Giant Cow Tipping by Savages](#). The emergence of new classes of investors, new financial instruments and new legal theories impaired the privilege of corporate survival and the economic structure which supported the mean reversion of returns on capital within a corporation over time. The time of buying statistically cheap stocks and holding them until someone fixed the problems in the business ended. Venture investors build new competing businesses from scratch that are not burdened by legacy liabilities or bureaucratic org structures. Distressed debt investors profit from bankruptcy through credit default swaps and make money again as the debt converts to equity, resulting in ownership in an unlevered company with lower costs after bankruptcy. There is a profit to be made by investing capital on the opposing sides of an incumbent business – in disrupting (venture) or dunking (distress). The excess returns in venture and distressed investing come from reducing the returns on capital of the incumbent – preventing mean reversion. This changing ecology of investors and capital markets instruments is one of two key factors which have killed the mean reversion of returns on capital.

The other factor is the acceleration of technology and innovation. Before the era of rapid technological change, the biggest plant would achieve economies of scale that formed an effective barrier to entry from new capital. In their 2005 book [Competition Demystified](#), Bruce Greenwald and Judd Kahn collapsed Porter's five competitive forces to one: scale. But now technological innovation frequently trumps scale. We keep getting [More from Less](#) as Andrew McAfee describes in his 2019 book. As a

result, the newest plant or product has the tendency to also have the lowest cost structure; the level of the cost advantage is increasingly a function of improvements in technology and process innovations rather than size. In cash flow terms, capital expenditures do not necessarily sustain profit margins and run the risk of impairment. Innovation creates stranded assets in legacy businesses because the capital cost of new capacity falls so fast that legacy capital investments cannot be recovered. Stranded assets are assets that will never earn back the capital invested in them. These assets revert to zero, not some historic average return.

Value investing is not simply mean reversion in returns on invested capital anymore. It is “*Gretzky at a discount*” – *businesses skating to where the puck is going to be, while market prices don’t reflect much of an expectation that they can skate*. To have an idea *where the puck is going to be*, we need a narrative or hypothesis about how different businesses or segments of the economy evolve and also how individual businesses adapt to economic shocks, competition and other slings and arrows of fortune.

In the narrative, numbers support a story describing a target market, management team, business franchise, financial model, and expectations in current market prices. We divide market price changes into shifts in (a) investor expectations and (b) company fundamentals on which those expectations are formed. Investor expectations change more frequently and dramatically than company fundamentals.

Value investing is about buying low expectations, and valuation is the work of measuring the level of expectations in the price of a security.

CLOSING

One of the greatest risks to an investor is being hypnotized by the numbers which are constantly dancing past us in economic releases, earning releases and price quotes. The way to avoid being hypnotized by numbers isn’t more numbers, it’s using the numbers to construct a narrative of how businesses and industries are evolving, in order to get in the way of and profit from changes in the economy. Each of the companies in the portfolio has its own story, which we are happy to discuss.

DECEMBER 2020

Management: Nothing Compounds without Leadership

No business ever accomplished exceptional results over time without quality leadership. Effective leaders build strong organizations and set a vision that endures. Quality management sounds qualitative, but the magic of good management is that a qualitative vision implemented through a set of practices, processes and norms in a firm end up showing up in the financial statements. The test for every vision is the future sales and profit growth which it helps create.

Opportunity: Developing & Developed World Growth

For the most part, innovation drives economic growth in the developed world, while demand growth, mainly from enfranchising new consumers, drives economic growth in the developing world.

Most innovation and productivity improvements now come from the digital transformation of businesses and economies. We think of the process of digital transformation as moving through four phases: digital model to digital shadow to digital twin to digital source⁹. The transformation starts with being able to build a digital model of a process or artifact and proceeds to where decisions and actions in the physical world are based on and coordinated by the digital version, the digital source.

We are still very early in this process, and there will be seemingly endless iterations. One easy way to think of digital transformation is to think of Tesla and Bitcoin as primitive prototypes or instances of digital shadows (2nd stage of 4) of a car and a currency. A lot of the cloud is an attempt to make a sort of digital shadow of your phone or your computer – but then as we made the digital shadow of the phone and the computer, those two merged into smart phones and tablets in different ways. The transformation is not just digitization or virtualization, it is that in the process of digitizing one thing it eats other items (phones displaced cameras and clocks and tape recorders and players and many other single use devices) and becomes something else. Part of the digital transformation is [refactoring](#) and re-engineering. The examples are all around us.

Opportunities created by digital transformation will impact every part of the economy and are at the core of economic progress: our ability to [do more with less](#). The challenge is to try to track what the digital transformation means for each industry and company. The rapid development of COVID vaccines illustrates the digital transformation at work: the rapid formulation of vaccines was enabled by sequencing the virus RNA and then synthesizing RNA sequences to “teach” the human immune system to protect against infection.

The opportunity of fundamental economic growth in the developing world is driven by individuals exercising increased agency in their lives because of increasing wealth: improving technology and reductions in coercion. [Gapminder](#) is an excellent resource on global health and how people live in different parts of the world at different levels of wealth. From 1970 to 2015, the global economy lifted 40% of humanity out of extreme poverty to just above that – existing on \$2 to \$8 per day, or what Gapminder refers to as income level two. The opportunity for demand growth in the developing world is that 40% of humanity is not going to stop and be satisfied with living on \$2 to \$8 a day. The rich world is only a little over 10% of humanity and should expect a lot more people to be joining them in the next 20 years. These two sources of growth are not mutually exclusive. The most compelling investments will benefit from both digital transformation and economic enfranchisement.

⁹ From discussions with James Anderson and Ivan Labra of Datagraph GmbH.

Once we find a business that is “in the way of the future” the next question is if we can get in the way at the right price. ***Investing is always about the price.***

Value in Variant Expectations: Prices, Profitability & Growth

When our expectations for a company diverge from what is discounted in the market price, we refer to this as “variant” expectations. Understanding how our expectations vary from the consensus is essential. Knowing how our expectations vary is an application of what we refer to as the value equation – or the way in which market prices discount sets of expectations for future growth and profitability. The work in research and valuation is constructing our own expectations for the future growth and profitability of businesses and comparing that to what is in the price.

What is a dollar of earnings in a business worth? The question is debated even though the way of calculating it is clear. The present value of a steady profit stream is defined by the value equation: $V = p / (r - g)$, where p = profits, r = required or target return and g = growth of profits.

For today, let us review a few notes on the value equation and how we use it:

1. The required or target return is our return -- not a weighted average of what everyone in the capital markets wants. It is the amount of money an investor requires to pursue an opportunity. *This rate has nothing to do with interest rates – we are not borrowing money and trying to pocket a spread.*
2. The price to earnings (price to profits) multiple a business is worth to you is defined by $1 / (r - g)$. If you know your required return and have an estimate of the sustainable growth rate, you can know in seconds what multiple to pay for a business and reach your target return.
3. Required return and growth are bounded. Our required return is always above 12% and our maximum growth rate for a steady state business at scale is 8%¹⁰. Therefore, our gap between required return and growth is always above 4%, meaning our target long-term P/E multiple will always be less than 25x earnings. These limits are part of building in a margin of safety.
4. Convert the P/E to a price to sales ratio with a "normalized" or "long-term" net margin. Since $\text{Profits} = \text{Sales} \times \text{Net Margin}$, a Fair Price/Sales = Net Margin / $(r - g)$. Convert a P/E to a price to book ratio using long-term ROE as the measure of profitability: Fair Price/Book = ROE / $(r - g)$.

A key value of this model is its simplicity. We can sit down as a team and discuss the drivers of value for each investment – everything collapses into a discussion of what management is driving the business toward and the growth and profitability of the business in that state.

The value equation is a simple way to organize the complex discussion of how different businesses create value and how the value of a business compares to prices in the markets. For each of our holdings we can explain how and why we expect greater growth and or profitability than the market. The model defines the relationship between growth, profitability, price, and value. *Growth is just part of the value equation.*

...The digital transformation of each business has the potential to drive higher future growth and wider margins than the market price of many stocks imply. But if every company transforms, we must also remember that the competition unleashed by digital transformation, in that case, could drive growth and margins down instead.

¹⁰ If your growth rate is above 8%, you are describing a company that will attract intense competition and probably is not in a steady state.

PRIMER LESSON

EXCERPTS FROM NOTES AND LETTERS FROM 2002 TO 2005

FEBRUARY 2002

FALSIFICATION, COMPETITION, VALUATION & INVESTMENT HYPOTHESES.

“Oz, left to himself, smiled to think of his success in giving the Scarecrow and the Tin Woodman and the Lion exactly what they thought they wanted. ‘How can I help being a humbug,’ he said, ‘when all these people make me do things that everybody knows can’t be done?’”

Frank Baum, The Wizard of Oz

The Wizard of Oz is a great work of American literature, one that I appreciate for putting Kansas (my home state) on the map. In the quote above, the wizard asks a good question, how he can help being a humbug, “when all these people make me do things everybody knows can’t be done?”

The markets demand companies have steadily growing earnings, even though earnings are a residual. *The media demands* single-state predictions of the future, although science indicates that our world is best described in terms of probabilities. It’s worth noting that what we actually know about the future is that it hasn’t happened yet. Therefore, it cannot be known or definitively predicted. There appears to be a constant and steady *demand for assurances* and knowledge that we know doesn’t exist, requiring an ample *supply of humbug*.

NOW WHAT EXACTLY ARE WE CALLING HUMBUGH?

For the definition we reach back to an argumentative Austrian, Karl Popper, in the autumn of 1934, when he published the Logic of Scientific Discovery. While all the other thoughtful folk were asking how to verify facts, Popper flipped the question on its head, and asked how to falsify them. Any statement that isn’t falsifiable (and involves opening your wallet) is humbug.

WHAT’S THE DIFFERENCE BETWEEN FALSIFYING & VERIFYING?

Verification is *positive*; falsification is *negative*. When we verify a theory we try to confirm it or prove it *positively* with observations. When we falsify a theory, we test whether the observations are consistent with the theory, and reject the theory only when the observations are *negative*. Falsification does not allow a theory to ever be proven, but provides the criteria for a theory or speculation to be rejected. The scientific method of testing hypotheses is centered on falsifiability – a theory is never proven, only disproved; it tells us that we “*only know what is not true*.”

WHAT DOES THIS HAVE TO DO WITH INVESTING?

Falsification also provides the foundation for long-term investing, because once you realize that you can “*only know what you don’t know*,” the goal becomes avoiding (a) making mistakes, (b) operating on false hypotheses, (c) talking when you should be listening. The goal is **negatively** defined, in terms of “***what not to do***.”

Warren Buffett of Berkshire Hathaway states this in two ways: (1) the first rule of investing is not to lose the client’s money and the second rule is never to forget the first rule, and (2) one must stay within one’s circle of competence. Peter Bernstein, the author of myriad investment tomes, observes that investing is a loser’s game, a game in which the winner is the one who makes the fewest mistakes. Both views are consistent with the concept of falsifiability or avoiding situations that don’t make sense.

The successful long-term investor will focus on what one doesn’t know and how one knows that. How one “knows” what one “doesn’t know” is realized by one’s investment framework.

Our investment framework incorporates two basic ideas:

(1.) **Competition** – Competitive analysis applies at every level of the economy and competition is the key engine behind improvements in productivity and growth that we rely on to improve the standard of living. We believe *businesses compete for capital and capital competes for returns*. A company's competitive plan is synonymous with its strategy – competition defines strategy.

(2.) **Valuation** – The value of an asset is determined by the present value of the future cash flows to that asset. The components for determining the value of an asset are well known and widely accepted:

i. the discount rate ii. the growth rate iii. the profits

The discount rate is our required rate of return on the investment. Company growth rates cannot deviate from industry growth rates for very long in a competitive industry. A return on capital framework may be used to reasonably estimate a *competitively* sustainable level of profitability. (A range of values may be calculated for the cash flows and growth rates to come up with a reasonable valuation range for the security. A margin of safety is built into the discount rate or our required rate of return.)

We believe that one can increase wealth over time, outperforming the relevant indices, by investing in companies with sustainable businesses that are selling at a discount to our estimate of the present value of their future cash flows. On particular names we may be right or wrong, but in aggregate, if we are falsifying our hypotheses and avoiding humbug, this should be the case.

HOW DOES THIS ALL COME TOGETHER INTO AN INVESTMENT HYPOTHESIS FOR A COMPANY?

The numbers that we use in the valuation of the security are said to constitute our investment hypothesis for a particular stock. The expected profitability of the company and its calculation method, the required rate of return and the sustainable growth rate of profits, are each part of the investment hypothesis regarding the value of the company. Whether the value is realized by shareholder will depend on whether the company is able to deliver the projected numbers and whether the market will discount the growth in the price of the stock, or some other combination of company performance and market expectation.

HOW DOES ONE FALSIFY AN INVESTMENT HYPOTHESIS?

Check your estimates of the value of a company by looking at the company's financial statements. In a sense, a company's financial statements are its credit application for a loan from our clients – a loan which the company has no obligation to ever repay, and is sold to us second-hand. In this “credit application” we first adjust the numbers to reflect the economics of the business¹¹, and then look for evidence that the adjusted numbers do in fact support the current valuation. For every valuation, or market price, there is a set of numbers (discount rate, growth, and profits) that justifies that price. If we look at the numbers reflected in the price of the stock, given our adjusted perspective on the company's financials: capital employed, return on capital, growth rate, etc. – and the numbers implied by the price are both credible and lower than our expectations for the company-- we may invest, but not yet. Next we look at a variety of other metrics over time. The metrics differ based on the industry, but we are looking for some quantification of what we qualitatively assessed as the company's competitive position. As the company and economy change, the investment hypothesis may be modified.

HOW DOES THIS ALL TIE BACK TO HUMBUG?

As an analyst one **qualifies**, **quantifies**, and **tests**. Each qualitative observation should be traceable back to a quantitative metric or proxy in the financial statements, otherwise it's just humbug that we can't test or falsify. *Numbers and testing -- “checking the tape” or falsification, under any of these names, is the antidote for humbug.* This is the antidote that underpins the scientific method and the process of discovery. Along with

¹¹ The credibility and reliability of financial statement information is an important issue, particularly given the recent bankruptcy of Enron, and the apparent excesses of the late 1990's.

Charles Mulford and Eugene Comiskey have a timely new book out entitled The Financial Numbers Game: Detecting Creative Accounting Practices. The book lives up to its title and gives lists of methods for detecting different accounting abuses, as well as understandable descriptions of the abuses. We recommend it.

competition, this approach to the world has led to our nation's extraordinary growth. The same basic logic should underpin the analysis of what goes a portfolio. *And the analysis is never done.*

APRIL 2002

THE USE OF MODELS & FRAMEWORKS

We use models to “climb up,” or quantify what we easily can, and then we “throw it away” and try to understand the business issues that are not easily quantified.

To steal a quote from Karl Popper's arch rival:

“My propositions serve as elucidation in the following way: anyone who understands me eventually recognizes them as nonsensical, when he has used them – as steps – to climb up beyond them. (He must, so to speak, throw away the ladder after he has climbed up it.) He must transcend these propositions, and then he will see the world aright.”

Ludwig Wittgenstein, Tractatus Logico-Philosophicus, 6.54

As investors, the conservative and prudent way to use the model is not to make projections, but to use the framework to identify companies for which the market has irrationally low expectations – companies that look like winners but are expected to lose. The purpose of this model is to act as a framework for highlighting these businesses.

COUNTING, SCORING, FINANCIAL STATEMENTS, AND FALSIFICATION

“The first step in any analytical approach to credit evaluation is to see precisely what the information on the application and credit report tell us. The way to do this is to identify what information exists and then to count.... The counting process is always most revealing.”

An Introduction to Credit Scoring, Edward M. Lewis, p.27

Two of the most important things we each know how to do are count and speak. Of these two, the one we most often forget – as we enjoy the sound of our own voices, is counting. Counting answers the questions how many, how often, how long etc. Any analytic approach to a problem will rely on data, numbers – what has been counted. In the case of companies, their credit application to us is their financial statements.

This approach is somewhat similar to what a bank might do before deciding to give an individual or institution a loan, and, just as the bank is the creditor, the investor is a creditor. Just as the bank tallies up a credit score for a company or individual before giving them a loan, we tally up a score for companies before investing, giving them a long-term loan without a commitment to repay, purchased second-hand.

If we rely on things that we can count, then we will miss some things. Arguably, the things we will miss will not be real – but the market has a way of turning perceptions into reality by giving companies currency in the form of their stock. This shadowboxing with perceptions and expectations is not a game we play, although we will take advantage of the game when it offers us securities at a discount.

We will stick with items that we can count in a company's financials. Add these counts up into a score, and then double-check. We will argue over what to count and how to add up the score, but not the value of counting and scoring. After all, without keeping score, how do we know if we won?

The purpose of the model is to provide a framework for testing the hypothesis that certain measurable, observable data from a company's financial statements, the company's competitive position and market prices

can provide useful investment information. This theory cannot be proven, but it can be tested and refined, again and again.

MAY 2002

EXPECTATIONS, SURPRISES AND VALUE

“Expect the Unexpected”

– from a Kansas City Kings bumper sticker in the early 1980’s

The last time the Kings were a contender, they were in Kansas City and their motto was “*expect the unexpected.*” The “backcourt boozers” never guessed that this meant that they would leave one small market for another, but nevertheless, the motto appears to have relevance in sports, life, and the market.

Who would have expected that, as the Enron hearings continue, and their accountant pleads guilty to obstruction of justice – PBS is showing Anthony Trollope’s *The Way We Live Now*, in which a financier swindles 1870’s London in an eerily familiar plot. The financier, Melmont, silences questions stating that “this is a delicate and complicated business matter” – the same quote used by **both** Mr. Skilling, and now Merrill Lynch.

As we sit in surprise and wonder at the way that life imitates art imitating life adhering to a basketball slogan, it’s important to point out that we are certain of at least one thing: *at this time next year we will have been surprised again.* This is the nature of the beast. So, we follow the accidental wisdom of the Kansas City Kings and *expect the unexpected.*

This *certainty of surprise* is an important reason that value investing makes sense to us. Why? We know that we will be surprised and would rather be surprised by good news than bad news. This is a temperamental disposition that also happens to have financial advantages: asset prices tend to increase after positive surprises, and decrease after negative ones.

Since one can estimate the expectations for growth and profitability that are implied by a stock price, one can select stocks that have low expectations in their price. Alfred Rappaport and Michael Mauboussin explain the math for calculating these “embedded” expectations in their book *Expectations Investing*.

The level of expectations in the price of a stock may imply improving profitability, or rapid growth, or both or **neither**. The important aspect to remember, as an investor, is that you may suffer a permanent loss of capital if the company fails to achieve the expectations implied in the price of the stock and fails to ever even achieve those expectations again. Since *we view the permanent loss of capital as an important risk*, the only way to minimize this risk is to buy low expectations.

The matrix to the below illustrates the problem. The “+” in represents a gain, the “-“ a loss, and the “0” signifies a draw. What the matrix illustrates is that the only time one gains is when the actual performance of the company exceeds the expectations in the price. While one may do somersaults to get around this fundamental fact, an investor who ignores it will suffer the same fate of the salmon swimming upstream who avoids the fishermen, escapes from the bears, but still never gets back to the ocean.

		<u>Expectations</u>	
		Low	High
<u>Actuals</u>	High	+	0
	Low	0	-

The matrix illustrates a one-dimensional example of surprises. In the real world, companies and the market have many dimensions and the problem is more complex. But the addition of different dimensions or “factors” such as industry competition, innovation, or economic growth does not change the nature of the problem.

The nature of the problem is that once one has high expectations there is less upside. High expectations are a set-up for disappointment, just as low expectations are a set-up for a positive surprise.

This way of viewing surprises and expectations is a temperamental disposition, investment style and wealth accumulation strategy which either makes sense to you or it doesn't. The temperamental issue is why value investors are born that way, not made.

FEBRUARY 2004

CUCKOO CLOCKS & BALD SPOTS: THE PERSISTENCE OF INNOVATION & INDECISION

And time yet for a hundred indecisions, And for a hundred visions and revisions, Before the taking of a toast and tea.

In the room the women come and go Talking of Michelangelo.

And indeed there will be time

To wonder, "Do I dare?" and, "Do I dare?"

Time to turn back and descend the stair,

With a bald spot in the middle of my hair—

From The Love Song of J. Alfred Prufrock,

by TS Eliot

We all sometimes worry about appearances, or chat idly and urgently about nothing. This is human nature. *The key is not to be frustrated by these biases, but learn to profit from them.*

As the War on Terrorism, War on Iraq, War on Afghanistan, War on Drugs and other “wars” drag on, we realize there will be time to worry about gossip and appearances. The backdrop reminds one of Orson Welles as Harry Lyme in The Third Man:

“Italy for 30 years under the Borgias they had warfare, terror, murder, and bloodshed, but they produced Michelangelo, Leonardo da Vinci, and the Renaissance. In Switzerland they had brotherly love - they had 500 years of democracy and peace, and what did that produce? The cuckoo clock.”

As a war criminal on the run, Lyme had a vested interest in romanticizing post WWII Vienna – a grotesque perspective. Warfare presented opportunity for men like him. But Lyme also had a point -- *never underestimate human ingenuity in the face of adversity.*

When faced with a challenge, people and organizations often surprise themselves with their ability to adapt. This ability to adapt is part of the fundamental process that ends up manifesting itself as competition and mean reversion of economic profits. In the United States this process of competition and mean reversion is more robust than in other parts of the world because we have fewer constraints on the flows of capital and labor.

Over time, Wall Street’s “hundred visions and revisions” of earnings become noise. As this noise fades into the background, we expect the most likely surprise to be the extent to which businesses have adapted to their respective markets.

This is not to say that we expect a sustained economic boom, but that because of this persistently surprising adaptability, the resiliency of the economy is much greater than most thought possible.

As investors, an understanding of competitive forces and innovation results in our being less optimistic than most in a boom and less pessimistic than most in the bust. In a boom, we expect high returns to attract capital and competition, thus reducing the returns. In a bust, we expect innovation to produce surprising efficiencies and support returns.

While Wall Street looks for the \$0.01 in next quarter’s earnings, at we will be trying to find the \$1.00’s which companies are adapting to earn three to five years from now, but which are not in the price today.

In this sense, our goal is to constantly be out-of-step with today’s gossip and appearances, or, in other words – to systematically find profit for our clients in the fear of bald spots and in the comfort of cuckoo clocks.

DECEMBER 2004

OBSERVATION & OWNERSHIP: AVOIDING THE “WYLIE COYOTE MOMENT”

Tolstoy “believed that only by patient empirical observation could any knowledge be obtained; that this knowledge is always inadequate, that simple people often know the truth better than learned men, because their observation of men and nature is less clouded by empty theories, not because they are inspired...” *Isaiah Berlin, The Hedgehog and the Fox, p.46*

“No one in the history of the world ever washed a rental car.” *Lawrence Summers*

Observation and ownership are two of the keys of a functioning business, economy or organization. This is obvious, but often overlooked, or underestimated. Berlin’s analysis of Tolstoy gets at the observation angle, and we’ve all rented a car, but never washed it.

A fundamental advantage of a market economy is that it solicits and incorporates many observations from different parties by allowing them ownership¹² (equity shares) of the profits or losses. The prospect of profit or loss (ownership) “solicits” the input, and an effective marketplace “incorporates” these observations into prices.

The incremental difference between the value of one individual observation and the informed collective observations of a group *has historically been* a key difference between the United States and much of the rest of the world.¹³ Ownership can be thought of as the difference between how folks treat a rental car and their own car, and in the markets this phenomenon is magnified millions of times.

A focus on observation and ownership informs our investment process. We look for companies that recognize their strengths and weaknesses, recognize the changes in the world around them, and take responsibility for adapting to, or shaping those changes.

The net result of observation and ownership, when they work, is constant learning and improvement within an organization. This improvement and learning is best understood in economic terms as extending the duration of any competitive advantages, or extending the “competitive advantage period.” The competitive advantage period is the period over which a company’s returns in excess of the industry average disappear, or the amount of time that it takes for the returns to mean revert.

The results of observation are not something that shows up as a separate line item on a financial statement. It shows up in the sustainability of a business and the stability of exceptional financial performance, which one would expect to be competed away.

Managements that do not observe, adapt, and take ownership (responsibility) for profits and losses are headed for a “Wylie Coyote moment.” That’s the moment when they run past the edge of a cliff, look down, only to find nothing there but air (often hot) supporting their existence.

This Wylie Coyote Moment can be thought of as when a competitive advantage unexpectedly disappears.

One way to protect against the “Wylie Coyote moment” is to find management that is an owner. In other words, where *ownership isn’t a perspective that they adopt in decision-making, it is who they are.*

¹² The ownership perspective simply acknowledges that in the long-term people respond to their incentives, whether you want to credit this observation to supply-side economics, B.F. Skinner, or others.

¹³ This “information” argument is the core of Hayek’s critique of Communism/Socialism, and also the core of micro-economic price theory.

Aligning Observation and Ownership, in this sense, is a principal-agent problem. Management is the agent that we want to behave as an owner -- driven by empirical observations about their business. If management is in fact an owner, then this lowers the cost of getting them to behave like an owner (dilution from options or complicated contracts) and solves one problem in analyzing the company and its incentive structure.

This facet of ownership, while not on the balance sheet, is in the annual proxy statement filed with the SEC. We refer to firms where management is also a significant owner as “owner-operators” and we tend to prefer owner-operators as investments.

Another, more traditional, way to identify companies whose business is driven by principles of observation and ownership is to listen. In the manic world of Mr. Market, simply listening, taking notes, and comparing what folks said they would do, with what they did is a powerful tool:

- How are shortfalls of surprises explained?
- Who takes responsibility/ownership?
- *How are plans modified for the unexpected, and does it make sense? Or would it only work in a cartoon?*

Understanding the *differences between perceptions and reality* both within the market and within organizations which produce surprises, price shifts or “Wylie Coyote moments” can be a key source of competitive advantage for an investor.

We spend a significant amount of our time observing companies and *doing our best* to either avoid or exploit the Wylie Coyote moments – as we endeavor to invest the core of the portfolio in businesses run by folks who understand the role of observation and ownership in the operation of an excellent business.

MARCH 2005

FIRST RATE FUNDAMENTAL ANALYSIS: PLUMBING LEAKY ABSTRACTIONS

“All non-trivial abstractions, to some degree, are leaky.

Abstractions fail. Sometimes a little, sometimes a lot. There's leakage. Things go wrong.

It happens all over the place when you have abstractions.”

Joel Spolsky, The Law of Leaky Abstractions

www.joelonsoftware.com/articles/LeakyAbstractions.html

“The test of a first-rate intelligence is the ability to hold two opposed ideas in the mind at the same time, and still retain the ability to function.”

F. Scott Fitzgerald, The crack up.

First rate fundamental analysis is about finding two or more opposing facts, and finding how it is that both are true. It is not just functioning while holding two opposing ideas in the mind at the same time but actually being energized and motivated by the paradox.

Fundamental analysis may be taught but good analysts aren't taught. They were made that way.

The paradoxes that the fundamental analyst seeks are often a result of leaky abstractions. When we abstract, we inevitably simplify, losing something of the description of a specific item to extend the description to a larger set of items. Computer programming relies on abstraction, just as accounting does. All languages – whether in the context of business transactions or bits and bytes, are based on systems of abstraction. Thus they all leak.

But you probably don't think – wow – what was leaky about that last sentence I uttered, or that last transaction I made. But in some fashion a good analyst is thinking about precisely what information the price mechanism, or other language, has missed.

The prices on the stock exchange, commodity and currency exchanges represent a mix of real and speculative activity riding on top of consumer, business and government transactions described in financial statements and economic data. If you think family gatherings can be a mess of miscommunication – think of the different economic participants as family members with a 2 word vocabulary: buy and sell – all shouting simultaneously.

SO WHERE DOES THE ANALYST BEGIN?

With a hypothesis, which is translated into a search of either qualitative or quantitative sources. For example, one might begin with simple screens, looking for two opposing ideas present in the data associated with one company, such as companies with rising dividend payments and decreasing or flat stock prices. Find me companies with lower expectations in the stock price than they have had in the past, but which still have strong cash flows, balance sheets, etc... The conflicting ideas are the expectations in the current price versus the company's historical or expected financial performance. Then we go through the list to sort the "probably will be's" from the "has been's."

But you can't be fixed to one screen or another. You have to be fixed to the structured, deliberate and unceasing search for apparent contradictions. The apparent contradictions hide information that the abstractions have lost – data that has leaked out.

Leaky abstractions in accounting come from the application of the matching principle. Matching is the principal underlying accrual accounting which seeks to match expenses and revenues with each other in time. This is different from cash accounting, which seeks to match the amount of cash in the drawer or account. There is no necessary wrong or right here. Just that what actually happens in the cash drawer or account will diverge from the accrual accounting – in depreciation, amortization, capitalization & expensing.

AND WHERE DOES THIS LEAD?

You never know. Generally, to some interesting reading, and conversations. Always to some boring reading and boring conversations. And occasionally to investable ideas. The hit rate is well below 50%.

WHAT MAKES AN IDEA "INVESTABLE"?

The classic line which we have used is "differentiating" our perspective from Wall Street. When our perspective on a company is different than what we are reading, and different than what's in the price, we lose some sleep, and take another look.

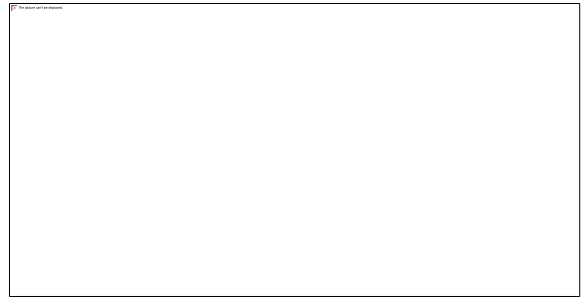
We are not looking for the "sure thing." We are looking for opportunities in which the potential and probable upside is significantly greater than the potential and probable downside. Quantitatively, if you can get a good sized set of economically independent companies which have more than twice as much upside potential as downside risk, one does just fine. So that's the basic probabilistic backdrop.

A FINAL PARADOX

Investable and profitable are two different things. The profitability of an investment is not in our control. We are price takers. That's the structure of the market. We don't control the prices that get flashed at us. We do control whether we take the price. This is another of those paradoxes. You don't control the price, but nothing matters more than controlling yourself in the face of prices -- price taking discipline.

So first-rate fundamental analysis is about dealing with and reconciling contradictions that begin and end with prices. Finding the leaks in the price mechanism. We are just plumbers after all.

We'll be here plumbing the depths of the markets and market prices – hoping to profit from and occasionally help fix the leaks.



PART 2

PRINCIPLES

&

PROCESSES

EXCELLENCE

Time will tell if the fund achieves excellence, but that is our unrelenting goal. Although he has passed away, Bloom left clear thoughts on mediocrity:

... we certainly owe mediocrity nothing, whatever collectivity it purports to advance or at least represent. (How to Read and Why).

Collective forms of mediocrity dominate the current business environment, from index funds to negative interest rates. We believe that one should be careful of these institutionalized forms of mediocrity. Blindly relying on them for your future wealth and investment returns might not work as well as the crowds suggest.

Excellence is idiosyncratic. There isn't a formula for it – it's another reverse Anna Karenina problem: "mediocrity is all alike; excellence emerges in its own way." The only path I know to excellence is to reject mediocrity. Rejecting mediocrity takes different forms, but there must be a persistence to it. A persistent and consistent drive for *what's next*.

The way to excellence brings us back to the original principles of focus and imagination. We use our shared imagination to catalog possibilities of what's next in business, the economy and specifically for our companies. Then we focus. Focus drives the persistence, day after day – learning about the companies and events, refining our knowledge and processes. We focus on both executing on our business and on what's next for the businesses we are invested in.

In the context of our work excellence is indicated by:

1. Being able to communicate consistent with the Johnny Cash Principle:

sharp like a razor, steady like a train.

High performance requires clear communication. Discipline and simple shared frameworks are the foundation of effective communication.

(build high performing teams)

2. Knowing the Map of the Portfolio and Companies,

from investment theses, to add/trim prices, to areas of innovation and convergence

Know the thesis and it's three components for each holding, and link that to the tactical add/trim prices through growth and profitability targets, then go back up to how each holding and thesis fits into how the economy and businesses are evolving. Insight comes after mastery.

(seeing what others don't)

3. Staying on the Front foot:

always learning and asking and listening to figure out what's next

excellence orients toward possibilities and engages, to pick possibilities to focus on understanding or making real.

(allocating time and attention)

FOCUS & IMAGINATION

Focus & imagination are the two mental modes we use in analyzing companies. When we zoom into the details – we are focused. Without focus we cannot see the granular details of events and calibrate our expectations. When we zoom out to look for what's not on the page or what's not in the price – that requires imagination.

I've been struggling with how to create a structured, shared imagination. It isn't enough to just say that we use our imagination, or we see what's not on the page. If we just do imagine we get the tower of babel problem (Bible) or private language problem (Wittgenstein) – where everyone is speaking in different languages or using similar words but assigning different definitions to them. Then we have the traditional logical problems too, which boil down to framing expectations in a falsifiable way so that we can test our assumptions (Popper). We can't generate useful insights if we can't test the possibilities we imagine in some way.

This last bit is the key: what actually defines any creative process is the set of limits or gates or constraints (an idea derived from Kauffman). In Kaufman's work on complexity and networks, order emerges from chaos through binary gates or binary reactions. In research, falsifiability is the structure we use for gating imagination, for testing hypotheses. Falsifiability or testability forces a structure of clearly defined terms and common objective – solving the other structural problems of not having a shared language for the shared imagination.

Because we are investors, falsifiable ideas need to be defined in the language of business: accounting. We quantify imagined futures in terms of how they would play out in the financial statements of companies. Our task is not to understand Popper or Wittgenstein or Kauffman here (although I would recommend it - because doing so results in being able to see things like Feynman in the last reading in the guidebook.). ***Our task is to imagine what “progress” may look like out 5 years, 10 years, 20 years and then walk it back analytically. We start with a broad imagination and keep working at it until parts of the puzzle come into focus as financial statement impacts.***

Seeing what's not on the page is the process of seeing where the future has already arrived and then thinking through how it will be “distributed” to the rest of the world, and then asking what may arrive next and repeating how it might be distributed, and so on. Imagination is iterative. We have the key advantage of living in one of the parts of the world where the future seems to arrive earlier than most places -- this is a critical advantage for us. If we combine the advantage of seeing the future arrive with valuation discipline and analytical skill -- we should be able to create an incredible amount of wealth for our clients over time.

This shifts our focus from short-term economic cycles or product cycles or political cycles to long-term progress. We must not get stuck thinking only of the world as it is. ***We are investing in the world as it will be, and we are attempting to study how it gets from “here” to “there.”***

The process of assembling sets of long-term expectations and studying how they might play out in the next 10-20 years and then walking them back to today is generally referred to as “futuring.” Building these scenarios is the imaginative part of our investment process focused on human progress.

HIGH PERFORMING TEAMS

All high performing teams have three things:

- Shared Goals: a common target or purpose

- Shared Language: a common way of making sense of the world and communicating

- Shared Practices: common ways of doing the work

We commonly refer to these “shared beliefs & behaviors” in an organization as corporate culture. Which is accurate, but I prefer the more traditional anthropological term for culture – a shared imagination. High performing teams and winning cultures have a shared imagination that gives them an advantage.

By creating shared sets of goals, language and practices – a group of people removes the normal costs to working together by sharing common goals or targets, sharing language or models of how to make sense of information, and by sharing practices or ways of doing work.

These normal costs of normal people interacting inspired Sartre to write the play No Exit, in which a character famously observes that “hell is other people.” The difference between having a functional shared imagination versus not having it can feel like the difference between heaven and hell.

Without an efficient shared imagination, no individual can do anything significant. Any significant work requires teams and teams require culture and shared imagination to perform well.

For Merchant Adventure to build a culture and organization of multiple inter-locking high performing teams, we need to establish and defend the shared goals, language and practices which will enable us to achieve excellence. This guidebook is a significant step in the process of developing our culture – shared language and shared practices. A complete Merchant Adventurer’s Guidebook will be a summary of our shared language and shared practices. This is an essential step in building a high-performance culture, and organizational excellence.

ETHICS – THREE SIMPLE PRINCIPLES

When I started in the investment business, I remember being intimidated, scared and confused by all the ethics rules. I understood their purpose – but the complexity of some of the details seemed illogical and confusing. And in all honesty, the CFA program was a promulgator and the industries bad actors seemed to be advocates of the complexity – which allowed them to break the spirit of the rules with bending their letters.

While we are required to observe the forms implemented by institutions with fundamentally compromised interested and captured boards – the key from my perspective is for us to adhere to our own key principles which are more prescriptive and definite. The clients interests always come first – but the clients interests are an abstract and legally defined concept and different clients have different interests – so this is complex.

In actual daily actions, simple principles are useful guides. Ethics gets a bad rap because of the way it has been strangled and bent by lawyers, but good ethics are essential and good ethics are pragmatic and made up of simple rules or principles.

Imagine an action “X”, and you aren’t sure if you should do “X.” The three specific rules are useful to keep in mind and which we should all check our conduct against are:

1. Values: Is “X” who I want to be? Is it consistent with my values and who I want to be?
2. Reciprocation: How would I feel if someone were doing “X” to me or in my name?
3. Reputation (Wall Street Journal Rule): Would you want it published in the WSJ that you did “X”?

The only firm which I have worked for which did a good job with ethics was the old pre-2000 Factset Research Systems. At that time, all new employees sat down with the CFO Ernest Wong and he explained the Wall Street Journal Rule, and explained that we would all be handling a lot of sensitive information for different clients and it would be impossible to anticipate all the potential situations we might find ourselves in – but we needed to be empowered to take care of the clients and along those lines Ernest emphasized that we should just keep in mind the Wall Street Journal Rule and if it felt like we were being asked to do something we wouldn’t want published in the WSJ, then don’t do it and the firm will have your back. This was the first and only experience I ever had with an executive explaining ethical principles as a way to empower a group of employees and also as a way to make a promise to the people in the firm. Everywhere else, I had seen ethics as a control structure and as an instrument that effectively created fear and confusion more than it guided any conduct in the interest of clients or business standards.

From my perspective, the proper understanding of ethics and implementation of an ethics program creates a set of rules which primarily empower employees. By centering on clear principles to guide conduct, ethics becomes empowering – while still serving its other purpose as a control structure.

COACHING [KATA]

The coaching kata is the habit, or practice, of coaching one another toward excellence.

Each of us has different strengths. This is the nature of any group, and as Ricardo famously observed, the groups that compete most effectively focus on their strengths (he didn't exactly say that – but that is my interpretation of why competitive advantage works).

If you have a strength, it is your responsibility to the group to coach others to help them understand and then either defer to you in that area or, the more difficult and more valuable path, for you to help build your strength in an area into a strength of others in the organization.

Coaching isn't a role – but a mode of interaction. One key aspect is that we don't believe that one person is the coach on all things. I may coach one skill and be coached on another. Coaching is a mode and the role each of us plays depends on the skill. We are not either “mentor” or “mentee” in a relationship – we are mentor or mentee in teaching or learning a specific skill or domain. The roles flip because we each have more weaknesses than strengths.

Mastery, or the ability to coach a skill, requires a lot of work and experience. For an individual to achieve the mastery to teach one skill well is itself exceptional. And that is just one skill, and it means that the “master” is mediocre at a host of other things. Being excellent in one area always means under developing other skills or missing other opportunities. That is how time works – we only have so much of it to allocate. The awareness of those sacrifices is humbling – a stark reminder of our limits.

Coaching is a mode of interaction, but to be a coach is to develop skill and awareness, striving for excellence and sacrificing other opportunities. To be coached is also to be aware of your weaknesses and open to improving them. The underlying mindset to be able to coach and be coached requires a desire for excellence, humility, and a curious open mind.

Coaching and learning are at the core of continuous improvement. The practice, habit or kata of continuous improvement is core to building a culture of excellence. Mike Rother describes the habit in the Toyota Kata reading and also in his books on how Toyota has built a culture of continuous improvement and industrial excellence.

CALLS. MEETINGS & INTERVIEWS

We have five principles for all calls, meetings and interviews – research oriented or otherwise. Our fundamental long-term orientation, curiosity about the world and people, and focus on possibilities from the present forward motivate these principles.

In each call and meeting we may have some tactical objective or important facts to check. These principles do not preclude fact-checking but highlight that fact-checking should never be the entire purpose or communication and although it may periodically be the primary or motivating purpose for a call – it shouldn't be the primary motivation for most calls or meetings.

Each conversation is an opportunity to learn, build relationships and differentiate who we are as investors. These principles are designed as reminders of that opportunity:

1. **Use your voice:** People have sensitive radar and the way in which you speak and how you ask specific questions all must be in your own voice. So while I offer examples – those are in my voice and the way you might ask questions with the same content and intent will probably be different. The goal isn't for us all to ask the same questions, but for us to have common principles and use those to shape questions we each ask in our own voice or own way.
2. **Establish Rapport & Context:** *Ask the person you are speaking to more broadly about their interests and their career.* You have to genuinely be curious, and doing some research on executives can be helpful here – but it is more important to be genuinely curious and listen actively. The benefits of this process are that we get some context for strengths and weaknesses in someone's perspective which is analytically useful and we also establish rapport and ideally begin building a friendly relationship and create a different tone for conversations with Merchant Adventure. For an early call with someone – this is more important than any facts they may tell you – because without context, how do you know what to make of what they tell us?
3. **Long-term Focus:** If we have a dialog going with a company, the real value of that dialog is in understanding what lessons they learned from the past and what their long-term plans are for the future. Our questions about the business will always avoid the near past or near future (narrow parts of the cone of possibility – where we can get a lot of that information from sell side analysts I want it, and what we can't it is probably illegal for us to ask directly from the company). *Part of how I introduce myself is to say that I really don't care about any of the next 6 months or last 6 months developments in the context of a call or meeting other than to quickly fact-check. If I have the chance to talk with companies and executives what I want to focus on with the time is who they are, how they think about the business and where they see the risks and opportunities in the long-term.*
4. **Scuttlebutt / Competitive Perspectives:** Once a dialog is established, ongoing conversations with companies often provide the best information on their competitors or adjacent industries (supplies and customers). One fun question to ask in different ways once we have a conversation going with a company, is to ask for their thoughts on competitors or what's they see happening around them. This can be done in different ways – it can be as simple as, "what's going on?" or be a creative combination drawing on each of the principles here and asking, "who or what businesses, current or present, do you most admire/respect/learn from? Which ones really surprised you but in a sort of 'wow, that was brilliant' way?" or something like that.
5. **Take notes:** Take notes on the call. But remember the notes are for communicating to the team following the Johnny Cash Principle for communications – *sharp like a razor, steady like a train.* Meeting notes should be a series of 4-5 bullet points. If, as an analyst you realize there is more to digest from the call,

then that is a longer separate note, which can be linked to the meeting note – but not part of the meeting notes.

It took me a number of years to realize that a key professional skill as an analyst and portfolio manager is interviewing. Interviewing is an art, not a science – and the only way to learn it is to practice. That's said, there are a number of excellent books on Interviewing which are useful for putting lessons into context. Useful resources on interviewing include (in no particular order):

1. The Craft of Interviewing, Brady (older but still good – focused on writers and journalists)
2. Interviewing, Sedorkin (recent – focused on writers and journalists)
3. Liespotting, Meyer (useful summary of signs of deception or evasion)
4. Influence, Cialdini (older, but updated – general principles of human psychology / interaction)
5. Make It Stick, Brown (focuses on learning and psychology, but useful - being remembered is important)

Developing our skills as interviewers and differentiating ourselves in the process is another of the areas where we can achieve a competitive advantage. To win in our competitive business we need to capitalize on every opportunity for advantage.

TEAM INVESTMENT DISCUSSIONS

As the team grows the way we structure team discussions will evolve.

For now, while we are a small team, there are three key principles for us to keep in mind – one for focus, one for imagination, and one for both:

1. Johnny Cash Rule: Sharp like a razor, Steady like a train.
2. Freestyling: Changing Directions without Judging.
3. Focused on the Possible, the Present: Cone of possibility.

In the way we discuss things we should always seek to share information and make our points as succinctly and powerfully as possible – sharp like a razor, steady like a train. That’s the Johnny Cash Rule.

When we explore and question each other’s ideas each step in the discussion should be an “and” – plussing. Discussion proceeds by taking an element of the previous idea and building on it. This structure comes from improv. The key to this is to deliberately follow a structure designed to naturally build a shared imagination or where our ideas can build off of each other and we can build off of each other’s ideas.

Finally, focus and imagination. We look at the possible from the point of view of the present. Not what could be or should or could have been done or should have been done – but what do we expect from today forward based on what we know now. And what would we like to know more about tomorrow?

COMPANY & MEETING NOTES

Notes are useful as communication tools and to archive what we were thinking at a point in time. Like our communications in meetings, the key principle to keep in mind comes from the man in black:

....sharp like a razor steady like a train.

The use of notes in our process is for
communication,
reflection and
calibration.

Notes should not be in the form of minutes, or a complete summary of attendees – unless that is specifically legally required. The standard model of notes that we all learned which was to summarize information and document events comes from legal requirements and due diligence and prioritizes a sort of comprehensiveness over being comprehensible and useful in practical decision-making.

Within our analytical framework for a company, note-worthy observations must be clearly linked to elements of the investment thesis (business, management) and any impact on expected growth or profitability (valuation), as well as expected future events (scenario process). This is the complete logical set and the structure of our approach.

Observations not directly and concisely linked are of unclear relevance. but more importantly when we don't make the connections we are being "sharp like a spoon, steady like a mule" ... which isn't an efficient or effective way to work.

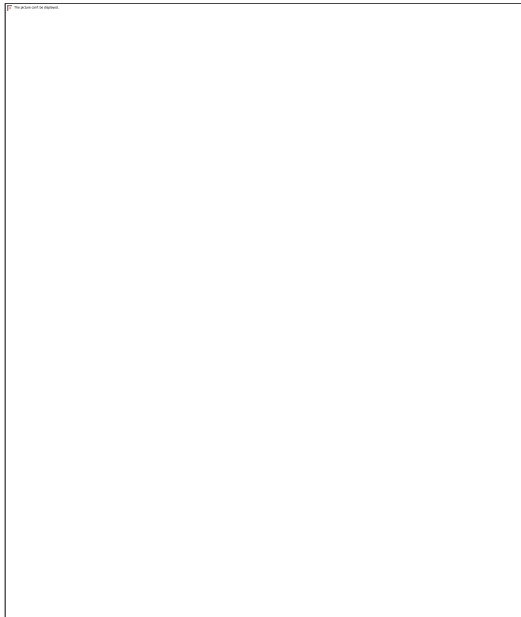
STRUCTURE OF POSSIBILITY & ITS IMPLICATIONS

A lot of our models are just about limiting our energy and research to the possible and probable. The most powerful model in doing this is just to understand the basic structure of possibilities always starts from now. This point in time. That's it. But think about how much time you hear people spending traveling through possibilities that are impossible because they are about the past. If we can focus our energy and analysis on the possible world – this can be a source of analytical advantage.

CONE OF POSSIBILITY

The basic idea in terms of the physical constraints of our universe are simply explained in Hawking's Brief History of Time in Chapter 2: Space and Time, where he introduces the idea of light cones and this idea is expanded on elsewhere and can be thought of in our context, as “cones of possibility.” Functionally, the light cone is the cone of possibility because the fastest that information can systematically travel is at the speed of light.

Hawking illustrated it as shown below.



There is an area in the past which can influence the present and then the present is a point, and events in the present can then go on and expand out to influence events in an expanding space in the future. It's a very simple idea and it's just how we understand reality works – with the speed of light as a hard limit. The key is that you can't even pretend to analyze the possible unless you make sure you are sitting in the present, and an accurate picture of the present is like a focal point for understanding the past and the future.

Now, I know thinking about the basic structure of the universe and the speed of light constraints seem like odd principles to introduce in a discussion of investment principles – but the next time you catch yourself thinking about what someone should have done – remember that you are asserting a strange ability to change the physical laws of the universe. Whereas if you were to ask about the impacts of what they did or what they learned and what they may do next, you would be asking logical questions of analytical value and without the absurdity of needing to change the physical laws of the universe. A simple awareness of the structure of reality offers useful boundaries & tools for every analyst.

SPEED LIMITS

The first question it should raise is that once we see an event, its impact ripples out in space and time at some rate. The rate it travels out into society and through human institutions is never the speed of light – there will be other constraints and limits based on the institutions and existing belief structures and how the new event or information relates to existing institutions and belief structures.

Some institutions, like regulatory institutions or self-regulatory organizations, are literally in place to control or limit the spread of impact from events in different areas of human endeavor, to protect the larger society. Many innovative businesses create economic value by evading these regulatory structures. This question about diffusion and the shape of the cone of possibility going out from a current event is there in the structure of reality and if we pay attention to this structure, it can lead us to the key questions to ask.

IMPLICATIONS – OPPORTUNITY COSTS & ALL CASH PORTFOLIOS

The structure of reality also tells us that our choice – our only choice – is what we do in the present. This is obvious but it is also not an explicit and required assumption of a lot of decision frameworks, which is really weird and wrong.

The most obvious example of this is all of continuous finance which assumes continuous access to credit. Note that to make the continuity assumption at all you assume that in some present where you begin your model all of the future in which you will be operating will not experience a discontinuity in any key markets. So you make a decision in the present based on present conditions being largely continuous in the future. It's just wrong – look at the cone. While reality is a relevant long-term constraint, the paradox of social institutions is that we have the ability to create our own legal fictions and shared imaginations which frequently deviate from and make assumptions that violate the structure of the physical world. This isn't actually a problem – but just part of being complex emergent phenomena in a creative universe – or that's probably how someone like Kauffman would look at it. Our job is to play around the edges of the possible.

In terms of our portfolio and our positions – the edge of the possible is to look at the opportunity costs of holding one position versus another based on everything we know today, and to aggregate all that information into a set of recommended weights for the entire portfolio based on what we know today. We frequently refer to this as our “all cash portfolio” exercise. In practical terms, the all cash portfolio exercise is a way to challenge all the positions in the portfolio which were based on past decisions which made sense at the time but which we may or may not have re-evaluated. What would we buy today if someone hired us to manage an account which was all cash? And comparing that to our portfolios which are all necessarily aggregations of past buy and sell or past decisions. The all-cash portfolio exercise is about pulling ourselves and our portfolios into the present – into the focal point of the cone, and a way to shed the past and get rid of analyses anchored in the past.

I know John's logic for the all-cash portfolio is practical, and I agree with the pragmatics. But it is also important to be able to see how best practices and pragmatic patterns also fit elegantly into a broader set of principles. Where we don't see these simple rules or frameworks and how they are guiding us, we don't understand what we are doing (Kelvin's exhortation to quantification).

THE PRESENT IS ALWAYS THE FOCAL POINT

One of our key principles as a firm is focus, and the structure of possibility is such that the present is always the focal point of possibilities. This doesn't mean that we focus on just what is happening in the present but it means that we are always looking at the impact of events in the past present and future through the lens of what we know in the present. The perspective and awareness of always operating in the present and thinking about how events diffuse through space and time gives us a structure to bring the whole picture into focus.

COMPOSITION OF ECONOMIC GROWTH

The rule of thumb I remember from early in my career was that economic growth was a mix of unit growth and productivity/innovation, and that mix was roughly 50/50. The unit growth was driven by demographics while the growth in excess of that was called innovation or productivity. Economists weren't really sure how to define innovation or productivity other than to say it was the part that wasn't explained by other numbers.

But if we look at the world today, population growth in most of the developed world is a thing of the past. Peak child for the world, the most children ever, was way back in 2007.

The doubling rate of global population, from 2 to 4 billion people was 25 years and 4 to 8 was then 42 years (slowing dramatically since the 1970s) and we are now at 8 billion and likely to peak around 8.8 billion people within 25 years (the 12 billion number is old bad math).

The composition of economic growth can't be the old 50/50 anymore. Looking at the math - for most of the world growth is now entirely an innovation game - the demographics are flat to negative.

Looking at the last decade in the US, we have been shifting to low population growth and gently aging. Arguably, the shift in capital to tech in the last decade was part of the markets reacting to this shift in the composition of economic growth.

The key is that the composition of economic growth going forward will not be what it was in the past. Since as investors we are looking for companies driving economic growth and profiting from it – this is a big deal.

A lot of our assumptions around the developing world are based on unit growth and income level convergence. If the unit growth isn't there and the convergence is mediated by digital platforms as part of a digital path to development, then where we look for opportunities in the developing world fundamentally shifts.

One avoids developing world places and developed world places with negative demographics because those societies are failing (see Moffett in the readings). This is Russia and East Asia.

One focuses on innovation as the necessary component since it will be the major driver of economic growth and the best innovators which achieve scale will then be in position to get unit growth - so unit growth comes for free with innovation but doesn't exist on its own in a stable population world. Innovation doesn't necessarily mean high tech - but it does mean best in class processes and efficiency and visionary leadership and empowered/engaged employees, which will involve digitization and the digital path in production, distribution and sales.

The composition of economic growth going forward won't look like the past. Most societies have stopped growing in population, so there can't be a balance between unit growth driven by population growth and growth driven by innovation. What does this mean? Our process should focus on the remaining factor driving economic growth -- if we want to get in the way of where the world is going / future economic growth.

ECONOMIC GEOGRAPHY

The way we think of economic geography leads us to focus on a surprisingly focused group of countries as attractive target markets and places where we can effectively deploy capital.

We have three basic criteria for a country to qualify as an attractive economic geography to invest: (1) that they have infrastructure for investment capital and connections to the global economy through a mega city, (2) that the country is large enough to support scale economies in its domestic markets, (3) that the country isn't shrinking in population.

1. Mega-cities, Domestic Scale, Capital Infrastructure: Economic growth and innovation tends to be clustered in mega-cities around the world. Open Capital Markets: pricing, interest rate and currency infrastructure which we can deploy capital through.
2. Domestic population need to be large enough to support economies of scale.
3. Population Growth – Immigration & Births: Successful societies add new members from neighboring states and/or new births. Failing societies create conditions where people don't have children and want to leave.

Countries with over 40 million in population, positive 5-year average population growth through 2021, open capital markets and at least one mega city are:

***Brazil, Indonesia, Mexico, South Africa,
Turkey, United Kingdom, United States.***

Most countries are shrinking, have smaller domestic markets or limited and conditional investment markets.

The countries below meet all the criteria above, except they have certain capital market or government restrictions that cause us to exclude them from our open markets list:

***Argentina, Bangladesh, Democratic Republic of Congo, Egypt, Ethiopia,
India, Iran, Iraq, Kenya, Nigeria, Pakistan, Philippines, Vietnam.***

Our hope is that over time the capital markets in this longer list of countries open up. In the near term, the best way to invest in countries with promising human geography, but closed or restricted capital markets, is to look for quality country fund managers, or country specialists.

(Country lists based on data from the World Bank – need to write up specific process)

IDEA GENERATION

There is no magic to investment idea generation and evaluation. There is just a little logic and a constant flow of work. The purpose of an idea generation discipline or process is to use simple logical rules to limit the scope and define the order of tasks in reviewing a constant flow of new ideas, while also keeping up with all the existing holdings, emerging technologies, and global trends. Logically, the idea generation is one-part unstructured imagination and two-parts structured focus.

UNSTRUCTURED & IMAGINATIVE

The unstructured and imaginative part of idea generation is easily described. There isn't much that can be said – these are the ideas that come from reading, or companies that keep coming up unexpectedly in conversations. We should allocate some time and attention to “looking out” for the unexpected in unstructured inquiry. We should be curious about the world and this unstructured part of the idea generation process can be thought of as curiosity time.

But since unstructured inquiry doesn't impose its own limits, time spent in this mode should normally be 20% of the time any of us spend doing research. While we don't allocate much time to this, ideas and observations from an unstructured inquiry are allowed to build up over whatever period of time is required for us to build a structured perspective and feed into the structured part of the process if appropriate.

Our interest in the role of nuclear energy is a recent example of something that knocked around in the unstructured inquiry space as being of interest but without structure for the last year, a time during we were just learning little by little as we watched events and read about the companies and the science. During that time, we were able to build enough fundamental knowledge of nuclear energy and merge that with our perspective of future energy demands as well as the companies operating in the space, such that going forward we have a structure for researching developments in nuclear energy and the companies in that space. Over time, themes and companies will emerge from the unstructured space and move into more structured systematic inquiry or they will get discarded.

STRUCTURED & FOCUSED

The structured part of idea generation is one part qualitative: ideas that come up while researching existing holdings (maintenance research) and quantitative: ideas that come from security screening. Each of these is discussed in its own section below.

MAINTENANCE RESEARCH

While keeping up with existing holdings or other companies of interest, we listen to quarterly calls and conference presentations; talk to analysts, management teams and experts; read trade journals and books and newspapers. And while we are learning about a subject company, we are also learning about its competitors, and suppliers and clients. Many new ideas come up while keeping up with or refreshing old ideas. There's not a single way this happens.

As I tell young analysts, just because you start researching one company doesn't mean that you will still be researching that company in the end. If you find another company that seems more interesting during your research, that isn't a mistake or a problem – it means you learned something. That's the point of research.

Ideas noticed during maintenance research are sort of like motion you see in your peripheral vision. You will be focused on learning about one company – but by focusing on that company, you will also start to see how everything is moving around it and relates to that company. At this point, if you notice that a supplier or

competitor or customer has power over the company you are focused on, it should be natural to shift focus to understanding how and why another business has more market power.

SECURITY SCREENING: ROADKILL & EXCELLENT COMPANIES

We utilize two different type of quantitative screens to look for new investment ideas:

1. **Roadkill with a Pulse (“RwP”):**

A significant decline in price (“roadkill”) combined with some set of financial characteristics which indicate some signs of financial life (“pulse”).

2. **Excellent Company Universe (“ECU”):**

Each analyst designs their own screen, producing a list of 200-300 companies with the “best” characteristics (“excellent companies”). Each analyst gets to define their own characteristics and can change their criteria for an excellent company any time. Valuation is not required as a factor but may be included. Each analyst must be able to explain the basis for their criteria.

These two different screening modes are deliberately intended to look at different sets of companies. The Roadkill screens focus a portion of our attention on the stocks which the market is currently punishing the most. We will normally be buying companies that would be showing up on a Roadkill screen. But stocks with significant recent price declines always have problems – either problems that caused the decline or problems that the stock decline caused. As analysts and portfolio managers, it is important that we don’t spend too much of our time looking at problems. If you spend too much time studying problem companies you begin to think problems are normal and forget what a normal company looks like, much less how an excellent company works.

The purpose of the Excellent Company Universe is to allocate time and attention to studying currently successful companies – even if the valuation isn’t attractive. As analysts we are students of excellence in business. Our investment approach is to seek businesses with the seeds of excellence or signs of it, but at a time when the market is focused on their problems. But we can’t recognize these signs of excellence if we aren’t studying it, and if we don’t have an ongoing discussion about what makes an excellent business in different industries.

FOCUS & FIT

Our portfolio management discipline excludes certain categories of “profitable” investment ideas. No portfolio includes all the “good” ideas.

In the Merchant Adventure Fund, we will miss the following:

- Short-term merger arbitrage opportunities.
- Nifty biotech ideas that are only pursuing modest goals.
- Cyclical companies of all types.
- Asset plays based on long-tail residual cash flows.
- Comebacks of overleveraged companies.
- Already massive companies with scale continuing.
- Private investment opportunities internationally.
- Private investment opportunities where we can’t learn directly from the founders.
- Public investment opportunities in focused on markets lacking domestic scale or population growth.
- Publicly traded opportunities in administratively complex markets.
- Opportunities where we can’t think of ourselves as owners and make sense of management plans.
- Etc.

The only way to be able to be structured, systematic and build expertise is through focus. The flip side of focus is that we will miss opportunities – what statistics refers to as Type 2 errors, or “false negatives.” In statistics, there is generally a trade-off between Type 1 errors (false positive) and Type 2 errors (false negatives). If we want to reduce the number of false positives (in our case buying a stock that goes down) then we have to accept an

increase in false negatives (or rejecting an investment that would have made money). This is useful for analysts to understand because the natural tendency for an analyst is to view success as getting their name in the portfolio, whereas the portfolio management discipline is designed to minimize false positives which means the portfolio manager will reject many profitable ideas (false negatives) and this isn't a bug or problem with the process – it is a feature of the process. You can't reduce false positives without increasing false negatives.

In your research, you will find a constant stream of “probably profitable” or “obviously awesome” investment opportunities -- but the portfolio only holds a few dozen positions, and most ideas won't fit for our portfolio discipline.

The brutal reality is that the portfolio averages around one new position and one exited position a month. Do the math. With five analysts on the team then each analyst should average around 2 new names in the portfolio a year, and that means a year could pass where an analyst wouldn't add any companies to the portfolio – and this would be a sign of doing good work. New ideas do not equal good work – it is often the opposite. Too many new ideas indicate the absence of a depth of understanding and an absence of work.

The inevitable and appropriate fate of most new ideas is to be shot down. That is fine. The point of the process isn't the ideas, but the lessons learned from constantly scanning the business landscape for disasters, opportunities, emerging themes and excellence.

The ultimate focus of studying business is to learn to recognize excellence.

COMPANY ANALYSIS

Plenty of ink has been spilled explaining how to analyze companies. The fact of the matter is that each company is idiosyncratic – or its success is likely to be idiosyncratic. Company analysis is a sort of reverse Anna Karenina problem: “Unsuccessful companies are all alike; every successful company is successful in its own way.”¹⁴

Therefore, the task of analyzing a company is not to find the ways in which it fits some specific form, but the process of making sure that it functions and then understanding what elements could “make it successful in its own way” – in a way that it different.

A more humorous way to look at the company analysis problem and the search for exceptional businesses which could be long term winners comes from This is Spinal Tap, “[these go to eleven](#).” So you are looking for that company where it can go to 10, but then when it needs something a little more, it can go to 11.

Note, with these statements I am trying to fit a form to what an exceptional company looks like – but the key is really that an exceptional company in a specific business will be the only one of its kind. I use these examples as illustrations, but that is all. *Analyzing an exceptional company the a process of getting to know and understand the company by studying it over time. It doesn't fit a single form; it is best described as an adventure.*

WHERE TO START

How the company sees themselves.... Company filings and reports and presentations – not sell side or media reports.

PUTTING IT DOWN & PICKING IT BACK UP

Most companies won't make sense the first time you look at them, and when you get to something that doesn't make sense – then you put it down. Being able to spend some time on a company and learn a little bit and then move on is its own skill.

But just because we have looked at something before and passed on investing doesn't mean we have rejected it permanently. If a company name keeps coming up or you keep coming across information on a company – you want to pick it back up and take another look and learn a little more.

This cycle of picking up and putting back down and picking up again can go on every few months for years in some cases. This is an excellent way to learn about a company in the long-term.

SEEING WHAT ISN'T THERE YET

Ultimately, when it is time for us to think about building a position is when we begin to be able to see what isn't on the page yet or what hasn't happened yet. Understanding an existing business is simple – but to be able to see the potential of a management team and the business they are running takes focus and imagination.

GETTING TO THE POINT

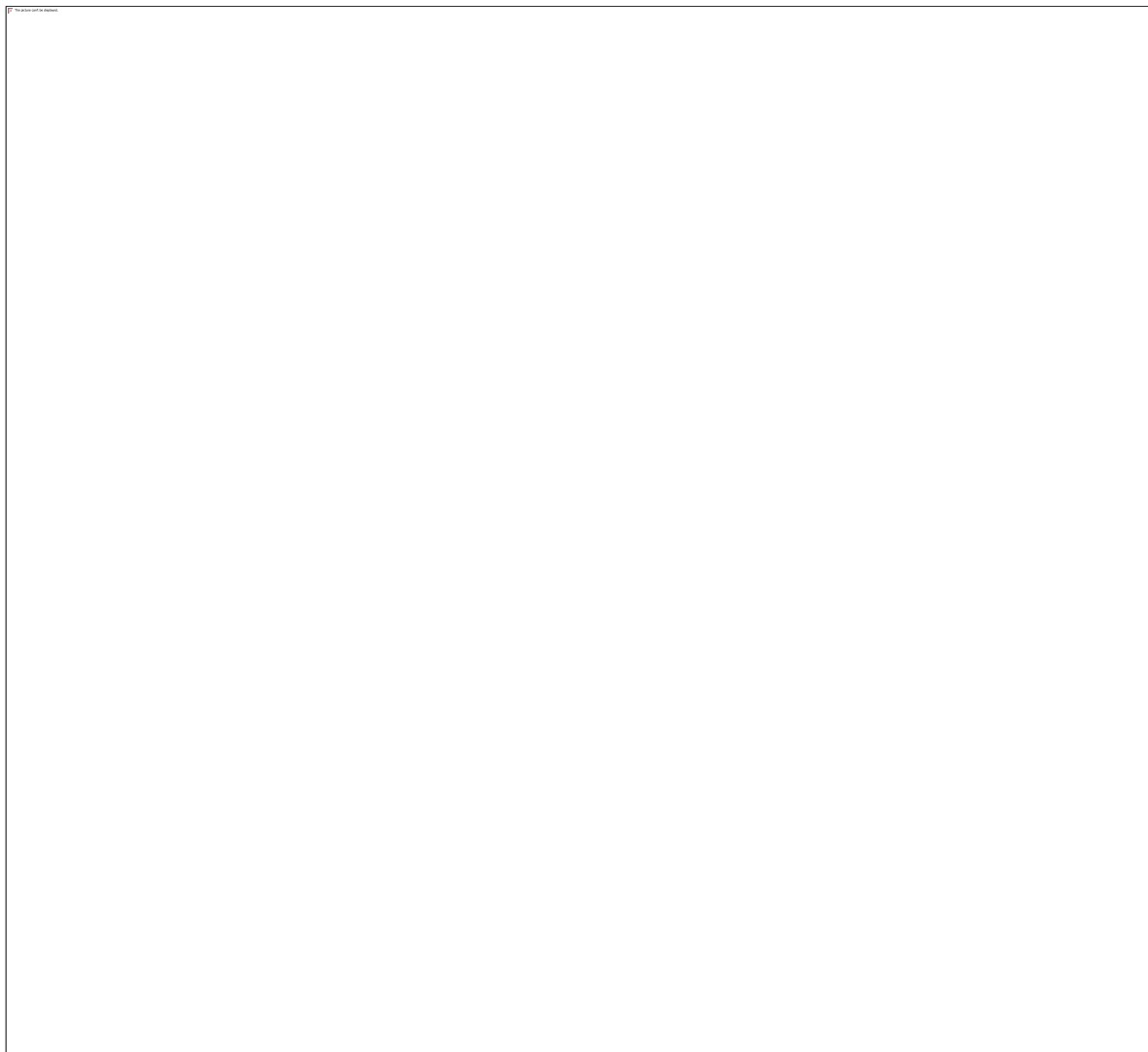
Pulling together the investment thesis. (need to write this – but its just something you learn by doing... need to model how to do it more.) Why is it a value gorilla?

¹⁴ Tolstoy opens Anna Karenina, ““Happy families are all alike; every unhappy family is unhappy in its own way.”

ALLOCATING YOUR TIME

The most valuable resource we all have is our time. In terms of making positive statements on how to allocate time in research – well that’s impossible because if you are really “doing” research, you are exploring something new – which requires focus and imagination as we have discussed. The process can’t be precisely defined – but the requirements for useful outputs can.

By focusing on what the endpoints need to be, we can establish some simple questions/rules/principles of when research is a waste of time – shown in the graphic below which originally came from the Dodge & Cox analyst handbook circa 1992.



Note how the list isn’t about “doing” research. It’s defining the characteristics of a “merely adequate” research product. This principle of maintaining a focus on achieving an adequate result should remind you of the May Sarton quote in the Prologue.

THINKING LIKE AN OWNER

Owning shares of a company is owning part of the company. That is what the shares are: ownership interests. So, the logic of it should be clear to everyone. An owner of a business should think like an owner. And the relevant mindset in analyzing whether to become an owner of a business is to think like an owner of that business.

This has a few simple but often ignored implications. An owner probably isn't worried about technical trading patterns. An owner is worried about growth and profitability. We think like owners and focus on growth and profitability – not technical trading patterns or any of the hundreds of other distractions the market provides.

IMAGINE YOU ARE RUNNING THE BUSINESS

A simple unstructured exercise is to imagine that you were running the business you are studying or analyzing. You will find that you ask questions differently in this mode and it is often useful in highlighting gaps in our knowledge or understanding of either how businesses work or elements of a specific business.

Lessons to be learned from this exercise:

1. If you really understand a business you will find that you understand why management is doing things without needing to read their new releases on it or talk with them about it... If you need to talk with management to understand why they did something you don't understand the business.
2. If you can't imagine running the business at all – either because something seems insane or doesn't make sense to you in some way – exactly what doesn't make sense or why you wouldn't want to be running the business is the analytical information we are looking for.

One of the most useful insights might be the simple fact that you wouldn't want to run a business because of legal or ethical or personal safety concerns you would feel if you were running that business. That feeling is worth doing some research on, because if those concerns aren't just the result of an overactive imagination, they represent material investment risks that should be part of your analysis.

OWNER OPERATORS

The classic analysis of why owner operators can do better has to do with what is referred to as “agency problems.” Agency problems come from the misalignment of interests – where management may be running the business for their interests based on their incentives and those interests and incentives may not be in the best interests of shareholders, or the long-term business.

Because no one has a time machine – debating agency issues can be an analytical morass. So instead, we tend to look at the incentives and culture of an organization to try to understand why the organization works or doesn't work.

In the background of all of this is another agency problem though – and that is on our side as investors. Are we really taking a view that is aligned with the long-term best interests of the company?

And the only way to answer that question is to go back and imagine you are running the business. Imagining we own the business and are running it is not only a key to testing our knowledge of a business – it also is a useful trick for getting around the agency issues in our organization and processes.

INVESTMENT THESIS

The investment thesis is the synthesis of the company research process. The thesis statement is deceptively simple and focuses us on the key risks and opportunities for the company in one to three short paragraphs.

The investment thesis offers a qualitative assessment of the business franchise and management quality, and a quantitative assessment of expected profitability and growth relative to current valuation. A well-structured investment thesis will only need to have the quantitative assessment updated periodically, while the qualitative assessment could remain accurate and relevant for years. The idea that the qualitative assessment of the business and management in an investment thesis should be durable and timeless is important to keep in mind. If a thesis included statements that won't be relevant or true in 3 years, you aren't done yet. You may have an important observation, but it isn't part of the investment thesis.

Note: It is tempting to include add and trim prices as well as expected events in the investment thesis – but these elements are not part of the investment thesis. The add/trim price lives in the portfolio management process and expected events are part of the scenario process – and these processes need to be where those data items are managed/updated.

VALUE GORILLAS

Our elevator pitch of each investment should be why each holding is a value gorilla. What is the economic opportunity in front of it and what are they doing to exploit this opportunity? Why is it likely to succeed while the market is pricing the stock as if they will not succeed? That's it.

Value Gorilla's are characterized by the following traits:

1. The unlocking of a large amount of value through applying a combination of multiple old known technology to a choke point in a key value chain (Gorilla large, inspired by the old Geoffrey Moore book - <https://www.harpercollins.com/products/the-gorilla-game-revised-edition-geoffrey-a-moore>).
2. Which is obvious, although not necessarily believable (Gorilla in the picture but did you notice it - http://theinvisiblegorilla.com/gorilla_experiment.html).

We can then look for companies which are working on these opportunities leveraging existing businesses and clients, and whose businesses trade as though the opportunities will not be realized (invisible gorilla / value stock / free real options).

The investment thesis for companies in the fund can be thought of as a brief explanation of what makes the business a value gorilla, with a minimal amount of structure to it – so that we make sure we touch on the key elements of business, management, and valuation.

SENTIEO THESIS TEMPLATE

The investment thesis should be kept update in the thesis template or thesis note for each company in Sentieo. The thesis note is made up of two parts: (1) the thesis statement, (2) a brief section of essential summary statistics.

ESSENTIAL STATISTICS

For our decision process and investment framework, the essential statistics are:

1. Current Price
2. Shares outstanding

3. Market Cap
4. Total Debt
5. Shareholder's Equity*
6. Cash
7. Sales*
8. Gross Margin
9. Net Margin
10. Price to Sales*
11. Price to Book*
12. Dividend Yield
13. Expected Growth FY0 to FY2: Book and Sales

**Question: Should we include Expected Book Growth and Expected Sales Growth calculated based on FY2 consensus analyst estimates compared to the most recent year end (so a 2 year compound near term growth rate) and then use FY2 Expected Sales and FY2 Expected Book in the valuation multiples? This implicitly creates a sort of 2 stage financial modelling structure where we “outsource” the near term modelling to the sell-side analysts who are focused on that, and then we use the estimate from the end of the time horizon they are focused on as the base point for anchoring our longer-term view...*

FINANCIAL SIGNPOSTS

One can easily get lost in all the financial signposts, ratios and indicators that different practitioners promote.

Our goal is to try to boil all the complexity down to simple frameworks. This doesn't get rid of the complexity – but it offers us a way to contain it, the same way that a line item in an accounting statement contains a mess of journal entries but can be understood without needing to worry about the journal entries.

In our valuation work, everything rolls up to measures of profitability and growth. This brief note is intended to introduce a few financial statement indicators of market power and management skill. The idea being that without market power and management skill it is difficult to have much confidence in the ability of a business to deliver financial performance over time. So finding indicators of market power and management skill in a company's financial statements is a useful check or confirmation of a company's capability of performing.

MARKET POWER

There are two classic signs of market power that are part of every set of financial statements:

1. Gross Margin Stability: a company which can maintain a stability of gross margins, or is able to hold steady the mark-up they get on a product over their cost to make the product (cost of good sold) displays a combination of discipline and market power. Long-term market power is stability through an economic cycle or a period including a strong economy and a weak economy. Market power is not the ability to increase gross margins, since that's something customers will notice and won't appreciate in the long run.
2. Negative Working Capital: If a company can require its customers to pay it for goods quickly while it can also negotiate with suppliers to give it a long time to pay, the company has market power. Quantitatively this shows up as days payables outstanding minus days sales outstanding being a large positive number.

Market power is a sign of skill but it is more complicated than just management skill, because in the case of market power we are observing a more complex phenomenon or set of circumstances that enables a business to be able to charge and hold steady their mark-up on their product through economic cycles, or where their customers are happy to pay quickly and their suppliers will settle for being paid later, Power is this ability to dictate price and to dictate payment.

There are other factors that are arguably signs of market power – but these two are the main ones. The only tricky factor is that everyone knows these metrics are being watched, so you also must learn to look for how companies play games. That will take you into the accounting literature.

MANAGEMENT SKILL

The way I quantify management skill is similar to the framework of Raef Raesendes and Dan Obrycki of the Applied Finance Group (“AFG”). A lot of the basic modelling assumptions and techniques in AFG's Economic Margin Framework are just best practices accounting adjustments to get at the underlying economics of the business. Then they add some other fancy stuff that I don't agree with – but all the basics are sound.

LONG-TERM CAPITAL ALLOCATION

The concept that I think I got from them was the idea that the CEO's most important job is long-term capital allocation. So you can score the CEO, or come up with a management score, based on long-term capital allocation. And the CEO's job in allocating capital is basically to drain capital from underperforming businesses and allocate additional capital to outperforming businesses. So the CEO is doing a good job if they are decreasing capital and returns are steady or improving, or if they are increasing capital and returns are improving. The no-no is increasing capital and decreasing returns.

Notice I haven't specified the time horizon over which to measure change in capital and change in returns on capital? Yeah – that's the difficult part with this measure. So it's not quite as simple as just running a financial model of a point in time, but you can graph the returns on capital and changes in invested capital over time and when you know a business you can walk through the story it tells and that understanding is a useful perspective on how the CEO has performed in allocating long-term capital.

To quickly step back, the importance of reinvestment of profits and long-term capital allocation is easy to forget. To make it simple, if a company has a return on equity of 20%, that means that in the next five years the CEO will allocate an amount of capital to projects in excess of the existing equity capital base of the company. So as long-term investors, the way that capital is allocated will be one of the key drivers of what the business is five years from now. And how profits are reinvested and long-term capital allocated is one of the CEO's key responsibilities.

As analysts, we might not be able to have a precise quantitative scorecard for the CEO, but we can get an EKG of sorts by looking at a long time series of relevant data.

SHORT-TERM CAPITAL ALLOCATION

While the CEO is focused on the long-term capital cycle, the CFO's daily job includes the way the company breathes cash in and out of its accounts. The CFO's scorecard is the short-term or the working capital cycle.

Now, negative working capital was also a sign of market power – but it is important to say that negative working capital isn't a positive sign of CFO skill. The structure or amount of working capital that a business may need to run efficiently isn't at the discretion of the CFO. The CFO's job is to run the business he is part of as efficiently as possible. The CFO scorecard is based on the stability or consistent incremental improvement of the working capital accounts over time. An exceptional CFO will maintain steady working capital accounts through different economic conditions, and you will see signs of incremental efficiencies in some of the numbers. But the key is the ability to maintain consistency.

The relationship of the working capital accounts is best visualized as a time series graph converting each account to "days balances" so the that graph is showing you a picture of the timing of how cash goes through the working capital accounts: payables, inventory, receivables, cash.

FUTURE WORK

We have old screens and templates for looking at these items, but a good task for an intern or junior analyst would be to start with a blank sheet of paper and think about how they would create tools to visualize or evaluate these and other financial signposts.

The process of building visualizations for financial ratios and metrics over time is a wonderful way to begin to link the lines in a chart with events or decisions driving a business.

TURD BLOSSOMS

While George W. Bush's term of endearment for his chief of staff Karl Rove is emotionally loaded for some or perhaps a little off color for others – the evocative phrase has a useful meaning to us.

A turd blossom is literally a flower that grows from a pile of manure. Rhetorically, it echoes Reagan's famous story of seeing a pile of manure and looking for the pony – or the more sublime observations that “out of the mud grows the lotus.” The lotus allusion is more polite, but it's also in the title of Goldie Hawn's autobiography – so we are sticking with the Texas derivation instead of either California versions of it.

Symbolically, the turd blossom is the good that comes after – after the deluge, if one is in a more biblical mood. The relevance of the turd blossom to us is, in its way like the rainbow to Noah. But the sign of better days for us is some company event, not a bending of light.

Technical Meaning in Company Research

We buy companies which we think are long term winners when the market thinks they aren't.

We are usually buying when a company has real problems – those problems are why the market has low expectations for the stock. In colloquial terms, the stock is priced like a steaming turd, and like it's future is just to get cold and being a cold turd is the only thing worse than a warm turd.

As analysts – the question at that point is whether the turd will blossom? What signs should we watch for and when and why would we give up hope for the turd blossom?

The early shoots of the turd blossom are the signs of improvement – of mediocre corporate performance being converted into something good, which could be a step to building an exceptional company in the long-term or at least a period of outperformance.

In the terms of this note – what we do is... invest in human progress by collecting “things” likely to blossom.

Usefulness of Humor

It's not fun and takes a certain temperament to see the potential in a messy or bad situation or underperforming company – just as it is an odd thing to look at a turd and watch for it to blossom. To continue mixing metaphors, it is like whistling through the graveyard and across the railroad tracks.

Taking risk is part of investing – but taking risk is stressful and investors aren't Vulcans. We feel the stress and risk and letting it off with humor is the best way to manage the stress. You want to recognize the stress, and then laugh at it and then step back and use the perspective laughter yields to analyze what was bothering you.

Turd blossoms are funny by nature – unexpected and foul and sublime in way too. Thinking of an underperforming company as a potential turd blossom offers a good way to acknowledge the problems while challenging everyone to look at the problems as “fertilizer” for what's next.

And what's next is always what we need to be focused on, even if that starts with imagining a steaming pile....

All true beginnings aren't just hard, they are also humble.

[DRAFT] VALUE EXPECTATIONS

<< this will get directly specified in the work we are doing with Ivan and James.>>

Value driver – Growth – Profitability - Adjustments

The ability to build a full model of the projected financial statements of a company is a useful skill to have. There is no better way to get an understanding of how financial statements are linked together and the way money flows through the different financial statements. That said – once one has built a number of these models, you can't help but notice that the value of the model is in this understanding of the flows and linkages – because the models inevitably fail to predict the future financials. The error rate in the model ends up being proportional to its level of precision.

In forecasting financial expectations our goal is to trade off precision for accuracy. To do this we make a series of simplifying assumptions. Each assumption may seem simple and has the effect of simplifying our analysis and math – but there is solid fundamental basis for each simplifying assumption and none of these things are actually shortcuts – they are more like tunnels burrowed through rock.

The simplifying assumptions are:

1. Stage 1 - One-Two Years Out: The Sell side Consensus is probably actually fairly accurate for the next 12-18 months, or we aren't likely to consistently pay more attention to the near future wiggles and wobbles than the street analysts – so the most efficient way to financially model the near future is to use the consensus numbers.
2. Stage 2 – Three – Five Years Out: There may be a midterm growth phase or adjustment phase – a period of normalization between the near-term stuff in the analysts model and the business settling into what we think is a likely long term growth and profitability range.
3. Stage 3: Five Years out – Business has stabilized or we just assume it will have to be able to model it – at a multiple which reflects our expectation of the long-term growth and profitability of the business.
4. We view growth and profitability as being driven either by sales or book/capital. We identify the value driver for the business as either sales or book and then quantify the long term growth as growth in sales or growth in book and long term profitability as net margins or ROE.
5. Using a 15% Target rate of return for all investments, we get a target terminal multiple 5 years out of price to book or price to sales.
6. We allow for adjustments, which usually would involve balance sheet adjustments -- adjusting for cash, stock issuance/buybacks, mergers or spin-offs or debt repayment.
7. We take the value driver at the end of Stage 1, with the analysts concerns about the near term issues baked in, then we make our adjustments, grow or shrink it in an intermediate period (stage 2) and show our math.
8. In its simplest form, the difference between the multiple today and the long-term multiple plus growth in the next 5 years is your return on the investment over 5 years. The return is expected growth plus multiple expansion.

[DRAFT] Scenario Process & Tools

<< also being defined in the application we are working on with Ivan & James.>>

Our scenario process helps us construct detailed event-based expectations for our companies and the markets they operate in. From an academic perspective, we build on the work of Tetlock in [Superforecasting](#). The process is designed as a disciplined way to prepare for the future and how the near future will differ from the recent past. The key analytical errors we are trying to adjust for are expressed by Peter Schwartz in his call for imagination and analytical realism. We must not assume that things will not change and we must be skeptical of boosters breathlessly overstating changes.

Operationally, our scenario process focuses on constructing sets of key future events, structured as yes or no questions (binomial outcomes) over fixed time horizons in different contexts. Mario Orellana and John Magruder are working on constructing an Observable Notebook building off of a [Tidy Tree Conditional Probability Notebook](#) created by another Observable user.

Our template will be set up to take as input the following:

1. Type of Scenario
2. Subject
3. Events
4. Order or Path Dependence
5. Future Considerations

The subject is the company or domain/issue which the events in the scenario apply to. The events should be a list of questions framed to fit the type of analysis and including a probability estimate. Functionally, an event is composed of (1) a name, (2) a yes or no question with includes a specific date / deadline, (3) if the event is dependent on another event, path dependence / order, (4) our probability estimate of the event, (5) basis for the probability estimate specifically in terms of either technical feasibility or scaling/distribution issues.

The four types of scenarios:

1. Company Update (Quarterly or other periodic updates)
2. Company Analysis (Investment Thesis)
3. Emerging Economic Potential (Short-term Disruption/Shift)
4. Economic Potential Analysis (Long-term Economic Drivers)

In the case of the first and second types of scenarios, the subject is a company – but the difference between the types is the time horizon under consideration. In the third and fourth type of scenario, the subject could be a country or an industry or the path of a specific technology and the difference is, time horizon. Each type of scenario forces us think expected events in different contexts at different time horizons. We are not predicting the future – but if we can more quickly see what has happened and rapidly discard things that are not possible – well, that’s how one gets an analytical advantage that isn’t based on luck.

The three time horizons for scenarios are:

1. Tactical (12 to 18 months),
2. Strategic (3-5 years),
3. Long-term (7-10 years).

The purpose of the tactical time horizon is to take the key challenges & opportunities we know a company is facing in the near term and frame them in terms of questions for us to watch. The key here isn’t really to get the probabilities right – strangely enough getting the probabilities right only begins to matter as we refine this process over time. The key here is to make sure we can articulate the key opportunities and challenges each of our companies faces in the next 12-18 months and have a framework for tracking their progress. Additionally – once we have events for all the companies, we can also look at the portfolio in terms of weights that are

dependent on future events. So the events could function as a forward looking basis for thinking about risks in the portfolio.

The strategic time horizon could be applied to a major economic shift which is under way and which we want to try to detail out more precisely or it would also apply to the construction of the investment thesis for a company. A specific example of a near term economic potential analysis which I think we should put together is the emergence of nuclear power. This is already happening because of Russia's recent actions and the climate beliefs of the millennials and gen-z. So what events should we be watching for as this gets started? In terms of the strategic company analysis – we should be able to articulate the events which would need to happen for our underlying investment thesis on a company to hold true. So the strategic company analysis should include a set of events pertaining to management, business franchise / financials and valuation which would map to our investment thesis for the company.

The longer-term time horizon only applies to long-term economic potential and general domains of economic opportunity. There is no real granularity to this sort of analysis – but a well-constructed long-term analysis of economic potential should inform what issues we need to dive into and understand in more detail and so while this is less granular and longer term, it is not disconnected from the other time horizons. A long-term view must be consistent with the shorter-term views.

The table below summarizes types of scenarios, subjects of each analysis and time horizons involved.

		Type of Scenario			
		Company Update	Company Investment Thesis	Emerging Economic Potential	Economic Potential Analysis
	<i>Subject</i>	<i>Company</i>	<i>Company</i>	<i>Not a Company</i>	<i>Not a Company</i>
Time Horizon	Tactical (12 – 18 mths)	#			
	Strategic (3-5 yrs)		#	#	
	Long-term (7-10 yrs)				#

The output of the scenario process is a set of key questions regarding the future of the subject, where the questions are structured around a subject, an event, a time horizon and a probability. In terms of estimating probabilities, we will need to use common principles with a shared basis so that we can have intelligent discussions about the probabilities. Ultimately, the probabilities are more useful for helping us rank order the likelihood of a series of events than they will be for precisely knowing in any fashion what will happen.

There are three general levels of certainty:

1. Proximate: It has already started to happen in some way, but could stop (67% or higher)
2. Probable: all the elements exist for it to happen (33% to 66%)
3. Possible: could happen but unlikely (less than 32%).

Thinking about the future is difficult. Most scenario planning processes are designed to drive strategic decisions about how businesses plan to create change or evolve – scenario planning is designed for actors in the real economy. But as investors we are not actors or managers, but observers like Drucker's bystander. As investors, we can use the overall scenario planning frameworks to mentally prepare ourselves, and structure our own expectations – allowing us to see changes in the economy as they are happening. Investing is a strange game. There is a stability to it because individual people, humans, don't really change – but our technologies, societies and cultures are constantly evolving. What I am trying to introduce as a structured part of our investment process is the use of scenario thinking to imagine or construct sets of key future events for us to watch for. These events should be steps in the evolution of business or technology or society from what it is now to something new, and a useful tool in observing progress.

PORTFOLIO GUIDELINES & TRADING

No one step in the portfolio guideline process is complicated. Each step, each piece is simple. But the guideline process brings together company research, portfolio management and trading. So, while each piece is simple, the ways in which all the pieces interact can be complex,

WHAT ARE GUIDELINES?

The guideline for a company is a static target weight for the company in the portfolio. The guidelines can add up to 100 or less. Frequently, the guideline weights in stocks will only add up to 96% to 99%.

Because the guideline is a static weight and the actual weights of companies in the portfolio constantly shifts as prices change, stocks which are declining in price or lagging the market will decline in weight and tend to fall behind their guidelines and stocks which increase in price or outperform tend to rise above their guidelines.

Part of the portfolio review process is to reset guidelines where appropriate to match lower or higher weights if after a shift in price we decide not to buy or sell a position to guideline. In this way, the static guideline process is a way of deliberately setting & resetting the weights in the portfolio and creates active decision and discussion points for the portfolio management team.

SETTING GUIDELINES

Active portfolio management breaks down to two things: getting the stocks right and getting the weights right. The decision isn't just to buy or sell, but also how much, when and at what price? Our process of working through the guidelines is designed to keep us focused on getting the stocks right, and the weights right – how much, when and at what price.

The guidelines for the Merchant Adventure Fund are reviewed during our weekly investment team meeting. All team members can recommend changes either at the meeting or during the week. The weekly investment team meeting is the only time guidelines are changed, so recommended changes can be queued up for the meeting or presented at the meeting. Research meetings focused on company updates happen more frequently during the week, so the research recommendation from an analysts would be tracked as a recommended change to be reviewed at the next investment team meeting.

When we are reviewing company weights in the portfolio, we have a simple rule of thumb explicitly and deliberately linking the investment thesis which we maintain as part of our research process with the “default” or “natural” weight for a company in the portfolio. The investment thesis has three parts: business franchise, management quality, and valuation. Confidence in each element of the thesis is worth 1% in guideline: powerful business model and competitive position (+1%), an exceptional inspirational management team (+1%), and a low valuation reflecting that very little value exists (+1%). In its simplest form, if we feel confident in all three parts of the thesis for a company, it's a 3% position. Confident in one of the three – that's a 1% position. Confident to the level of feeling certain and we've been following or owned the company for years – well the guideline can go higher than 3%.

This linkage of the guideline to the investment thesis naturally translates shifts in an investor thesis into the guideline discussion. Does the weight match our outlook and confidence in the three elements of the investment thesis? Based on that there is a default weight of 1% or 2% or 3% and then the question is if that is the right weight for the portfolio, or if there are other considerations that would cause us to hold more or less of it?

With a 20-40 stock portfolio, our average position size will always be above 2%. The maximum guideline weight for a position is 10%.

Beyond the simple logical structure to link the investment thesis to the guideline weight, which serves as a useful long-term reference point – there will be shorter-term tactical considerations that come into play in adjusting

guidelines. Part of the guideline process is also making “floating changes” to the guidelines when a stock may have moved up or down, but we are waiting for some additional information or news before buying or selling. This is a functional accommodation to the realities of managing a portfolio – the goal is to be trading the portfolio to the guideline weights and so we need to take into account percentages of the portfolio which we aren’t using because we aren’t buying right now or which are being used because we aren’t selling.

BUYING OR SELLING TO GUIDELINES

Whenever we are buying or selling shares, we refer to it as buying or selling to guideline. That’s just how it works. The guideline in a stock has to be above or below the current weight of the stock in the portfolio for us to trade, but this is only one of three conditions or triggers that need to be in place for us to buy or sell.

The other two conditions are:

1. The Price is Right: Above the Trim Price for Selling / Below the Add Price for Buying
2. More than a Bite: To get to guideline we need to buy or sell more than 0.25%

When all three conditions are met, we may buy or sell a position to guideline. The process of buying or selling a position to guideline and how that is broken into steps or timed is at the discretion of the trader.

SETTING THE ADD AND TRIM PRICES

Add and trim prices start out as buy and sell target prices set in the original security recommendation. But once a position has been established these prices can and will be adjusted to help manage the position in a disciplined way.

Add and trim prices were created as a tool for averaging into a stock as it declines in price or averaging out of a position as it appreciates. Philosophical, the add and trim price reflect the fact that we aren’t trying to predict the perfect buy or sell price. We are pragmatists. We have developed a process over time that naturally encourages us to average our costs on a holding down as prices decline and to average our realizations up as it appreciates.

The process is simple. Let’s assume that a stock is appreciating and rises above its trim price, and that as a result of the appreciation the weight of the holding in the portfolio is 1% over its guideline. Now the investment team has a two decision to make:

1. Getting an update from the analyst asking if the trim price was still “right”.
2. Deciding if the Guideline should be changed or “floated” to the position.

If the trim price is raised or the guideline is raised such that the deviation from the guideline is less than 0.25% - then the team has decided not to trade. But if the trim price isn’t changed and the guideline isn’t floated, or maybe the team just increases the guideline 0.5%, then part of the position can be sold. The team is effectively deciding to trim the position but doing it indirectly by checking elements of a process.

For our example here – let’s assume we trim the holding. Once the trades are made bringing the holding to the guideline, the trader will increase the trim price between 15% to 30%, such that if nothing else changed but the price the stock would be more than 0.25% above guideline if it were to appreciate to the updated trim price. This process of updating the trim price upward mechanically after a round of trades is a mechanical ratchet that sets the default behavior to averaging out of an appreciating position.

The same thing is done but “in reverse” for stocks where we are buying because the price has declined. The ratchet there works to force us to talk about averaging down on positions or to admit that we got it wrong and need to exit the position.

INTERNAL DATA FIELDS

In making estimates and creating a data frame for this process, we track a set of our own data items.

1. Lead Analyst
2. Shares held
3. Guideline
4. Recommended Changes
5. Add Price
6. Trim Price
7. Value Driver
8. Profitability
9. Growth
10. Target Multiple
11. Notes

FUTURE WORK

Currently the guidelines and associated information as stored in a spreadsheet which works – but can be improved on. We should be looking at better ways to do this – ideally through a secure web interface.

The quick and dirty valuation framework in the spreadsheet: value driver, profitability, growth and resulting target multiple -- needs to be written up and implemented in a way that these fields link to the investment thesis for a company.

The notes field is used for a variety of different things – from reminders for dates for wash sales to expected next events to next research tasks. The notes field should be written up and in the process we should link that to the actual data it contains and this process will also highlight additional functionality we may want in the next iteration of the guidelines “interface”...

There is always room to learn and improve.

Private Company Investments

In the summer of 2022, Merchant Adventure made its first private company investment in Reduct Inc. The founders of Reduct, Robert and Prabhas, are promising young leaders in programming and design. Reduct is working on re-imagining how people process and curate video – which is economically significant. All the normal analytical details lined up. But with smaller companies the real risks come from a whole host of factors which cannot be anticipated at this early stage. In Kauffman’s terms, the early-stage private company is exploring the adjacent possible and functionally testing a hypothesis that is un-pre-statable.

Although the goal is for private investments to be profitable, the prospective investment return is a variable that we cannot control, and our standard public market risk control tools: low valuations and long time horizons, don’t work. We are also not investing in enough promising early-stage companies for the functionally random risks of scaling to be averaged out through many positions, as has become the norm at successful venture investment firms.

The real return which Merchant Adventure can be sure to earn from investing in private companies is to gain domain knowledge on innovations in economically significant areas. Our private investment discipline and differentiation is our focus on learning. We invest in companies advancing human progress, and our primary focus is on public companies. We invest in private companies to help understand emerging threats and opportunities to public companies. Given time constraints and limits to the amount of the portfolio invested in private companies, we will invest in no more than 2 new private companies per calendar year. Most of our activities in the private company space over time will be follow-on deals with existing investments and learning from those management teams.

Therefore, the key criteria for Merchant Adventure in reviewing private investments are:

1. DOMAIN: Company pursuing a large target market in a domain of interest to Merchant.
2. OVERLAP: No other existing private company investments with overlapping markets.
3. MODEL: Target company has modelled the problem in their own way that offers new insight (like seeing Gravity and realizing it holds up the moon and pulls down the rain, etc.). The *model* creates new tools to add to a society’s or an economy’s *arsenal* for solving a socially and economically important issue¹⁵.
4. DEPTH: Management team has a depth of knowledge and experience in the area – should have been working on problems in the domain for at least a decade such that we can check references informally and also review past work in academic journals, GitHub or elsewhere.
5. ALLOCATION: We can see allocating \$1MM or more to the investment over time and that capital can be deployed in a way that would be meaningful to the target company. We are not trying to compete to write them checks. We look to allocate our capital in ways that maximize our impact and learning.
6. RECIPROCATION: Management needs to be natural reciprocators. Reciprocation is the flywheel for human relationships and creates the back and forth where learning takes place. It is easy and natural to learn from and collaborate with reciprocators.
7. CAPACITY: Have we already done 2 new company deals in the last 12 months or calendar year? Who will be committed to advising the company and communicating the lessons learned?
8. ALTERNATIVES: All the alternatives compete for the fund’s capital and our time. Is adding a new company “better” than adding to a position in an existing private company or public company? Exactly how do we think it’s better?

¹⁵ An arsenal of models is the human mental component of how we explore Kauffman’s adjacent possible.

ROLE OF MACROECONOMICS & CURRENCIES

Our investment approach focuses on three key factors: businesses, management, and valuation. Macroeconomics and currencies are not independent factors in our framework. This doesn't mean that we ignore macroeconomics and currencies – it is just that to treat them as analytical entities on their own is a form of double counting. Let me explain with a few simple questions focused on the areas of our analytical focus.

BUSINESS

When we analyze a business, how do we come up with expected growth rates and margins? We will be factoring in macroeconomic expectations there because the company will be taking those conditions into account in their plans. The company will also consider expected currency issues and may or may not hedge. Additionally, we will likely adjust our expectations based on non-consensus macro outlooks for primary markets for the company. Macroeconomic expectations are embedded in our assessment of the business.

MANAGEMENT

Macroeconomic conditions are never as “macro” as they seem to be. The devil is always in the details and economies are incomprehensibly complex. The role of management is to understand the details of the specific positioning, assets, suppliers, markets, and so on – which their company must negotiate, navigate and profit from. In our assessment of management, we either believe they have the skills to navigate those details, or we should not be investing in the company. Part of our process is to assess management's ability to adapt and profit from macro-economic shocks. So we expect our companies to act based on changing macro-economics on their own. Companies are not static entities. For us to act and adapt when the companies are acting, when part of our initial thesis on management should have been that they have better information and have an impressive capability to make decisions and drive execution – well it seems like our action might be redundant or counterproductive.

VALUATION

Finally, doesn't most of the market seem to act based on perceived macro-economic information like inflation or rising rates? This indisputable fact has two important implications: (1) often the macro-economic information is already in the price, while (2) management teams reactions and adaptations are not in the price. Thinking through these simple ideas on how macro-economic “information” impacts the market is useful. The profit is always in focusing on what's not in the price – and that is the company specifics. That happens to be what our process is designed to focus on.

That's the basic story. But the basic story always leaves something out. You may remember the September 2019 letter:

Shakespeare was a master of ellipsis, of leaving things out...

Bloom, Harold. Cleopatra: I Am Fire and Air

THE ELLIPSIS

We are not masters of ellipsis, and the goal of defining these processes is not to leave anything out – but that is an impossible goal. When we leave things out – it is an error of omission. Errors of omission are the most dangerous.

In the case of macro-economics and currencies, what this means is that we don't ignore macro-economics. That would be a serious error of omission. Many investors confidently assert they ignore macro when what they actually do is delegate responsibility to adapting to macro conditions to company management. We want to be explicit about that.

The ellipsis, or analytical error, in macro analysis is to fail to explicitly assess how management adapts to different expected and unexpected changes in their markets – of which currency shifts and macro-economic shocks and shifts are categories of changes which management must be prepared to adapt to.

I highlight this omission because it is an error I have made repeatedly in my career and something I remind myself of regularly. It isn't enough to find an excellent business with visionary management at a low valuation – the business and management also have to be prepared to adapt to changes in the macro conditions.

ALIEN PHILOSOPHIES & ANTIBODIES

This one is very simple. We must defend against foreign ways of thinking – thinking that is not defined in our framework or within our shared language. We must defend against it like a disease and develop organizational antibodies to reject alien investment philosophies.

This isn't because ours is the only way of making money, or because we somehow "see the truth" with some vision of "intrinsic value" or other nonsense. It is because to achieve excellence we need to focus, using our shared imagination and shared language to do more as a team than any of us could as individuals.

There are so many alien philosophies, that it is always easier to come back to Polonius' advice to Laertes – the part after neither lender nor borrower be: "this above all else, to thine own self be true." But replace self with investment philosophy, approach, or discipline.

Alien philosophies can be very successful, but this does not mean that we can succeed being someone else. A hawk does not decide to try hunting like a cat, nor does a cat try to hunt like a hawk – cats and hawks each hunt in their own way, in the way they have adapted or evolved to hunt. We are not macro investors; we are not short-term traders; we are not merger arbitrage players; we neither buy nor sell volatility as a thing in itself; we do not short; we do not borrow on margin; we have no use for technical analysis; the patterns we see involve profitability, growth and company financials, or they are the patterns of innovation and diffusion which haven't diffused to valuation.

As Laertes lamented to Hamlet, "there are more things on heaven and earth than are dreamt of in your philosophie." And that is the point. We are curious about all those things, but we will always be asking how to understand in our terms – in terms of business, management, and valuation, in terms of expected events and valuation expectations, in terms of gorilla sized opportunities and invisible gorilla, in terms of complex interactions and simple rules, from the owner's perspective, in terms of human progress and the digital path we are now on.



PART 3

READINGS & REFERENCES

TOYOTA KATA

Rother, Mike. Toyota Kata: Managing People for Improvement, Adaptiveness and Superior Results

RELEVANCE:

Toyota is an exceptional company and its long-term excellence in industrial processes is worth studying. Excerpts below are from Mike Rother's excellent and succinct book Toyota Kata. As you read the passages below from Rother describing industrial process excellence, pay attention for the ways in which the simple rules in a kata are a behavioral model of the simple rules in biology or physics or elsewhere – and the value of thinking in systems, capabilities, and complexity – rather than in static states.

This guidebook is the beginning of Merchant Adventure defining its own set of “kata” or practices. While I reviewed the literature on knowledge organizations and investment businesses, I believe the best guide in what we are trying to build is as a set of disciplines or practices which are “habits of mind” – like kata and implement them to build Merchant Adventure.

If an organization wants to thrive by continually improving and evolving, then it needs systematic procedures and routines—methods—that channel our human capabilities and achieve the potential. Such routines would guide and support everyone in the organization by giving them a specific pattern for how they should go about sensing, adapting, and improving.

Toyota has a method, or means, to do exactly that. At Toyota, improvement and adaptation are systematic and the method is a fundamental component of every task performed, not an add-on or a special initiative. Everyone at Toyota is taught to operate in this standard way, and it is applied to almost every situation. This goes well beyond just problem-solving techniques, to encompass a firm-specific behavior routine. Developing and maintaining this behavior in the organization, then, is what defines the task of management.

Upon closer inspection, Toyota's way, as it is sometimes called, is characterized less by its tools or principles than by sets of procedural sequences—thinking and behavior patterns—that when repeated over and over in daily work lead to the desired outcome. These patterns are the context within which Toyota's tools and principles are developed and function. If there is one thing to look at in trying to understand and perhaps emulate Toyota's success, then these behavior patterns and how they are taught may well be it.

Kata

In Japan such patterns or routines are called kata (noun). The word stems from basic forms of movement in martial arts, which are handed down from master to student over generations. Some common translations or definitions are:

- A way of doing something; a method or routine
- A pattern
- A standard form of movement
- A predefined, or choreographed, sequence of movements
- The customary procedure
- A training method or drill

... there is a further definition and translation for the word:

- A way of keeping two things in alignment or synchronization with one another

.....

The Management Challenge

Based on what I have been learning, the challenge we face is not to turn the heads of executives and managers toward implementing new production or management techniques or adopting new principles, but to achieving systematic continuous evolution and improvement across the organization by developing repeatedly and consistently applied behavioral routines: kata. Note that this challenge is significantly different than what we have been working on so far in our lean implementation efforts, and is primarily an issue of how we manage and lead people. Some adjustment in how we have been trying to adopt “lean manufacturing” will be necessary.

Before we go on I should mention that ***the idea of standardized behavioral routines often generates a prognosis that they will disable our creativity and limit our potential. What if, however, we can be even more creative, competitive, smart, out-of-the-box, and successful precisely because we have a routine that does a better job of tapping and channeling our human capabilities? A difference lies in what we define as the routine. Notably, Toyota’s improvement kata does not specify a content—it cannot—since that varies from time to time and situation to situation, but instead only the form that our thinking and behavior should take as we react to a situation.***

Humans derive a lot of their sense of security and confidence—what psychologist Albert Bandura calls “self-efficacy,” from predictable routines: from doing things the same way again and again. However, it’s not possible for the content of what we do to stay the same, and if we try to artificially maintain it, it causes problems, because we are then adjusting to reality far too late and in a jerky manner. Any organization whose members can face unpredictable and uncertain situations (which are the norm) with confidence and effective action, because they have learned a behavioral routine for doing that, can enjoy a competitive advantage.

Toyota’s improvement kata is an excellent example of this second kind of routine. It tells us how to proceed, but not the content, and thus gives members of the organization an approach, a means, for handling an infinite variety of situations and being successful. We may be standing before a different way of operating our organizations, which can take us toward nearly any achievement we might envision.

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Stay Home

One lesson implicit in this discussion is that we should not spend too much time benchmarking what others—including Toyota—are doing. You yourself are the benchmark:

Where are you now?

Where do you want to be next?

What obstacles are preventing you from getting there?

For example, if you find that your technical support staff cannot respond quickly enough to machine problems, you might think, “I wonder how Toyota handles this?” Or you could stay home and ask, “How fast do we want our technical support to respond? What is preventing that from happening? What do we need to do to achieve the desired condition?”

Remember, the ability of your company to be competitive and survive lies not so much in solutions themselves, but in the capability of the people in your organization to understand a situation and develop

solutions. And you don't have to be perfect, just ahead of your competitors in aspects of your product or service.

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Focusing On Process Instead of Blame

Toyota's approach of not stigmatizing failures, but instead utilizing them to learn and move forward, has an interesting effect: thinking that an abnormality or problem is neither positive nor negative shifts the focus from the individual to the process. We know that most problems are caused by the system within which people work, rather than by the individuals themselves. Therefore, Toyota maintains a no-blame focus on the process, instead of on the people around the problem. The assumptions are:

- *People are doing their best.*
- *A problem is a system problem, not an operator problem.*
- *There is a reason for everything, and we can work together to understand the reason for a problem.*

An elegant question in this respect that I learned from Toyota is, "What is preventing the operators from working according to the standard?" I encourage you to utilize this question as you strive for a target condition, because it alters your thinking and changes where you look when a problem occurs.

Be hard on the process, but soft on the operators.

....

The Five Questions

The five questions are a summary of Toyota's approach for moving toward a target condition and are perhaps the most useful information in this book, now that you know what they mean.

They are highly effective in practice:

- 1. What is the target condition? (The challenge)***
- 2. What is the actual condition now?***
- 3. What obstacles are preventing you from reaching the target condition? Which one are you addressing now?***
- 4. What is your next step? (Start of next PDCA cycle)***
- 5. When can we go and see what we have learned from taking that step?***

The five questions come into play once you are "on the staircase," that is, in the Plan-Do-Check-Act ("PDCA") phase of the improvement kata, after a target condition has been established. The questions build upon one another. The better you've defined the target condition, the better you'll be able to assess the current condition. The better you assess the current condition, the better you can recognize obstacles. The better you recognize obstacles, the better you can define your next step. Note that before a target condition has been established, the order of questions 1 and 2 is reversed from what is shown here.

This sequence of five questions is a device to give you a routine and mental pattern for approaching any process or situation, and to help you learn the improvement kata. The questions distill part of the improvement kata down to a point where it becomes accessible and usable by anyone. They are a "minikata," if you will, perfect for practicing. I keep the five questions in mind any time I visit a process and apply them to many other activities as well. I highly recommend that you use and internalize them.

What Toyota Emphasizes in Problem-Solving

Despite what the words “problem-solving” might lead us to think, the primary focus in problem solving at Toyota is not solutions but understanding the current situation in a work system so deeply, firsthand, that the right solution (called a countermeasure) becomes obvious and practically falls in your lap. Most of the effort of problem solving at Toyota is placed in grasping the situation—deeply understanding the conditions that led to the problem—as opposed to hunting for solutions.

We often mistakenly think that good problem-solving means solving the problem, that is, applying countermeasures, and we may even propose and apply several countermeasures in the hope that one of them will stop the problem. **In contrast, in Toyota’s way of thinking if the solution to a problem is not yet obvious, it means we have not yet understood the situation sufficiently. Time to go and see again**

An example: A factory that makes precision-cast turbine blades for aircraft engines was experiencing a quality problem. One of the last processes in the turbine-blade value stream is a spray-coating line, much like a paint line, and some blades were coming out of the coating process with dents from banging against one another. Due to the damage, these expensive parts would have to be scrapped. Engineers quickly put forth a number of potential countermeasures, such as hanging the blades farther apart on the coating line chain conveyor, putting a protective shield between each blade, and so on.

One engineer took a different approach and simply observed the coating process in action. After about three hours of watching, he noticed something at a point in the process where the chain conveyor makes a 90-degree turn. As the turbine blades went around this corner, some of them would rotate counterclockwise a little and slightly unscrew the hook upon which they were hanging. When the hook became unscrewed far enough, it allowed the blade to swing and on occasion contact the neighboring blade. Once the engineer understood the problem, then the right countermeasure became obvious: prevent the hooks from unscrewing.

Few of us take the time to keep observing a process until the cause of a problem becomes clear. We tend instead to reward firefighters and expeditors who temporarily fix a problem.....

It Keeps Going

Once you begin working with the improvement kata at a process, there is no end. If the target condition is achieved with some consistency day in and day out, it may be time to develop the next target condition for this process. Without a target condition (challenge) to strive for, a process will tend to slip back.

This is the time to make an overall reflection, to summarize what was learned in this complete improvement kata cycle in preparation for the next. While you are working to achieve the current target condition, you will usually begin to see elements of the next target condition (challenge). If not, then you’re probably not struggling enough with process details.

You may not arrive at a target condition 100 percent. For example, it is unlikely that a production process can ever be 100 percent stable. At production processes you may reach a state where you are just reacting to deviations and abnormalities, rather than still striving to reach a challenging target condition. A question I sometimes ask myself is: “Are we still working under a challenge here?” If not, then it may be time to define the next target condition.

Occasionally you will not achieve a target condition on time, but this is sometimes acceptable. Why? Because we learn the most from failures.

For a few years I chaired a manufacturing conference in Munich, and one year several speakers, in presenting improvements they had made, ended their presentations with a photograph of the award—a trophy or plaque—

they had won. After this happened a few times in a row, I felt compelled to point out that sure, Toyota too would show its awards, but this would not be the last slide in its presentation. Toyota's last slide would describe the next challenge. ***It is okay to celebrate successes, but we should always be looking ahead and focusing on a target condition and the next step. If we decide to use awards, then they should not be seen as an end, but rather as a beginning, a doorway to more learning.***

The benchmark to beat is yourself and your current condition.

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Adaptive Persistence

By embedding the improvement kata into daily work, Toyota has done something elegant: it has developed a practical and universal method for evolving along unforeseeable routes toward only generally defined long-term visions. This could be called "Adaptive Persistence," a fitting phrase coined by Richard T. Pascale in his famous 1984 California Management Review article.

To paraphrase Mr. Pascale, Toyota's continued success is not due to perfect up front decisions and plans (that is, perfect aim). Many priorities become clear only as you strive to move toward something, rather than through advance planning. Thousands of PDCA cycles toward target conditions contribute incrementally and cumulatively to Toyota's cost, quality, and market position. Toyota finds the path along the way based on what is being learned along the way. In hindsight, then, what seems to be strategy emerges.

Toyota does not really have any solutions to offer us, but rather a means for us to sense situations and develop appropriate, smart responses. Toyota's executives, managers, and leaders are operating on the basis that organization survival arises from adaptation to unfolding events, on the way to a desired condition. They do not think of good versus bad situations, but of problems as something to be expected and as opportunities to more deeply understand and further develop our work processes. Toyota's strategy for moving toward a vision is target conditions + PDCA; which is to say, the improvement kata. Furthermore, Toyota's executives, managers, and leaders see as perhaps their main task teaching people the improvement kata in a learn-by-doing mode, which will be the subject of Part IV.

A Way of Thinking and Acting

It is important to realize that the improvement kata is about behavior routines. It is a routine of thinking and acting that harnesses our human capability to improve and to solve problems. When we view and interpret what Toyota is doing in this light, it becomes easier to grasp, and we can go further in our own efforts to compete on a similar basis.

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Coaching Is Required

Suppose we would like to teach a team of athletes some new skills and have them do well in competition. In this case, we would certainly not expect that simply explaining a different way of doing something, no matter how well that explaining is done, will be sufficient for altering their behavior. It is not possible to objectively assess one's own performance and see what skills you need to work on. This is because we tend not to perceive our own habits and do not know what we do not know. In the case of athletes, we naturally expect them to practice under the observation and guidance of an experienced coach, and that the necessity of having a coach will not go away. If no one observes them and provides feedback, they can end up internalizing the wrong routines.

It is the same at Toyota, where the improvement kata does not happen automatically or autonomously. Toyota's managers and leaders work hard every day to both teach it and to keep improvement going in an effective manner. Since improvement and adaptation are central to Toyota's business philosophy (normal daily management = process improvement), it is not surprising that managers and leaders there work to ensure that improvement happens. What is surprising, however, is how they go about it.

The Role of Leaders at Toyota

The primary task of Toyota's managers and leaders does not revolve around improvement per se, but around increasing the improvement capability of people. That capability is what, in Toyota's view, strengthens the company. Toyota's managers and leaders develop people who in turn improve processes through the improvement kata.

Developing the improvement capability of people at Toyota is not relegated to the human resources or the training and development departments. It is part of every day's work in every area, and it is managers and supervisors who are expected to teach their people the improvement kata. The improvement kata is part of how people are managed day to day. This means, of course, that the managers and leaders must themselves be experienced in using the improvement kata.

Because the improvement kata is a set of behavioral guidelines, it is something that we learn through repeated practice. It takes conditioning to make behavioral routines become second nature, and consequently a lot of Toyota's managerial activities involve having people practice the improvement kata with their guidance. For team leaders and group leaders, this teaching occupies more than 50 percent of their time, and for higher-level managers it can also occupy up to 50 percent. Developing people in this manner is part of a manager's or supervisor's evaluation, bonus, promotion, and salary.

Training While Working On the Real Thing

We may think of taking classes and seminars to learn, but Toyota thinks of working through actual improvement challenges as how one learns. There are some classroom training courses at Toyota, but to ensure that improvement happens, and that people internalize the improvement kata the primary emphasis is on doing: managers and leaders at Toyota teach people the improvement kata by guiding them in making real improvements in real processes. This approach is not unlike skills training in sports, where "training" means athletes performing an actual activity over and over under the observation and guidance of an experienced coach.

By comparison, in many of our companies the concept of training seems to have devolved to classroom teaching and simulation exercises. Unfortunately, classroom training and simulations cannot ensure change, mastery, and consistency. Classroom training alone, even if it includes simulations, at best only achieves awareness. We can only discern what people are learning and how they are thinking—and hence what they need to learn and practice next—as they try to apply in real life what they are being taught.

Toyota does not make a distinction between learning the improvement kata and improving processes. Toyota's teachers—that is, managers and leaders—observe and work with their students as those students are doing the actual activity day to day. Of course, as you might expect, at Toyota there is a kata for how managers and leaders do this.

The Coaching Kata

Toyota's kata for teaching the improvement kata is a mentor/mentee dialogue, which probably has its roots in the Buddhist master/apprentice teaching method. Like the improvement kata, the coaching kata is often not directly visible to visitors and benchmarks. Yet the development of continuous improvement and adaptation

at Toyota, through application of the improvement kata, has depended to considerable degree on such coaching.

Everyone at Toyota Has a Mentor

Mastery is the objective of any kata, and even people at higher levels in Toyota are honing their skills and working toward that goal. Like the improvement kata, the pattern of the coaching kata is also practiced at all levels throughout Toyota. Each employee is assigned a more experienced employee—a mentor—who provides active guidance through the process of making actual improvements or dealing with work-related situations. That mentor, in turn, has his or her own mentor who is doing the same. It is the buddy system, or two-man rule, with one buddy being the mentor.

These mentor/mentee relationships are not necessarily linked to the organizational hierarchy. For shop-floor operators, their mentor is their team leader, who is in turn supported by the group leader. However, above these levels in the organization you may get mentored not only by your immediate superior,¹ but also by someone else, who may be assigned to you based on your current activities and development needs. Over the years, you will have different mentors.

It easily takes 10 years of practicing the improvement kata and the coaching kata before both become so ingrained that one can be a good mentor. This is one reason why Toyota has in the past avoided bringing managers in from outside the company, and instead preferred to develop them from within. One of Toyota's key challenges while growing rapidly is to have enough mentors.

A person's need for coaching never goes away. Regardless of how much experience one has gained, it is unlikely that anyone can become so good at discerning the reality of a situation and applying the improvement kata that coaching will no longer be necessary. The intention is that both the improvement kata and the coaching kata increasingly become second nature (automatic and reflexive) as a person rises in the organization.

The Mentor/Mentee Dialogue

The mentor (coach) guides the mentee in applying the improvement kata through a back-and-forth dialogue over time, which has sometimes been compared to a game of catch:

Mentor→Mentee→Mentor→Mentee ...

The mentor/mentee dialogue is utilized, for example, when a current situation is being assessed, when a target condition is being developed, and then when the routine of the five questions comes into play.

One key element in Toyota's mentor/mentee dialogue is that the intention is for the mentee to figure things out for him- or herself under guidance, which is a well-known teaching method. The dialogue often begins with the mentor giving the mentee a purposely vague assignment, need, or challenge. For example, the mentor may ask the mentee to have a look at a problem or situation, or suggest something like, "We should wash parts within the one-by-one assembly flow at this process, rather than in batches." The mentor then asks what the mentee proposes. The mentee's answer helps the mentor discern how the mentee is thinking and what input the mentor should give next. That is why the mentor's assignment or challenge is often vague at the start—so he can see how the mentee is approaching the situation.

The mentee gives her observations or proposal, sometimes in bullet form in a single-page document (more on this later). After this is presented to the mentor, it is often pushed back to the mentee in the first cycle, with a suggestion such as, "Please think about this some more," or simply, "Why?" This is only the first of several catch cycles, through which the mentee's analysis and proposal become progressively more developed and detailed.

Once the current situation has been analyzed and the target condition defined and detailed to the mentor's satisfaction, then the mentee's role becomes planning and carrying out PDCA cycles, also with the oversight of the mentor. In doing this, the mentee often has to justify actions to her mentor, and to define in advance the expected result of an action.

The point to remember here is that the mentor is asking questions of the mentee not to direct the mentee to a particular solution—although it may appear that way—but rather to learn what and how the mentee is thinking and how the mentee is approaching the situation. The mentor works to teach the mentee the routine of the improvement kata by providing step-by-step guidance, based on the mentee's reactions and responses along the way. The mentor guides the mentee through the improvement kata, but in a manner that has the mentee learn for him- or herself the routine and thinking inherent in that kata. The highest, though somewhat bittersweet, praise for a mentor is if the mentee feels he or she learned and achieved the target condition independently. ...

A second, and clever, element of mentor/mentee dialogues at Toyota is that while the mentee is responsible for the doing, the mentor bears considerable responsibility for the results but should not give solutions to the mentee. This overlap of responsibility creates a bond between mentor and mentee, because if a mentee fails, then it is the mentor who will get the scrutiny.

If the learner hasn't learned, the teacher hasn't taught.

—Common expression, which is frequently used at Toyota

The mentee is the person who works on the problem, whereas the mentor's task is to keep the mentee "in the corridor" of the improvement kata routine. ... Of course, to be able to guide the mentee in this way, the mentor must also be looking at the situation and often thinking one step ahead of the mentee; but no more than a step ahead. The mentor works to bring the mentee into the behavior corridor prescribed by the improvement kata, but ultimately the mentor must accept the solution that the mentee develops. Leading the mentee to a solution would block the development of the mentee's capability, that is, the purpose of the mentor/mentee dialogue approach.

Although Toyota mentors are usually not directive about solutions, they can, however, be directive about how to go about understanding a situation and developing solutions. For example, once he has heard from the mentee in one cycle, the mentor may, at times, be directive about the next step.

A third element of the mentor/mentee dialogue is that it is not just learning by doing, but that people learn when they make discoveries through small errors. The mentor expects the mentee to make small mistakes in applying the improvement kata, and it is especially at these points that the mentee will learn, and that the mentor can see what coaching input is required. ***In other words, the mentor lets the mentee make small missteps rather than giving the mentee answers up front.***

With the mentor/mentee teaching approach, Toyota has developed a smart answer to the question of how an organization can teach its members the improvement kata and keep improvement going. The mentor/mentee approach has several benefits:

- Leaders can discern how a mentee is thinking and thereby determine the appropriate next step and what skills the mentee needs to practice to become a better problem-solver. We do not recognize people's current skill-development needs when we tell them what to do.
- This approach—learning by doing with trial and error, under guidance—is more effective than relying on written documentation, classroom training, or telling someone what to do for passing on organizational culture—that is, developing specific behavior patterns.

- This approach develops alignment between company goals and workplace behaviors. It provides focus, direction, and control, but with a considerable amount of leeway that helps people develop their own capability. It is not top-down or bottom-up; it is both, simultaneously. The mentor/mentee approach develops individual responsibility and initiative while also providing common direction and approach. This is what I mean when I refer to “operating in the desired corridor.”
- The needs of the mentee and the situation determine the next mentoring and training that the mentor provides. This means that information is flowing both down and up in the organization. As a result, strategic decisions can be more in sync with the actual situation at the process level.

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CONCLUSION

.... Toyota’s improvement kata and coaching kata are largely invisible when we benchmark Toyota. Yet these two kata play perhaps the major role in Toyota’s ability to achieve ambitious targets, keep improving, and adapt. I have worked with these kata extensively now and I am intrigued by their capacity to help us move through the unpredictable paths ahead and achieve beyond what we can see. When you look behind the curtain at how Toyota manages itself, you realize that Toyota has achieved not only a commercial but also an intellectual accomplishment....

Permanent pressure to adapt can keep an organization fit and healthy, if it has a systematic way—a kata—of responding to that pressure. This book does not describe everything about Toyota, but it provides more than enough information and detail for you to begin developing—through experimentation and practice—your own continuous improvement system like Toyota’s. ***You can even see your organization as part of human history through your efforts to bring continuous improvement and adaptation into it. This is because each step in that direction not only benefits your company, it also helps move our society forward because it mobilizes our capability.***

Does the way ahead for developing improvement kata behavior in your organization seem unclear? Are you unsure about what you will need to do to achieve successful culture change? Well, that is exactly how it should be, and if so, I can assure you that you are already on the right track. We cannot know what the path ahead will be, but the improvement kata shows us a way to deal with and perhaps even enjoy that unpredictable aspect of life. That latter sentiment is my wish for all of us, and, with that in mind, I will end with a question:

What is your improvement kata?

THE COMPANY

Micklethwait, John; Wooldridge, Adrian. *The Company: A Short History of a Revolutionary Idea* (Modern Library Chronicles Series Book 12). Random House Publishing Group. Kindle Edition.

RELEVANCE:

Micklethwait's 2003 *History of the Revolutionary Idea of the Company* correctly emphasize the role of the company in creating modernity. Micklethwait fails to make the anthropological connection to group competition – he does a shallow history in this way. But he does accurately identify off the bat the two key group forms in modern human societies: families and companies. In an oversimplified but roughly accurate form: families are the units of consumption in the West; companies are the units of production. This is a different alignment of production and consumption than human societies prior to 1900 and outside of the West adhere to.

The elements of what makes a company are also essential for us to remember – because the failure of companies in many countries to actually legally be differentiated from the state means that they aren't actually companies in Micklethwait's definition – and this is the issue which we have been reminded of in Russia and China recently.

Note that because Micklethwait doesn't reach all the way back to anchor the company as a form of group competition, when he gets to the future of the company or where its going, I would argue he gets it wrong – so the book is a good recent history of a key form of group competition which plays an essential role in modern western societies. You can feel Micklethwait's lack of the anthropological lens of why companies appeal to humans in the final paragraph of the introduction – he suggests a mystery where none exists: companies form winning groups in an economy and people will always fundamentally desire to be on the winning team. That human agency and human will and human desire, anchored in what we are (anthropology) is the key element of why companies work that Micklethwait misses – but he pretty much nails the rest of it.

Below is part of the introduction which captures the general message of the book.

INTRODUCTION

.... Nowadays, the influence of this unsettling organization is even more pervasive. Hegel predicted that the basic unit of modern society would be the state, Marx that it would be the commune, Lenin and Hitler that it would be the political party. Before that, a succession of saints and sages claimed the same for the parish church, the feudal manor, and the monarchy. The big contention of this small book is that they have all been proved wrong. The most important organization in the world is the company: the basis of the prosperity of the West and the best hope for the future of the rest of the world. Indeed, for most of us, the company's only real rival for our time and energy is the one that is taken for granted—the family. (Meanwhile, in a nice reversal of fortune, the world's best-known family, the British monarchy, on whose whims and favors many of the earliest English joint-stock companies depended, now refers to itself as “the firm.”) That does not mean that the role of the company has been appreciated, least of all by political historians. The great Companies Acts of the mid-nineteenth century get barely a sentence in most recent biographies of William Gladstone, one of their political champions; the intellectual godfather of the modern company, Robert Lowe, is more remembered for his work on education and his grumpy opposition to universal suffrage. The relevant volume of the *New Oxford History of England* that covers 1846 to 1886 does not find room to discuss the invention of the company in its 720 pages.

In fact, the history of the company is an extraordinary tale. What often began as a state-sponsored charity has sprawled into all sorts of fields, reconfiguring geography, warfare, the arts, science, and, sadly, the language. Companies have proved enormously powerful not just because they improve productivity, but also because they possess most of the legal rights of a human being, without the attendant disadvantages of biology: they are not condemned to die of old age and they can create progeny pretty much at will. This privilege of immortality, not to mention the protection that the artificial corporate form has afforded various venal people down the ages, has often infuriated the rest of society—particularly governments. Hence, there has been a lengthy series of somewhat bad-tempered laws trying to tamper with the concession—from the Statute of Mortmain, which Edward I issued in 1279 to stem the flow of assets being transferred beyond his feudal writ to the “dead hand” of corporate bodies (particularly monasteries), to the 2002 Sarbanes-Oxley Act, through which Congress tried to make bosses more accountable for the sins of “their” companies.

There are two ways to define a company. The first is merely as an organization engaged in business: this definition, as we shall see, includes everything from informal Assyrian trading arrangements to modern leveraged buyouts. The second is more specific: the limited-liability joint-stock company is a distinct legal entity (so distinct, in fact, that its shareholders can sue it), endowed by government with certain collective rights and responsibilities. This was the institution that the Utopians’ “Astonishing Fact,” the Companies Act of 1862, unleashed, and which is still spreading around the world, conquering such obstinate refuseniks as the Chinese Communist Party and the partners of Goldman Sachs.

Though this is primarily a book about the joint-stock company, it unapologetically strays into broader territory. ***From the beginning of economic life, businesspeople have looked for ways to share the risks and rewards of their activities.*** One of the fundamental ideas of medieval law was that “bodies corporate”—towns, universities, guilds—had a life beyond that of their members. In the sixteenth and seventeenth centuries, European monarchs created chartered companies to pursue their dreams of imperial expansion. One of these, the East India Company, wound up ruling India with a private army of 260,000 native troops (twice the size of the British army). Another, the Virginia Company, helped to introduce the revolutionary concept of democracy to the American colonies, to the fury of James I, who called it “a seminary for a seditious parliament.” Yet another, John Law’s Mississippi Company, wrecked the economy of France, Europe’s richest country in the eighteenth century.

Yet William Gilbert was right to think that something fundamental changed in nineteenth-century Britain. ***The most powerful economic power of the day finally brought together the three big ideas behind the modern company: that it could be an “artificial person,” with the same ability to do business as a real person; that it could issue tradable shares to any number of investors; and that those investors could have limited liability (so they could lose only the money they had committed to the firm).*** Just as important, the Victorians changed the point of companies. It was no longer necessary to seek special sanction from parliament to set one up or to limit its business to a specific worthy aim (like building a railway between two cities); now it was possible to set up general purpose corporations at the drop of a hat. All that was necessary was for seven people (“If possible, all Peers and Baronets,” Goldbury mischievously advised the Utopians) to sign a memorandum of association for the company to be registered and for it to use the word “limited” to warn creditors that they would have no recourse to the company’s owners.

The Companies Acts, which were rapidly copied in other countries, unleashed entrepreneurs to raise money, safe in the knowledge that investors could lose only what they had put in. They also gave birth to an organization that soon seemed to acquire a life of its own, swiftly mutating from one shape to another, with government usually unable to restrain it. Nowadays, nobody finds it odd that, a century after its foundation, the

Minnesota Mining and Manufacturing Company makes Post-it notes, or that the world's biggest mobile-phone company, Nokia, used to be in the paper business.

The Victorians also gave us many of the most profound arguments that swirl around companies. Nowadays it is assumed that the causes of capitalism and companies are inseparable. Yet many of the earliest critics of the joint-stock company and the “subsidy” of limited liability were economic liberals, taking their cue from Adam Smith, who had derided them as antiquated and inefficient. One noted Victorian thinker, A. V. Dicey, fretted that the company would become the harbinger of a new age of collectivism: “one trade after another” would pass from the “management of private persons into the hands of corporate bodies created by the state.” (Karl Marx gave a grudging welcome to companies for much the same reason.)

For the company's early critics, it was not just a question of allowing investors to repudiate responsibility for their debts; many Victorian liberals also worried whether professional managers could be trusted to act in the interests of the owner shareholders. They had a point: the potential conflict of interest between the “principals” who own companies and their “agents,” who run them, which was later dubbed the ***agency problem***, has dogged the history of the company, from the mills of Lancashire to software start-ups in Palo Alto, with shareholders repeatedly trying to find ways to make managers' interests the same as their own (most recently with share options) and managers usually wriggling out of them. John Stuart Mill settled his own doubts on this score only by wearily concluding that for new capital-hungry businesses, like railways, the only alternative to the joint-stock system was direct state control.

Even after the Companies Acts, Victorians were still prey to the traditional cultural prejudices against these soulless institutions. The Morning Post ran a xenophobic campaign against the railway companies on the grounds that they were exporting British jobs. At the same time, American populists denounced the very same companies on the grounds that the British were trying to recolonize the country by stealth. In Anthony Trollope's *The Way We Live Now* (1875), a company, supposed to build a great railway linking Mexico and the United States, is hijacked by an unscrupulous continental financier, Augustus Melmotte, and his American partner, Hamilton K. Fisk. Its board, consisting of know-nothing aristocrats and unscrupulous politicians, meets for a perfunctory fifteen minutes. (“There was not one of them then present who had not after some fashion been given to understand that his fortune was to be made, not by the construction of the railway, but by the floating of the railway shares.”) And the whole enterprise predictably results in a speculative crash.

In America, where the influence of companies was greatest, the howls against “the malefactors of great wealth” reached a crescendo at the beginning of the twentieth century. Yet no sooner had society reined in the robber barons than it discovered that a still less accountable villain had seized control of the company: the faceless manager. ***The rise and fall of the juggernauts of corporate America forms a large part of our story.*** Of course, not everybody worked for them—but it sure felt that way. Until 1975, the big American corporation was the model against which all other sorts of company were measured. Yet, since then, Company Man, too, has been chased out. Companies have become flatter, less hierarchical organizations.

Throughout the twentieth century, the company jostled with the state that spawned it. European and Asian governments tried to run companies of their own—and failed spectacularly. Many on the Left would argue that companies tried to set up governments of their own—and succeeded equally spectacularly. Meanwhile, the ways that companies have subtly influenced our lives have multiplied. It was a company—Lever Brothers—that introduced us to the concept of “BO.” (“It was not enough to produce satisfactory soap,” Joseph Schumpeter once observed. “It was also necessary to induce people to wash.”) It is a company—McDonald's—that is credited with teaching the Chinese how to queue.

Three themes stand out in our story. First, the company's past is often more dramatic than its present. Modern business books may have macho titles such as *Barbarians at the Gate* and *Only the Paranoid Survive*, but early businessmen took risks with their lives as well as their fortunes. Send a fleet to the Spice Islands at the beginning of the seventeenth century, and you might be lucky if a third of the men came back alive. This was a time when competitive advantage meant blowing your opponents out of the water, when marketing meant supplying an English rose for the sultan's harem (a London merchant of "honorable parentage" selflessly offered his daughter), and when your suppliers might put your head on a stick.

The second point is to some extent a correlation of the first. In general, companies have become more ethical: more honest, more humane, more socially responsible. The early history of companies was often one of imperialism and speculation, of appalling rip-offs and even massacres. People who now protest about the new evil of global commerce plainly have not read much about slavery or opium. People who talk in terrified tones about the unprecedented skulduggery at WorldCom seem to have forgotten about the South Sea Bubble. Those who fear the unparalleled might of Bill Gates could do with a little reading on J. P. Morgan. Today, the number of private-sector companies that a country boasts—the United States had 5½ million corporations in 2001, North Korea, as far as we can tell, none—is a better guide to its status than the number of battleships it can muster. It is also not a bad guide to its political freedom.

This leads to the third point. The company has been one of the West's great competitive advantages. Of course, the West's success owes much to technological prowess and liberal values. But Lowe and Gladstone ushered in an organization that has been uniquely effective in rendering human effort productive. ***The idea that the company itself was an enabling technology is something that liberal thinkers once understood instinctively.*** "The limited liability corporation is the greatest single discovery of modern times," proclaimed Nicholas Murray Butler, one of the great sages of the Progressive Era; "even steam and electricity would be reduced to comparative impotence without it."

Economists have elaborated on why such institutions are crucial to economic development. Companies increase the pool of capital available for productive investment. They allow investors to spread their risk by purchasing small and easily marketable shares in several enterprises. And they provide a way of imposing effective management structures on large organizations. Of course, companies can ossify, but the fact that investors can simply put their money elsewhere is a powerful rejuvenator.

A cluster of competing companies makes for a remarkably innovative economy. Nowadays, you only have to look at Silicon Valley to grasp this point. But in the mid-nineteenth century, the effect of Western governments delegating key decisions about which ideas to back to independent firms was revolutionary. Rather than being trapped in government monopolies, capital began to search for the most efficient and flexible companies; and rather than being limited by family partnerships, it came together in bigger and bigger conglomerations. By contrast, civilizations that once outstripped the West yet failed to develop private-sector companies—notably China and the Islamic world—fell farther and farther behind. It cannot be just coincidence that Asia's most conspicuous economic success is also the country that most obviously embraced companies—Japan.

This book is an attempt to chart the rise of this remarkable institution. But we have also taken the liberty to spend a little time puzzling about its future. At first sight, that future should be assured. Nation-states are on the defensive. Churches are struggling to find recruits. Trade unions are a mere shadow of their former selves. But companies are going from strength to strength. Most people in the West now work for companies, which also produce the bulk of the world's products. Any young Napoleon who yearns for the scent of global conquest would be better off joining a company than running for political office or joining the army.

Yet, the company is much less powerful than it seems. Although the influence of companies as a species has never been more widespread, the clout of individual big companies has arguably declined. The much-vaunted idea that companies are now bigger than mere governments is, as we shall see, statistically fraudulent. Big companies are giving way to small ones, so much so, in fact, that an old question is now more pressing:

What is the point of companies?

That question was most succinctly answered back in 1937 by Ronald Coase, a young British economist. In an article called “The Nature of the Firm,” he argued that the main reason why a company exists (as opposed to individual buyers and sellers making ad hoc deals at every stage of production) is because it minimizes the transaction costs of coordinating a particular economic activity. Bring all the people in-house, and you reduce the costs of “negotiating and concluding a separate contract for each exchange transaction.”

But the gains from reducing transaction costs that companies deliver have to be balanced against “hierarchy costs”—the costs of central managers ignoring dispersed information. In the nineteenth century, the gains to be had from integrating mass production with mass distribution were enormous—as Alfred Chandler, the doyen of business historians, puts it, the “visible hand of managerial direction” replaced “the invisible hand of market mechanisms.” In the twenty-first century, technology and globalization are helping to reduce barriers to entry—and thus helping to unbundle the corporate package. At the touch of a button, a mere journalist can get access to more information than a corporate giant could amass a decade ago. The fashion nowadays is for virtual companies—for airlines that do not own their own planes, for banks that do not have branches, for the invisible hand to claw back ground from the visible one.

That should not imply that the company is beginning a slow, inevitable decline. Despite the seductive charm of frictionless capitalism, most people seem to like being in companies. (We should admit that through luck, absence of opportunity, laziness, and, especially, the charity of others, we have both remained at the same organization for most of our working lives.) The economic argument has also deepened since Coase, with some economists preferring to look at the firm as a network of contracts and others seeing it as a bundle of organizational capabilities. But ***the basic questions being asked by modern investors, managers, and workers—What does this company do? Why do I work here? Will it make money? —are worth remembering*** as we head back into the past.

GREAT MENTAL MODELS VOLUME 1

Parrish, Shane; Beaubien, Rhiannon. The Great Mental Models Volume 1: General Thinking Concepts . Latticework Publishing Inc.. Kindle Edition.

via Shane Parrish & Farnam Street

RELEVANCE:

The introduction from Shane Parrish’s series of books on Mental Models is below. While some of the language is cliché and makes me cringe – the general logic is all solid. Starting from addressing blind spots (which is what the David Dunning reading is about) to the idea of seeing reality (I wouldn’t call it “reality” – because that implies a whole host of other things, it’s really just being “pragmatic” – or what works). Parrish is sloppy and uses cliché in many places where there is precise language – so what he is saying is wrong, but its intent is right.

Notice the focus on interconnections is logically identical to complexity theory, and the need to shift perspectives is core to why we work as teams and central to our work.

Acquiring Wisdom

In life and business, the person with the fewest blind spots wins. Removing blind spots means we see, interact with, and move closer to understanding reality. We think better. And thinking better is about finding simple processes that help us work through problems from multiple dimensions and perspectives, allowing us to better choose solutions that fit what matters to us. The skill for finding the right solutions for the right problems is one form of wisdom.

This book is about the pursuit of that wisdom, the pursuit of uncovering how things work, the pursuit of going to bed smarter than when we woke up. It is a book about getting out of our own way so we can understand how the world really is. Decisions based on improved understanding will be better than ones based on ignorance. While we can’t predict which problems will inevitably crop up in life, we can learn time-tested ideas that help us prepare for whatever the world throws at us.

Perhaps more importantly, this book is about avoiding problems. This often comes down to understanding a problem accurately and seeing the secondary and subsequent consequences of any proposed action. The author and explorer of mental models, Peter Bevelin, put it best: “I don’t want to be a great problem solver. I want to avoid problems—prevent them from happening and doing it right from the beginning.”

How can we do things right from the beginning?

We must understand how the world works and adjust our behavior accordingly. Contrary to what we’re led to believe, thinking better isn’t about being a genius. It is about the processes we use to uncover reality and the choices we make once we do.

How this book can help you

This is the first of a series of volumes aimed at defining and exploring the Great Mental Models—those that have the broadest utility across our lives. Mental models describe the way the world works. They shape how we think,

how we understand, and how we form beliefs. Largely subconscious, mental models operate below the surface. We're not generally aware of them and yet they're the reason when we look at a problem we consider some factors relevant and others irrelevant. They are how we infer causality, match patterns, and draw analogies. They are how we think and reason.

A mental model is simply a representation of how something works. We cannot keep all of the details of the world in our brains, so we use models to simplify the complex into understandable and organizable chunks. Whether we realize it or not, we then use these models every day to think, decide, and understand our world. While there are millions of mental models, some true and some false, these volumes will focus on the ones with the greatest utility—the all-star team of mental models.

Volume One presents the first nine models, general thinking concepts. Although these models are hiding in plain sight, they are useful tools that you were likely never directly taught. Put to proper use, they will improve your understanding of the world we live in and improve your ability to look at a situation through different lenses, each of which reveals a different layer. They can be used in a wide variety of situations and are essential to making rational decisions, even when there is no clear path. Collectively they will allow you to walk around any problem in a three-dimensional way.

Our approach to the Great Mental Models rests on the idea that the fundamentals of knowledge are available to everyone. There is no discipline that is off limits—the core ideas from all fields of study contain principles that reveal how the universe works, and are therefore essential to navigating it. Our models come from fundamental disciplines that most of us have never studied, but no prior knowledge is required—only a sharp mind with a desire to learn.

Why mental models?

There is no system that can prepare us for all risks. Factors of chance introduce a level of complexity that is not entirely predictable. But being able to draw on a repertoire of mental models can help us minimize risk by understanding the forces that are at play. Likely consequences don't have to be a mystery.

Not having the ability to shift perspective by applying knowledge from multiple disciplines makes us vulnerable. Mistakes can become catastrophes whose effects keep compounding, creating stress and limiting our choices. Multidisciplinary thinking, learning these mental models and applying them across our lives, creates less stress and more freedom. The more we can draw on the diverse knowledge contained in these models, the more solutions will present themselves.

Understanding reality

Understanding reality is a vague phrase, one you've already encountered as you've read this book. Of course we want to understand reality, but how? And why is it important?

In order to see a problem for what it is, we must first break it down into its substantive parts so the interconnections can reveal themselves. This bottom-up perspective allows us to expose what we believe to be the causal relationships and how they will govern the situation both now and in the future. Being able to accurately describe the full scope of a situation is the first step to understanding it.

Using the lenses of our mental models helps us illuminate these interconnections. The more lenses used on a given problem, the more of reality reveals itself. The more of reality we see, the more we understand. The more we understand, the more we know what to do.

Simple and well-defined problems won't need many lenses, as the variables that matter are known. So too are the interactions between them. In such cases we generally know what to do to get the intended results with the

fewest side effects possible. When problems are more complicated, however, the value of having a brain full of lenses becomes readily apparent.

That's not to say all lenses (or models) apply to all problems. They don't. And it's not to say that having more lenses (or models) will be an advantage in all problems. It won't. This is why learning and applying the Great Mental Models is a process that takes some work. But the truth is, most problems are multidimensional, and thus having more lenses often offers significant help with the problems we are facing.

Keeping your feet on the ground

In Greek mythology, Antaeus was the human-giant son of Poseidon, god of the sea, and Gaia, Mother Earth. Antaeus had a strange habit. He would challenge all those who passed through his country to a wrestling match. Greek wrestling isn't much different from what we think of today when we think of wrestling. The goal is to force the opponent to the ground. Antaeus always won and his opponents' skulls were used to build a temple to his father. While Antaeus was undefeated and nearly undefeatable, there was a catch to his invulnerability. His epic strength depended on constant contact with the earth. When he lost touch with earth, he lost all of his strength.

On the way to the Garden of the Hesperides, Heracles was to fight Antaeus as one of his 12 labors. After a few rounds in which Heracles flung the giant to the ground only to watch him revive, he realized he could not win by using traditional wrestling techniques. Instead, Heracles fought to lift him off the ground. Away from contact with his mother, Antaeus lost his strength and Heracles crushed him.

When understanding is separated from reality, we lose our powers. Understanding must constantly be tested against reality and updated accordingly. This isn't a box we can tick, a task with a definite beginning and end, but a continuous process.

You all know the person who has all the answers on how to improve your organization, or the friend who has the cure to world hunger. While pontificating with friends over a bottle of wine at dinner is fun, it won't help you improve. The only way you'll know the extent to which you understand reality is to put your ideas and understanding into action. ***If you don't test your ideas against the real world—keep contact with the earth—how can you be sure you understand?***

Getting in our own way

The biggest barrier to learning from contact with reality is ourselves. It's hard to understand a system that we are part of because we have blind spots, where we can't see what we aren't looking for, and don't notice what we don't notice.

There are these two young fish swimming along and they happen to meet an older fish swimming the other way, who nods at them and says "Morning, boys. How's the water?" And the two young fish swim on for a bit, and then eventually one of them looks over at the other and goes "What the hell is water?"

- David Foster Wallace

Our failures to update from interacting with reality spring primarily from three things: not having the right perspective or vantage point, ego-induced denial, and distance from the consequences of our decisions. As we will learn in greater detail throughout the volumes on mental models, these can all get in the way. They make it easier to keep our existing and flawed beliefs than to update them accordingly. Let's briefly flesh these out:

The first flaw is perspective. ***We have a hard time seeing any system that we are in.*** Galileo had a great analogy to describe the limits of our default perspective. Imagine you are on a ship that has reached constant velocity

(meaning without a change in speed or direction). You are below decks and there are no portholes. You drop a ball from your raised hand to the floor. To you, it looks as if the ball is dropping straight down, thereby confirming gravity is at work.

Now imagine you are a fish (with special x-ray vision) and you are watching this ship go past. You see the scientist inside, dropping a ball. You register the vertical change in the position of the ball. But you are also able to see a horizontal change. As the ball was pulled down by gravity it also shifted its position east by about 20 feet. The ship moved through the water and therefore so did the ball. The scientist on board, with no external point of reference, was not able to perceive this horizontal shift.

This analogy shows us the limits of our perception. We must be open to other perspectives if we truly want to understand the results of our actions. Despite feeling that we've got all the information, if we're on the ship, the fish in the ocean has more he can share.

The second flaw is ego. Many of us tend to have too much invested in our opinions of ourselves to see the world's feedback—the feedback we need to update our beliefs about reality. This creates a profound ignorance that keeps us banging our head against the wall over and over again. Our inability to learn from the world because of our ego happens for many reasons, but two are worth mentioning here. First, we're so afraid about what others will say about us that we fail to put our ideas out there and subject them to criticism. This way we can always be right. Second, if we do put our ideas out there and they are criticized, our ego steps in to protect us. ***We become invested in defending instead of upgrading our ideas.***

The third flaw is distance. The further we are from the results of our decisions, the easier it is to keep our current views rather than update them. When you put your hand on a hot stove, you quickly learn the natural consequence. You pay the price for your mistakes. Since you are a pain-avoiding creature, you update your view. Before you touch another stove, you check to see if it's hot. But you don't just learn a micro lesson that applies in one situation. Instead, you draw a general abstraction, one that tells you to check before touching anything that could potentially be hot.

Organizations over a certain size often remove us from the direct consequences of our decisions. When we make decisions that other people carry out, we are one or more levels removed and may not immediately be able to update our understanding. We come a little off the ground, if you will. The further we are from the feedback of the decisions, the easier it is to convince ourselves that we are right and avoid the challenge, the pain, of updating our views.

Admitting that we're wrong is tough. It's easier to fool ourselves that we're right at a high level than at the micro level, because at the micro level we see and feel the immediate consequences. When we touch that hot stove, the feedback is powerful and instantaneous. At a high or macro level we are removed from the immediacy of the situation, and our ego steps in to create a narrative that suits what we want to believe, instead of what really happened.

These flaws are the main reasons we keep repeating the same mistakes, and why we need to keep our feet on the ground as much as we can. As Confucius said, "A man who has committed a mistake and doesn't correct it, is committing another mistake."

The majority of the time, we don't even perceive what conflicts with our beliefs. It's much easier to go on thinking what we've already been thinking than go through the pain of updating our existing, false beliefs. When it comes to seeing what is—that is, understanding reality—we can follow Charles Darwin's advice to notice things "which easily escape attention," and ask why things happened.

We also tend to undervalue the elementary ideas and overvalue the complicated ones. Most of us get jobs based on some form of specialized knowledge, so this makes sense. We don't think we have much value if we know the things everyone else does, so we focus our effort on developing unique expertise to set ourselves apart. The problem is then that we reject the simple to make sure what we offer can't be contributed by someone else. But simple ideas are of great value because they can help us prevent complex problems.

In identifying the Great Mental Models we have looked for elementary principles, the ideas from multiple disciplines that form a time-tested foundation. It may seem counterintuitive, to work on developing knowledge that is available to everyone, but the universe works in the same way no matter where you are in it. What you need is to understand the principles, so that when the details change you are still able to identify what is really going on. This is part of what makes the Great Mental Models so valuable—understanding the principles, you can easily change tactics to apply the ones you need.

Most geniuses—especially those who lead others—prosper not by deconstructing intricate complexities but by exploiting unrecognized simplicities.

- Andy Benoit

These elementary ideas, so often overlooked, are from multiple disciplines: biology, physics, chemistry, and more. These help us understand the interconnections of the world, and see it for how it really is. This understanding allows us to develop causal relationships, which allow us to match patterns, which allow us to draw analogies. All of this so we can navigate reality with more clarity and comprehension of the real dynamics involved.

Understanding is not enough

However, understanding reality is not everything. The pursuit of understanding fuels meaning and adaptation, but this understanding, by itself, is not enough. Understanding only becomes useful when we adjust our behavior and actions accordingly. The Great Models are not just theory. They are actionable insights that can be used to effect positive change in your life. What good is it to know that you constantly interrupt people if you fail to adjust your behavior in light of this? In fact, if you know and don't change your behavior it often has a negative effect. People around you will tell themselves the simplest story that makes sense to them given what they see: that you just don't care. Worse still, because you understand that you interrupt people, you're surprised when you get the same results over and over. Why? You've failed to reflect on your new understanding and adjust your behavior.

In the real world you will either understand and adapt to find success or you will fail.

Now you can see how we make suboptimal decisions and repeat mistakes. We are afraid to learn and admit when we don't know enough. This is the mindset that leads to poor decisions. They are a source of stress and anxiety, and consume massive amounts of time. Not when we're making them—no, when we're making them they seem natural because they align with our view of how we want things to work. We get tripped up when the world doesn't work the way we want it to or when we fail to see what is. Rather than update our views, we double down our effort, accelerating our frustrations and anxiety. It's only weeks or months later, when we're spending massive amounts of time fixing our mistakes, that they start to increase their burden on us. Then we wonder why we have no time for family and friends and why we're so consumed by things outside of our control.

We are passive, thinking these things just happened to us and not that we did something to cause them. This passivity means that we rarely reflect on our decisions and the outcomes. Without reflection we cannot learn.

Without learning we are doomed to repeat mistakes, become frustrated when the world doesn't work the way we want it to, and wonder why we are so busy. The cycle goes on.

But we are not passive participants in our decisions. ***The world does not act on us as much as it reveals itself to us and we respond.*** Ego gets in the way, locking reality behind a door that it controls with a gating mechanism. Only through persistence in the face of having it slammed on us over and over can we begin to see the light on the other side.

Ego, of course, is more than the enemy. It's also our friend. If we had a perfect view of the world and made decisions rationally, we would never attempt to do the amazing things that make us human. Ego propels us. Why, without ego, would we even attempt to travel to Mars? After all, it's never been done before. We'd never start a business because most of them fail. We need to learn to understand when ego serves us and when it hinders us. Wrapping ego up in outcomes instead of in ourselves makes it easier to update our views.

We optimize for short-term ego protection over long-term happiness. Increasingly, our understanding of things becomes black and white rather than shades of grey. When things happen in accord with our view of the world we naturally think they are good for us and others. When they conflict with our views, they are wrong and bad. ***But the world is smarter than we are and it will teach us all we need to know if we're open to its feedback***—if we keep our feet on the ground.

Mental models and how to use them

Perhaps an example will help illustrate the mental models approach. Think of gravity, something we learned about as kids and perhaps studied more formally in university as adults. We each have a mental model about gravity, whether we know it or not. And that model helps us to understand how gravity works. Of course we don't need to know all of the details, but we know what's important. We know, for instance, that if we drop a pen it will fall to the floor. If we see a pen on the floor we come to a probabilistic conclusion that gravity played a role.

This model plays a fundamental role in our lives. It explains the movement of the Earth around the sun. It informs the design of bridges and airplanes. It's one of the models we use to evaluate the safety of leaning on a guard rail or repairing a roof. But we also apply our understanding of gravity in other, less obvious ways. We use the model as a metaphor to explain the influence of strong personalities, as when we say, "He was pulled into her orbit." This is a reference to our basic understanding of the role of mass in gravity—the more there is the stronger the pull. It also informs some classic sales techniques. Gravity diminishes with distance, and so too does your propensity to make an impulse buy. Good salespeople know that the more distance you get, in time or geography, between yourself and the object of desire, the less likely you are to buy. Salespeople try to keep the pressure on to get you to buy right away.

Gravity has been around since before humans, so we can consider it to be time-tested, reliable, and representing reality. And yet, can you explain gravity with a ton of detail? I highly doubt it. And you don't need to for the model to be useful to you. Our understanding of gravity, in other words, our mental model, lets us anticipate what will happen and also helps us explain what has happened. We don't need to be able to describe the physics in detail for the model to be useful.

However, not every model is as reliable as gravity, and all models are flawed in some way. Some are reliable in some situations but useless in others. Some are too limited in their scope to be of much use. Others are unreliable because they haven't been tested and challenged, and yet others are just plain wrong. In every situation, we need to figure out which models are reliable and useful. We must also discard or update the unreliable ones, because unreliable or flawed models come with a cost.

For a long time people believed that bloodletting cured many different illnesses. This mistaken belief actually led doctors to contribute to the deaths of many of their patients. **When we use flawed models we are more likely to misunderstand the situation, the variables that matter, and the cause and effect relationships between them.** Because of such misunderstandings we often take suboptimal actions, like draining so much blood out of patients that they die from it.

Better models mean better thinking. The degree to which our models accurately explain reality is the degree to which they improve our thinking. Understanding reality is the name of the game. Understanding not only helps us decide which actions to take but helps us remove or avoid actions that have a big downside that we would otherwise not be aware of. Not only do we understand the immediate problem with more accuracy, but we can begin to see the second-, third-, and higher-order consequences. This understanding helps us eliminate avoidable errors. Sometimes making good decisions boils down to avoiding bad ones.

Flawed models, regardless of intentions, cause harm when they are put to use. When it comes to applying mental models we tend to run into trouble either when our model of reality is wrong, that is, it doesn't survive real world experience, or when our model is right and we apply it to a situation where it doesn't belong.

Models that don't hold up to reality cause massive mistakes. Consider that the model of bloodletting as a cure for disease caused unnecessary death because it weakened patients when they needed all their strength to fight their illnesses. It hung around for such a long time because it was part of a package of flawed models, such as those explaining the causes of sickness and how the human body worked, that made it difficult to determine exactly where it didn't fit with reality.

We compound the problem of flawed models when we fail to update our models when evidence indicates they are wrong. Only by repeated testing of our models against reality and being open to feedback can we update our understanding of the world and change our thinking. We need to look at the results of applying the model over the largest sample size possible to be able to refine it so that it aligns with how the world actually works.

The power of acquiring new models

The quality of our thinking is largely influenced by the mental models in our heads. While we want accurate models, we also want a wide variety of models to uncover what's really happening. The key here is variety. Most of us study something specific and don't get exposure to the big ideas of other disciplines. We don't develop the multidisciplinary mindset that we need to accurately see a problem. And because we don't have the right models to understand the situation, we overuse the models we do have and use them even when they don't belong.

You've likely experienced this first hand. An engineer will often think in terms of systems by default. A psychologist will think in terms of incentives. A business person might think in terms of opportunity cost and risk-reward. Through their disciplines, each of these people sees part of the situation, the part of the world that makes sense to them. None of them, however, see the entire situation unless they are thinking in a multidisciplinary way. In short, they have blind spots. Big blind spots. And they're not aware of their blind spots. There is an old adage that encapsulates this: "To the man with only a hammer, everything starts looking like a nail." Not every problem is a nail. **The world is full of complications and interconnections that can only be explained through understanding of multiple models.**

What Can the Three Buckets of Knowledge Teach Us About History?

"Every statistician knows that a large, relevant sample size is their best friend. What are the three largest, most relevant sample sizes for identifying universal principles? Bucket number one is inorganic systems, which are 13.7 billion years in size. It's all the laws of math and physics, the

entire physical universe. Bucket number two is organic systems, 3.5 billion years of biology on Earth. And bucket number three is human history, you can pick your own number, I picked 20,000 years of recorded human behavior. Those are the three largest sample sizes we can access and the most relevant.” —Peter Kaufman

The larger and more relevant the sample size, the more reliable the model based on it is. But the key to sample sizes is to look for them not just over space, but over time. You need to reach back into the past as far as you can to contribute to your sample. We have a tendency to think that how the world is, is how it always was. And so we get caught up validating our assumptions from what we find in the here and now. But the continents used to be pushed against each other, dinosaurs walked the planet for millions of years, and we are not the only hominid to evolve. Looking to the past can provide essential context for understanding where we are now.

Removing blind spots means thinking through the problem using different lenses or models. When we do this the blind spots slowly go away and we gain an understanding of the problem.

We’re much like the blind men in the classic parable of the elephant, going through life trying to explain everything through one limited lens. Too often that lens is driven by our particular field, be it economics, engineering, physics, mathematics, biology, chemistry, or something else entirely. Each of these disciplines holds some truth and yet none of them contain the whole truth.

Here’s another way to look at it: think of a forest. When a botanist looks at it they may focus on the ecosystem, an environmentalist sees the impact of climate change, a forestry engineer the state of the tree growth, a business person the value of the land. None are wrong, but neither are any of them able to describe the full scope of the forest. Sharing knowledge, or learning the basics of the other disciplines, would lead to a more well-rounded understanding that would allow for better initial decisions about managing the forest.

Relying on only a few models is like having a 400-horsepower brain that’s only generating 50 horsepower of output. To increase your mental efficiency and reach your 400-horsepower potential, you need to use a latticework of mental models. Exactly the same sort of pattern that graces backyards everywhere, a lattice is a series of points that connect to and reinforce each other. The Great Models can be understood in the same way—models influence and interact with each other to create a structure that can be used to evaluate and understand ideas.

In a famous speech in the 1990s, Charlie Munger summed up this approach to practical wisdom: “Well, the first rule is that you can’t really know anything if you just remember isolated facts and try and bang ‘em back. If the facts don’t hang together on a latticework of theory, you don’t have them in a usable form. You’ve got to have models in your head. And you’ve got to array your experience both vicarious and direct on this latticework of models. You may have noticed students who just try to remember and pound back what is remembered. Well, they fail in school and in life. You’ve got to hang experience on a latticework of models in your head.”

The chief enemy of good decisions is a lack of sufficient perspectives on a problem.

- Alain de Botton

Expanding your latticework of mental models

A latticework is an excellent way to conceptualize mental models, because it demonstrates the reality and value of interconnecting knowledge. The world does not isolate itself into discrete disciplines. We only break it down that way because it makes it easier to study it. But once we learn something, we need to put it back into the

complex system in which it occurs. We need to see where it connects to other bits of knowledge, to build our understanding of the whole. This is the value of putting the knowledge contained in mental models into a latticework.

It reduces the blind spots that limit our view of not only the immediate problem, but the second and subsequent order effects of our potential solutions. Without a latticework of the Great Models our decisions become harder, slower, and less creative. But by using a mental models approach, we can complement our specializations by being curious about how the rest of the world works. A quick glance at the Nobel Prize winners list show that many of them, obviously extreme specialists in something, had multidisciplinary interests that supported their achievements.

To help you build your latticework of mental models, this book, and the books that follow, attempt to arm you with the big models from multiple disciplines. We'll take a look at biology, physics, chemistry, economics, and even psychology. We don't need to master all the details from these disciplines, just the fundamentals. To quote Charlie Munger, "80 or 90 important models will carry about 90 percent of the freight in making you a worldly-wise person. And, of those, only a mere handful really carry very heavy freight."

These books attempt to collect and make accessible organized common sense—the 80 to 90 mental models you need to get started. To help you understand the models, we will relate them to historical examples and stories. Our website fs.blog will have even more practical examples.

The more high-quality mental models you have in your mental toolbox, the more likely you will have the ones needed to understand the problem. And understanding is everything. The better you understand, the better the potential actions you can take. The better the potential actions, the fewer problems you'll encounter down the road. Better models make better decisions.

I think it is undeniably true that the human brain must work in models. The trick is to have your brain work better than the other person's brain because it understands the most fundamental models: ones that will do most work per unit. If you get into the mental habit of relating what you're reading to the basic structure of the underlying ideas being demonstrated, you gradually accumulate some wisdom.

- Charlie Munger

It takes time, but the benefits are enormous. What successful people do is file away a massive, but finite, amount of fundamental, established, essentially unchanging knowledge that can be used in evaluating the infinite number of unique scenarios which show up in the real world.

It's not just knowing the mental models that is important. First you must learn them, but then you must use them. Each decision presents an opportunity to comb through your repertoire and try one out, so you can also learn how to use them. This will slow you down at first, and you won't always choose the right models, but you will get better and more efficient at using mental models as time progresses.

We need to work hard at synthesizing across the borders of our knowledge, and most importantly, synthesizing all of the ideas we learn with reality itself. No model contains the entire truth, whatever that may be. What good are math and biology and psychology unless we know how they fit together in reality itself, and how to use them to make our lives better? It would be like dying of hunger because we don't know how to combine and cook any of the foods in our pantry.

Disciplines, like nations, are a necessary evil that enable human beings of bounded rationality to simplify their goals and reduce their choices to calculable limits. But parochialism is everywhere,

and the world badly needs international and interdisciplinary travelers to carry new knowledge from one enclave to another.

- Herbert Simon

You won't always get it right. Sometimes the model, or models, you choose to use won't be the best ones for that situation. That's okay. The more you use them, the more you will be able to build the knowledge of indicators that can trigger the use of the most appropriate model. Using and failing, as long as you acknowledge, reflect, and learn from it, is also how you build your repertoire.

You need to be deliberate about choosing the models you will use in a situation. As you use them, a great practice is to record and reflect. That way you can get better at both choosing models and applying them. Take the time to notice how you applied them, what the process was like, and what the results were. Over time you will develop your knowledge of which situations are best tackled through which models. Don't give up on a model if it doesn't help you right away. Learn more about it, and try to figure out exactly why it didn't work. It may be that you have to improve your understanding. Or there were aspects of the situation that you did not consider. Or that your focus was on the wrong variable. ***So keep a journal. Write your experiences down. When you identify a model at work in the world, write that down too. Then you can explore the applications you've observed, and start being more in control of the models you use every day.*** For instance, instead of falling victim to confirmation bias, you will be able to step back and see it at work in yourself and others. Once you get practice, you will start to naturally apply models as you go through your life, from reading the news to contemplating a career move.

As we have seen, we can run into problems when we apply models to situations in which they don't fit. If a model works, we must invest the time and energy into understanding why it worked so we know when to use it again. At the beginning the process is more important than the outcome. As you use the models, stay open to the feedback loops. Reflect and learn. You will get better. It will become easier. Results will become more profoundly useful, more broadly applicable, and more memorable. While this book isn't intended to be a book specifically about making better decisions, it will help you make better decisions. Mental models are not an excuse to create a lengthy decision process but rather to help you move away from seeing things the way you think they should be to the way they are. Uncovering this knowledge will naturally help your decision-making. Right now you are only touching one part of the elephant, so you are making all decisions based on your understanding that it's a wall or a rope, not an animal. As soon as you begin to take in the knowledge that other people have of the world, like learning the perspectives others have of the elephant, you will start having more success because your decisions will be aligned with how the world really is.

When you start to understand the world better, when the whys seem less mysterious, you gain confidence in how you navigate it. The successes will accrue. And more success means more time, less stress, and ultimately a more meaningful life.

Time to dive in.

LITTLE BETS

Sims, Peter. Little Bets (pp. 69-72)

RELEVANCE:

The core practice, based on improvisation, which Pixar used for creative discussions, was called “plus-ing.” I dislike the term because it sounds too fluffy – and it is a serious matter. What improvisation does is create a structure where participants build on each other.

In the improv tradition and what I think the creative process of building off of each other is more accurately called is freestyling – a nod to [Freestyle Love Supreme](#) and the fact that building can also be a little bit like battling in ways too. In the passage below I replaced plus-ing with freestyling to put it in our language.

CHANGING DIRECTIONS WITHOUT JUDGING

Throughout the Pixar creative process, they rely heavily on what they call freestyling; it is likely the most-used concept around the company. The point of freestyling is to build upon and improve ideas without using judgmental language. Creating an atmosphere where ideas are constantly being freestyled, while maintaining a sense of humor and playfulness, is a central element of Pixar’s magic. The practice of freestyling draws upon those core principles from improvisation: accepting every offer and making your partner look good. Rather than criticize an idea in its entirety (even if they don’t think it’s good), people accept the starting point before suggesting improvements.

To understand how freestyling works, imagine that you were an animator working on the movie Toy Story 3. The script might call for a scene to last a few seconds (also called a “shot”) with the main character, Woody, talking with Buzz Lightyear. The director or supervising animators will assign one or two shots, or a sequence of up to five or seven shots, to animators at a time based on the shot needs and the animator’s strengths. Pixar animators produce about one hundred frames per week on average (about four seconds of footage).

After getting a new shot assignment, animators are free to sketch out the scene as they see fit, called “blocking,” keeping in mind their deadlines and other constraints such as what comes before or after their shot. Storyboards help animators understand the story flow, including what specific shots come before and after theirs, as well as what feelings the director would like to evoke... Animators will then take a first pass at their shot. It will be very rough.

At this point, the animator will share her rough sketches and ideas with the film’s supervising animator or the director. Pixar’s directors are the final decision makers, but not even they will have the final vision for the shot

clearly in mind. ***This is where freestyling comes in. Instead of criticizing the sketch or saying "no," the director will build on the starting point by saying something like, "I like Woody's eyes, and what if we ..."*** Again, notice the use of the word and rather than a word that implies a judgment...

So, the director will take the rough material and say something like, "I like Woody's eyes, and what if his eyes rolled left?" He'll know what he likes when he sees it, at least directionally, and that's the point of freestyling. "You always want to present your ideas in a constructive manner and be respectful of the other animator's feelings," Pixar animator Victor Navone says. "I usually start my suggestions with 'what if' or 'would it be clearer if' [the character] did it this way." As Pete Docter, director of Monsters Inc. and Up, puts it, "I think ***everyone [at Pixar] has gotten very good at freestyling ideas or changing directions without judging.***"

Imagine how intimidating it might be to work as an animator at Pixar if freestyling weren't the prevailing philosophy. Remember, when animators are deep into working on a film, they submit their draft work each day to a central computer system (regardless of how complete or incomplete), where colleagues can review it. Animation team members will then gather in a conference room each day with the supervising animator and (often) the film's director for dailies. Certain shots are then projected onto a screen, and everyone (including the newest animator) is encouraged to share their opinions, ideas, and solutions. Without an atmosphere of offering constructive, nonjudgmental criticism, every day could be an ordeal.

The Pygmalion Effect: Proving Them Right

From the Farnam Street Blog, by Shane Parrish: <https://fs.blog/pygmalion-effect/>

RELEVANCE:

A few notes here – Shane Parrish has a considered perspective and there is a lot that can be learned from his blog. He's an ex-intelligence agency analyst and has useful summaries of on a lot of topics. His branding is all around mental models. The blog post includes a number of great links from Parrish, not me.

A lot of what we identify as excellence can be thought of as minimizing golem effects and maximizing Pygmalion effects. Catmull's plus-ing or what we are calling freestyling is just minimizing Golem and maximizing Pygmalion in a systemic but simple way in team discussions. To maximize team performance and maximize our own performance, we need to constantly seek to minimize Golem effects and maximize Pygmalion effects.

The Pygmalion Effect is a powerful secret weapon. Without even realizing it, we can nudge others towards success. In this article, discover how expectations can influence performance for better or worse.

How Expectations Influence Performance

Many people believe that their pets or children are of unusual intelligence or can understand everything they say. Some people have stories of abnormal feats. In the late 19th century, one man claimed that about his horse and appeared to have evidence. [William Von Osten](#) was a teacher and horse trainer. He believed that animals could learn to read or count. Von Osten's initial attempts with dogs and a bear were unsuccessful, but when he began working with an unusual horse, he changed our understanding of psychology. Known as [Clever Hans](#), the animal could answer questions, with 90% accuracy, by tapping his hoof. He could add, subtract, multiply, divide, and tell the time and the date.

Clever Hans could also read and understand questions written or asked in German. Crowds flocked to see the horse, and the scientific community soon grew interested. Researchers studied the horse, looking for signs of trickery. Yet they found none. The horse could answer questions asked by anyone, even if Von Osten was absent. This indicated that no signaling was at play. For a while, the world believed the horse was truly clever.

Then psychologist [Oskar Pfungst](#) turned his attention to Clever Hans. Assisted by a team of researchers, he uncovered two anomalies. When blinkered or behind a screen, the horse could not answer questions. Likewise, he could respond only if the questioner knew the answer. From these observations, Pfungst deduced that Clever Hans was not making any mental calculations. Nor did he understand numbers or language in the human sense. Although Von Osten had intended no trickery, the act was false.

Instead, Clever Hans had learned to detect subtle, yet consistent nonverbal cues. When someone asked a question, Clever Hans responded to their body language with a degree of accuracy many poker players

would envy. For example, when someone asked Clever Hans to make a calculation, he would begin tapping his hoof. Once he reached the correct answer, the questioner would show involuntary signs. Pfungst found that many people tilted their head at this point. Clever Hans would recognize this behavior and stop. When blinkered or when the questioner did not know the answer, the horse didn't have a clue. When he couldn't see the cues, he had no answer.

The Pygmalion Effect

Von Osten died in 1909 and Clever Hans disappeared from record. But his legacy lives on in a particular branch of psychology.

The case of Clever Hans is of less interest than the research it went on to provoke. Psychologists working in the decades following began to study how the expectations of others affect us. If someone expected Clever Hans to answer a question and ensured that he knew it, could the same thing occur elsewhere?

Could we be, at times, responding to subtle cues? Decades of research have provided consistent, robust evidence that the answer is yes. It comes down to the concepts of the self-fulfilling prophecy and the Pygmalion effect.

The Pygmalion effect is a psychological phenomenon wherein high expectations lead to improved performance in a given area. Its name comes from the story of Pygmalion, [a mythical Greek sculptor](#). Pygmalion carved a statue of a woman and then became enamored with it. Unable to love a human, Pygmalion appealed to Aphrodite, the goddess of love. She took pity and brought the statue to life. The couple married and went on to have a daughter, Paphos.

False Beliefs Come True Over Time

In the same way Pygmalion's fixation on the statue brought it to life, our focus on a belief or assumption can do the same. The flipside is the [Golem effect](#), wherein low expectations lead to decreased performance. Both effects come under the category of self-fulfilling prophecies. Whether the expectation comes from us or others, the effect manifests in the same way.

The Pygmalion effect has profound ramifications in schools and organizations and with regard to social class and stereotypes. By some estimations, it is the result of our brains' poorly distinguishing between perception and expectation. Although many people purport to want to prove their critics wrong, we often merely end up proving our supporters right.

Understanding the Pygmalion effect is a powerful way to positively affect those around us, from our children and friends to employees and leaders. If we don't take into account the ramifications of our expectations, we may miss out on the dramatic benefits of holding high standards.

The concept of a self-fulfilling prophecy is attributed to sociologist [Robert K. Merton](#). In 1948, Merton published the first paper on the topic. In it, he described the phenomenon as a false belief that becomes true over time. Once this occurs, it creates a feedback loop. We assume we were always correct because it seems so in hindsight. Merton described a self-fulfilling prophecy as self-hypnosis through our own propaganda.

As with many psychological concepts, people had a vague awareness of its existence long before research confirmed anything. Renowned orator and theologian Jacques Benigne Bossuet declared in the 17th century that “The greatest weakness of all weaknesses is to fear too much to appear weak.”

Even Sigmund Freud was aware of self-fulfilling prophecies. In *A Childhood Memory of Goethe*, Freud wrote: “If a man has been his mother’s undisputed darling he retains throughout life the triumphant feeling, the confidence in success, which not seldom brings actual success with it.”

The IQ of Students

Research by Robert Rosenthal and Lenore Jacobson examined the influence of teachers’ expectations on students’ performance. Their [subsequent paper](#) is one of the most cited and discussed psychological studies ever conducted.

Rosenthal and Jacobson began by testing the IQ of elementary school students. Teachers were told that the IQ test showed around one-fifth of their students to be unusually intelligent. For ethical reasons, they did not label an alternate group as unintelligent and instead used unlabeled classmates as the control group. It will doubtless come as no surprise that the “gifted” students were chosen at random. They should not have had a significant statistical advantage over their peers. As the study period ended, all students had their IQs retested. Both groups showed an improvement. Yet those who were described as intelligent experienced much greater gains in their IQ points. Rosenthal and Jacobson attributed this result to the Pygmalion effect. Teachers paid more attention to “gifted” students, offering more support and encouragement than they would otherwise. Picked at random, those children ended up excelling. Sadly, no follow-up studies were ever conducted, so we do not know the long-term impact on the children involved.

Prior to studying the effect on children, Rosenthal performed preliminary research on animals. Students were given rats from two groups, one described as “maze dull” and the other as “maze bright.” Researchers claimed that the former group could not learn to properly negotiate a maze, but the latter could with ease. As you might expect, the groups of rats were the same. Like the gifted and nongifted children, they were chosen at random. Yet by the time the study finished, the “maze-bright” rats appeared to have learned faster. The students considered them tamer and more pleasant to work with than the “maze-dull” rats.

In general, authority figures have the power to influence how the people subordinate to them behave by holding high expectations. Whether consciously or not, leaders facilitate changes in behavior, such as by giving people more responsibility or setting stretch goals. Like the subtle cues that allowed Clever Hans to make calculations, these small changes in treatment can promote learning and growth. If a leader thinks an employee is competent, they will treat them as such. The employee then gets more opportunities to develop their competence, and their performance improves in a positive feedback loop.

This works both ways. When we expect an authority figure to be competent or successful, we tend to be attentive and supportive. In the process, we bolster their performance, too. Students who act interested in lectures create interesting lecturers.

In [*Pygmalion in Management*](#), J. Sterling Livingston writes,

Some managers always treat their subordinates in a way that leads to superior performance. But most ... unintentionally treat their subordinates in a way that leads to lower performance than they are capable of achieving. The way managers treat their subordinates is subtly influenced by what they expect of them. If manager's expectations are high, productivity is likely to be excellent. If their expectations are low, productivity is likely to be poor. It is as though there were a law that caused subordinates' performance to rise or fall to meet managers' expectations.

The Pygmalion effect shows us that our reality is negotiable and can be manipulated by others — on purpose or by accident. What we achieve, how we think, how we act, and how we perceive our capabilities can be influenced by the expectations of those around us. Those expectations may be the result of biased or irrational thinking, but they have the power to affect us and change what happens. While cognitive biases distort only what we perceive, self-fulfilling prophecies alter what happens.

Of course, the Pygmalion effect works only when we are physically capable of achieving what is expected of us. After Rosenthal and Jacobson published their initial research, many people were entranced by the implication that we are all capable of more than we think. Although that can be true, we have no indication that any of us can do anything if someone believes we can. Instead, the Pygmalion effect seems to involve us leveraging our full capabilities and avoiding the obstacles created by low expectations.

Clever Hans truly was an intelligent horse, but he was smart because he could read almost imperceptible nonverbal cues, not because he could do math. So, he did have unusual capabilities, as shown by the fact that few other animals have done what he did.

We can't do anything just because someone expects us to. Overly high expectations can also be stressful. When someone sets the bar too high, we can get discouraged and not even bother trying. Stretch goals and high expectations are beneficial, up to the point of diminishing returns. Research by McClelland and Atkinson indicates that the Pygmalion effect drops off if we see our chance of success as being less than 50%. If an endeavor seems either certain or completely uncertain, the Pygmalion effect does not hold. When we are stretched but confident, high expectations can help us achieve more.

Check Your Assumptions

In [*Self-Fulfilling Prophecy: A Practical Guide to Its Use in Education*](#), Robert T. Tauber describes an exercise in which people are asked to list their assumptions about people with certain descriptions. These included a cheerleader, “a minority woman with four kids at the market using food stamps,” and a “person standing outside smoking on a cold February day.” An anonymous survey of undergraduate students revealed mostly negative assumptions. Tauber asks the reader to consider how being exposed to these types of assumptions might affect someone's day-to-day life.

The expectations people have of us affect us in countless subtle ways each day. Although we rarely notice it (unless we are on the receiving end of overt racism, sexism, and other forms of bias), those expectations dictate the opportunities we are offered, how we are spoken to, and the praise and criticism we receive. Individually, these knocks and nudges have minimal impact. In the long run, they might dictate whether we succeed or fail or fall somewhere on the spectrum in between.

The important point to note about the Pygmalion effect is that it creates a literal change in what occurs. There is nothing mystical about the effect. When we expect someone to perform well in any capacity, we treat them in a different way. Teachers tend to show more positive body language towards students they expect to be gifted. They may teach them more challenging material, offer more chances to ask questions, and provide personalized feedback. As Carl Sagan declared, “The visions we offer our children shape the future. It matters what those visions are. Often they become self-fulfilling prophecies. Dreams are maps.”

A perfect illustration is the case of James Sweeney and George Johnson, as described in [Pygmalion in Management](#). Sweeney was a teacher at Tulane University, where Johnson worked as a porter. Aware of the Pygmalion effect, Sweeney had a hunch that he could teach anyone to be a competent computer operator. He began his experiment, offering Johnson lessons each afternoon. Other university staff were dubious, especially as Johnson appeared to have a low IQ. But the Pygmalion effect won out and the former janitor eventually became responsible for training new computer operators.

The Pygmalion effect is a powerful secret weapon. Who wouldn't want to help their children get smarter, help employees and leaders be more competent, and generally push others to do well? That's possible if we raise our standards and see others in the best possible light. It is not necessary to actively attempt to intervene. Without even realizing it, we can nudge others towards success. If that sounds too good to be true, remember that the effect holds up for everything from rats to CEOs.

The Trouble of Not Knowing What You Do Not Know

Psychological, Philosophical, and Societal Implications

By David Dunning, from Reason, Bias, and Inquiry (pp. 205-226). Oxford University Press.

RELEVANCE:

The Dunning Kruger effect is included below and worth understanding, but so are other issues that bridge psychology and decision-making. But remember – the psychology is WEIRD psychology – as Henrich notes 90+% of the participants in these studies are western college students,... so it's not "human psychology"...

But the implications of developing meta-cognitive skills to notice or adjust for our own errors and to build and work with teams and tools to extend the scope of what we can know and understand – slipping some of the limits of individual human cognitions... well that's what I really want us all to be thinking about every day.

The Master said, "Yu, shall I teach you what knowledge is? When you know a thing, to hold that you know it; and when you do not know a thing, to allow that you do not know it;-this is knowledge." —Confucius (551–479 BCE)

In this quotation, Confucius (2015) makes a sage observation. Any assessment of one's condition, any decision of what course of action to take, requires not one but two judgments. The first is the direct judgment itself, but the second might be more important.

That second judgment is assessing whether one's first judgments should be considered valid or tentative, ones held with certainty or accompanied by doubt. A military general, for example, may design a battle plan that looks like it has the best chance to defeat the enemy, but whether the general goes into battle may depend not so much on the details of the plan but rather whether the general is confident it will succeed. Does the general feel as though every contingency has been accounted for? Is the general sure that the plan is simple enough or instead too complex to execute? Does the plan contain the necessary flexibility in order to weather any number of surprises that might be discovered only after the engagement starts?

In psychology, coming up with a battle plan in the first place can be called the cognitive task facing the general. ***The second task, one of assessing the battle plan's soundness, is the meta-cognitive task the general must execute (Dunlosky & Metcalfe, 2009; Metcalfe & Shimamura, 1994). This second task contains many components, but one central component is evaluating the worth of one's reasoning.*** Is the reasoning accurate, or does it contain errors? Does it neglect important aspects of the task? Does the reasoning flow from complete and accurate information or instead from imperfect data and assumptions? In the example, did the general have all the information needed? Was intelligence about the enemy sound? Are there any unknowns that must be addressed? Does the general have sufficient expertise to draw up a plan? Is their reasoning coherent? Ultimately, should the general be confident or wait instead for another day?

Or, as Confucius put it, is the general wise enough to lead the troops to battle not only having a good plan but knowing it is good?

Recognizing One's Own Ignorance

Here, however, is where Confucius (2015) enters the picture again. ***Wise knowledge is not only knowing what we know—it also and importantly requires having a reasonable grasp of what we are ignorant of. This chapter is a discussion of how well people have that grasp. Its central contention is that knowing what one doesn't know, having a reasonable understanding of the shape and scope of one's ignorance, makes knowing a hard task. Herein lies the problem not only for the general but for the rest of us as well.***

Psychological research over the past few decades has repeatedly demonstrated that people have deep and persistent problems when it comes to recognizing the extent of their ignorance. They often fail to see the line where the solid ground of their knowledge ends and the shifting and dangerous soil of unknowing begins. Real knowledge, the way that Confucius defined it, is something that people have much less than they realize (Dunning, 2005; Dunning et al., 2004).

In this chapter, I discuss the intrinsic difficulty people have in grasping the shape and scope of their ignorance. *I discuss three phenomena from empirical psychology—overclaiming knowledge, the illusion of understanding, and the Dunning-Kruger effect—that show that people often claim expertise they do not have, thus revealing their inability to separate their knowledge from their ignorance. I then discuss the reasons why people have such a hard time identifying the boundary between knowledge and ignorance. I describe just how invisible people's ignorance is of them, argue further that people tend to actively neglect what they do not know, and often suffer from false knowledge that has the look and feel of true knowledge. I end the chapter by discussing both the personal and societal implications of this ignorance of ignorance.* At the personal level, people often fail to ask for advice when they need it and fail to recognize true expertise in others. At the societal level, ignorance of ignorance may cause people to dismiss experts on social issues and may leave them vulnerable to the arguments of false prophets bearing enticing but misleading information.

Three Demonstrations Illustrating Ignorance of Ignorance

Many studies in psychology provide ample evidence that people cannot tell where their expertise ends and their ignorance begins. In three different paradigms, researchers easily catch people crossing the boundary between knowledge and ignorance without their ever noticing it.

Overclaiming Knowledge

In our own lab, we have demonstrated this unknowing step into ignorance by giving people simple surveys. We might, for example, give them a list of common financial terms, such as stock options, revolving credit, and whole life insurance, and ask them how familiar they are with the terms (Atir et al., 2015). People show a good deal of familiarity with these terms, which is good.

However, what is interesting is the familiarity they report with a special class of terms, such as fixed-rate deductions or pre-rated stocks. In one survey, over 90% of respondents reported familiarity with at least one of these items. However, this presents a problem: These items are ones we have simply invented among ourselves in our office. These concepts do not exist. We have made sure they appear nowhere, for example, on the internet. Thus, there is no possible way for respondents to have any familiarity with them.

Yet, respondents report knowledge of and familiarity with items that are impossible to know because these items do not exist. This phenomenon is known as overclaiming, and its existence has been known for many decades in behavioral science (Paulhus et al., 2003). In our studies, participants overclaim knowledge for “concepts” falling into philosophy, biology, and social science (Atir et al., 2015). They report familiarity with fictitious cities such as Cashmere, Oregon, and nonexistent politicians such as Michael Merrington. Consumers report some acquaintance with food (e.g., Barjolet cheese) and industrial products (e.g., Thompson drill bits) that do not exist (Graeff, 2003).

In a similar vein, survey respondents express opinions about recent governmental actions, such as the International Monetary Act, that have never been proposed (Schuman & Presser, 1980). In 2015, 30% of Republicans versus 19% of Democrats favored bombing the Middle Eastern kingdom of Agrabah—the only problem being that the kingdom existed only in the child’s tale of Aladdin (Jensen, 2015).

The calendar year 2017 was a banner year for overclaiming in the political realm. In February, 51% of people who had voted for Donald Trump for president the preceding year cited the “Bowling Green Massacre,” a completely fictitious event, as a rationale for banning Muslims from entering the United States (England, 2017). In October, two South Carolina legislators proposed a statue to honor Confederate African American soldiers, although history provides no record of any, for obvious reasons (Criss, 2018). Later in December, two Russian radio comedians prompted United Nations Ambassador Nikki Haley to condemn Russian interference in elections taking place in the island nation of Binomo, which existed only in the context of that radio conversation (Mortimer, 2017).

Illusions of Understanding

Overclaiming is not the only way that people demonstrate failures to locate the line between their knowledge and ignorance. Ask people to explain how everyday items work, such as a ballpoint pen or a coat zipper, and they profess confidence in their ability to do so. However, if then asked to go ahead and explain how such items function, people retreat from their confidence to a more humble position. ***This phenomenon of intellectual confidence that evaporates to humility is known as the illusion of explanatory depth (Fisher & Keil, 2016; Rozenblit & Keil, 2002).***

The illusion even extends to the political and social arena (Fernbach et al., 2013). Political partisans often proclaim to have a clear idea about how certain social policies work, such as national healthcare or sanctions against a foreign country. However, once asked to explain in detail how these policies operate, partisans realize they do not know as much as they thought. They retreat to a position of more intellectual modesty—and, perhaps more important, moderate their partisanship.

The Dunning-Kruger Effect

The most extensive demonstration, however, that people know not where the geography of their ignorance begins comes from studies on the phenomenon that has come to be known as the Dunning-Kruger effect (Dunning, 2011; Kruger & Dunning, 1999). ***The effect asserts that people of poor expertise fail to recognize just how poor their expertise is. The theoretical claim is stronger. The claim is that people with poor expertise lack what they need to be able to recognize their shortcomings. It is not that they fail to recognize their deficits; instead, they are simply not in a position to recognize those deficits and should not be expected to do so.***

The logic for this claim is rather transparent. No matter whether they are called inexpert, naïve, untrained, or unskilled, people who suffer from the Dunning-Kruger effect simply lack the expertise they need to be able to recognize the expertise they lack. To recognize their deficits requires the exact knowledge they fail to have. As such, they suffer a double curse: Not only are they incompetent but they are also too incompetent to recognize just how deep their incompetence runs. Thus, poor performers often think they are doing just fine when they are doing anything but (Dunning, 2019).

By now, demonstrations of the Dunning-Kruger effect are varied and numerous. People performing badly on tests of logic, grammar, financial literacy, physics, and emotional intelligence all dramatically overestimate how well they think they are doing—often being almost but not quite as positive about their performance as top achievers (Kruger & Dunning, 1999; Sheldon et al., 2014; Williams et al., 2013; for a review, see Dunning, 2011). Students failing or nearly failing an exam in a college course and who in reality are performing at only the 13th percentile relative to their peers think on average that their performance puts them in the 60th to 65th percentile (Dunning et al., 2003; Schlösser et al., 2013). None of this is due to careless judgment on the part of respondents. Offering to pay participants up to \$100 for accurate estimates of performance (in this case, on a logical reasoning quiz) does nothing to enhance how accurately respondents report their performances (Ehrlinger et al., 2008).

The Dunning-Kruger effect is also evident outside our laboratory. Gun enthusiasts, participating in a trap-and-skeet shooting competition, who did badly on a quiz about firearm care and safety were just as confident in their skill as were those who did the best on the quiz (Ehrlinger et al., 2008). In medicine, residents completing their rotation in obstetrics/gynecology at the bottom of their class (and so receive an F on their final exam and a C– for the rotation) think they are achieving a B or B– on the exam and a B+ for the rotation (Edwards et al., 2003). In a similar vein, of 95 first-year medical students learning basic CPR, only three thought they had failed the class when a full 36 did (Vnuk et al., 2006). In more leisurely pursuits, the worst players in debate teams, bridge clubs, chess tournaments, and poker competitions all overestimate how well they are doing (Ehrlinger et al., 2008; Park & Santo-Pinto, 2010; Simons, 2013).

Why Ignorance Is Invisible

Why is human ignorance so invisible to those who possess it? Why do people fail to recognize what they do not know?

Any discussion of this issue must start with a description of the human mind. First, one must concede that what the typical mind knows is truly impressive. By age 60, the typical English speaker knows the equivalent of 48,000 words and their dictionary definitions (Brysbaert et al., 2016). Research subjects can view 2500 photographs, and then distinguish the ones they have seen from photographs taken at a

different angle with 87% accuracy—a testament to just how much storage capacity the brain has for detailed memories of visual stimuli (Brady et al., 2008). One cognitive psychologist, looking at the rate at which people add pictures, words, or music to memory, has estimated that the human brain can carry 109 bits of learned information over the course of a lifetime, which is more than 50,000 times the text contained in the US Library of Congress (Landauer, 1986).

Moreover, people have a remarkable capacity for taking these bits of information and flexibly applying and combining them to understand the situations they face or to creatively mold new ones (Marsh et al., 2016). Such a capacity to use information to understand situations is essential, in that people never really encounter the exact same situation twice. The cat they see slinking in the bushes is one they have never seen, but they know enough to know it is a cat. Or they need to leave a message outside a friend's house but have no pen or pencil. They know enough to take a piece of charcoal from a nearby grill and write the message on the concrete landing outside their friend's door.

But this capacity for storing, applying, and weaving knowledge—no matter how impressive—pales against the capacity of the world to produce what people could know. People may know nearly 50,000 words by age 60, but there are over 600,000 word definitions contained in the Oxford English Dictionary—with a massive number of other ideas contained in other languages. Just to give one example, there are at least 216 words related to “well-being” in other languages that are untranslatable to English (Lomas, 2016). People may also remember the details of thousands of photographs, but there are an infinite number of images possible on planet Earth. People may know 109 bits of information, but that equates to just a few hundred megabytes, well within the capacity of their ordinary desktop computer (Marois & Ivanoff, 2005). *It is as Michel de Montaigne (1877) once observed, “there escapes us a hundred times more than comes to our knowledge” and that “if we saw as much of the world as we do not see, we would perceive . . . a perpetual multiplication and vicissitude of forms.”*

Unknowns Beyond One's Ken

The problem people have is perceiving, or even conceiving of, the multiplicities and vicissitudes that they do not know. In economics, it is customary to place the unknown into three separate categories (Zeckhauser, 2006), two of which are much easier to think about than the third. First, there are things that are unknown, but the probability that they will appear is well defined. As an example, you flip a coin in the air. While in the air, it is unknown whether the coin will land heads or tails, but those two options define a finite and easily decidable set of outcomes (coins rarely, if ever, land and stick on their edge) such that one can assign a 50–50 probability to each outcome. This, according to economists, is a scenario involving risk.

But, second, unknowns can be more unruly. We know the stock market will go up or down over the course of the next year, and this again defines the entire outcome set. However, in this case, we cannot assign exact probabilities to either outcome. This, according to economists, is uncertainty and represents a type of problem that people typically face in the world but which is rarely represented in economic textbooks (Zeckhauser, 2006). People may know which of two teams may win the National Football League's Super Bowl but differ in the specific odds they assign to either team winning.

Unknown Unknowns

Third, unknowns can be even more unruly, in that we fail to recognize all the possible outcomes themselves that may occur. As such, we cannot assign probabilities to them because we simply do not

know that they exist. In this, we are in a state of ignorance (Zeckhauser, 2006). We move from the world of unknowns that are known to the murky and complicated world of unknown unknowns—risks, events, or even opportunities that are so unknown to us that we do not know that we do not know them.

If known unknowns are questions we do not have the answers to, unknown unknowns are the questions we do not even know we need to ask. Unknown unknowns can importantly shape the fate of human events. Napoleon once set 25,000 soldiers in Haiti to put down a slave rebellion. He was ultimately defeated not by his enemy but by yellow fever, which claimed so many lives that he had to retreat from the island with the 3000 men he had left (Keyes, 2014).

In our work, we have shown that people fail to possess adequate intuitions about what they do not know. For example, consider the word spontaneity. How many other English words can you create, from 2 to 11 letters long, from picking and rearranging the letters in this seed word? Some words are easy to see, such as opt, pen, and tape. But how many are there total? It turns out that there are 718 English words that one can generate. However, to the typical person, many of these words remain in the unknown unknown category. These words exist, but a person staring at spontaneity has no knowledge of them and thus would never generate them, words such as pentosan (a polysaccharide widely distributed in plants), pontine (relating to the pons of the brain), or saponite (a trioctahedral mineral of the smectite group).

In a word game not unlike this word search, we gave participants puzzles that contained from 100 to 254 solutions. On average, participants thought they had missed 18 possible solutions, but the real figure was 154. In another study, we asked psychology graduate students to go over an experiment and report the methodological flaws it contained. The best students found 69% of the flaws; the worst performers found only 21% of them. Participants thought their errors of omission mattered—in that once those errors were revealed to them, they lowered the self-ratings of performance. In yet another study involving a word game, they bet less money that they had beaten another student at the game once their errors of omission were pointed out to them (Caputo & Dunning, 2005).

Thus, it appears that people are blind to their errors of omission, presumably because some of those errors lie in the realm of the unknown unknown. In a study of medical malpractice suits, the leading cause could be traced to errors in diagnosis, of which 40% resulted in death and 17% in permanent injury. Of those errors, a majority involved missed diagnoses (54%) rather than delayed (20%) or incorrect (10%) diagnoses (Tehrani et al., 2013). It was errors of omission that proved to be the most common and lethal.

Hypocognition

One important variant of unknown unknowns falls into the category of hypocognition. If you do not know what hypocognition is, then you have just experienced it. Hypocognition is lacking a linguistic or cognitive representation for some object, emotion, category, or idea (Wu & Dunning, 2018).

The term was first discussed extensively by anthropologist Robert Levy (1973) in his fieldwork with Tahitians of the Society Islands. During his work, Levy observed that although Tahitians were quite expert in some emotions (in particular, anger), they had no conception of long-term grief. If a person experienced a death in the family, the initial anguish was well understood and public; but any continuing distress seemed unrecognized and unexamined. Islanders would feel grief but not completely understand it, describing themselves instead as feeling “sick” or “strange.”

In a similar vein, non-Arabic speakers, when looking at Arabic letters, cluster letters together according to their similarity in a different way than fluent Arabic speakers. Each group experiences the visual nature of the letters differently. Non-speakers just see the physical letter, whereas speakers also internally experience a number of associations, such as sounds and the brushstrokes needed to create those letters. These associations not only change what they experience but make them more accurate and efficient in judging the physical similarity of different letters (Wiley et al., 2016).

Much of human action, thus, might depend not on what people know but rather on what they do not know. Without knowledge of grief, for example, there can be no grief work. In finance, many people remain hypocognitive to the idea of compound interest and so fail to recognize just how much saving money could benefit them or how much debt can penalize them. They simply do not anticipate how much a compounding interest rate can cause any amount of money to grow and grow quickly (Lin et al., 2016). People ignorant of compound interest also refuse to use financial decision aids, although their use would obviously benefit them (Levy & Tasoff, 2017). In a similar vein, it is estimated that a full third of people suffering from type II diabetes do not know it and do not seek out medical help because they do not recognize that seemingly isolated symptoms (chronic fatigue, numbness, blurry vision, frequent urination) actually indicate a single serious underlying medical condition (Cowie et al., 2006).

Corralling the Unknown

Although so much information lies beyond the ken of the individual human, people could do a lot better dealing with unknowns. The real problem is that not all of the unknown lies in the realm of the unknown unknown but can be brought into the realm of the known unknown or even the known—if people would simply pay more attention to what they might be ignorant of.

That is, people make decisions based on what they know and set aside or do not consider what they do not know, even when knowledge of it is within their grasp. They act as though they have complete information even when information outside their knowledge could change their decision. For example, college students can often remain undecided between a \$700 stereo and a \$1000 one of somewhat better quality. However, if they are simply reminded that they could also buy \$300 worth of music if they bought the cheaper stereo, students quickly break for that option (Frederick et al., 2009).

In a similar vein, people tend to ignore information that is missing, even though it should be relevant to the decision at hand. For example, when buying a car, people fail to ask for the safety record of the car's brand if that record is never mentioned, even though they think it is relevant if mentioned. In addition, when judging the quality of a camera, people make just as confident assessments of the camera's quality if shown only four attributes of the camera as when shown eight (Sanbonmatsu et al., 1992, 2003). People also make more extreme judgments about an object (e.g., a bicycle) when thinking about it in memory, after much information has been forgotten, than when the information is fresh in their mind, again neglecting what they have failed to retain in memory (Sanbonmatsu et al., 1991).

To be sure, people who are expert are better at attending to information that is missing (Sanbonmatsu et al., 1992). Apparently, ridding one's self of hypocognition is useful in aiding people to recognize and weight omissions in information, thus aiding their choices. Blatantly pointing out to people that there is information they miss and asking them to list what that information is also prompts them to be less overconfident in their decisions (Feduzi & Runde, 2014; Walters et al., 2017).

The Veil of False Belief

The second reason why people fail to recognize the scope of their ignorance is that it is often veiled by beliefs and opinions that are false but which have the look and feel of the truth. Recall that a person learns up to 109 bits of information as they navigate life. It would be a surprise, and quite arrogant, to assume that all of those bits of information are accurate. Some of them must be wrong or misleading. This is the reason why, for example, so many people believe Toronto is the capital of Canada (sorry, Ottawa), that the Sahara is the largest desert in the world (actually, the driest spot on the globe is Antarctica), and that Pluto is the farthest planet from the sun (sadly, it has been demoted to a less-than-planetary status). People know enough to come up with an answer to any question posed; those answers do not necessarily have to be true (Tauber et al., 2013).

Consider a survey conducted in my lab the day after the US congressional midterm elections took place in November 2014 (Dunning & Roh, 2018). Participants were asked about their politics and then asked what they believed to be true about social, economic, and political conditions in the country. They were asked, for example, whether the poverty rate had gone down under the administration of President Obama, whether teenage pregnancies were at an all-time high, and whether the budget deficit was shrinking in line with Obama's promises. Half the statements were conservative-friendly; half were more congenial to liberals.

Not surprisingly, conservatives and liberals differed in which facts they claimed were true of the country. Conservatives were more likely to think teenage pregnancies were an epidemic; liberals were more likely to believe that the Obama presidency had lowered the poverty rate. Of key interest, however, 35% to 40% of what each side endorsed as true was actually false (or what they believed to be false was actually true). Teenage pregnancies in 2014 were actually at a several-decade low; the poverty rate had risen in the early days of Obama. ***In short, conservatives and liberals lived in different factual worlds—and this divergence occurred even though every respondent on every question had the option of saying “I don’t know”; they did not have to guess.***

Everyday Paralogia

We have termed this phenomenon of generating false answers everyday paralogia, borrowing the terminology from clinical psychology for beliefs that are unsound, illogical, or delusional, and have traced their implications (Dunning & Roh, 2018).

These paralogic beliefs, for example, matter for self-perception. We asked respondents how “well-informed” they considered themselves about civic and national affairs. Not surprisingly, respondents who answered our quiz more correctly also rated themselves as more informed, as they should. Those who opted to say “I don’t know” more often rated themselves as less informed—again, appropriate. What was important, however, was the relationship between giving false answers and self-perception. The more respondents gave false answers to our questions, the more they rated themselves as well informed. In short, when it came to a positive self-view, people gave themselves credit for both their right and wrong answers. Clearly, the ability to generate those wrong answers—presumably fed by up to 109 bits of material in their brain—hid from people how little they actually knew (Dunning & Roh, 2018; see Dunlosky & Lipko, 2007, for similar effects in self-ratings of text comprehension).

Such divergences in belief mattered for behavior as well. In a follow-up analysis, we examined how sensitive respondents' true-false responses were (a) to the truth or (b) to their partisan leanings, in that they tended to endorse statements friendly to their politics as true. Then we looked at who reported they

had voted the day before. Voters, relative to non-voters, were not clearly more sensitive to the truth in their responses. However, respondents whose responses were sensitive to their partisan leanings were much more likely to vote than those whose answers were more even-handed (Dunning & Roh, 2018).

The pattern of people giving themselves positive credit for paralogic answers is one we have seen in a number of areas—from civics to beliefs about financial literacy. Those who think they know something that is wrong are as confident in their ability as those giving right answers. Consider the following financial question: You invest \$100 in a bond that pays 6% per year, compounded annually. If you leave it untouched, how long will it take for the value of the bond to double?

If you said 12 years, you have a good handle on compound interest and exponential growth. That is the correct answer. Many people, however, fail to understand how compound interest works and believe, instead, that investments grow in a more incremental and linear manner. They are likely to say it will take at least 16 years for your investment to double. In that, they are wrong, but people who consistently follow a more “linear” model of investment are likely to be as certain in their answers as are their peers who are consistently right (Williams et al., 2013).

Application to Beginners

This ability to form paralogical beliefs also explains how the Dunning-Kruger effect tends to be so widespread among beginners at a task. People do not necessarily approach new tasks with overconfidence. However, give them just a modicum of experience and feedback, and they display a burst of confidence that far outstrips any level of performance they achieve at the task. Put differently, a little learning might not necessarily be a dangerous thing, but it is a thing that feeds undue confidence in one’s ability.

In our lab, we have demonstrated this beginner’s bubble of overconfidence by asking people to tackle a computer game in which they must diagnose who is infected with a zombie disease in a post-apocalyptic world (Sanchez & Dunning, 2018). At the very beginning of the task, participants are quite cautious in the evaluations they make of their diagnoses, but it only takes a few diagnoses before their confidence explodes into something running far ahead of their accuracy rate. After a while, this inflation of confidence cools down considerably, but accuracy never catches up to it.

Further, we have found that the burst occurs because people take small scraps of experience and feedback as they begin the task and spin elaborate and self-assured theories about how to diagnose zombie diseases. In effect, they give their early experience too much credit, in that any small sample of experience is likely to contain a lot of noise and misleading signals (Sanchez & Dunning, 2018). It is only after extensive experience that one can begin to separate true patterns from chaotic noise. People learn this fact only after a while, while the degree of their overconfidence cools off; but they never completely learn the lesson that the information they have in their head, and their creativity at weaving those pieces together into plausible theories, makes it much too easy to reach an answer to any question, just not necessarily the right answer.

Implications for Self and Society

Naturally, all these difficulties in identifying the shape and scope of one’s ignorance carry many implications for human life. Some of them are personal and affect the individual. Some of them

ultimately affect society at large. Many of them, interestingly, intersect with philosophical topics and issues.

Self

Take the issue, for example, of rational ignorance (Downs, 1957). It is impossible for a person to know everything there is to know about the world. Much too much information now exists to allow a person to become a contemporary renaissance person. Further, it is imprudent to try to be that renaissance person, in that it is just not worth the effort to become expert in some topics. My life is not diminished, for example, by not knowing the history of the Baltic States or basic ballistics or the poems of Middle English. As such, one can make the argument that there are clear circumstances where it is rational to be ignorant of a topic, times when the person is better off being uninformed than expending the effort to become well informed.

No doubt, there is some truth to this argument, but the invisibility of personal ignorance makes it intrinsically difficult to know whether one is achieving rational ignorance or not. ***Essentially, blind to an area, we do not know enough to know if we are better off being blind. To make such a proper call about whether we can remain uneducated about an area, we would have to put in the effort of gaining an education in that area. As such, we are not in a position to make a call about whether our ignorance is adaptive or maladaptive.***

To be sure, in some areas, one could imagine the call would be easy to make. My work would likely not improve if I spent a few years studying the science and aesthetics of different typefaces. But how does one really know? For example, in my line of psychology, it would seem clear that I would never have to learn Bose-Einstein statistics, an esoteric strand of analysis developed in physics to account for the quantum behavior of small particles in various energy states. However, such a call might be preliminary and wrong—in that other psychological researchers have begun successfully applying such statistics to the errors people make in everyday decisions (Busemeyer et al., 2011).

Asking for Advice

Beyond deciding when it might be appropriate to be ignorant, people often face a much simpler question: Are they ignorant, and should they turn to the expertise of others. It is inefficient for each person to know or try to learn everything. A society benefits if people trade off which areas they will be expert in so that everyone can gain access to necessary expertise when they need it.

The problem is that the invisibility of ignorance suggests that people have substantial trouble knowing when they need to rely on someone else's expertise rather than their own. Any academic instructor who has had student advisees has seen the syndrome: Students with the most need of advice never come into the office to get it until it is too late. Work on in our lab demonstrates this advisor's paradox, in that people with the most need for advice are no more likely to seek it out than those who do not need it.

In these studies, we give participants a quiz that should be of some interest and familiarity to them. For example, it may be a survey about household hazards given to parents with at least one child under the age of 6 at home. They are given small monetary bonuses for each correct answer, but they can also ask for advice—to see how another person has responded to the same question (with no guarantee that this other person got the question right)—for a reduced bonus. We find that respondents rarely ask for advice

(13% of the time for the household safety questionnaire) and that poor performers ask for no more advice than do top performers (Yan & Dunning, 2018).

Judging Others

There is a further problem, however, that follows from the Dunning-Kruger effect: Flawed expertise not only prevents people from seeing their own ignorance; it also prevents them from recognizing superior competence in others. We have termed this problem the Cassandra quandary, after the Greek myth of the princess who was given the gift of true prophecy but also cursed by the gods to be disbelieved by all other humans. This inability to evaluate the expertise of others is more severe among poor performers, who cannot accurately identify which individuals are best to approach for advice. In one demonstration of this, we gave participants the answers that other people had given to a quiz on financial literacy. We asked who they would want to approach for financial advice. Of those with perfect scores on the quiz themselves, 91% chose the other person with a perfect score. Only 9% of those posting the worst score on the quiz chose similarly (Dunning & Cone, 2018).

Society

This theme of expertise carries over when thinking about societal implications. If people cannot identify either the expert or the worth of their expertise, society will be worse off in the aggregate.

Dismissing Expertise

The invisibility of ignorance promotes problems in dealing with expertise in two ways, which on the surface may seem contradictory. On the one hand, not being able to identify expertise may cause the less able to dismiss or discount it. At an extreme, nonexperts may mistakenly think their knowledge rivals that of experts. One sees signs of this in disparate areas of human life. A recent Pew survey showed that 83% of Americans felt they “understood the challenges police face on the job,” but only 13% of police officers agreed that citizens had sufficient understanding. It is likely that the public does not know how often the police have to deal with verbal abuse (two-thirds of officers reported experiencing such abuse within the last month) or how often they enter situations of uncertain safety (Morin et al., 2017).

Or, in the area of medicine, people may substitute their judgment for that of the doctor, an action called epistemic trespassing (Ballantyne, 2019). A study that examined 50 years of data revealed that patients failed to substantively adhere to their doctor’s instructions in roughly 45% of cases, depending mostly on the specific malady in question (DiMatteo, 2004). In the case of high chronic blood pressure, misbelief and ignorance are the main reasons people fail to adhere to the regimen their physician gives them. Patients may misunderstand the rationale for the treatment, misunderstand their condition, or overestimate their skill at controlling their blood pressure without medication (Kirscht & Rosenstock, 1977; Patel & Taylor, 2002; Svensson et al., 2000).

Ignorance and misunderstanding may also underlie, at least in part, distrust of scientific expertise. A significant proportion of the American public distrusts scientific conclusions, for example, about climate change and the safety of vaccines. Underlying this distrust is a belief that scientists are free to conclude and say whatever they wish to, but every scientist knows that is not the case. To make a statement to scientific standards, one must follow the strict and constraining rules of the scientific method and generate data that support the scientist’s conclusion. If the data fail to cooperate, the scientist cannot make the claim. In our work, we have shown that people who believe scientists can say whatever they please fail to show adequate knowledge of the scientific method, its constraining rules, and how those rules limit scientific claims. ***In short, hypocognitive of the scientific method, people misunderstand the***

basis for scientific claims. Those with fuller knowledge of scientific rules and their constraints turn out to trust scientists more (De Oliveira Chen & Dunning, 2018).

Gullibility to False Information

If the invisibility of ignorance leads to improper dismissal of true and valuable expertise, it also creates a cross-cutting problem—a gullibility to false information that purports to be fact or expertise. Upon its release, a deeply flawed 2014 study linking water fluoridation to lower IQs in children generated Twitter views and shares numbering in the tens of thousands (Vogel, 2017). The National Science Foundation has recently had to publicly deny a report that it was running a child prostitution ring on Mars (Holley, 2017).

The human inability to tell true from false but plausible information may lead people to be too credulous to such fake news and false information. Actually, the psychological situation may be worse than that. People show a truth bias, assuming that any new information they encounter is more likely to be true than false (DePaulo et al., 1997; Vrij, 2000; Zuckerman et al., 1981). Such a bias is reasonable for human activities like conversation—imagine a world in which people routinely disbelieved every single thing other people told them while chatting—but a truth bias may lead to costly gullibility in a digital world where the other person (if it is another person) shows no concern about maintaining honesty.

Concluding Remarks

*Better light a candle than curse the darkness.
—William L. Watkinson (2009)*

I began this chapter with Confucius. Let me end with the sermons of William L. Watkinson. As Confucius initially pointed out, real knowledge involves not only awareness of a fact but also the confidence to act on that fact. Much of this confidence depends on how people evaluate their expertise, but this chapter has pointed out all the ways in which that evaluation can go awry. Knowing what you know intrinsically requires also knowing what you do not know, but the problem here is the inherent invisibility of ignorance. People often fail to have a correct and authentic understanding of the shape and scope of what they do not know.

I have taken as my mission in this chapter to light a candle and expose just how difficult it is to see one's own ignorance. The candle reveals just how much darkness there is to curse. A reader might think exposing all this darkness may not be worth it, but I believe the both the old master and Pastor Watkinson would disagree. As Confucius himself further admonished, when you have faults, do not fear to abandon them. Learning how difficult it is to recognize one's ignorance may be an important initial step toward achieving knowledge as Confucius defined it. Lighting that candle, thus, may be an essential first step for finding the way.

BEHAVE

Sapolsky, Robert M.. Behave: The Biology of Humans at Our Best and Worst

RELEVANCE:

Best summary of what we are as humans in a few pages. Sapolsky is a brilliant, wonderful man.

Everyone should read Behave cover to cover a few times.

EPILOGUE

We've covered lots of ground, and some themes have arisen repeatedly. It's worth reviewing them before considering two final points.

As the single most important of them, virtually every scientific fact presented in this book concerns the average of what's being measured. There is always variation, and it's often the most interesting thing about a fact. Not every person activates the amygdala when seeing the face of a Them; not every yeast adheres to another one bearing the same surface protein marker. Instead, on the average, both do. Reflecting this, I've just discovered that this book contains variations on "average," "typically," "usually," "often," "tend to," and "generally" more than five hundred times. And I probably should have inserted them even more as reminders. There are individual differences and interesting exceptions everywhere you look in science.

Now, in no particular order:

- It's great if your frontal cortex lets you avoid temptation, allowing you to do the harder, better thing. But it's usually more effective if doing that better thing has become so automatic that it isn't hard. And it's often easiest to avoid temptation with distraction and reappraisal rather than willpower.
- While it's cool that there's so much plasticity in the brain, it's no surprise— it has to work that way.
- Childhood adversity can scar everything from our DNA to our cultures, and effects can be lifelong, even multigenerational. However, more adverse consequences can be reversed than used to be thought. But the longer you wait to intervene, the harder it will be.
- Brains and cultures coevolve.

- Things that seem morally obvious and intuitive now weren't necessarily so in the past; many started with nonconforming reasoning.
- Repeatedly, biological factors (e.g., hormones) don't so much cause a behavior as modulate and sensitize, lowering thresholds for environmental stimuli to cause it.
- Cognition and affect always interact. What's interesting is when one dominates.
- Genes have different effects in different environments; a hormone can make you nicer or crummier, depending on your values; we haven't evolved to be "selfish" or "altruistic" or anything else— we've evolved to be particular ways in particular settings. Context, context, context.
- Biologically, intense love and intense hate aren't opposites. The opposite of each is indifference.
- Adolescence shows us that the most interesting part of the brain evolved to be shaped minimally by genes and maximally by experience; that's how we learn— context, context, context.
- Arbitrary boundaries on continua can be helpful. But never forget that they are arbitrary.
- Often, we're more about the anticipation and pursuit of pleasure than about the experience of it.
- You can't understand aggression without understanding fear (and what the amygdala has to do with both).
- Genes aren't about inevitabilities; they're about potentials and vulnerabilities. And they don't determine anything on their own. Gene/ environment interactions are everywhere. Evolution is most consequential when altering regulation of genes, rather than genes themselves.
- We implicitly divide the world into Us and Them, and prefer the former. We are easily manipulated, even subliminally and within seconds, as to who counts as each.
- We aren't chimps, and we aren't bonobos. We're not a classic pair-bonding species or a tournament species. We've evolved to be somewhere in between in these and other categories that are clear-cut in other animals. It makes us a much more malleable and resilient species. It also makes our social lives much more confusing and messy, filled with imperfection and wrong turns.
- The homunculus has no clothes.

- While traditional nomadic hunter-gatherer life over hundreds of thousands of years might have been a little on the boring side, it certainly wasn't ceaselessly bloody. In the years since most humans abandoned a hunter-gatherer lifestyle, we've obviously invented many things. One of the most interesting and challenging is social systems where we can be surrounded by strangers and can act anonymously.
- Saying a biological system works "well" is a value-free assessment; it can take discipline, hard work, and willpower to accomplish either something wondrous or something appalling. "Doing the right thing" is always context dependent.
- Many of our best moments of morality and compassion have roots far deeper and older than being mere products of human civilization.
- Be dubious about someone who suggests that other types of people are like little crawly, infectious things.
- When humans invented socioeconomic status, they invented a way to subordinate like nothing that hierarchical primates had ever seen before.
- "Me" versus "us" (being prosocial within your group) is easier than "us" versus "them" (prosociality between groups).
- It's not great if someone believes it's okay for people to do some horrible, damaging act. But more of the world's misery arises from people who, of course, oppose that horrible act . . . but cite some particular circumstances that should make them exceptions. The road to hell is paved with rationalization.
- The certainty with which we act now might seem ghastly not only to future generations but to our future selves as well.
- Neither the capacity for fancy, rarefied moral reasoning nor for feeling great empathy necessarily translates into actually doing something difficult, brave, and compassionate.
- People kill and are willing to be killed for symbolic sacred values. Negotiations can make peace with Them; understanding and respecting the intensity of their sacred values can make lasting peace.
- We are constantly being shaped by seemingly irrelevant stimuli, subliminal information, and internal forces we don't know a thing about.
- Our worst behaviors, ones we condemn and punish, are the products of our biology. But don't forget that the same applies to our best behaviors.
- Individuals no more exceptional than the rest of us provide stunning examples of our finest moments as humans.

Two Last Thoughts

- If you had to boil this book down to a single phrase, it would be “It’s complicated.” Nothing seems to cause anything; instead everything just modulates something else. Scientists keep saying, “We used to think X, but now we realize that . . .” Fixing one thing often messes up ten more, as the law of unintended consequences reigns. On any big, important issue it seems like 51 percent of the scientific studies conclude one thing, and 49 percent conclude the opposite. And so on. Eventually it can seem hopeless that you can actually fix something, can make things better. But we have no choice but to try. And if you are reading this, you are probably ideally suited to do so. You’ve amply proven you have intellectual tenacity. You probably also have running water, a home, adequate calories, and low odds of festering with a bad parasitic disease. You probably don’t have to worry about Ebola virus, warlords, or being invisible in your world. And you’ve been educated. In other words, you’re one of the lucky humans. So try.
- Finally, you don’t have to choose between being scientific and being compassionate.

Protein Basics

Sapolsky, Robert M.. Behave: The Biology of Humans at Our Best and Worst

RELEVANCE:

Best summary description of the importance of proteins – this is an appendix that Sapolsky has in his book Behave. Proteins matter because they are molecular machines which are at the center of biology, but which are beginning to be utilized as nano-motors and nano-machines as the scale of human industrial activities and biology converge.

The potential role of proteins in physical engineering as well as in biological engineering is why we need to understand them. In the industrial context, when people talk about synthetic biology or using cells as factories – they are using cells to make proteins.

Proteins are a class of organic compounds that are the most abundant molecules in living systems. They are hugely important, since numerous hormones, neurotransmitters, and messengers of the immune system are made of protein; ditto for the receptors that respond to those messengers, the enzymes that construct or degrade them, the scaffolding that shapes a cell, and so on.

A key feature of proteins is their shape, since the shape of a protein determines its function. Proteins that form the scaffolding of a cell have the shape of the different crossbars in scaffolding at construction sites (sort of). A protein hormone will have a distinctive shape that is unique and distinctively different from the shape of a hormone that has different effects. And a protein receptor must have a shape that is complementary to the shape of the hormone or neurotransmitter that it binds (back to the time-honored cliché of appendix 1, namely that a messenger like a hormone fits into its receptor like a key into a lock).

Some proteins change their shape, typically moving between two conformations. Suppose you have an enzyme (again, a protein) that synthesizes a molecule of sucrose by linking a molecule of glucose to a molecule of fructose. The enzyme must have one conformation that resembles the letter V, where one end binds a glucose molecule at a particular angle, the other fructose. The binding of both triggers the enzyme to shift to its other conformation, where the two ends of the V move close enough for the glucose and fructose to be linked. The sucrose floats off, and the enzyme flips back to its original conformation.

What determines the shape and function of a protein? Any given protein is made of a string of amino acids. There are about 20 different types of amino acids— including some familiar ones like tryptophan and glutamate. Each protein's string of amino acids is unique— like the string of letters that composes a word. Your typical protein is about 300 amino acids long, and with 20 different amino acid types, there are nearly 10400 possible sequences (that's ten followed by four hundred zeros)— more atoms than there are in the universe.* The amino acid sequence of a protein influences the unique shape(s) of that protein. Dogma used to be that amino acid sequence determines the shape(s) of that protein, but it turns out that the shape is also subtly altered by things like temperature and acidity— in other words, environmental influences.

And what determines the sequence of amino acids that are strung together to form a particular protein? A particular gene.

DNA AS THE BLUEPRINT FOR CONSTRUCTING PROTEINS

DNA is another class of organic compounds, and just as there are roughly 20 different types of amino acids, there are 4 different “letters” (called nucleotides) that make up DNA. A sequence of 3 nucleotides (called a codon) codes for a single amino acid. If there are 4 different types of nucleotides, and each codon is 3 nucleotides long, there can be a total of 64 different codons (4 possibilities in the first place $\times 4$ in the second $\times 4$ in the third = 64). A few of those 64 are reserved to signal the end of a gene, and after eliminating those “stop codons,” there are 61 different codons coding for 20 different amino acids. Therefore, there is “redundancy”—almost all amino acids can be specified by more than one unique codon (an average of about 3, i.e., $61/20$). Typically the different codons coding for the same amino acid differ by only a single nucleotide. For example, four different codon sequences code for the amino acid alanine: GCA, GCC, GCG, and GCT (A, C, G, and T are the abbreviations for the four types of nucleotides).^{*} Redundancy will be important for understanding gene evolution.

The entire stretch of nucleotides that codes for a single type of protein is called a gene. The entire collection of DNA is called the genome, coding for all of the tens of thousands of genes in an organism; “sequencing” the genome means determining the unique sequence of the billions of nucleotides that make up that organism’s genome. That stretch of DNA is so long (containing roughly twenty thousand genes in humans) that it has to be broken into separate volumes, called chromosomes.

This produces a spatial problem. The DNA library is found in the center of the cell, in the nucleus. Proteins, however, occur all over the cell, are constructed all over it (just think of proteins in the axon terminals of a spinal neuron in a blue whale, terminals that are light-years away from that neuron’s nucleus). How do you get the DNA information out to where the protein is made? There is an intermediary that completes the picture. The unique nucleotide sequence in DNA that codes for a particular gene is copied into a string of similar nucleotide letters in a related compound called RNA. Any given chromosome contains a staggeringly long stretch of DNA, coding for one gene after another; in contrast, this stretch of RNA is only as long as the particular gene. In other words, a more manageable length. That RNA is then shipped to wherever it is supposed to be in the cell, where it then directs which amino acids are strung together in which sequence to form a protein (and there are amino acids floating around in a cell, ready to be grabbed for the protein-construction project). Think of RNA as a photocopy of a single page out of this vast twenty-thousand-page-long DNA encyclopedia. (And multiple copies of the cognate protein can be made from the instructions in a photocopy page of RNA. This sure helps in circumstances where copies of the protein must wind up in each of the thousands of a single neuron’s axon terminals.)

This produces what is termed the “central dogma” of life, a concept first framed in the early 1960s by Francis Crick, half of the renowned Watson and Crick, who discovered the “double helix” structure of DNA (with more than a little purloined help from Rosalind Franklin, but that’s another story). Crick’s central dogma holds that the nucleotide sequence of DNA that composes a gene determines how a unique stretch of RNA is put together . . . which determines how a unique stretch of amino acids are put together . . . which determines the shape(s) of the resulting protein . . . which determines that protein’s function. DNA determines RNA determines protein. And implicit in that central dogma is another critical point: one type of gene specifies one type of protein.

Just for everyone’s sanity, I’m going to mostly ignore RNA. For our purposes, what is interesting is what genes, the starting point, have to do with their end products—proteins and their functions.

MUTATIONS AND POLYMORPHISMS

Genes are inherited from your parents (half the genes from each [not entirely true, as covered in the main text]). Suppose that when someone's DNA genome is being copied for inclusion in their egg or sperm, a mistake is made in the copying of one single nucleotide; with billions of nucleotides, that's bound to happen sometimes. As a result, unless corrected, the gene, now with its nucleotide sequence erroneously differing in one spot, is passed on to an offspring. This is a mutation.

In classical genetics there are three types of mutations that can occur. The first is called a point mutation. One single nucleotide is copied incorrectly. Will this change the amino acid sequence of the protein coded for? It depends. Back to redundancy in the DNA code, from a few paragraphs ago. Suppose there is a codon in a gene with the sequence GCT, coding for alanine. But there has been a mutation, yielding GCA instead. No problem—that still codes for alanine. It's an inconsequential, "neutral" mutation. But suppose the mutation instead was GAT. This codes for a completely different amino acid called asparagine. Uh-oh.

In actuality, though, this may not be a big deal, if the new amino acid looks a lot like the one that was lost. Suppose you have a nucleotide sequence coding for the following metaphorical amino acid sequence: "I/ am/ now/ going/ to/ do/ the/ following".

Thanks to a subtle mutation, there is a change of one amino acid, but one without a ton of consequences: "I/ am/ now/ going/ ta/ do/ the/ following". This would still be comprehensible to most people; the protein would merely be perceived as coming from New York. Translated into protein-ese, the protein has a slightly different shape and does its usual task a bit differently (maybe a little slower or faster). Not the end of the world.

But if the mutation codes for an amino acid that produces a protein with a dramatically different shape, the consequences can be enormous (even fatal).

.... Point, deletion, and insertion mutations are most of what mutations are about. Deletion and insertion mutations often have major consequences, usually deleterious, but sometimes produce a new, interesting protein.

Back to point mutations. Consider one that results in the substitution of a single amino acid in the protein, one that works a bit differently from the correct amino acid. As noted above, as a result, the protein still does its old job, but maybe does it a bit faster or slower. This could be the grist for evolutionary change—if the new version is disadvantageous, reducing the reproductive success of anyone who carries it, it will be gradually selected against, removing it from a population. If instead the new version is more advantageous, it will gradually replace the old one in a population. Or if the new version works better than the original in some circumstances but worse in others, it may reach equilibrium in the population with the original version, where a certain percentage of people have the old version, the remainder the new. In this case the particular gene would be described as coming in two different forms or variants, as coming with two different "alleles." Most genes come with multiple alleles. And the result is individual variation in the functioning of genes (this is covered in far more complexity in chapter 8).

MORE FROM LESS

McAfee, Andrew. More from Less: The Surprising Story of How We Learned to Prosper Using Fewer Resources—and What Happens Next.

RELEVANCE:

There's a lot one can take from McAfee's book, More From Less, and the excerpts below.

He offers a nice framework in his four horsemen of optimism and also his four paths to dematerialization. The basic idea that technology enabled dematerialization is the core of getting more from less – but if we connect more from less with ideas of convergence and acceleration – then what happens when we digitize any area is that it both dematerializes, gets more efficient and the innovation cycles accelerate as something that we physical can be modelled and iterated and shared, etc., virtually.

Note that McAfee's discussion of recombination is logically identical to all the complexity folks – they are all talking about the same things – same elephant, different blind man.. Also – worth noting McAfee's co-author on a few of his previous books, Erik Brynjolfsson, is at Stanford now.

So while McAfee expounds on where and why we are getting more from less – we should analytically flip this on its head and note that the sign of a company which has what it will take to prosper in the future will already be getting more from less today.

INTRODUCTION: README

*Listen! I will be honest with you,
I do not offer the old smooth prizes, but offer rough new prizes
—Walt Whitman, "Song of the Open Road," 1856*

We have finally learned how to tread more lightly on our planet. It's about time.

For just about all of human history our prosperity has been tightly coupled to our ability to take resources from the earth. So as we became more numerous and prosperous, we inevitably took more: more minerals, more fossil fuels, more land for crops, more trees, more water, and so on.

But not anymore. In recent years we've seen a different pattern emerge: the pattern of more from less. In America—a large, rich country that accounts for about 25 percent of the global economy—we're now generally using less of most resources year after year, even as our economy and population continue to grow. What's more, we're also polluting the air and water less, emitting fewer greenhouse gases, and seeing population increases in many animals that had almost vanished. America, in short, is post-peak in its exploitation of the earth. The situation is similar in many other rich countries, and even developing countries such as China are now taking better care of the planet in important ways.

This book is about how we turned the corner and started getting more from less, and what happens from here forward. I want to make one thing clear at the start: my argument is not that things are good enough now, or that there's nothing to be concerned about. Those claims would be absurd. Human-caused global warming is both real and bad, and we urgently need to take action to deal with it. We also need to reduce pollution levels around the world and bring back the species we've pushed to the brink of extinction. And we have to keep fighting poverty, disease, malnutrition, fraying communities, and other roadblocks to human flourishing.

So, we have plenty of work ahead. The broad point I want to make is that we know how to succeed with this work. In large parts of the world we've already turned the corner and are now improving both the human condition and the state of nature. The trade-off between the two has ended, and I'm confident it's never going to reappear if we play our cards right. In these pages I'll explain where this confidence comes from and try to get you to share it.

The Thread of the Argument

This book shows that we've started getting more from less and tells how we reached this critical milestone. The strangest aspect of the story is that we didn't make many radical course changes to eliminate the trade-off between human prosperity and planetary health. Instead, we just got a lot better at doing the things we'd already been doing.

In particular, we got better at combining technological progress with capitalism to satisfy human wants and needs. That conclusion will strike many people as bizarre, and for good reason. After all, it's exactly this combination that caused us to massively increase our resource use and environmental harms starting with the Industrial Revolution in the late eighteenth century. The Industrial Era was a time of startlingly large and fast improvements in human prosperity, but these improvements came at the expense of our planet. We dug out resources, chopped down forests, killed animals, fouled the air and water with pollution, and committed countless other offenses against the earth. We committed more and more of them year after year, apparently without end.

The twin forces of tech progress and capitalism unleashed during the Industrial Era seemed to be impelling us in one direction: that of increasing human population and consumption while degrading our planet. By the time of the first Earth Day festival in 1970, it was obvious to many that these two forces would push us to our doom, since we couldn't continue to abuse our planet indefinitely.

And what actually happened? Something completely different, which is the subject of this book. As I'll show, capitalism continued and spread (just look around you), but tech progress changed. ***We invented the computer, the Internet, and a suite of other digital technologies that let us dematerialize our consumption: over time they allowed us to consume more and more while taking less and less from the planet. This happened because digital technologies offered the cost savings that come from substituting bits for atoms, and the intense cost pressures of capitalism caused companies to accept this offer over and over. Think, for example, how many devices have been replaced by your smartphone.***

In addition to capitalism and tech progress, two other forces have also been essential for allowing us to get more from less. These are public awareness of the harms we're doing our planet (such as pollution and species loss) and responsive governments, which act on the desires of their people and put in place sound measures to counteract these harms. Both public awareness and responsive government were greatly accelerated by Earth Day and the environmental movement in the United States and around the world.

I call tech progress, capitalism, public awareness, and responsive government the "four horsemen of the optimist." I When all four are in place, countries can improve both the human condition and the state of nature. When the four horsemen don't all ride together, people and the environment suffer.

The good news is that all four are at present advancing around the world. So we don't need to make radical changes; instead, we need to do more of the good things that we're already doing. Let me switch, metaphorically, from horses to cars: we don't need to yank the steering wheel of our economies and societies in a different direction; we just need to step on the accelerator.

Something for Everyone to Dislike

As you read this book, it'll be important to keep an open mind because you're likely to come across at least a few ideas and conclusions that won't seem right at first. I've found that the book's fundamental concept—that capitalism and tech progress are now allowing us to tread more lightly on the earth instead of stripping it bare—is hard for many people to accept.

It was hard for me to accept when I came across it for the first time—in [Jesse Ausubel's amazing essay "The Return of Nature: How Technology Liberates the Environment," published in 2015](#) in the Breakthrough Journal. When I encountered that headline, I had to click on it, which led me to one of the most interesting things I'd ever read.

Ausubel documented the dematerialization of the American economy. Even though he did it carefully and thoroughly, I found myself thinking, "Well, that can't be right." It was hard to let go of the notion that as economies grow they must consume more resources. Ausubel's work started me down a path of questioning that notion, and eventually rejecting it.

An important part of traveling that research path was coming up with an explanation of how we started getting more from less. What caused economic growth to become decoupled from resource consumption? What caused dematerialization to take over? As I've already mentioned and as you'll see in the chapters ahead, capitalism is a big part of my explanation.

This is not a universally popular conclusion. Ever since Marx, capitalism has been passionately opposed by many—and viewed with much skepticism by many more. So my cheerleading for it is going to strike many as ignorant, or worse. If you're one of these people, I'm glad you're reading this book. I hope that you'll listen to what I mean when I talk about capitalism and evaluate my arguments based on the evidence and logic I present.

And if you're a fan of capitalism, you might not like that I argue here in favor of new taxes (on carbon) and strict regulation (on pollution and trade in products from endangered animals). Many ardent capitalists will dislike these ideas. I also propose more nuclear power and genetically modified organisms, both of which are adamantly opposed by many people.

So just about any reader will probably initially feel that something in this book is wrong. Again, I just ask that you approach the book's ideas with an open mind. I hope you'll believe that I'm arguing in good faith. My intention here is not to write a polemic or start a flame war. I'm not trying to troll or dunk on anyone (in other words, I'm not trying to provoke anyone into losing their temper or to demonstrate my superiority). I'm just trying to highlight a phenomenon that I find fascinating and deeply encouraging, explain how it came about, and discuss its implications. I hope you'll come along for the journey.

Chapter 7: What Causes Dematerialization? Markets and Marvels

The triumph of the industrial arts will advance the cause of civilization more rapidly than its warmest advocates could have hoped.

—Charles Babbage, *The Exposition of 1851; or, Views of the Industry, the Science, and the Government of England, 1851*

If CRIB strategies aren't responsible for the large-scale dematerialization of the American economy that has taken place since Earth Day, then what is? How have we got more from less? I believe that four main forces are

responsible, and that it's helpful to think of them as two pairs. In this chapter we'll look at the first pair, then take up the second in chapter 9.

Capitalism and technological progress are the first pair of forces driving dematerialization. This statement will come as a surprise to many, and for good reason. After all, it's exactly this combination that caused us to massively increase our resource consumption throughout the Industrial Era. As we saw in chapter 3, the ideas of William Jevons and Alfred Marshall point to the distressing conclusion that capitalism and tech progress always lead to more from more: more economic growth, but also more resource consumption.

So what changed? How are capitalism and tech progress now getting us more from less? To get answers to these important questions, let's start by looking at a few recent examples of dematerialization.

Fertile Farms

America has long been an agricultural juggernaut. In 1982, after more than a decade of steady expansion due in part to rising grain prices, total cropland in the country stood at approximately 380 million acres. Over the next ten years, however, almost all of this increase was reversed. So much acreage was abandoned by farmers and given back to nature that cropland in 1992 was almost back to where it had been almost twenty-five years before. This decline had several causes, including falling grain prices, a severe recession, over-indebted farmers, and increased international competition.

A final factor, though, was the ability to get ever-more corn, wheat, soybeans, and other crops from an acre of land, pound of fertilizer and pesticide, and gallon of water. The material productivity of agriculture in the United States has improved dramatically in recent decades, as we saw in chapter 5. ***Between 1982 and 2015 over 45 million acres—an amount of cropland equal in size to the state of Washington—was returned to nature. Over the same time potassium, phosphate, and nitrogen (the three main fertilizers) all saw declines in absolute use. Meanwhile, the total tonnage of crops produced in the country increased by more than 35 percent.***

As impressive as this is, it's dwarfed by the productivity improvements of American dairy cows. In 1950 we got 117 billion pounds of milk from 22 million cows. In 2015 we got 209 billion pounds from just 9 million animals. The average milk cow's productivity thus improved by over 330 percent during that time.

Thin Cans

Tin cans are made of steel coated with a thin layer of tin to improve corrosion resistance. They've been used since the nineteenth century to store food. Starting in the 1930s, they began also to be used to hold beer and soft drinks

In 1959 Coors pioneered beer cans made of aluminum, which is lighter and more corrosion resistant than steel. Royal Crown Cola followed suit for soda five years later. As Vaclav Smil relates, "A decade later steel cans were on the way out, and none of them have been used for beer since 1994 and for soft drinks since 1996.... At 85 g the first aluminum cans were surprisingly heavy; by 1972 the weight of a two-piece can dropped to just below 21 g, by 1988 it was less than 16 g, a decade later it averaged 13.6 g, and by 2011 it was reduced to 12.75 g."

Manufacturers accomplished these reductions by making aluminum cans' walls thinner, and by making the sides and bottom from a single sheet of metal so that only one comparatively heavy seam was needed (to join the top to the rest of the can). Smil points out that if all beverage cans used in 2010 weighed what they did in 1980, they would have required an extra 580,000 tons of aluminum. And aluminum cans kept getting lighter. In 2012 Ball packaging introduced into the European market a 330 ml can that held 7.5 percent less than the US standard, yet at 9.5 g weighed 25 percent less.

Gone Gizmos

In 2014 Steve Cichon, a “writer, historian, and retired radio newsmen in Buffalo, NY,” paid \$3 for a large stack of front sections of the Buffalo News newspaper from the early months of 1991. On the back page of the Saturday, February 16, issue was an ad from the electronics retailer Radio Shack. Cichon noticed something striking about the ad: “There are 15 electronic gizmo type items on this page.... 13 of the 15 you now always have in your pocket.”

The “gizmo type items” that had vanished into the iPhone Cichon kept in his pocket included a calculator, camcorder, clock radio, mobile telephone, and tape recorder. While the ad didn’t include a compass, camera, barometer, altimeter, accelerometer, or GPS device, these, too, have vanished into the iPhone and other smartphones, as have countless atlases and compact discs.

The success of the iPhone was almost totally unanticipated. A November 2007 cover story in Forbes magazine touted that the Finnish mobile phone maker Nokia had over a billion customers around the world and asked, “Can anyone catch the cell phone king?”

Yes. Apple sold more than a billion iPhones within a decade of its June 2007 launch and became the most valuable publicly traded company in history. Nokia, meanwhile, sold its mobile phone business to Microsoft in 2013 for \$7.2 billion to get “more combined muscle to truly break through with consumers,” as the Finnish company’s CEO Stephen Elop said at the time of the deal.

It didn’t work. Microsoft sold what remained of Nokia’s mobile phone business and brand to a subsidiary of the Taiwanese electronics manufacturer Foxconn for \$350 million in May of 2016. Radio Shack filed for bankruptcy in 2015, and again in 2017.

From Peak Oil to... Peak Oil

In 2007 US coal consumption reached a new high of 1,128 million short tons, over 90 percent of which was burned to generate electricity. Total coal use had increased by more than 35 percent since 1990, and the US Energy Information Administration (the official energy statisticians of the US government) forecast further growth of up to 65 percent by 2030.

Also in 2007 the US Government Accountability Office (GAO), a federal agency known as “the congressional watchdog,” published a report with an admirably explanatory title: “Crude Oil: Uncertainty about Future Oil Supply Makes It Important to Develop a Strategy for Addressing a Peak and Decline in Oil Production.” It took seriously the idea of “peak oil,” a phrase coined in 1956 by M. King Hubbert, a geologist working for Shell Oil. As originally conceived, peak oil referred to the maximum amount of oil that we could annually produce for all of humanity’s needs.

The first oil wells pumped out the crude oil that was closest to the earth’s surface or otherwise easiest to access. As those wells dried up, we had to drill deeper ones, both on land and at sea. As the world’s economies kept growing, so did total demand for oil, which kept getting harder and harder to obtain. Peak oil captured the idea that despite our best efforts and ample incentive, we would come to a time after which we would only be able to extract less and less oil year after year from the earth. Most of the estimates summarized in the GAO report found that peak oil would occur no later than 2040.

The report did not mention fracking, which in retrospect looks like a serious omission. Fracking is short for “hydraulic fracturing” and is a means of obtaining oil and natural gas from rock formations lying deep underground. It uses a high-pressure fluid to cause fractures in the rock, through which oil and gas can flow and be extracted.

The United States and other countries have long been known to have huge reserves of hydrocarbons in deep rock formations, which are often called shales. Companies had been experimenting with fracking to get at them since the middle of the twentieth century but had made little progress. In 2000 fracking accounted for just 2 percent of US oil production.

That figure began to increase quickly right around the time of the GAO report. Not because of any single breakthrough, but instead because the suite of tools and techniques needed for profitable fracking had all improved enough. A gusher of shale oil and gas ensued.

Thanks to fracking, US crude oil production almost doubled between 2007 and 2017, when it approached the benchmark of 10 million barrels per day. By September of 2018 America had surpassed Saudi Arabia to become the world's largest producer of oil. American natural gas production, which had been essentially flat since the mid-1970s, jumped by nearly 43 percent between 2007 and 2017.

As a result of the fracking boom the United States has experienced peak coal rather than peak oil. And the peak in coal is not in total annual supply, but instead in demand. Fracking made natural gas cheap enough that it became preferred over coal for much electricity generation. By 2017 total US coal consumption was down 36 percent from its 2007 high point.

The phrase peak oil is still around, but, as is the case with coal, it usually no longer refers to supply. As a 2017 Bloomberg headline put it, "Remember Peak Oil? Demand May Top Out Before Supply Does." Even though the extra supply from fracking has helped push down oil and gas prices, many observers now believe that energy from other sources—the sun, wind, and the nuclei of uranium atoms—is getting cheaper faster and becoming much more widely available. So much so that, as a 2018 article in Fortune about the future of oil hypothesized, "This wouldn't be just another oil-price cycle, a familiar roller coaster in which every down is followed by an up. It would be the start of a decades-long decline of the Oil Age itself—an uncharted world in which... oil prices might be 'lower forever.' " Analysts at Shell, the company from which the phrase peak oil originated, now estimate that global peak oil demand might come as soon as 2028.

Taking Stock of Rolling Stock

My friend Bo Cutter started his career in 1968 working for Northwest Industries, a conglomerate that owned the Chicago and North Western Railway. One of his first assignments was to help a team tasked with solving a problem that sounds odd to modern ears: figuring out where CNW's railcars were.

These cars are massive metal assemblies, each weighing thirty tons or more. In the late 1960s CNW owned thousands of them, representing a huge commitment of both material and money. Across the railroad industry, the rule of thumb then was that about 5 percent of a company's railcars moved on any given day. This was not because the other 95 percent needed to rest. It was because their owners didn't know where they were.

CNW owned thousands of miles of track in places as far from Chicago as North Dakota and Wyoming. Its rolling stock (as locomotives and railcars are called) could also travel outside the company's network on tracks owned by other railroads. So these assets could be almost anywhere in the country.

When the railcars weren't moving, they sat in freight yards. At the time Cutter started his job, freight yards didn't keep up-to-date records of the idle rolling stock they contained because, in the days before widespread digital computers, sensors, and networks, there was no way to cost-effectively know or communicate the location of each car. So it was impossible for CNW or any other railroad to systematically track its most important inventory, even though doing so would be hugely beneficial to the company's bottom line. For example, Cutter's team knew that if they could increase the percentage of cars moving each day from 5 percent

to 10 percent, they would need only half as many of them. Even a single percentage point increase in freight-car use would yield major financial benefits.

When Cutter started his assignment, CNW and all other railroads employed spotters, who visited yards and watched trains pass, then telegraphed their findings to the head office. Other railroads passed on similar information to collect the demurrage charges they were owed for each CNW car on their tracks and in their yards. Cutter's team improved on these methods by making them more systematic and efficient. They put in place a better baseline audit of where railcars were, employed more spotters, painted CNW cars differently so they were easier to see, and explored how to make more use of a new tool for businesses: the digital computer.

That tool and its kin are now pervasive in the railroad industry. In the early 1990s, for example, companies started putting radio-frequency identification tags on each piece of rolling stock. These tags would be read by trackside sensors, thus automating the work of spotting. At present over 5 million messages about railcar status and location are generated and sent throughout the American railway system every day, and the country's more than 450 railroads have nearly real-time visibility over all their rolling stock.

....

What's Going On?

There is no shortage of examples of dematerialization. I chose the ones in this chapter because they illustrate a set of fundamental principles at the intersection of business, economics, innovation, and our impact on our planet. They are:

- **We do want more all the time, but not more resources.**

Alfred Marshall was right, but William Jevons was wrong. Our wants and desires keep growing, evidently without end, and therefore so do our economies. But our use of the earth's resources does not. We do want more beverage options, but we don't want to keep using more aluminum in drink cans. We want to communicate and compute and listen to music, but we don't want an arsenal of gadgets; we're happy with a single smartphone. As our population increases, we want more food, but we don't have any desire to consume more fertilizer or use more land for crops.

Jevons was correct at the time he wrote that total British demand for coal was increasing even though steam engines were becoming much more efficient. He was right, in other words, that the price elasticity of demand for coal-supplied power was greater than one in the 1860s. But he was wrong to conclude that this would be permanent. Elasticities of demand can change over time for several reasons, the most fundamental of which is technological change. Coal provides a clear example of this. When fracking made natural gas much cheaper, total demand for coal in the United States went down even though its price decreased.

With the help of innovation and new technologies, economic growth in America and other rich countries—growth in all of the wants and needs that we spend money on—has become decoupled from resource consumption. This is a recent development and a profound one.

- ***Materials cost money that companies locked in competition would rather not spend.***

The root of Jevons's mistake is simple and boring: resources cost money. He realized this, of course. What he didn't sufficiently realize was how strong the incentive is for a company in a contested market to reduce its spending on resources (or anything else) and so eke out a bit more profit. After all, a penny saved is a penny earned.

Monopolists can just pass costs on to their customers, but companies with a lot of competitors can't. So American farmers who battle with each other (and increasingly with tough rivals in other countries) are eager to cut their spending on land, water, and fertilizer. Beer and soda companies want

to minimize their aluminum purchases. Producers of magnets and high-tech gear run away from REE as soon as prices start to spike. In the United States, the 1980 Staggers Act removed government subsidies for freight-hauling railroads, forcing them into competition and cost cutting and making them all the more eager to not have expensive railcars sit idle. Again and again, we see that competition spurs dematerialization.

- **There are multiple paths to dematerialization.**

As profit-hungry companies seek to use fewer resources, they can go down four main paths. First, they can simply find ways to use less of a given material. This is what happened as beverage companies and the companies that supply them with cans teamed up to use less aluminum. It's also the story with American farmers, who keep getting bigger harvests while using less land, water, and fertilizer. Magnet makers found ways to use fewer rare earth metals when it looked as if China might cut off their supply.

Second, it often becomes possible to substitute one resource for another. Total US coal consumption started to decrease after 2007 because fracking made natural gas more attractive to electricity generators. **If nuclear power becomes more popular in the United States (a topic we'll take up in chapter 15), we could use both less coal and less gas and generate our electricity from a small amount of material indeed. A kilogram of uranium-235 fuel contains approximately 2–3 million times as much energy as the same mass of coal or oil.** According to one estimate, the total amount of energy that humans consume each year could be supplied by just seven thousand tons of uranium fuel.

Third, companies can use fewer molecules overall by making better use of the materials they already own. Improving CNW's railcar utilization from 5 percent to 10 percent would mean that the company could cut its stock of these thirty-ton behemoths in half. Companies that own expensive physical assets tend to be fanatics about getting as much use as possible out of them, for clear and compelling financial reasons. For example, the world's commercial airlines have improved their load factors—essentially the percentage of seats occupied on flights—from 56 percent in 1971 to more than 81 percent in 2018.

Finally, some materials get replaced by nothing at all. When a telephone, camcorder, and tape recorder are separate devices, three total microphones are needed. When they all collapse into a smartphone, only one microphone is necessary. That smartphone also uses no audiotapes, videotapes, compact discs, or camera film. The iPhone and its descendants are among the world champions of dematerialization. They use vastly less metal, plastic, glass, and silicon than did the devices they have replaced and don't need media such as paper, discs, tape, or film.

If we use more renewable energy, we'll be replacing coal, gas, oil, and uranium with photons from the sun (solar power) and the movement of air (wind power) and water (hydroelectric power) on the earth. All three of these types of power are also among dematerialization's champions, since they use up essentially no resources once they're up and running.

I call these four paths to dematerialization slim, swap, optimize, and evaporate. They're not mutually exclusive. Companies can and do pursue all four at the same time, and all four are going on all the time in ways both obvious and subtle.

- **Innovation is hard to foresee.**

Neither the fracking revolution nor the world-changing impact of the iPhone's introduction were well understood in advance. Both continued to be underestimated even after they occurred. The iPhone was introduced in June of 2007, with no shortage of fanfare from Apple and Steve Jobs. Yet several months later the cover of Forbes was still asking if anyone could catch Nokia.

Innovation is not steady and predictable like the orbit of the Moon or the accumulation of interest on a certificate of deposit. It's instead inherently jumpy, uneven, and random. It's also combinatorial, as Erik Brynjolfsson and I discussed in our book The Second Machine Age. Most new technologies and other innovations, we argued, are combinations or recombinations of preexisting elements.

The iPhone was “just” a cellular telephone plus a bunch of sensors plus a touch screen plus an operating system and population of programs, or apps. All these elements had been around for a while before 2007. It took the vision of Steve Jobs to see what they could become when combined. Fracking was the combination of multiple abilities: to “see” where hydrocarbons were to be found in rock formations deep underground; to pump down pressurized liquid to fracture the rock; to pump up the oil and gas once they were released by the fracturing; and so on. Again, none of these was new. Their effective combination was what changed the world's energy situation.

Erik and I described the set of innovations and technologies available at any time as building blocks that ingenious people could combine and recombine into useful new configurations. These new configurations then serve as more blocks that later innovators can use. Combinatorial innovation is exciting because it's unpredictable. It's not easy to foresee when or where powerful new combinations are going to appear, or who's going to come up with them. But as the number of both building blocks and innovators increases, we should have confidence that more breakthroughs such as fracking and smartphones are ahead. Innovation is highly decentralized and largely uncoordinated, occurring as the result of interactions among complex and interlocking social, technological, and economic systems. So it's going to keep surprising us.

- ***As the Second Machine Age progresses, dematerialization accelerates.***

Erik and I coined the phrase Second Machine Age to draw a contrast with the Industrial Era, which as we've seen transformed the planet by allowing us to overcome the limitations of muscle power. Our current time of great progress with all things related to computing is allowing us to overcome the limitations of our mental power and is transformative in a different way: it's allowing us to reverse the Industrial Era's bad habit of taking more and more from the earth every year.

Computer-aided design tools help engineers at packaging companies design generations of aluminum cans that keep getting lighter. Fracking took off in part because oil and gas exploration companies learned how to build accurate computer models of the rock formations that lay deep underground—models that predicted where hydrocarbons were to be found.

Smartphones took the place of many separate pieces of gear. Because they serve as GPS devices, they've also led us to print out many fewer maps and so contributed to our current trend of using less paper. It's easy to look at generations of computer paper, from 1960s punch cards to the eleven-by-seventeen-inch fanfold paper of the 1980s and conclude that the Second Machine Age has caused us to chop down ever more trees. The year of peak paper consumption in the United States, however, was 1990. As our devices have become more capable and interconnected, always on and always with us, we've sharply turned away from paper. Humanity probably hit peak paper in 2013.

As these examples indicate, computers and their kin help us with all four paths to dematerialization. Hardware, software, and networks let us slim, swap, optimize, and evaporate. I contend that they're the best tools we've ever invented for letting us tread more lightly on our planet.

All of these principles are about the combination of technological progress and capitalism, which are the first of the two pairs of forces causing dematerialization.

Technology: The Human Interface with the Material World

One of my favorite definitions of technology comes from the philosopher Emmanuel Mesthene, who called it “the organization of knowledge for the achievement of practical purposes.” Sometimes that knowledge is crystallized into products such as hammers and iPhones, and sometimes it exists as techniques such as those for fracking or precision agriculture.

Like knowledge itself, technologies accumulate. We haven’t forgotten about the lever, the plow, or the steam engine in the Second Machine Age, and we haven’t had to give them up to use cloud computing or drones. Like innovation itself, technologies are combinatorial; most of them are combinations or recombinations of existing things. This implies that the number of potentially powerful new technologies increases over time because the number of available building blocks does.

These facts help me understand why we didn’t start to dematerialize sooner. It could simply be that we didn’t have the right technologies, or enough building blocks, to allow large-scale dematerialization. We had technologies that made it feasible and profitable for us to grow by taking more and more from the earth—more and more metals, fuels, water, fertilizers, and so on—but not ones that made it possible to profitably grow while taking less and less. In the Second Machine Age, that has changed.

My other preferred definition of technology comes from the great science fiction author Ursula K. Le Guin, who wrote, “Technology is the active human interface with the material world. Its technology is how a society copes with physical reality: how people get and keep and cook food, how they clothe themselves, what their power sources are (animal? human? water? wind? electricity? other?), what they build with and what they build, their medicine—and so on and on. Perhaps very ethereal people aren’t interested in these mundane, bodily matters, but I’m fascinated by them.” So am I, because these “mundane matters” have twice reshaped the world—first during the Industrial Era, when technological progress allowed us to prosper by taking more from the planet, and now in the Second Machine Age, when we’ve finally figured out how to prosper while taking less.

Capitalism: Means of Production

Capitalism and religion are the two subjects that leave the fewest people on the sidelines. People have very firmly held opinions on both topics, and few change their minds no matter what evidence and arguments are presented to them. Yet despite this clear history of intransigence, many thinkers and writers have tried to bring others around to their point of view on both topics. Most have failed.

I’m going to join this long sad parade by arguing in favor of capitalism. Before I do that, though, I want to define what I’m talking about. **Even more than is the case with technology, clear definitions are important with capitalism because it’s such a triggering word. As the psychologist Jonathan Haidt has pointed out, some hear it as a synonym for liberation, others for exploitation.**

But let me put the dictionary before the thesaurus and offer a definition of what capitalism is before suggesting what it’s like. For our purposes, capitalism is a way to come up with goods and services and get them to people. Every society that doesn’t want its people to starve or die of exposure has to accomplish this task; capitalism is simply one approach to doing it. The important features of this approach are:

- **Profit-seeking companies.** Under capitalism, most goods and services are produced by for-profit companies rather than nonprofits, the government, or individuals. Companies can be owned by only a few people (such as the partners in a law firm) or a great many (publicly traded companies have shareholders all over the world) and are assumed to last over time; they don’t have a predefined end date.

- **Free market entry and competition.** Companies can go after one another's markets and customers; there are few if any protected monopolies. It might not be legal to completely copy a rival's patented product, but it's perfectly legal to try to come up with something better. In economist-speak, markets are contested. Similarly, people can take their skills from one market to another; they're not tied to a single geography or job.
- **Strong property rights and contract enforcement.** Patents are a form of intellectual property. They can be bought and sold just as other kinds of property—from land to houses to cars—can. Laws and courts ensure that none of these kinds of property can be stolen or destroyed, even by large, powerful entities such as billionaires, giant corporations, or the government. Similarly, if a small company and a big one sign a contract to work together, neither party gets to unilaterally walk away from the agreement without fear of getting sued.
- **Absence of central planning, control, and price setting.** The government does not decide what goods and services are needed by people, or which companies should be allowed to produce them. No central body decides if there is “enough” volume and variety in smartphones, caffeinated beverages, steel girders, and so on. The prices of these and most other goods and services are allowed to vary based on the balance of supply and demand, rather than being set in advance or adjusted by any central authority.
- **Private ownership of most things.** Smartphones, cups of coffee, steel girders, and most other products are owned by the people or companies that bought them. The companies that produced these things are also owned by people. Many shares of Apple, Starbucks, US Steel, and other public companies are held by mutual funds, pension funds, and hedge funds, but all these funds are themselves ultimately owned by people. Most houses, cars, land, gold, Bitcoin, and other assets are also owned by people rather than the government.
- **Voluntary exchange.** The phrase most closely associated with capitalism is voluntary exchange. People can't be forced to buy specific products, take a certain job, or move across the country. Companies don't have to sell themselves if they don't want to. They also don't have to make some products and not others, or stay within specific markets. The Waffle House chain doesn't have any of its breakfast restaurants in my state of Massachusetts, but that's not because lawmakers there are keeping it out. The legislature in Boston doesn't have that power.

I want to highlight a couple of things about this definition. First, capitalism is not without oversight. The government has clear roles to play in establishing laws and settling disputes (to say nothing of setting tax rates, controlling the money supply, and doing other things of critical economic importance). As we'll see in the next two chapters, every sane advocate of capitalism also recognizes that while voluntary exchange and free market entry are great, they don't create utopia. Some important “market failures” need to be corrected by government action.

The second thing I want to point out is that all of today's rich countries are capitalist, by this definition. This is not to say that all capitalist countries are alike. Denmark, South Korea, and the United States are very different places. They have dissimilar trade policies, tax systems, social safety nets, industrial structures, and so on. But they all have all of the things listed above; they are all inherently capitalist. Denmark's economy is not planned and controlled out of Copenhagen, people in Korea own their own houses and furniture, and contracts in America are generally respected and enforced.

Today's poorer countries, in sharp contrast, reliably do not have all the things listed above. Their governments tend to run such things as airlines and telephone networks that are run by private companies in rich countries. It's generally much harder to start a company in less affluent countries, so free market entry and competition are constrained. According to the World Bank, in 2017 it took less than six days to start a business in America,

Denmark, Singapore, Australia, and Canada, and seventy days or more in Somalia, Brazil, and Cambodia. The world champion of entrepreneurial sclerosis was Venezuela (a country we'll talk more about in the next chapter), at two hundred and thirty days. In poorer countries, it's also often not clear who owns what. Things that are taken for granted in the rich world, such as unambiguous land registries and clear title to houses and other property, are problematic in many developing countries.

The biggest difference between rich and poor countries might be whether laws are clearly and consistently enforced. Poorer countries don't lack laws; they often have extensive legal codes. What's in short supply is justice for all. Officials are corrupt; the elite get special treatment and rarely lose in court; police, regulators, and inspectors can expect bribes; and contested markets, property rights, and voluntary exchange suffer in countless other ways. It's not that these abuses don't occur in rich countries, but they occur much, much less often.

I'll make some more points about capitalism in the next chapter. To wrap up this one, I want to emphasize how well technological progress and capitalism work together.

Overcoming the Limits

A great way to see what happens when capitalism and tech progress combine is to look back at 1972's *The Limits to Growth*, which we first came across in chapter 4. It's a fascinating document for two reasons. First, it's one of the most Malthusian books written since Malthus. It's far gloomier than anything Jevons came up with. The team behind *The Limits to Growth* tried to model the future of the exponentially growing world economy and concluded, "We can thus say with some confidence that, under the assumption of no major change in the present system, population and industrial growth will certainly stop within the [twenty-first] century, at the latest. The system... collapses because of a resource crisis."

Second, *The Limits to Growth* provided an invaluable service by recording what the known global reserves of important resources were in 1972. "Known global reserves" are the deposits of a resource that can be profitably extracted given the prevailing knowledge and state of technology. The authors of *The Limits to Growth* included the known reserves of many resources to show how inadequate they were in the face of exponential growth of both output and resource consumption. The authors had little reason to suppose in the early 1970s that either kind of growth would stop on its own. As we saw in chapter 4, resource consumption went up in lockstep with overall economic output all throughout the twentieth century up to Earth Day. Few people expected that to change. The team behind *The Limits to Growth* certainly didn't.

The most generous estimate of future resource availability included in *The Limits to Growth* assumed that exponential consumption would continue, and that proven reserves were five times greater than commonly assumed. Under these conditions, the team's computer models showed that the planet would run out of gold within twenty-nine years of 1972; silver within forty-two years; copper and petroleum within fifty; and aluminum within fifty-five.

These weren't accurate predictions.

We still have gold and silver, and we still have large reserves of them. In fact, the reserves of both are actually much bigger than in 1972, despite almost half a century of additional consumption. Known global reserves of gold are almost 400 percent larger today than in 1972, and silver reserves are more than 200 percent larger. And it's probably not too early to say that we're not going to run out of copper, aluminum, and petroleum as quickly as estimated in *The Limits to Growth*. Known reserves of all are much larger than they were when the book was published. Known aluminum reserves are almost twenty-five times what they were in the early 1970s.

How could these predictions about resource availability, which were taken seriously when they were released, have been so wrong? Because the Limits to Growth team clearly underestimated both dematerialization and the endless search for new reserves. Capitalism and tech progress combine to drive both trends—the use of fewer resources and the hunt for more of them—and neither of these two drivers is about to become less powerful. So we’ll continue to innovate our way to greater dematerialization while we keep finding more reserves.

The counterintuitive conclusion from this line of reasoning is that resource scarcity isn’t something we need to worry about. The earth is finite, so the total quantity of resources such as gold and petroleum is limited. But the earth is also very, very big—big enough to contain all we need of these and other resources, for as long as we’ll need them. The image of a thinly supplied Spaceship Earth hurtling through the cosmos with us aboard is compelling, but deeply misleading. Our planet has amply supplied us for our journey. Especially since we’re quickly slimming, swapping, optimizing, and evaporating our way to dematerialization.

The Second Enlightenment

Abraham Lincoln, the only US president to hold a patent, had a deep insight about capitalism. He wrote that the patent system “added the fuel of interest to the fire of genius in the discovery and production of new and useful things.” “The fire of genius” is a wonderful label for technological progress. “The fuel of interest” is equally good as a summary of capitalism. They interact in a self-reinforcing and ever-expanding cycle, and they’re now creating a dematerializing world.

Innovators come up with new and useful technologies. They then partner with entrepreneurs or become entrepreneurs themselves as James Watt did. A new company is the result. Investors such as steam-engine backer Matthew Boulton often join in to provide the capital needed for growth in its early days. The start-up enters a market and takes on incumbents like the Newcomen steam engine. Customers like the new technology better and are free to choose it. Rivals can’t just copy the new technology because it’s protected by patents. So they either have to license it or come up with innovations themselves. The start-up grows and prospers and eventually becomes the new incumbent. Its success inspires the next round of innovators, entrepreneurs, and investors, who once again take aim at the incumbent by offering something better to their customers.

Because of free market entry, the next innovators and start-ups can come from anywhere. And because innovation is such a distributed, dynamic, and unpredictable activity, it often comes from an unexpected place. It’s not necessary to plan this process. In fact, it’s a terrible idea to try to do so. Any central planner will miss many of the actual innovators or actively try to squelch them to protect the status quo of which the planners themselves are a part.

This cycle of capitalist, technology-rich “creative destruction” was beautifully described in the middle of the twentieth century by the Austrian economist Joseph Schumpeter. But since the late nineteenth century and the work of Alfred Marshall and William Jevons, we’ve believed that this cycle would cause us to use up more and more of our planet’s resources. This was true throughout the Industrial Era, and especially in the years around Earth Day and the birth of the modern environmental movement. Environmentalists’ urgent cautions about resource use and planetary depletion were born out of an awareness of how powerfully technological progress and capitalism interacted.

But then, for the reasons described in this chapter, that interaction changed. ***Tech progress and capitalism continued to reinforce each other, and to cause economies to get bigger and people to become more prosperous. But instead of also causing greater use of natural resources, they instead sparked***

dematerialization, something truly new under the sun. The fuel of interest in eliminating costs was added to the fire of the computer revolution, and the world began to dematerialize.

The economic historian Joel Mokyr argues that the Industrial Era was made possible by the values of the Enlightenment. This intellectual movement began in the second half of the eighteenth century with many societies in the West embracing what Steven Pinker characterizes as four values: reason, science, humanism, and progress. According to Mokyr, the Enlightenment created a “culture of growth” that let both capitalism and technological progress flourish.

I see an interesting inversion taking place now. If the Enlightenment led to the Industrial Era, then the Second Machine Age has led to a Second Enlightenment—a more literal one. We are now lightening our total consumption and treading more lightly on our planet. ***In America, the United Kingdom, and other rich countries, we are past “peak stuff” and are now using fewer total resources year after year. We’re accomplishing this because of the combination of technological progress and capitalism, which now let us get more from less.***

A complete Second Enlightenment requires more. It also requires that we not only use fewer resources but also generate less pollution and take better care of our fellow creatures. As powerful as they are, tech progress and capitalism won’t accomplish these goals on their own. To accomplish them, we need another pair of forces. Before we examine them, though, we need to talk about capitalism a bit more because it’s so important and so widely misunderstood.

INVISIBLE WOMEN

Criado Perez, Caroline. Invisible Women: Data Bias in a World Designed for Men

RELEVANCE:

Half the world population is women. Most consumer decisions in advanced economies are made by women. But women are under-represented or ignored systematically, both intentionally and unintentionally, in Western society and many other societies. Medical testing excludes women because hormone cycles are too problematic. Crash test dummies come in the man and small man size so all cars are designed such that they injure women disproportionately. As we learned during covid, the crash test dummies also designed medical gear with masks and gowns and gloves made for men and small men despite the fact that women obviously work in health care.

One of the greatest economic opportunities is for companies to design for women or to incorporate a female perspective in their business practices and product design.

Also, Caroline Criado Perez is wicked smart and articulate.

Note – the gender blindness of the venture community is one of the reasons I'm not worried they will or have figured out innovation at all. And the gender blindness of healthcare is just part of how I know we are still closer to the end of the beginning than anything else. How can we be at the end of solving anything when we are just getting started collecting data on the fact that humans come in male and female genders?

Afterword

The quarrels of popes and kings, with wars and pestilences, in every page; the men so good for nothing and hardly any women at all – it is very tiresome.

Jane Austen

It took about two hours for Daina Taimina to find the solution that had eluded mathematicians for over a century. It was 1997, and the Latvian mathematician was participating in a geometry workshop at Cornell University. David Henderson, the professor leading the workshop, was modelling a hyperbolic plane constructed out of thin, circular strips of paper taped together. 'It was disgusting,' laughed Taimina in an interview.

A hyperbolic plane is 'the geometric opposite' of a sphere, explains Henderson in an interview with arts and culture magazine Cabinet. 'On a sphere, the surface curves in on itself and is closed. A hyperbolic plane is a surface in which the space curves away from itself at every point.' It exists in nature in ruffled lettuce leaves, in coral leaf, in sea slugs, in cancer cells. Hyperbolic geometry is used by statisticians when they work with multidimensional data, by Pixar animators when they want to simulate realistic cloth, by auto-industry engineers to design aerodynamic cars, by acoustic engineers to design concert halls. It's the foundation of the theory of relativity, and 'thus the closest thing we have to an understanding of the shape of the universe'.³ In short, hyperbolic space is a pretty big deal.

But for thousands of years, hyperbolic space didn't exist. At least it didn't according to mathematicians, who believed that there were only two types of space: Euclidean, or flat space, like a table, and spherical space, like a ball. In the nineteenth century, hyperbolic space was discovered – but only in principle.

And although mathematicians tried for over a century to find a way to successfully represent this space physically, no one managed it – until Taimina attended that workshop at Cornell. Because as well as being a professor of mathematics, Taimina also liked to crochet.

Taimina learnt to crochet as a schoolgirl. Growing up in Latvia, part of the former Soviet Union, ‘you fix your own car, you fix your own faucet – anything’, she explains. ‘When I was growing up, knitting or any other handiwork meant you could make a dress or a sweater different from everybody else’s.’ But while she had always seen patterns and algorithms in knitting and crochet, Taimina had never connected this traditional, domestic, feminine skill with her professional work in maths. Until that workshop in 1997. When she saw the battered paper approximation Henderson was using to explain hyperbolic space, she realised: I can make this out of crochet.

And so that’s what she did. She spent her summer ‘crocheting a classroom set of hyperbolic forms’ by the swimming pool. ‘People walked by, and they asked me, “What are you doing?” And I answered, “Oh, I’m crocheting the hyperbolic plane.”’ She has now created hundreds of models and explains that in the process of making them ‘you get a very concrete sense of the space expanding exponentially. The first rows take no time, but the later rows can take literally hours, they have so many stitches. You get a visceral sense of what “hyperbolic” really means.’ Just looking at her models did the same for others: in an interview with the New York Times Taimina recalled a professor who had taught hyperbolic space for years seeing one and saying, ‘Oh, so that’s how they look.’ Now her creations are the standard model for explaining hyperbolic space.

Taimina’s fundamental contribution to the study of the hyperbolic plane does not, of course, close a data gap that directly relates to women. What this story shows instead is that the case for closing the gender data gap extends beyond women’s rights. Closing the data gap, as we’ve seen from the impact women have in politics, in peace talks, in design and urban planning, is good for everyone. Even mathematicians.

When we exclude half of humanity from the production of knowledge we lose out on potentially transformative insights. Would male mathematicians have come up with Taimina’s elegantly simple solution on their own? Unlikely, given how few men are keen crocheters. But in Taimina the traditionally feminine skill of crochet collided with the traditionally masculine sphere of maths. And it was this collision that led to the problem that many mathematicians had given up on as a lost cause finally being solved. Taimina provided the link the male mathematicians were missing.

All too often, however, we don’t allow women to provide that link. And so, we continue to treat too many of the world’s problems as insoluble. Like Freud, we continue to ‘knock our heads’ against what seem like riddles. But what if, like representing the hyperbolic plane, these problems aren’t insoluble? What if, like the problems in broadcast science competitions, all they are missing is a female perspective? The data that we do have is unarguable: as we continue to build, plan and develop our world, we have to start taking account of women’s lives. We have to start accounting for the three themes that define women’s relationship with that world.

The first of these themes is the female body – or to be precise – its invisibility. Routinely forgetting to accommodate the female body in design – whether medical, technological, or architectural – has led to a world that is less hospitable and more dangerous for women to navigate. It leads to us injuring ourselves in jobs and cars that weren’t designed for our bodies. It leads to us dying from drugs that don’t work. It has led to the creation of a world where women just don’t fit very well.

There is an irony in how the female body is apparently invisible when it comes to collecting data, because when it comes to the second trend that defines women's lives, the visibility of the female body is key. That trend is male sexual violence against women – how we don't measure it, don't design our world to account for it, and in so doing, allow it to limit women's liberty. Female biology is not the reason women are raped. It is not the reason women are intimidated and violated as they navigate public spaces. This happens not because of sex, but because of gender: the social meanings we have imposed on male and female bodies. In order for gender to work, it must be obvious which bodies elicit which treatment. And, clearly, it is: as we've seen, 'the mere sight of a woman' is enough for the viewer to 'immediately elicit a specific set of associated traits and attributions'. To immediately class her as someone to speak over. Someone to cat call. Someone to follow. Someone to rape.

Or maybe just someone to make the tea. Which is where we run into the third trend, which is perhaps the most significant in terms of its impact on women's lives worldwide: unpaid care work. Women are doing far and away more than our fair share of this work – this necessary work without which our lives would all fall apart. And, as with male violence against women, female biology is not the reason women are the bum-wiping class. But recognising a child as female is the reason she will be brought up to expect and accept that as her role. Recognising a woman as female is the reason she will be seen as the appropriate person to clear up after everyone in the office. To write the Christmas and birthday cards to her husband's family – and look after them when they get sick. To be paid less. To go part-time when they have kids.

Failing to collect data on women and their lives means that we continue to naturalise sex and gender discrimination – while at the same time somehow not seeing any of this discrimination. Or really, we don't see it because we naturalise it – it is too obvious, too commonplace, too much just the way things are to bother commenting on. It's the irony of being a woman: at once hyper-visible when it comes to being treated as the subservient sex class, and invisible when it counts – when it comes to being counted.

There is one more trend I kept coming across while writing this book: the excuses. Chief amongst these is that women are just too complicated to measure. Everyone was saying this, from transport planners, to medical researchers, to tech developers: they were all knocking their heads up against Freud's riddle of femininity and coming away baffled and defeated. Female bodies are too unharmonious, too menstrual and too hormonal. Women's travel patterns are too messy, their work schedules are too aberrant, their voices are too high. Even when, in the early twentieth century, influential Swiss architect Le Corbusier was devising a standard human model for use in architecture, the female body was 'only belatedly considered and rejected as a source of proportional harmony' with humanity instead represented by a six-foot man with his arm raised (to reach that top shelf I can never reach).

The consensus is clear: women are abnormal, atypical, just plain wrong. Why can't a woman be more like a man? Well, apologies on behalf of the female sex for being so mysterious, but no, we aren't and no we can't. And that is a reality that scientists, politicians and tech bros just need to face up to. Yes, simple is easier. Simple is cheaper. But simple doesn't reflect reality.

Back in 2008, Chris Anderson, then editor of tech magazine Wired, penned an article headlined 'The End of Theory: The Data Deluge Makes the Scientific Model Obsolete'.¹⁰ We can 'stop looking for

models’, Anderson claimed. There is now a better way. Petabytes [that’s 1,000 million million bytes to you and me] allow us to say: ‘Correlation is enough.’ We didn’t need to hypothesise about anything, we just needed to crunch the numbers – or, more accurately, ‘let statistical algorithms’ crunch the numbers. In the era of Trump, Brexit and Cambridge Analytica, this seems Pollyanna-ish to say the least, but even before these data scandals it should have been obvious that his claims were hubristic, because back in 2008 we had even less data on women than we have now. And when you’re missing out half the global population in the numbers you feed your statistical algorithms, what you’re actually creating is just a big mess.

Anderson holds up Google as an exemplar of what he dubbed ‘The Petabyte Age’, singing the praises of its ‘founding philosophy’ that ‘we don’t know why this page is better than that one: If the statistics of incoming links say it is, that’s good enough. No semantic or causal analysis is required. That’s why Google can translate languages without actually knowing them (given equal corpus data, Google can translate Klingon into Farsi as easily as it can translate French into German).’ Except, as we’ve seen, Google actually can’t translate very well at all, even ten years later. That is, if you care about women being erased from language.

So. Not so simple after all.

Anderson is right about one thing though. There is a better way. And it’s a pretty simple one: we must increase female representation in all spheres of life. Because as more women move into positions of power or influence, there’s another pattern that is becoming even more apparent: women simply don’t forget that women exist as easily as men often seem to.

Women in the film industry are more likely to employ women. Female journalists are significantly more likely to centre a female perspective and to quote women. Female authors do the same: 69% of US female biographers wrote about female subjects in 2015, compared to 6% of male biographers. The emphasis by women on female voices and perspectives extends to the academy. Between 1980 and 2007, female history faculty in the US rose from 15% to 35% – meanwhile across a similar time period (1975-2015), US history faculty specialising in women’s history rose from 1% to 10%¹⁵ – a tenfold increase. Female academics are also more likely to assign female authors to their students.

Then there’s how women might interpret history: in a 2004 Guardian article comedian Sandi Toksvig wrote about how when she was studying anthropology at university one of her female professors held up a photograph of an antler bone with twenty-eight markings on it. ‘This,’ she said, ‘is alleged to be man’s first attempt at a calendar.’ We all looked at the bone in admiration. ‘Tell me,’ she continued, ‘what man needs to know when 28 days have passed? I suspect that this is woman’s first attempt at a calendar.’

When Britain’s EU Withdrawal Bill was announced in 2017, the Human Rights Act was explicitly excluded from alteration – but it took a woman, Maria Miller, the Conservative MP for Basingstoke, to force the government to agree to make a statement requiring that Brexit is also compatible with the Equalities Act. Without this concession, a whole range of women’s rights could be scrapped after Brexit, with no avenue for legal redress. In the workplace it is often women, like developmental biologist Christiane Nusslein-Volhard with her foundation to help female PhD students with children, who are putting in place solutions to structural male bias – a bias which male leaders have overlooked and ignored for decades.

Women are also leading the way when it comes to closing the gender data gap. A recent analysis of 1.5 million papers published between 2008-15 found that the likelihood of a study involving gender and sex analysis ‘increases with the proportion of women among its authors’. The effect is particularly pronounced if a woman serves as a leader of the author group. This concern for women’s health also extends to the political sphere: it took a woman (Paula Sherriff, the Labour MP for Dewsbury) to set up the UK’s first All-Party Parliamentary group for women’s health in 2016. It was two rogue female Republicans who scotched Donald Trump’s attempts to repeal Obamacare (which would have disproportionately impacted on women), voting three times against his proposals.

And women are making a difference in politics more generally. It was two women, Melinda Gates and Hillary Clinton, who spearheaded the UN-backed organisation Data2x that is aimed specifically at closing the global gender data gap. It was a woman, Hillary Clinton, who insisted on going to Beijing in 1995 to make the now famous declaration that ‘Human rights are women’s rights, and women’s rights are human rights.’

And when the worst happens, women are there too, filling in the gaps left by male-biased disaster relief. Researchers found that the ‘masculine and muscular image[s] of relief workers’ that dominated the media post-Katrina were belied by women who were ‘working tirelessly and courageously’ behind the scenes. The same thing has happened in Puerto Rico, all but abandoned by the US government after Hurricane Maria devastated the region in 2017. ‘The reality is that when you go to communities, mostly it is women as leaders and as community organizers,’ Adi Martinez-Roman, executive director for a non-profit that provides legal assistance to low-income families, told journalist Justine Calma. These women have collected data by ‘wad[ing] into flooded neighbourhoods’ and canvassing the abandoned communities. And they have developed and provided evidence-based solutions. They’ve set up soup kitchens. They’ve raised money and rebuilt roads. They’ve distributed ‘solar-powered lights, generators, gas, clothes, shoes, tampons, batteries, medication, mattresses, water’. They set up ‘free legal aid societies to help families navigate the confusing and ill-designed processes required to file FEMA claims’. They’ve even managed to source some communal, solar-powered washing machines.

The solution to the sex and gender data gap is clear: we have to close the female representation gap. When women are involved in decision-making, in research, in knowledge production, women do not get forgotten. Female lives and perspectives are brought out of the shadows. This is to the benefit of women everywhere, and as the story of Taimina, the crocheting maths professor shows, it is often to the benefit of humanity as a whole. And so, to return to Freud’s ‘riddle of femininity’, it turns out that the answer was staring us in the face all along. All ‘people’ needed to do was to ask women.

THE FUTURE IS FASTER THAN YOU THINK

Diamandis, Peter H.; Kotler, Steven. The Future Is Faster Than You Think: How Converging Technologies Are Transforming Business, Industries, and Our Lives (Exponential Technology Series)

RELEVANCE:

We are focused on domains of innovation: biological, physical, digital. But the real areas of interest are where these things converge. Diamandis offers a useful business focused perspective on the same phenomenon that Kauffman and Domingos and others are also observing,

CONVERGING TECHNOLOGY

If you want to understand convergence, it helps to start at the beginning. In our earlier books, *Abundance* and *BOLD*, we introduced the notion of exponentially accelerating technology; that is, any technology that doubles in power while dropping in price on a regular basis. Moore's Law is the classic example. In 1965, Intel founder Gordon Moore noticed that the number of transistors on an integrated circuit had been doubling every eighteen months. This meant every year-and-a-half computers got twice as powerful, yet their cost stayed the same.

Moore thought this was pretty astounding. He predicted this trend might last a few more years, maybe five, possibly ten. Well, it's been twenty, forty, going on sixty years. Moore's Law is the reason the smartphone in your pocket is a thousand times smaller, a thousand times cheaper, and a million times more powerful than a supercomputer from the 1970s.

And it's not slowing down.

Despite reports that we are approaching the heat death of Moore's Law—which we'll address in the next chapter—in 2023 the average thousand-dollar laptop will have the same computing power as a human brain (roughly 10¹⁶ cycles per second). Twenty-five years after that, that same average laptop will have the power of all the human brains currently on Earth.

More critically, it's not just integrated circuits that are progressing at this rate. In the 1990s, Ray Kurzweil, the director of engineering at Google and Peter's cofounding partner in Singularity University, discovered ***that once a technology becomes digital—that is, once it can be programmed in the ones and zeroes of computer code—it hops on the back of Moore's Law and begins accelerating exponentially.***

In simple terms, we use our new computers to design even faster new computers, and this creates a positive feedback loop that further accelerates our acceleration—what Kurzweil calls the “Law of Accelerating Returns.” The technologies now accelerating at this rate include some of the most potent innovations we have yet dreamed up: quantum computers, artificial intelligence, robotics, nanotechnology, biotechnology, material science, networks, sensors, 3-D printing, augmented reality, virtual reality, blockchain, and more.

But this progress is old news. The new news is that formerly independent waves of exponentially accelerating technology are beginning to converge with other independent waves of exponentially accelerating technology. For example, the speed of drug development is accelerating, not only because biotechnology is progressing at an exponential rate, but because artificial intelligence, quantum computing, and a couple other exponentials are converging on the field. In other words, these waves are

starting to overlap, stacking atop one another, producing tsunami-sized behemoths that threaten to wash away most everything in their path.

When an innovation creates a new market and washes away an existing one, we use the term “disruptive innovation” to describe it. When silicon chips replaced vacuum tubes at the beginning of the digital age, this was a disruptive innovation. Yet, as exponential technologies converge, their potential for disruption increases in scale. Solitary exponentials disrupt products, services, and markets—like when Netflix ate Blockbuster for lunch—while convergent exponentials wash away products, services, and markets, as well as the structures that support them.

But we’re getting ahead of ourselves. The rest of this book is devoted to these forces and their rapid and revolutionary impact. Before we dive deeper into that tale, let’s first examine convergence through a more manageable lens, returning to our initial question about flying cars: Why now?

To answer that, let’s examine the three basic requirements any Uber eVTOL will have to meet: safety, noise, and price. Helicopters, which are the closest model anyone has for a flying car, have been around for nearly eighty years—Igor Sikorsky built the world’s first one in 1939—yet they can’t come close to satisfying these requirements. Besides being loud and expensive, they have that bad habit of falling out of the sky. So why are Bell, Uber, Airbus, Boeing, and Embraer—just to name a few—bringing aerial taxis to market today?

Once again: Convergence.

Helicopters are loud and dangerous because they use a single gargantuan rotor to generate lift. Unfortunately, the tip-speed of that single rotor produces exactly the right thud-thud-thud frequency to annoy pretty much anyone with ears. And they’re dangerous because, if that rotor fails, well, gravity plays for keeps.

Now imagine, instead of one main rotor overhead, a bunch of smaller rotors—like a row of small fans beneath a plane’s wing—whose combination generates enough lift to fly but pumps out a lot less noise. Better yet, imagine if this multi-rotor system could fail gracefully, landing safely even if a couple rotors stopped working at once. Add to this design a single wing that enables speeds of 150 mph or more. All great ideas, except, thanks to their terrible power-to-weight ratios, gasoline-powered engines make none of this possible.

Enter distributed electric propulsion, or DEP for short. Over the past decade, a surge in demand for commercial and military drones has pushed roboticists (and drones are just flying robots) to envision a new kind of electromagnetic motor: extremely light, stealthily quiet, and capable of carrying heavy loads. To design that motor, engineers relied on a trilogy of converging techs: first, machine learning advances that allowed them to run enormously complicated flight simulations, then materials science breakthroughs that let them create parts both light enough for flying and durable enough for safety, and last, new manufacturing techniques—3-D printing—that can create these motors and rotors at any scale. And talk about functionality: These electric engines are 95 percent efficient compared to gasoline’s 28 percent.

But flying a DEP system is another story. Adjusting a dozen motors in microsecond intervals is beyond a human pilot’s skill. DEP systems are “fly-by-wire”—that is, computer controlled. And what produces that level of control? Another swarm of converging technologies.

First, an AI revolution gave us the computational processing horsepower to take in an ungodly amount of data, make sense of it in microseconds, and manipulate a multitude of electric motors and aircraft control surfaces accordingly, in real time. Second, to sweep in all that data, you need to replace the pilot’s eyes and ears with sensors capable of processing gigabits of information at once. That means GPS, LIDAR, radar, an advanced visual imaging suite, and a plethora of microscopic accelerometers—many of which are the dividends of a decade of smartphone wars.

Finally, you'll need batteries. They'll have to last long enough to overcome range anxiety—or the fear of running out of juice while running errands—and generate enough oomph, or what engineers call “power density,” to lift the vehicle, a pilot, and four passengers off the ground. To achieve this lift, the minimum requirement is 350-kilowatt hours per kilogram. This was out of reach, until recently. Thanks to the explosive growth of both solar power and electric cars, there's now a bigger need for better energy storage systems, resulting in a next generation of lithium-ion batteries with increased range, and, as an added bonus, enough power to lift flying cars.

In the aerial ridesharing equation, we've solved safety and noise, but price still requires a few more innovations. There's also the not small issue of manufacturing enough eVTOLs for Uber's program. To be able to meet Uber's outsized demand at an affordable price would require suppliers to produce aircraft faster than during WWII, when a still unbroken record of eighteen thousand B24 fighters were pumped out over two years—or, at its peak, one plane every sixty-three minutes.

For this to happen—which is what it would take to make flying cars a mainstream reality, not an elitist luxury—we need another trio of convergences. To start, computer-aided design and simulation need to become deft enough to draft the airfoils, wings, and fuselages required for commercial flight. At the same time, material science has to produce carbon fiber composites and complex metal alloys that are light enough for flight yet durable enough for safety. Finally, 3-D printers have to become fast enough to turn these new materials into usable parts so that all previous aircraft manufacturing records are shattered. In other words, exactly where we are today.

Sure, you can play this game with any new technology. Socks couldn't be invented until a materials revolution turned plant fibers into soft fabrics and a tool-making revolution turned animal bones into sewing needles. This is progress, of course, but linear in nature. It took thousands of years to get from these first steps in sock-dom to the next major innovation: the domestication of animals (which gave us sheep's wool). And thousands more years for electricity to bring sock-making to scale.

But the blurry acceleration we're witnessing today—that is, the answer to “Why now?”—is the result of a dozen different technologies converging. It's progress at a rate that we've not seen before. And this is a problem for us.

The human brain evolved in an environment that was local and linear. Local, meaning most everything that we interacted with was less than a day's walk away. Linear, meaning the rate of change was exceptionally slow. Your great-great-great-grandfather's life was roughly the same as his great-great-grandson's life. But now we live in a world that is global and exponential. Global, meaning if it happens on the other side of the planet, we hear about it seconds later (and our computers hear about it only milliseconds later). Exponential, meanwhile, refers to today's blitzkrieg speed of development. Forget about the difference between generations, currently mere months can bring a revolution. Yet our brain—which hasn't really had a hardware update in two hundred thousand years—wasn't designed for this scale or speed.

And if we struggle to track the growth of singular innovations, we're downright helpless in the face of converging ones. Put it this way, in “The Law of Accelerating Returns,” Ray Kurzweil did the math and found that we're going to experience twenty thousand years of technological change over the next one hundred years. Essentially, we're going from the birth of agriculture to the birth of the internet twice in the next century. This means paradigm-shifting, game-changing, nothing-is-ever-the-same-again breakthroughs—such as affordable aerial ridesharing—will not be an occasional affair. They'll be happening all the time.

It means, of course, that flying cars are just the beginning.

SUPERFORECASTING

Tetlock, Philip E.; Gardner, Dan. Superforecasting

RELEVANCE:

Good set of simple principles for forecasting drawn from Tetlock's classic book. The section below is one of the appendices from his book.

Ten Commandments for Aspiring Superforecasters

The guidelines sketched here distill key themes in this book and in training systems that have been experimentally demonstrated to boost accuracy in real-world forecasting contests. For more details, visit www.goodjudgment.com.

(1) Triage.

Focus on questions where your hard work is likely to pay off. Don't waste time either on easy "clocklike" questions (where simple rules of thumb can get you close to the right answer) or on impenetrable "cloud-like" questions (where even fancy statistical models can't beat the dart-throwing chimp). Concentrate on questions in the Goldilocks zone of difficulty, where effort pays off the most.

(2) Break seemingly intractable problems into tractable sub-problems.

Channel the playful but disciplined spirit of Enrico Fermi who—when he wasn't designing the world's first atomic reactor—loved ballparking answers to head-scratchers such as "How many extraterrestrial civilizations exist in the universe?" Decompose the problem into its knowable and unknowable parts. Flush ignorance into the open. Expose and examine your assumptions. Dare to be wrong by making your best guesses. Better to discover errors quickly than to hide them behind vague verbiage.

Superforecasters see Fermi-izing as part of the job. How else could they generate quantitative answers to seemingly impossible-to-quantify questions...

We find this Fermi-izing spirit at work even in the quest for love, the ultimate unquantifiable. Consider Peter Backus, a lonely guy in London, who guesstimated the number of potential female partners in his vicinity by starting with the population of London (approximately six million) and winnowing that number down by the proportion of women in the population (about 50%), by the proportion of singles (about 50%), by the proportion in the right age range (about 20%), by the proportion of university

graduates (about 26%), by the proportion he finds attractive (only 5%), by the proportion likely to find him attractive (only 5%), and by the proportion likely to be compatible with him (about 10%). Conclusion: roughly twenty-six women in the pool, a daunting but not impossible search task.

There are no objectively correct answers to true-love questions, but we can score the accuracy of the Fermi estimates... The surprise is how often remarkably good probability estimates arise from a remarkably crude series of assumptions and guesstimates.

(3) Strike the right balance between inside and outside views.

... Nothing is 100% "unique." ... So superforecasters conduct creative searches for comparison classes even for seemingly unique events, such as the outcome of a hunt for a high-profile terrorist (Joseph Kony) or the standoff between a new socialist government in Athens and Greece's creditors. Superforecasters are in the habit of posing the outside-view question: How often do things of this sort happen in situations of this sort?

So too apparently is Larry Summers, a Harvard professor and former Treasury secretary. He knows about the planning fallacy: when bosses ask employees how long it will take to finish a project, employees tend to underestimate the time they need, often by factors of two or three. Summers suspects his own employees are no different. One former employee, Greg Mankiw, himself now a famous economist, recalls Summers's strategy: he doubled the employee's estimate, then moved to the next higher time unit. "So, if the research assistant says the task will take an hour, it will take two days. If he says two days, it will take four weeks."² It's a nerd joke: Summers corrected for employees' failure to take the outside view in making estimates by taking the outside view toward employees' estimates, and then inventing a funny correction factor.

Of course, Summers would adjust his correction factor if an employee astonished him and delivered on time. He would balance his outside-view expectation of tardiness against the new inside-view evidence that a particular employee is an exception to the rule. Because each of us is, to some degree, unique.

(4) Strike the right balance between under- and overreacting to evidence.

Belief updating is to good forecasting as brushing and flossing are to good dental hygiene. It can be boring, occasionally uncomfortable, but it pays off in the long term. That said, don't suppose that belief updating is always easy because it sometimes is. Skillful updating requires teasing subtle signals from noisy news flows—all the while resisting the lure of wishful thinking.

Savvy forecasters learn to ferret out telltale clues before the rest of us. They snoop for nonobvious lead indicators, about what would have to happen before X could, where X might be anything from an expansion of Arctic sea ice to a nuclear war in the Korean peninsula. Note the fine line here between picking up subtle clues before everyone else and getting suckered by misleading clues. Does the appearance of an article critical of North Korea in the official Chinese press signal that China is about to squeeze Pyongyang hard—or was it just a quirky error in editorial judgment? The best forecasters tend to be incremental belief updaters, often moving from probabilities of, say, 0.4 to 0.35 or from 0.6 to 0.65, distinctions too subtle to capture with vague verbiage, like “might” or “maybe,” but distinctions that, in the long run, define the difference between good and great forecasters.

Yet superforecasters also know how to jump, to move their probability estimates fast in response to diagnostic signals. Superforecasters are not perfect Bayesian updaters but they are better than most of us. And that is largely because they value this skill and work hard at cultivating it.

(5) Look for the clashing causal forces at work in each problem.

For every good policy argument, there is typically a counterargument that is at least worth acknowledging. For instance, if you are a devout dove who believes that threatening military action never brings peace, be open to the possibility that you might be wrong about Iran. And the same advice applies if you are a devout hawk who believes that soft “appeasement” policies never pay off. Each side should list, in advance, the signs that would nudge them toward the other.

Now here comes the hard part. In classical dialectics, thesis meets antithesis, producing synthesis. In dragonfly eye, one view meets another and another and another—all of which must be synthesized into a single image. There are no paint-by-number rules here. Synthesis is an art that requires reconciling irreducibly subjective judgments. If you do it well, engaging in this process of synthesizing should transform you from a cookie-cutter dove or hawk into an odd hybrid creature, a dove-hawk, with a nuanced view of when tougher or softer policies are likelier to work.

(6) Strive to distinguish as many degrees of doubt as the problem permits but no more.

Few things are either certain or impossible. And “maybe” isn’t all that informative. So your uncertainty dial needs more than three settings. Nuance matters. The more degrees of uncertainty you can distinguish, the better a forecaster you are likely to be. As in poker, you have an advantage if you are better than your competitors at separating 60/40 bets from 40/60—or 55/45 from 45/55. Translating vague-verbiage hunches into numeric probabilities feels unnatural at first but it can be done. It just requires patience and practice...

Most of us could learn, quite quickly, to think in more granular ways about uncertainty. Recall the episode in which President Obama was trying to figure out whether Osama bin Laden was the mystery occupant of the walled-in compound in Abbottabad. And recall the probability estimates of his intelligence officers and the president's reaction to their estimates: "This is fifty-fifty...a flip of the coin." Now suppose that President Obama had been shooting the breeze with basketball buddies and each one offered probability estimates on the outcome of a college game—and those estimates corresponded exactly to those offered by intelligence officers on the whereabouts of Osama bin Laden. Would the president still have shrugged and said, "This is fifty-fifty," or would he have said, "Sounds like the odds fall between three to one and four to one"? I bet on the latter. The president is accustomed to granular thinking in the domain of sports. Every year, he enjoys trying to predict the winners of the March Madness basketball tournament, a probability puzzle that draws the attention of serious statisticians. But, like his Democratic and Republican predecessors, he does not apply the same rigor to national security decisions. Why? Because different norms govern different thought processes. Reducing complex hunches to scorable probabilities is de rigueur in sports but not in national security.

So, don't reserve rigorous reasoning for trivial pursuits...

(7) Strike the right balance between under- and overconfidence, between prudence and decisiveness.

Understand the risks both of rushing to judgment and of dawdling too long near "maybe." Manage the trade-off between the need to take decisive stands (who wants to listen to a waffler?) and the need to qualify their stands (who wants to listen to a blowhard?). Long-term accuracy requires getting good scores on both calibration and resolution—which requires moving beyond blame-game ping-pong. It is not enough just to avoid the most recent mistake. Find creative ways to tamp down both types of forecasting errors—misses and false alarms—to the degree a fickle world permits such uncontroversial improvements in accuracy.

(8) Look for the errors behind your mistakes but beware of rearview-mirror hindsight biases.

Don't try to justify or excuse your failures. Own them! Conduct unflinching postmortems: Where exactly did I go wrong? And remember that although the more common error is to learn too little from failure and to overlook flaws in your basic assumptions, it is also possible to learn too much (you may have been basically on the right track but made a minor technical mistake that had big ramifications). Also don't forget to do postmortems on your successes too. Not all successes imply that your reasoning was right. You may have just lucked out by making offsetting errors. And if you keep confidently reasoning along the same lines, you are setting yourself up for a nasty surprise.

(9) Bring out the best in others and let others bring out the best in you.

Master the fine arts of team management, especially perspective taking (understanding the arguments of the other side so well that you can reproduce them to the other's satisfaction), precision questioning (helping others to clarify their arguments so they are not misunderstood), and constructive confrontation (learning to disagree without being disagreeable).

Wise leaders know how fine the line can be between a helpful suggestion and micro-managerial meddling or between a rigid group and a decisive one or between a scatterbrained group and an open-minded one. Tommy Lasorda, the former coach of the Los Angeles Dodgers, got it roughly right: "Managing is like holding a dove in your hand. If you hold it too tightly you kill it, but if you hold it too loosely, you lose it."

(10) Master the error-balancing bicycle.

Implementing each commandment requires balancing opposing errors. Just as you can't learn to ride a bicycle by reading a physics textbook, you can't become a superforecaster by reading training manuals.

Learning requires doing, with good feedback that leaves no ambiguity about whether you are succeeding—"I'm rolling along smoothly!"—or whether you are failing—"crash!" Remember that practice is not just going through the motions of making forecasts, or casually reading the news and tossing out probabilities. Like all other known forms of expertise, superforecasting is the product of deliberate practice.

(11) Don't treat commandments as commandments.

"It is impossible to lay down binding rules," Helmuth von Moltke warned, "because two cases will never be exactly the same."⁵ As in war, so in all things. Guidelines are the best we can do in a world where nothing is certain or exactly repeatable. Superforecasting requires constant mindfulness, even when—perhaps especially when—you are dutifully trying to follow these commandments.

MOVE

Khanna, Parag. Move: The Forces Uprooting Us

RELEVANCE:

One can always question the results of a thought experiment, but Khanna's way of thinking about the future, building on his toolkit as an economic geographer, is useful to understand.

His 2016 book Connectography, focused on the built environment connecting all of us – our infrastructure. His more recent book Move focuses on the changes we will face and the role of migration in adapting to change.

Khanna's core point applies to any map - maps change over time. New countries form, borders shift, climate changes. It all changes. We must think about is how it will change. One place to start is to ask, "Where will you be in 2050?" or "What will the world look like in 2050?"

WHERE WILL YOU LIVE IN 2050?

April 2020 will forever be remembered as the month the world stood still. Never in human history has the global population simultaneously coordinated a single act: the Great Lockdown. Almost all offices and stores shut. Streets and parks were empty. Cars, trains, and planes sat idle. Goats, deer, foxes, boars, ducks, kangaroos, and even penguins roamed freely through normally bustling cities from Edinburgh and Paris to Cape Town and Canberra. The Economist summed it up in one word: "Closed."

The year of rolling global lockdowns that followed took an excruciating toll on billions of lives. Amid the chaos, one of the greatest ironies that surfaced was just how accustomed we had become to almost frictionless global movement. The year 2019 was a record year for tourism, with international arrivals topping 1.5 billion, the highest figure ever. More than 275 million people were classified as international migrants—from Indian construction workers and Filipino maids in Dubai to American executives and English teachers across Asia—also the largest number ever recorded. Then stopped.

Instead of migration and travel surging ahead, the lockdowns provoked a sudden reset of the world's population. From every corner of the world, tourists, students, and expats returned to their country of birth or nationality. European countries dispatched planes to Africa and Latin America to repatriate their citizens. Asian students bought one-way tickets back home from the US, UK, and Australia. More than two hundred thousand Indian laborers were flown back from Gulf countries such as Saudi Arabia and the United Arab Emirates. This unprecedented repatriation of peoples artificially realigned location and citizenship. For the first time anyone could remember, nearly the entire world population was "home." But for how long?

A staggering share of our personal and professional lives hinges on mobility: the movement of people, goods, money, and data within cities and countries as well as internationally. Society only functions normally if we can move. Once you stop pedaling a bicycle, it quickly falls over. Our civilization is that bicycle. And move we will.

In the early 2010s, my colleague Greg Lindsay and I set out to answer the question, "Where will you live in 2050?" The answer could simply have been "high-tech cities," but which ones? Some will be sites of predatory surveillance while others will allow residents to preserve some privacy. Some will be in areas resilient to climate change, while others may well have been submerged by then. Some will

have thriving service economies and lively culture, while others will have become the discarded “factory towns” such as those littered across Michigan. ***As we scanned the world for geographies that offer abundant freshwater, progressive governance, and could attract talent to innovative industries, we decided on... Michigan.***

More broadly, we pointed to the emergence of a “New North,” a collection of geographies such as the Great Lakes region and Scandinavia that are making significant investments in renewable energy, food production, and economic diversification. Not too long after living through Hurricane Sandy in 2012, Greg and his family moved from New York City to Montreal.

This seemingly simple thought experiment holds some valuable lessons. First, you don’t get to choose your crisis: Covid, climate change, an economic crash, and political unrest can unfold at the same time—and even amplify one another in a downward spiral. Another takeaway is that places abandoned yesterday could be rejuvenated tomorrow. Rust belt cities across the Great Lakes region are the epitome of dystopian blight: Michigan today is still losing twice as many people per year as it gains (and losing congressional seats in the process). But Detroit may well be tomorrow’s hot property market. Early signs of its comeback are already visible: a light railway, art museums, boutique hotels, artisan fashion, and sumptuous Arab and Asian foods. In the heart of Detroit there’s now a sandy urban beach where young professionals kick back to enjoy lunch and drinks. Industrial firms are retooling Michigan’s factories to crank out electric cars, and Alphabet’s Sidewalk Labs is building a highway catering to autonomous vehicles between Detroit and Ann Arbor. Next may come plants for 3D-printed housing. Within a couple of decades, US-Canada relations may have progressed such that Detroit serves as the midway point in a seamless and thriving Chicago-Toronto corridor.

The opposite extreme could be represented by Hong Kong, a thriving global city being violently squeezed by Beijing. The former British colony has degenerated from Asia’s premier capitalist hub into a battleground between native Hong Kong youth who cherish their freedom and a Chinese government that demands submission to its National Security Law. Long before 2047—the date Hong Kong was officially supposed to be fully integrated with mainland China—many of the city’s youth will have taken their talents elsewhere and been replaced by millions of obedient mainland Chinese citizens.

To forecast which places will succeed or fail in the decades ahead requires taking a holistic look at political, economic, technological, social, and environmental factors, projecting how they intersect with one another, and building scenarios for how each geography may adapt to this unending complexity. Plenty of twists and turns lie ahead: lockdowns today, mass migrations tomorrow; populist democracy today, data-driven governance tomorrow; national identity today, global solidarity tomorrow—or the reverse in some places, and flip-flopping in others. You may not know if you’ve made the right move—or moves—until 2050.

History is replete with seismic global disruptions—pandemics and plagues, wars and genocides, famines and volcanic eruptions. And time and again after great catastrophes, our survival instinct compels us to move. Humankind is embarking on the most extensive experiment it has ever run on itself: The pandemic is slowly beginning to pass, borders are reopening, and people are moving again. Which places will they leave and where will they relocate? What is the best way for all of us to cope with the complex interplay of political upheavals and economic crises, technological disruption and climate change, demographic imbalances and pandemic paranoia? The answer to these questions can be summed up in one word: ***Move.***

HIDDEN CHAMPIONS OF THE 21st CENTURY

Simon, Hermann. Hidden Champions of the Twenty-First Century

RELEVANCE:

Simon's work on Hidden Champions, or companies with leading market positions that few people know about, highlights a set of high performing firms and draws lessons from the group. A summary of the lessons Simon drew from his sequel, Hidden Champions of the 21st Century, is included below.

The list of lessons is useful in thinking about things to look for in exceptional companies.

Market & Focus - Summary

Most hidden champions of the twenty-first century remain true to their focus and continue to define their markets narrowly. However, we increasingly find cases of soft diversification, particularly among the larger hidden champions. With regard to market definition and the associated focus or diversification, the following points should be noted:

- *Hidden champions typically define their markets narrowly and build up strong market positions in these markets.*
- *The narrow market definition means that the hidden champions' world markets are and remain relatively small.*
- *However, these markets have predominantly grown strongly and have on average more than doubled their volume in the last ten years.*
- *In spite of the fragmentation and fuzziness of many of their markets, the hidden champions are relatively well informed about them. As a result of their specialization and pronounced closeness to their customers, the hidden champions typically have an excellent intuitive understanding of their markets—even in the absence of precise data.*
- *In defining their markets, the hidden champions primarily apply customer-oriented criteria such as application and needs. Product and technology are often additional elements of market definition. Typically, several criteria are used to define a market.*
- *Hidden champions often do not accept the market boundaries customary in their industry, but view market definition as a strategic parameter that they apply in an autonomous way.*
- *Focus leads to deep problem solutions or value chains for a narrow market. The value chain served is accordingly narrow and deep. Out of this focus specialized competencies and perfection evolve that are difficult for competitors to match.*

- *Many hidden champions are super-nichists operating in very small markets with global market shares as high as 70%– 100%. This can limit growth but, at the same time, establishes high entry barriers for new competitors.*
- *For better or worse, hidden champions strongly depend on their core markets. This dependence is not one-sided. It applies to the customers as well. Dependence on a single market increases the market risk but reduces the competitive risk, due to the full concentration of all resources on the core market. The balance between both risks must be judged for each individual case.*
- *Once the hidden champions have selected a market, they show a strong and lasting commitment to it. Fundamental redefinitions of the market and changes in basic technology are rare and occur only every 10– 15 years.*
- *The most important motive for soft diversification is not the desire to minimize the risk stemming from concentration on a narrow market, but to overcome growth barriers, to better employ existing competencies or to realize greater entrepreneurial ambitions.*
- *Although the jury is still out, hidden champions seem to diversify most successfully when they stay true to their philosophy. The new units should have a decentralized management and in turn strive for leadership in their markets.*

Finding the right market definition and focus are demanding challenges. The hidden champions have become and have remained market leaders with a narrow market definition and a strong focus. World-class performance is not the result of spreading resources broadly, but of sustained concentration. Every company should bear this in mind for its own strategy development. The risk of specialization seems less serious than the risk of dissipating talent and energy. The specialist frequently beats the generalist. However, we also must face the fact that narrow markets coupled with high market shares limit further growth and only confine investment opportunities as of a certain point. A promising route to continued growth in this situation is soft diversification. The new units should again have a narrow market definition and a clear focus. In this way, the hidden champion strengths can be most effectively transferred to new businesses and open future growth avenues.

THE LESSONS OF HIDDEN CHAMPIONS

What do the hidden champions teach us? According to modern management literature, we mainly— or only— learn from large successful corporations. In this book we have changed this perspective radically. We have discovered that many of the most consistently successful, longest-surviving, most competitive companies operate behind a veil of discretion and remain unknown— even to business experts. Consequently, their strategies and leadership methods are not finding their way into the textbooks, business magazines, and heads of the management community. However, the case studies of the hidden champions teach us extraordinary and valuable lessons that often differ from the patterns of large corporations and the management fads of the day.

Leadership and Goals

Ambitious goals are the foundation of outstanding success. Extraordinary success is never based on pure luck or coincidence. In the beginning there is always a vision of what the person wants to achieve, or at least there is a state of preparedness. Luck and windfall incidents may contribute, although luck is but a “by-product of the desire for perfection.” Most human beings follow more or less well-trodden paths and many have no particular ambitions at all. And there are few with a goal, a mission—the monomaniacs, as Peter Drucker called them. The founders and managers of the hidden champions belong to this second category. They know what they want and they have the willpower and the energy, sometimes even the obsession, to realize their goals. The issue of will hardly ever comes up in management literature, but it is of fundamental importance for the development of companies. We need not consider here whether willpower and ambition can be learned or whether these characteristics are hereditary. In any case, we encounter the core of the personality here.

However, one individual’s willpower and ambition are insufficient to create a company, let alone a world market leader. This may work for an artist, but it takes a lot of peers to develop a world market leader—people from all over the world. The entrepreneur must ignite the fire that burns inside him in many others of different nationalities and cultures. That’s leadership. Willpower and leadership combined ensure that everyone’s energy is directed toward becoming and remaining the best. This is the true goal that must be pursued consistently. If this is achieved, world market leadership results practically as a side effect.

Lesson 1. *Willpower and goals always come first. For hidden champions, leadership means inspiring employees from all over the world to be the best, to become a world market leader.*

High-Performance Employees

High performance can only be accomplished with a team that is highly motivated and strongly identifies with the company. The selection of the right employees is the foundation. Hidden champions select their employees very rigorously. This choice is accomplished primarily through social control, less through top-down intervention. Intolerance toward poor performance and shirking is a necessary part of a performance-oriented culture. It is helpful to have more work than people. This does not lead to employee discontent, on the contrary. The hidden champions create—and profit from—conditions that lead to an extremely low employee turnover. In addition, comebacks are frequent and an indicator for attractive corporate cultures. Low attrition retains expertise and makes investments in education profitable. It is impressive that the hidden champions successfully globalize these demanding values.

Lesson 2. *High performance requires intolerance against shirking and swift dismissal of employees who do not pull their weight. The low employee turnover proves that the remaining employees appreciate the hidden champions’ corporate cultures and reciprocate with due commitment and high performance.*

Depth

A striking difference between the hidden champions and normal companies is the depth of the value chain and the high vertical integration. Time and again I was amazed to observe that the best enterprises do far more themselves than average companies. “We do everything ourselves,” is the frequent statement made by many of the best. Conversely, I heard average companies say “we don’t do that ourselves,” “we outsource that,” “we leave that to others,” and so forth. I believe the word “depth”

describes this attitude and phenomenon precisely. The greatest ones have an incredible depth in their value chain, a very high vertical integration in manufacturing, and even stronger in R&D. They are very reserved when it comes to outsourcing and strategic alliances, especially where their core competencies are concerned. Is that old-fashioned? Or is this the core of their superiority? There is no straightforward answer.

Lesson 3. *Uniqueness can only come from within and cannot be bought on the market. It therefore requires depth and a certain reserve toward outsourcing.*

Decentralization

In view of the strong leadership personalities, we could have expected centralists practicing authoritarian leadership styles at the hidden champions. Leadership is in fact authoritarian when it comes to the principles. Yet, as we have discovered, the hidden champions are also fervent decentralizers and leave more freedom for execution and implementation than large corporations do. Some of these companies are confronted with growth restraints because their markets are narrow and their market shares are high. The reinvestment opportunities in their traditional business segments are therefore limited. They choose the route of soft diversification, fully aware that this could threaten their traditional strengths. Their answer is rigorous decentralization, usually up to the point of legally independent business units. New, smaller hidden champions thus emerge from older businesses. The importance of decentralization as a means of mobilizing entrepreneurial energies can hardly be overestimated. Decentralization must naturally be accompanied by responsibility and accountability.

Lesson 4. *Decentralization is the most effective way to retain the strengths of the hidden champions, even in larger and more complex structures. Decentralization should be put into practice wherever possible.*

Focus

There may be those who are top performers in different fields. There are even a few who have won two Nobel prizes. However, such people are extremely rare exceptions and hardly suitable as role models for the average entrepreneur or manager. More normal people and companies wanting to achieve something are well advised to concentrate on one area. “We only do one thing, but we do it right,” as the hidden champion Uhlmann says. Such focus is realistically the only way to attain world class. The hidden champions exemplify that it can be done. They are generally not geniuses, but they focus their limited resources better than others and stick with the direction they have chosen until they reach the top position. Focus also includes knowing what you do not want to do. This is just as important as knowing what you do want to do. It’s the only way to avoid the ineffective use of scarce resources. Likewise, the hidden champions teach us that the definition of the “playing field” is critical. Markets, target groups, and applications are not predefined by nature; an astute market definition often forms the foundation for superiority. The business and the market selection must take into consideration the external opportunities and the company’s internal competencies. It is advisable— at least at the outset— to avoid stiff competition and confrontation.

Lesson 5. *Ambitious goals can only be achieved by focusing one’s resources. The definition of the playing field itself is an essential means of getting the focus right.*

Globalization

Nothing in our era and in the decades to come will change our world more than globalization. Extraordinary growth opportunities present themselves to companies that take advantage of this secular trend. Global markets are many times larger than national markets. In global dimensions even niche markets reach sizes making adequate economies of scale possible. The Internet and modern transportation make the world market accessible for companies large and small. Yet the establishment of worldwide sales and manufacturing networks typically takes several generations and requires never-ending perseverance. Revenues are the first to internationalize, followed by personnel and eventually management. Most hidden champions are in the middle of this process. However, there will still be sectors and companies in the future that continue to operate only locally. The decision between local or international business must be well considered.

Lesson 6. *Globalization opens up unprecedented growth opportunities, even for small companies. In order to use these opportunities, leaders and employees must put aside their national and cultural boundaries. Incessant stamina and perseverance are required to survive the multigenerational globalization process. The greatest challenge is the internationalization of the people.*

Innovation

The massive increase of the hidden champions' relative market shares is primarily attributable to their innovativeness in recent years. 6 Whatever may have triggered this wave, the hidden champions are currently in a phase of significant innovative activities. They integrate market and technology as equal driving forces. Few large corporations achieve this balance. For hidden champions, innovation is less a question of budget than of heads. They can obtain a patent in approximately one fifth of the expenditure required by large corporations. And they have five times more patents per employee. Continuous improvements are more typical than breakthrough innovations. Customers are closely involved in the innovation activities.

Lesson 7. *Innovation is the only effective long-term means of succeeding in competition. Innovation is primarily a question of creativity and quality, less so a matter of money.*

Closeness to Customer

The hidden champions teach us that customer orientation is more important than competitive orientation. They consider the long-standing relationships with their customers their greatest strength, even ahead of their technical competencies. Closeness to customer is most effectively achieved with small units. Closeness to customer is more important than marketing professionalism, although the latter also grows with the increasing size of the company. High performance for the customer automatically leads to competitive advantages. Top customers, and likewise top competitors, should be considered performance drivers and consciously used as such. Multiple complex competitive advantages rooted in inner values and competencies create customer commitment and raise the barriers to entry.

Lesson 8. *Closeness to customer almost automatically creates competitive advantages. Top customers, like top competitors, should be employed systematically as drivers of performance.*

WORKING IN PUBLIC

Eghbal, Nadia. *Working in Public: The Making and Maintenance of Open Source Software*

RELEVANCE:

Eghbal's survey of the open-source world is recommended reading. Her concluding chapter is included below. I can't help but see Eghbal's analysis of emerging creatives and interaction patterns and managing time and attention as a sort of successor to or evolution of Clay Shirky's Cognitive Surplus (2010).

This can be read from different perspectives. First, we are creators ourselves – so it can be read personally. Second, software creation is a major industrial process, and it can be read for insight into how that production functions. Third, the employment model for knowledge workers is being renegotiated, and creator communities can be seen as knowledge workers who choose to maintain ownership of their intellectual capital.

This is all less than 10 years old. Without recent advances in network infrastructure and computing capacity (distributed compute) – this world didn't exist. But it seems to be more efficient, agile and more engaging – elicits human agency effects, and is rapidly crowding out legacy economic models.

We would be wise to watch how the “creatives” evolve and what they are trying because the rest of the culture and business tend to follow.

CONCLUSION

“Social media is in a pre-Newtonian moment, where we all understand that it works, but not how it works.” —KEVIN SYSTROM, cofounder, Instagram

Our online social spaces are littered with the artifacts of our creative endeavors. Today, “content” is better understood not as a thing we set out to make—as an automaker might exist solely to produce cars—but, as a friend wrote in an email to me, as “an externality from [our] existing social systems.” Content is a snapshot of our civilization.

The history of software gives us insight into how our attitudes toward content are changing. In the twentieth century, code was bundled into physical formats—a book, a floppy disk, a CD—which made it easier to price and sell. As code became liberated from these formats, and eventually distributed under open-source licenses, it became harder to directly charge for. With millions of lines of code freely available today, the focus has shifted from what developers make to who they are.

Python developer Shauna Gordon-McKeon once posed a hypothetical question to me: “Take a platform you love. Would you rather lose access to all the past content your connections have posted, or lose the connections themselves?” Her point was that the value created on these platforms doesn't lie in the content itself so much as in the underlying social graph. Our relationship to content matters less than our relationships to the people who make it. As a result, we're starting to treat content not as a private economic good but as the externalization of our social infrastructure.

Platforms have helped bring about this shift more quickly. By reducing the costs of production and distribution, they've made it easier for creators to function as one-man operations. Stratechery's Ben Thompson calls this the “faceless publisher” model, wherein “atomized content creators, fueled by

social media, build their own brands and develop their own audiences; the publisher, meanwhile, builds scale on the backside, across infrastructure, monetization, and even human-resource type functions.”

What’s more, by bringing creators themselves to one place, platforms have turned content production into a status game. Eugene Wei calls these “status as a service (StaaS)” businesses, which flourished after Facebook’s invention of the News Feed, and other platforms’ launches of their own social feeds, “unleashed a gold rush for social capital accumulation.”

How, then, should we think about the production of content today, in light of platform-creator relationships? How does the “atomization” of production affect prior theories about how people produce, whether Coase’s theory of the firm, Ostrom’s common pool resources, or Benkler’s peer production?

By treating content as a commodity, we risk solving the wrong puzzle. Finding answers means returning to the essential questions. Chapters 2 and 3 looked at the social dynamics between creators and the communities that form around them. I suggested that one-to-many models, typical among online creators, are centralized communities, with hidden roles played by both platforms and the creator’s audience; these communities stand in contrast to the distributed, many-to-many online communities we’re used to.

Chapter 4 revisited the question of marginal cost. We tend to assume that content doesn’t incur significant marginal costs, thanks to platforms that now absorb most of the distribution costs for creators. However, it’s the maintenance of content that incurs hidden costs with time and use.

A creator’s reputation also requires a certain type of maintenance. Reputation has a half-life on any social platform; successful creators accumulate reputation, which serves as a “battery” that helps them store consumer attention. But if they don’t keep producing new work, that battery will degrade, and eventually get depleted.

Finally, in Chapter 5 I suggested that there are two economic goods in open-source masquerading as one. Open-source code is consumed like a public good, because when code is viewed as a static commodity the cost of additional users is nearly zero. But open-source code is produced like a commons, where a maintainer’s attention is the limited resource. Maintainers must be mindful of how they allocate their attention, avoiding extractive contributors.

“Public” does not imply “participatory”; it’s possible to freely distribute content without depleting producer attention. The remainder of Chapter 5 examined various strategies that developers employ to manage their attention, whether it’s relying upon automation, limiting their availability, or distributing costs onto users.

Historically, most of our questions about the value of content have focused on the distribution side, rather than the production side. Today, the most interesting questions we can ask will focus on how content is made and maintained, and by whom. We’ve previously treated content as a first-copy cost problem, and have developed solutions like patents, intellectual property, and copyright to incentivize its creation. But these solutions don’t address the costs of maintenance, which accrue over time. The challenges facing online creators today derive from the fact that they are playing a repeated game, not a single one.

Maintenance costs are partly caused by poorly defined boundaries around online participation, which haven't scaled to the way we interact with one another today. Creators face a problem of over-participation, not under-participation, in managing expectations from their audience.

Another type of maintenance cost comes from the accumulation of reputation. Every platform that creates social capital for creators must also account for the cost of maintenance, because a creator's reputation degrades over time, unless they continue to invest in it.

I've spent most of this book examining this behavior in open source, which helped me understand the problems of scale that affect our online world more broadly. I'll spend the final pages zooming out in order to explore how what we've learned can be applied not just to open-source developers but other online creators. I'll focus on two areas in particular: the problem of managing over-participation and the problem of making money.

MANAGING OVER-PARTICIPATION

When people talk about the "attention economy," they're usually referring to the consumer's limited attention, as when multiple apps compete for a user's time. But a producer's limited attention is just as important to consider.

As audience engagement grows, more consumers appropriate attention from creators. So, it's critical to find ways to manage this demand, especially extractive demand, which includes comments, reactions, direct messages, and other requests for interaction. A tragedy of the commons occurs not from consumers over-appropriating the content itself, but from consumers over-appropriating a creator's attention.

Why is this problem so hard to solve for? Part of it comes from lingering social norms around the expectations of unfettered participation in our online social spaces. Another problem is that we currently lack the infrastructure to manage this demand. As social norms change, our platforms are evolving to address them. Like a bridge that needs to support more traffic than it was built for, every social platform is scrambling to upgrade its infrastructure to accommodate the volume of social interactions we're dealing with today. Platforms need to build skyscrapers where there were once villages.

Our social platforms were built for distributed, small-scale, many-to-many use cases: the quaint social world of yesteryear. They were modeled after internet forums, chat groups, and mailing lists, because these were the only blueprints that we had for our online social infrastructure.

As more people flocked to these platforms, we saw a few bridges collapse, because, it turns out, we had no idea how to build bridges for today's world. The aftermath of the 2016 United States presidential election saw Facebook wholly unprepared for its new role as a steward of civil society. But platforms are learning from those mistakes and rebuilding their services with new knowledge that we're still uncovering today.

If you imagine a music concert that keeps adding attendees until it becomes a rousing crowd, you can understand the effect of adding more people to our social platforms. When six people are gathered, they can all talk to one another; everyone is engaged in the same conversation. ***When a thousand people are gathered, they divide into two types of interaction. There's the broadcasting effect, when someone climbs onstage to control the crowd, and everyone turns to watch. And then there's the small-group effect, when people strike-up side conversations with their neighbors, ignoring the main stage.***

Social platforms must rebuild their infrastructure to accommodate these two use cases. Yancey Strickler, who cofounded Kickstarter, calls this the “dark forest theory” of the internet: “an increasing number of the population has scurried into their dark forests” to avoid the mainstream web, which has become “a relentless competition for power.” He points to newsletters, podcasts, and group chats as examples of dark forests, while Facebook and Twitter are examples of the mainstream, which will continue to exist alongside more private channels. Mark Zuckerberg himself declared to a crowd at Facebook’s 2019 F8 conference that “the future is private,” insisting, “Over time, I believe that a private social platform will be even more important to our lives than our digital town squares.”

The public stage increasingly reflects a one-way mirror pattern, where anybody can consume content but interactions with its creator are limited. Twitter has begun to experiment with giving its creators more moderation control, such as hiding replies to tweets and limiting who can reply to tweets. The Instagram Stories feature (a format initially invented by Snapchat) is designed for broadcasting rather than conversing with one’s audience. Kevin Systrom, Instagram’s CEO when Stories launched, noted at the time that one reason for introducing Stories was to give creators a place to share without the pressure of likes and comments. Podcasts are surging in popularity, with nearly one in three Americans saying that they listen to podcasts at least once per month—a number that doubled from 2014 to 2019. And newsletters are another, curiously reemerging format, where creators can broadcast in longform to large audiences, without expectation of a dialogue.

In the private sphere, small group chats are having a moment on messaging apps like Facebook Messenger, Instagram, iMessage, WhatsApp, Signal, and Telegram. Group chats are the embodiment of Ostrom’s thesis: a way of drawing boundaries around one’s community, excluding extractive contributors, in order to allow for higher-context interactions. They, too, are an adaptive measure to the problem of over-participation.

Finally, there are still distributed, many-to-many social patterns, such as on Reddit and Facebook groups, that are reminiscent of old online forums. These still work in limited contexts, according to the conditions identified by Benkler. When social interactions are modular and granular, with fungible membership and low coordination costs, they can scale costlessly. Blogger Kicks Condor, in an email to me, describes Reddit as “this massive N:N conversation that almost starts to feel like a singular voice because the names and identities are so muted.” As a result, these communities are frequently oriented around specific interests, whether it’s a local moms’ Facebook group or a subreddit about dating and relationships. Like small group chats, they are less dependent upon a specific platform, compared to creators who broadcast in one-to-many formats. ***Fundamentally, social platforms like Twitter and Facebook still follow the post-and-comment model of an earlier era, where interactions with the author are frictionless, even encouraged. These platforms struggle, more than any other, to adapt to modern social needs.***

One problem is that these platforms assume that all users are interchangeable, whereas in a one-to-many broadcasting format creators are a particular, non-fungible type of user. The experience of an Instagram user with 2 million followers is different from that of an Instagram user with 200 followers. The former is the epicenter of a centralized community, whereas the latter is either passively consuming 1:N content or participating in a small group. Instagram has started to recognize these differences with designated “creator accounts,” which allow their creators to better manage their inbound requests using multiple inboxes, hidden contact information, and automated replies.

While creators are the focus of attention, they don't do it all alone. There are a number of supporting roles that enable them to do their work, whether it's fans who serve as "moderators" and weed out bad actors or those who curate a creator's work to make it more digestible to others. As discussed in previous chapters, the 1:N social pattern is itself a community, even though it doesn't fit our typical definition of an online community. Therefore, we need to take a holistic approach to supporting creators, enabling dedicated supporters to pitch in, instead of treating the creator as a single unit. They're a tree in a forest, not a potted plant.

Designing for one-to-many interactions isn't an entirely new challenge—these types of creators have spiritual roots in television or radio broadcasting—but the social aspect makes things murky. By sharing so much of themselves, these creators occupy a strange space between the intimate and the pure broadcaster experience.

The platform-creator relationship makes things even more confusing, because it's not always clear whether a creator is operating in the platform's space or in their own. Moderation, for example, requires that communities be empowered to self-organize and manage their own rules, as we discovered from Ostrom's work. Platforms ought to give creators the tools they need to manage their own communities. But platforms also face enormous public pressure to develop their own moderation policies, which are difficult to apply indiscriminately to every creator and their respective community.

Many platforms, including Instagram and YouTube, are experimenting with hiding like counts and defaulting to hidden comments, in order to reduce demand from extractive contributors. But Instagram's experiment hides like counts from the public—not from creators themselves. This approach runs the danger of destroying the reputation economy they've created. Early on, aspiring creators who are building an audience want—even need—these social signals, and this isn't necessarily a bad thing. These reputational systems help us measure and assign value. It's only later on, with the costs that come with maintaining one's reputation, that a high volume of likes and comments can become exhausting.

The ability to hide, mute, or turn off these interactions is important to creators at later stages. Taylor Swift explains how she maintains her sanity while occupying a highly visible position: "One thing I do to lessen this weird insecurity laser beam is to turn off comments."...

Another helpful approach is to encourage the role of curators, who can serve as highly complementary and beneficial counterparts to creators, as we saw in the example of Twitch streamers working with their moderators, or Rich Hickey, the author of Clojure, working with Alex Miller, a Clojure maintainer. ***Curators help moderate the public space for a creator, filtering out extractive requests and only surfacing the things that most require a creator's attention. As we saw in Chapter 5, creators and platforms can't manage all their moderation needs manually, so they need to lean heavily upon user-to-user solutions...***

Finally, we might think of the "like," and its subsequent permutations, as a form of automation. By making social interactions more lightweight, platforms can reduce the amount of work that creators need to do.

The "like" was an adaptive mechanism to reduce the attention cost of comments. While it reduces the friction for anyone to participate, it also reduces a creator's mental overhead, similarly to how continuous integration and testing have reduced the amount of review that maintainers need to do for each pull request. (Relatedly: one of the features requested by GitHub's open source maintainers in their

“Dear GitHub” letter was a voting mechanism that would replace the overwhelming volume of “+1” comments they received, “which serve only to spam the maintainers and any others subscribed to the issue.” GitHub added emoji reactions in 2016, which function similarly to likes, and helped reduce comment overhead.)

Emoji, which took off in the 2010s, are another example of social micro-interactions that helped meet growing demand for people’s attention. It’s easier to send an emoji to convey a certain emotion than to type out full sentences. Likes and emojis fit Benkler’s condition of granularity, which makes it possible for distributed social interactions to scale. Apple has added tapbacks to iMessage, and Facebook Messenger has added reactions to group chats, both of which make it easier to acknowledge and respond to messages.

When it comes to internet fame, we have no notion of “organization building” to support high-demand creators as their audiences grow. Nor do creators always have the financial resources to manage growing demand on their own, since online fame doesn’t always translate into money. Big-name celebrities and CEOs eventually hire security details and administrative support. Similarly, platforms must provide the support systems that creators need to be able to thrive.

While we, as users, can try to devise our own solutions, we’re often limited by what platforms allow us to implement. Platforms provide the infrastructure for our online social spaces, and their design decisions massively shape our daily experiences. To borrow from anthropologist Edward T. Hall, who wrote about the social influence of “fixed-feature” spaces like buildings: platforms are “the mold into which a great deal of behavior is cast.”

MAKING MONEY

If the one-way mirror—where anybody can consume but only some people participate—is an adaptive response to over-participation, it’s unsurprising that these smaller, focused communities are giving rise to a particular set of models for funding creative work.

The first wave of monetization was about making meaning of the big. Display advertising, for example, operates as a function of viewership. Pricing is typically based on CPM, or cost per thousand impressions, and CPC, or cost per click, both of which require a lot of views to make these numbers meaningful. *These days, as consumers are faced with near-infinite options, making money as a creator is about making meaning of the small again: Kevin Kelly’s “1,000 true fans,” finally come to pass.* We’re seeing renewed interest in subscription models, sponsorships, and merchandise, all of which operate as a function of para-social, or one-sided, intimacy.

It’s hard to write about the economics of content without discussing advertising further, but because it’s such a big topic on its own, I’m not explicitly covering it in this book. Suffice it to say that the rise of subscription models shouldn’t be read as a death knell for advertising. These two models aren’t mutually exclusive; advertising is still alive and well, and it will continue to work for a very long time. And product sponsorships will probably only become more valuable, because they’re driven by the strength of a creator’s reputation. The writer Tim Ferriss, for one, found that advertising worked better than subscriptions for his podcast, because his listeners especially trust his product recommendations.

As for subscriptions, paywalls might seem like a way to monetize content, but, really, what creators are monetizing is their community, whether by offering a closer connection to the creator, like-minded people to talk to, or a haven from extractive contributors.

A paywall is more like the ticket kiosk at a theme park than a price tag on a car. Most comments aren't useful, so charging readers to leave a comment helps ensure that participants have meaningful skin in the game. Creators charge money in order to make social interactions meaningful again.

Something Awful, an internet forum that was especially popular in the mid-2000s, charged its users an "activation fee." While anybody could read the site's forums (minus an occasional sign-up banner blocking their view), only paying members could post. Kevin Bowen, a former administrator, explained in a Vice interview that "when [Richard Kyanka, who created Something Awful] started charging for forums accounts registrations, he wasn't doing that to make money. He was doing that because he was sick of banning people from the forums."

Jon Hendren, another former Something Awful administrator, chimed in, "When Rich put the paywall in effect, it kept idiots out to an enormous degree. . . . You have to put in a little investment if you want to participate."

With this principle in mind, it's easier to see how paywalls are frequently misapplied. Putting all content behind a paywall isn't likely to work well for creators or publishers, with the exception of a few heavy hitters (say, The New York Times). Similarly, micropayments—where consumers pay small sums for a specific piece of content, such as paying to unlock a specific article—don't usually add up to much either (again, unless you're The New York Times, and can play upon a reader's fear of missing out). *Micropayments make the transaction about content, rather than about creators, but because there is so much freely available, highly substitutable content they create decision fatigue for consumers.*

Subscription models can operate like a freemium model, but they get even more interesting as a two-sided market. In a freemium model, a creator gives away some of their content for free, but restricts other content to those with paid subscriptions. The free content helps creators grow their reputation via public network effects.

In a two-sided market, paying subscribers subsidize all of the content for nonpaying readers, under the assumption that creators aren't actually selling content but a sense of membership and identity. Instead of charging, say, all 100,000 readers ten cents to read an article, creators can instead give away the article for free, but charge 1,000 extra-dedicated subscribers ten dollars per year. Journalist Tim Carmody, for example, subsidizes his newsletter Amazon Chronicles this way. Paying subscribers make it possible for him to offer the newsletter for free to everyone:

The most powerful and interesting media model will remain raising money from members who don't just permit but insist that the product be given away for free. The value comes not just what [sic] they're buying, but who they're buying it from and who gets to enjoy it.

The bigger those two pools get—the bigger the membership, and the bigger the audience—the better it gets for everyone. . . . PBS as a service.

Matthew Butterick, who wrote the popular book Practical Typography, also found success by targeting a subset of his readers. Matthew offers his book for free online, but he didn't want to plaster it with

“nonstop wheedling and begging” for money, as “that would’ve made the book permanently ugly & annoying.” After noticing that much of his unpaid traffic came from just fifteen to twenty websites, he decided to target only visitors from those websites, suggesting that they pay for the book. As a result, the number of direct payments doubled that year, which helped him continue to offer the book for free to more than 600,000 readers annually.

Finally, the news industry is a useful case study to demonstrate how different funding models might apply to different types of content creators. Like “open source,” “news” doesn’t refer to just one type of content. Also like open source, there are a few big newspapers (The Washington Post, The Wall Street Journal, USA Today, and others), which don’t fully capture the experience of the rest of the news industry, much like Linux doesn’t reflect the experience of most open source projects. With the continued unbundling of newspapers, we’ll likely see different models that work for different types of writers. As Damon Kiesow, a professor at the Missouri School of Journalism, puts it, “National and local journalism are not the same business. The New York Times can chase scale for profit, The Centre Daily Times can’t. Because what the New York Times has is an audience. In College Park, they have a community.”

Social media is arguably supplanting the need for “breaking news” reporters. Like casual contributors to an open-source project, eyewitnesses post about a breaking news event because they are intrinsically motivated to do so. The fact that local news outlets frequently appear in eyewitnesses’ comments, asking them for the right to use their images, tells us that average citizens are becoming a better breaking news source for major media outlets, rather than the other way around.

Breaking news is the equivalent of casual contributions in open source: those who are closest to the problem have an intrinsic desire to “contribute” with no expectation of financial reward, but also have no desire to stick around after making their contribution. Most people who post about a breaking news event don’t want to become news reporters; they just want to share and move on. Similarly, most casual contributors have no desire to join a “contributor community” or become maintainers; they’re there to fix their own problem and move on.

For news reporting that focuses on a specific vertical, we’ll see a shift that looks like the shift from “funding projects” to “funding people,” which we explored in Chapter 5. As media companies shift toward “atomized” creators, journalists will find it easier to independently monetize their own reputations, much like any other content creator. ***Readers will pay for a curated, high-quality take on their interests, as well as, sometimes, a sense of community with other like-minded readers.***

For example, former journalist Azeem Azhar brings in six figures of annual recurring revenue with his technology newsletter, [Exponential View](#). The Athletic built its subscription platform, focused on sports, by recruiting high-reputation sports writers from local newspapers. Even reporting on the local news is a form of “niche journalism,” as in the case of Darrell Todd Maurina, who single-handedly curates the local news for Pulaski County, Missouri, on his Facebook page.

Overall, I’d expect to see in the news industry something similar to what happened in open-source: a “dumbbell”-shaped distribution of contributors. On one end are purely casual contributors (breaking news, casual punditry, and commentary on the news), who post on social media because it’s low-effort with the potential for a bit of upside, and no expectation of maintenance, funding, or having to make ongoing contributions. On the other end are the maintainers (news columns, investigative journalism,

and features): those with in-depth knowledge of their topic, who make highly non-fungible contributions, and who will make money off their reputations.

For those in the latter category, the name of the game is building highly targeted audiences and optimizing for fewer people paying more money, rather than shilling for page views. Podcasts, newsletters, and longform writing are attractive these days because they help creators filter for niche audiences, rather than losing potential value to broad reach on public channels. Instead of maximizing for likes, these creators maximize for meaning.

An interesting implication here is that a creator’s relevance (to some niche audience) matters more than quality or trust on a global scale. Early newspaper and media brands built their reputations on the promise of truth and “objectivity,” whereas it seems more likely that future media brands will build their reputations on the promise of relevance.

I’d argue that the reason this funding shift has taken so much longer in journalism—much like in open-source—compared to other forms of content creation is because it requires sloughing off preexisting social norms. Twitch and YouTube creators, for example, essentially created their medium along with the platform, whereas open source and journalism are still grappling with the inherited models of their predecessors. Transitions are hard, but when it comes to making money as a creator online, I feel more optimistic about these opportunities than ever. We’re moving toward a future where rewards are heavily influenced by the quality of one’s audience more than its size. This affords creators an enormous degree of freedom and helps perpetuate the renaissance of ideas that is already well underway.

We don’t have all the answers yet, but I’m hoping this book helps point us toward the right questions.

BUILDING A SECOND BRAIN

Forte, Tiago. Building a Second Brain: A Proven Method to Organize Your Digital Life and Unlock Your Creative Potential

RELEVANCE:

It is too easy to just use the tools we use and not think about it. The promise of a lot of computing in decision making and decision support has always been as “brain extension” technology in different ways. Tiago Forte is self-promotional, and I don’t agree with everything he says and not everything applies to what we are doing – but in terms of how he frames the problem and his nuts-and-bolts approach to drilling down to specific software technologies and practices for knowledge work – he’s worth paying attention to.

I believe it is useful for us to step back and look at our tools as elements that allow us to extend our mental capabilities and ask if the tools we use are doing that or if we are just working with what was state of the art when we learned the work... We have too much information trapped in spreadsheets and documents and emails. To achieve excellence, we need to construct more efficient ways of working.

How a Second Brain Works

It is in the power of remembering that the self’s ultimate freedom consists. I am free because I remember. —Abhinavagupta, tenth-century Kashmiri philosopher and mystic

Think of your Second Brain as the world’s best personal assistant.

It is perfectly reliable and totally consistent. It is always ready and waiting to capture any bit of information that might be of value to you. It follows directions, makes helpful suggestions, and reminds you of what’s important to you.

What would the job description for such a personal assistant look like? What “jobs” would you hire them to do for you? The same way you would hold your assistant accountable to a certain standard of performance, the same is true for your Second Brain. You need to know what it should be doing for you so you know if it’s worth keeping around.

In this chapter we’ll see how the four main capabilities of a Second Brain will actively work for you—immediately and over time; the one basic tool you’ll need to get started; how your Second Brain will evolve to serve what is most important to you; and, finally, an introduction to the four steps of the CODE Method that lies at the heart of it all.

The Superpowers of a Second Brain

There are four essential capabilities that we can rely on a Second Brain to perform for us:

- Making our ideas concrete.
- Revealing new associations between ideas.
- Incubating our ideas over time.
- Sharpening our unique perspectives.

Let’s examine each of these.

Second Brain Superpower #1: Make Our Ideas Concrete

Before we do anything with our ideas, we must “off-load” them from our minds and put them into concrete form. Only when we declutter our brain of complex ideas can we think clearly and start to work with those ideas effectively.

In 1953, American biologist James Watson and English physicist Francis Crick made a profound discovery: the structure of DNA was a double helix. Their discovery was built on the groundwork laid by other pioneers, including advancements in X-ray crystallography by Rosalind Franklin and Maurice Wilkins, and ushered in a golden age in molecular biology and genetics.

Watson and Crick’s breakthrough is well recognized, but there is a part of their story that is much less well known. An important tool of the researchers was building physical models, an approach they borrowed from American biochemist Linus Pauling. They made cardboard cutouts to approximate the shapes of the molecules they knew were part of DNA’s makeup and, like a puzzle, experimented with different ways of putting them together. They would shift around their models on their desktops, trying to find a shape that fit everything they knew about how the molecules were arranged. The double helix structure seemed to fit all known constraints, allowing the complementary base pairs to fit together perfectly while respecting the ratios between elements that had been measured previously.

This is a remarkable aspect of one of the most famous scientific discoveries of the last century: at the decisive moment, even highly trained scientists deeply familiar with mathematical and abstract thinking turned to the most basic, ancient medium available: physical stuff.

Digital notes aren’t physical, but they are visual. They turn vague concepts into tangible entities that can be observed, rearranged, edited, and combined together. They may exist only in virtual form, but we can still see them with our eyes and move them around with our fingers. As researchers Deborah Chambers and Daniel Reisberg found in their research on the limits of mental visualization, “The skills we have developed for dealing with the external world go beyond those we have for dealing with the internal world.”

Second Brain Superpower #2: Reveal New Associations Between Ideas

In its most practical form, creativity is about connecting ideas together, especially ideas that don’t seem to be connected.

Neuroscientist Nancy C. Andreasen, in her extensive research on highly creative people including accomplished scientists, mathematicians, artists, and writers, came to the conclusion that “Creative people are better at recognizing relationships, making associations and connections.”

By keeping diverse kinds of material in one place, we facilitate this connectivity and increase the likelihood that we’ll notice an unusual association.

Quotes from a philosophy book written in ancient times might sit next to the latest clever tweet. Screenshots from an interesting YouTube video can live right by scenes from classic movies. An audio memo might be saved alongside project plans, a link to a helpful website, and a PDF with the latest research findings. All these formats can be combined in a way that would be impossible in the physical world.

If you’ve ever played the word-tile game Scrabble, you know the best way to come up with new words is to mix up the letters in different combinations until a word jumps out at you. In our Second Brain we can do the same: mix up the order of our ideas until something unexpected emerges. The more diverse

and unusual the material you put into it in the first place, the more original the connections that will emerge.

Second Brain Superpower #3: Incubate Our Ideas Over Time

Too often when we take on a task—planning an event, designing a product, or leading an initiative—we draw only on the ideas we have access to right in that moment. I call this approach a “heavy lift”—demanding instantaneous results from our brains without the benefit of a support system.

Even when we do a brainstorm, that still relies only on ideas that we can think up right now. ***What are the chances that the most creative, most innovative approaches will instantly be top of mind? What are the odds that the best way to move forward is one of the first ways we come up with?***

This tendency is known as recency bias. We tend to favor the ideas, solutions, and influences that occurred to us most recently, regardless of whether they are the best ones. Now imagine if you were able to unshackle yourself from the limits of the present moment, and draw on weeks, months, or even years of accumulated imagination.

I call this approach the “slow burn”—allowing bits of thought matter to slowly simmer like a delicious pot of stew brewing on the stove. It is a calmer, more sustainable approach to creativity that relies on the gradual accumulation of ideas, instead of all-out binges of manic hustle. Having a Second Brain where lots of ideas can be permanently saved for the long term turns the passage of time into your friend, instead of your enemy.

Second Brain Superpower #4: Sharpen Our Unique Perspectives

Until now we’ve talked mostly about gathering the ideas of others, but the ultimate purpose of a Second Brain is to allow your own thinking to shine.

A recent study from Princeton University found that there is a certain kind of job that is least likely to be automated by machines in coming years. Surprisingly, it wasn’t jobs that required advanced skills or years of training that were predicted to fare best. It was jobs that required the ability to convey “not just information but a particular interpretation of information.”

In other words, the jobs that are most likely to stick around are those that involve promoting or defending a particular perspective. Think of a fundraising organizer sharing stories of the impact their nonprofit has made, a researcher using data to back up their interpretation of an experiment, or a project manager citing a couple of key precedents to support a decision. Our careers and businesses depend more than ever on our ability to advance a particular point of view and persuade others to adopt it as well.

Advocating for a particular point of view isn’t just a matter of sparkling charisma or irresistible charm. It takes supporting material. American journalist, author, and filmmaker Sebastian Junger once wrote on the subject of “writer’s block”: “It’s not that I’m blocked. It’s that I don’t have enough research to write with power and knowledge about that topic. It always means, not that I can’t find the right words, [but rather] that I don’t have the ammunition.”

When you feel stuck in your creative pursuits, it doesn’t mean that there’s something wrong with you. You haven’t lost your touch or run out of creative juice. It just means you don’t yet have enough raw material to work with. If it feels like the well of inspiration has run dry, it’s because you need a deeper well full of examples, illustrations, stories, statistics, diagrams, analogies, metaphors, photos, mindmaps,

conversation notes, quotes—anything that will help you argue for your perspective or fight for a cause you believe in.

Choosing a Notetaking App: The Neural Center of Your Second Brain

The same technology that has fueled an explosion in the volume of information coming our way has also provided the tools to help us manage it.

While your Second Brain is made up of all the tools you use to interact with information, including a to-do list, a calendar, email, and reading apps, for example, there is one category of software I recommend as the centerpiece of your Second Brain: a digital notetaking app. There are plentiful options available, from the free notes app that came preinstalled on your smartphone to third-party software you can download with exactly the features you need.

From Microsoft OneNote, Google Keep, and Apple Notes to Notion and Evernote, digital notes apps have four powerful characteristics that make them ideal for building a Second Brain. They are:

- ***Multimedia:*** Just like a paper notebook might contain drawings and sketches, quotes and ideas, and even a pasted photo or Post-it, a notes app can store a wide variety of different kinds of content in one place, so you never need to wonder where to put something.
- ***Informal:*** Notes are inherently messy, so there's no need for perfect spelling or polished presentation. This makes it as easy and frictionless as possible to jot things down as soon as they occur to you, which is essential to allow nascent ideas to grow.
- ***Open-ended:*** Taking notes is a continuous process that never really ends, and you don't always know where it might lead. Unlike more specialized kinds of software that are designed to produce a specific kind of output (such as slide decks, spreadsheets, graphics, or videos), notes are ideal for free-form exploration before you have a goal in mind.
- ***Action-oriented:*** Unlike a library or research database, personal notes don't need to be comprehensive or precise. They are designed to help you quickly capture stray thoughts so you can remain focused on the task at hand.

All four of the above qualities are shared by paper notes, but when we make them digital, we can supercharge these timeless benefits with the incredible capabilities of technology—searching, sharing, backups, editing, linking, syncing between devices, and many others. Digital notes combine the casual artistry of a daily sketchbook with the scientific power of modern software.

Choosing which apps and tools you'll use is a personal decision, and depends on which mobile device you use, the needs of your job or business, and even your own temperament and taste. The software landscape is dynamic and constantly changing. New apps regularly come out, and existing ones release innovative new features. You can find a free, continually updated guide to choosing your notes app and other Second Brain tools at <https://www.buildingasecondbrain.com/resources>.

Although you will always use many different software programs to manage information and do your work—from word processors to messaging platforms and project management tools used within your organization—your notes app is uniquely designed to facilitate personal knowledge management.

A good place to start is to look at the apps you already have and perhaps are already using. You can always start now with a basic option and upgrade later as your needs get more sophisticated.

Most important of all, don't get caught in the trap of perfectionism: insisting that you have to have the "perfect" app with a precise set of features before you take a single note. It's not about having the perfect tools—it's about having a reliable set of tools you can depend on, knowing you can always change them later.

Remembering, Connecting, Creating: Stages of Personal Knowledge Management

As people set out on their Second Brain journey, there are three stages of progress I often observe—and even encourage. Those stages are remembering, connecting, and creating. It takes time to fully unlock the value of using digital tools to enhance and extend what our minds are capable of, but there are also distinct benefits at every step along the way.

The first way that people tend to use their Second Brain is as a memory aid. They use their digital notes to save facts and ideas that they would have trouble recalling otherwise: takeaways from meetings, quotes from interviews, or the details of a project, for example.

Camille is the cofounder and lead designer of a start-up in Quebec, Canada. She uses her Second Brain to save excerpts from the many research reports and studies she reads as part of her work designing electric vehicle charging stations for large residential buildings. Most of these reports are published as PDFs, a notoriously inflexible and difficult-to-use format, but by importing the findings most relevant to her work into her notes, she can add as many annotations and comments as she wants.

The second way that people use their Second Brain is to connect ideas together. Their Second Brain evolves from being primarily a memory tool to becoming a thinking tool. A piece of advice from a mentor comes in handy as they encounter a similar situation on a different team. An illuminating metaphor from a book finds its way into a presentation they're delivering. The ideas they've captured begin gravitating toward each other and cross-pollinating.

Fernando is an oncologist at a world-renowned hospital who uses his Second Brain to organize his patient notes. He summarizes the key points from each patient's health history with a special focus on how long they've had the condition, which treatments they've received, and key features of their tumor. Fernando uses his Second Brain to connect what he knows from his training and research to the needs of his patients, to be able to treat them more effectively.

Eventually, the third and final way that people use their Second Brain is for creating new things. They realize that they have a lot of knowledge on a subject and decide to turn it into something concrete and shareable. Seeing so much supporting material ready and waiting gives them the courage to put their own ideas out there and have a positive impact on others.

Terrell is a young father of three working a demanding job at a large tech company in Texas. After taking my course, he used his Second Brain to start a YouTube channel where he shares stories and tips on parenting. For example, he's made videos on how to travel internationally with kids and how to request paternity leave, and shared clips from weekend trips he's taken with his family.

Being able to keep track of all the video ideas and production details outside his head has been crucial in allowing Terrell to balance his side gig with his full-time career, while still having enough time with his kids. He is using his Second Brain to express himself and create the content he wants to see in the world.

Each of these individuals has leveraged technology to remember, connect, and create far more effectively than if they had to do it on their own. They use their Second Brain in ways that complement

the current season of their lives. As those seasons change, they will be able to adapt how they use their notes so that they remain relevant and useful.

Introducing The CODE Method: The Four Steps to Remembering What Matters

To guide you in the process of creating your own Second Brain, I've developed a simple, intuitive four-part method called "CODE"—Capture; Organize; Distill; Express.

These are the steps not only to build your Second Brain in the first place, but also to work with it going forward. Each of these steps represents a timeless principle found throughout humanity's history, from the earliest cave paintings to the artisan workshops of the Renaissance to the most cutting-edge modern fields. ***They are flexible and agnostic for any profession, role, or career and for whatever notetaking methods and platforms you prefer. I'm even willing to bet that you're doing them already in some form, whether you realize it or not.***

CODE is a map for navigating the endless streams of information we are now faced with every day. It is a modern approach to creating a commonplace book, adapted to the needs of the Information Age.

The same way we have a genetic code that determines our height and eye color, we also have a creative code that is hardwired into our imagination. It shapes how we think and how we interact with the world. It is mirrored in the software code that runs the apps we use to handle information. It's also been a secret code for most of history—now it's finally time to reveal how it works.

Let's preview each of the four steps of the CODE Method—Capture, Organize, Distill, and Express—and then we'll dive into the details in the following chapters.

Capture: Keep What Resonates

Every time we turn on our smartphone or computer, we are immediately immersed in the flow of juicy content they present. Much of this information is useful and interesting—how-to articles that could make us more productive, podcasts with experts sharing hard-won lessons, or inspiring photos of travel destinations we might want to visit.

Here's the problem: we can't consume every bit of this information stream. We will quickly be exhausted and overwhelmed if we try. We need to adopt the perspective of a curator, stepping back from the raging river and starting to make intentional decisions about what information we want to fill our minds.

Like a scientist capturing only the rarest butterflies to take back to the lab, our goal should be to "capture" only the ideas and insights we think are truly noteworthy. Content tends to pile up all around us even without our involvement. There are probably emails filling your inbox, updates popping up in your social media feeds, and notifications proliferating on your smartphone as you're reading this.

It's already there, but we tend to capture it haphazardly, at best. You might email yourself a quick note, brainstorm some ideas in a document, or highlight quotes in a book you're reading, but that information probably remains disconnected and scattered. The insights you uncovered through serious mental effort remain hidden in forgotten folders or drifting in the cloud.

The solution is to keep only what resonates in a trusted place that you control, and to leave the rest aside.

When something resonates, it moves you on an intuitive level. Often, the ideas that resonate are the ones that are most unusual, counterintuitive, interesting, or potentially useful. Don't make it an analytical

decision, and don't worry about why exactly it resonates—just look inside for a feeling of pleasure, curiosity, wonder, or excitement, and let that be your signal for when it's time to capture a passage, an image, a quote, or a fact.

By training ourselves to notice when something resonates with us, we can improve not only our ability to take better notes, but also our understanding of ourselves and what makes us tick. It is a way of turning up the volume on our intuition so we can hear the wisdom it offers us. Adopting the habit of knowledge capture has immediate benefits for our mental health and peace of mind. We can let go of the fear that our memory will fail us at a crucial moment. Instead of jumping at every new headline and notification, we can choose to consume information that adds value to our lives and consciously let go of the rest.

Organize: Save for Actionability

Once you've begun capturing notes with the ideas that resonate with you, you'll eventually feel the need to organize them.

It is tempting to try to create a perfect hierarchy of folders up front to contain every possible note you might ever want to capture. Even if it was possible, that approach would be incredibly time-consuming and require way too much effort, pulling you away from what interests you today.

Most people tend to organize information by subject, like the Dewey decimal system you've probably seen at the library. For example, you might find a book under a broad subject category like "Architecture," "Business," "History," or "Geology."

When it comes to digital notes, we can use much easier and lighter ways of organizing. Because our priorities and goals can change at a moment's notice, and probably will, we want to avoid organizing methods that are overly rigid and prescriptive. The best way to organize your notes is to organize for action, according to the active projects you are working on right now. Consider new information in terms of its utility, asking, "How is this going to help me move forward one of my current projects?"

Surprisingly, when you focus on taking action, the vast amount of information out there gets radically streamlined and simplified. There are relatively few things that are actionable and relevant at any given time, which means you have a clear filter for ignoring everything else.

Organizing for action gives you a sense of tremendous clarity, because you know that everything you're keeping actually has a purpose. You know that it aligns with your goals and priorities. Instead of organizing being an obstacle to your productivity, it becomes a contributor to it.

Distill: Find the Essence

Once you start capturing ideas in a central place and organizing them for action, you'll inevitably begin to notice patterns and connections between them.

An article you read about gardening will give you an insight into growing your customer base. An offhand testimonial from a client will give you the idea to create a web page with all your client testimonials. A business card will remind you of a fascinating conversation you had with someone who you could reach out to for coffee.

The human mind is like a sizzling-hot frying pan of associations—throw a handful of seeds in there and they'll explode into new ideas like popcorn. Every note is the seed of an idea, reminding you of what you already know and already think about a topic.

There is a powerful way to facilitate and speed up this process of rapid association: distill your notes down to their essence.

Every idea has an “essence”: the heart and soul of what it is trying to communicate. It might take hundreds of pages and thousands of words to fully explain a complex insight, but there is always a way to convey the core message in just a sentence or two.

Einstein famously summarized his revolutionary new theory of physics with the equation $E=mc^2$. If he can distill his thinking into such an elegant equation, you can surely summarize the main points of any article, book, video, or presentation so that the main point is easy to identify.

Why is it so important to be able to easily find the main point of a note? Because in the midst of a busy workday, you won't have time to review ten pages of notes on a book you read last year—you need to be able to quickly find just the main takeaways.

If you've highlighted those takeaways in the flow of the reading you were already doing anyway, you'll be able to remind yourself of what the book contains without having to spend hours rereading it.

Every time you take a note, ask yourself, “How can I make this as useful as possible for my future self?” That question will lead you to annotate the words and phrases that explain why you saved a note, what you were thinking, and what exactly caught your attention.

Your notes will be useless if you can't decipher them in the future, or if they're so long that you don't even try. Think of yourself not just as a taker of notes, but as a giver of notes—you are giving your future self the gift of knowledge that is easy to find and understand.

[Express: Show Your Work](#)

All the previous steps—capturing, organizing, and distilling—are geared toward one ultimate purpose: sharing your own ideas, your own story, and your own knowledge with others.

What is the point of knowledge if it doesn't help anyone or produce anything?

Whether your goal is to lose weight, get a promotion at work, start a side business, or strengthen your local community, personal knowledge management exists to support taking action—anything else is a distraction.

A common challenge for people who are curious and love to learn is that we can fall into the habit of continuously force-feeding ourselves more and more information, but never actually take the next step and apply it. We compile tons of research, but never put forward our own proposal. We gather untold business case studies, but never pitch one potential client. We study every piece of relationship advice available, but never ask anyone out on a date.

It's so easy to endlessly delay and postpone the experiences that would so enrich our lives. We think we're not ready. We fear we're not prepared. We cannot stand the thought that there is one little piece of information we're missing that, if we had it, would make all the difference.

I'm here to tell you that that is no way to live your life. Information becomes knowledge—personal, embodied, verified—only when we put it to use. You gain confidence in what you know only when you know that it works. Until you do, it's just a theory.

This is why I recommend you shift as much of your time and effort as possible from consuming to creating. We all naturally have a desire to create—to bring to life something good, true, or beautiful. It's

a part of our essential nature. Creating new things is not only one of the most deeply fulfilling things we can do, it can also have a positive impact on others—by inspiring, entertaining, or educating them.

What should you create?

It depends on your skills, interests, and personality. If you are highly analytical, you could evaluate the many options for camping gear and create a list of recommended products to share with your friends. If you like to teach, you could record your favorite dessert recipe and post it on social media or a blog. If you care about a local cause such as public parks, you could create a plan to lobby the city council for more funding.

All these actions—evaluate, share, teach, record, post, and lobby—are synonyms for the act of expression. They all draw on outside sources for raw material, they all involve a practical process of refinement over time, and they all end up making an impact on someone or something that matters to you.

Information is always in flux, and it is always a work in progress. Since nothing is ever truly final, there is no need to wait to get started. You can publish a simple website now, and slowly add additional pages over time. You can send out a draft of a piece of writing now and make revisions later when you have more time. The sooner you begin, the sooner you start on the path of improvement.

I've introduced a lot of new concepts and terms, and I know at this point it can seem a little overwhelming. It may feel like you have to learn and do a lot of new things to be able to build a Second Brain.

Here's the surprising truth: you are already doing most of the work required.

You are already learning new things—you couldn't stop if you wanted to. You already consume interesting ideas—note the dozens of tabs you likely have open in your browser. You already put in so much effort to keep track of all the information you need for your studies, your job, or your business. All you need is a slightly more intentional, more deliberate way to manage that information, plus a few practical habits to ensure it gets done.

CREATING CAPABILITIES: DEVELOPMENT AS FREEDOM

Sen, Amartya. DEVELOPMENT AS FREEDOM. Oxford University Press.

RELEVANCE:

Nussbaum and Sen have expanded on the Capabilities approach to human development which links to Capabilities computing which evolved from erights and is instantiated in crypto systems and also gmail anything using protocol buffers. The way the digital path becomes real is by extending capabilities to people through distributed applications. This has already started and it is changing everything. We need to understand it. To do this, part of what we all need to do is to go back and understand Sen and Nussbaum.

INTRODUCTION

Development can be seen, it is argued here, as a process of expanding the real freedoms that people enjoy. Focusing on human freedoms contrasts with narrower views of development, such as identifying development with the growth of gross national product, or with the rise in personal incomes, or with industrialization, or with technological advance, or with social modernization. Growth of GNP or of individual incomes can, of course, be very important as means to expanding the freedoms enjoyed by the members of the society. But freedoms depend also on other determinants, such as social and economic arrangements (for example, facilities for education and health care) as well as political and civil rights (for example, the liberty to participate in public discussion and scrutiny). Similarly, industrialization or technological progress or social modernization can substantially contribute to expanding human freedom, but freedom depends on other influences as well. If freedom is what development advances, then there is a major argument for concentrating on that overarching objective, rather than on some particular means, or some specially chosen list of instruments. Viewing development in terms of expanding substantive freedoms directs attention to the ends that make development important, rather than merely to some of the means that, inter alia, play a prominent part in the process.

Development requires the removal of major sources of unfreedom: poverty as well as tyranny, poor economic opportunities as well as systematic social deprivation, neglect of public facilities as well as intolerance or overactivity of repressive states. Despite unprecedented increases in overall opulence, the contemporary world denies elementary freedoms to vast numbers—perhaps even the majority—of people. Sometimes the lack of substantive freedoms relates directly to economic poverty, which robs people of the freedom to satisfy hunger, or to achieve sufficient nutrition, or to obtain remedies for treatable illnesses, or the opportunity to be adequately clothed or sheltered, or to enjoy clean water or sanitary facilities. In other cases, the unfreedom links closely to the lack of public facilities and social care, such as the absence of epidemiological programs, or of organized arrangements for health care or educational facilities, or of effective institutions for the maintenance of local peace and order. ***In still other cases, the violation of freedom results directly from a denial of political and civil liberties by authoritarian regimes and from imposed restrictions on the freedom to participate in the social, political and economic life of the community.***

EFFECTIVENESS AND INTERCONNECTIONS

Freedom is central to the process of development for two distinct reasons:

- 1) The evaluative reason: assessment of progress has to be done primarily in terms of whether the freedoms that people have are enhanced;***
- 2) The effectiveness reason: achievement of development is thoroughly dependent on the free agency of people.***

I have already signaled the first motivation: the evaluative reason for concentrating on freedom. In pursuing the second, that of effectiveness, we have to look at the relevant empirical connections, in particular at the mutually reinforcing connections between freedoms of different kinds. It is because of these interconnections, which are explored in some detail in this book, that free and sustainable agency emerges as a major engine of development. ***Not only is free agency itself a “constitutive” part of development, it also contributes to the strengthening of free agencies of other kinds. The empirical connections that are extensively explored in this study link the two aspects of the idea of “development as freedom.”***

The relation between individual freedom and the achievement of social development goes well beyond the constitutive connection -- important as it is. What people can positively achieve is influenced by economic opportunities, political liberties, social powers, and the enabling conditions of good health, basic education, and the encouragement and cultivation of initiatives. The institutional arrangements for these opportunities are also influenced by the exercise of people’s freedoms, through the liberty to participate in social choice and in the making of public decisions that impel the progress of these opportunities. These interconnections are also investigated here.

SOME ILLUSTRATIONS: POLITICAL FREEDOM AND QUALITY OF LIFE

The difference that is made by seeing freedom as the principal ends of development can be illustrated with a few simple examples. Even though the full reach of this perspective can only emerge from a much more extensive analysis (attempted in the chapters to follow), the radical nature of the idea of “development as freedom” can easily be illustrated with some elementary examples.

First, in the context of the narrower views of development in terms of GNP growth or industrialization, it is often asked whether certain political or social freedoms, such as the liberty of political participation and dissent, or opportunities to receive basic education, are or are not “conducive to development.” ***In the light of the more foundational view of development as freedom, this way of posing the question tends to miss the important understanding that these substantive freedoms (that is, the liberty of political participation or the opportunity to receive basic education or health care) are among the constituent components of development.*** Their relevance for development does not have to be freshly established through their indirect contribution to the growth of GNP or to the promotion of industrialization. As it happens, these freedoms and rights are also very effective in contributing to economic progress; this connection will receive extensive attention in this book. But while the causal relation is indeed significant, the vindication of freedoms and rights provided by this causal linkage is over and above the directly constitutive role of these freedoms in development.

...

TRANSACTIONS, MARKETS AND ECONOMIC UNFREEDOM

A third illustration relates to the role of markets as part of the process of development. The ability of the market mechanism to contribute to high economic growth and to overall economic progress has been widely—and rightly—acknowledged in the contemporary development literature. But it would be a mistake to understand the place of the market mechanism only in derivative terms. As Adam Smith noted, freedom of exchange and transaction is itself part and parcel of the basic liberties that people have reason to value.

To be generically against markets would be almost as odd as being generically against conversations between people (even though some conversations are clearly foul and cause problems for others—or even for the conversationalists themselves). The freedom to exchange words, or goods, or gifts does not need defensive justification in terms of their favorable but distant effects; they are part of the way human beings in society live and interact with each other (unless stopped by regulation or fiat). The contribution of the market mechanism to economic growth is, of course, important, but this comes only after the direct significance of the freedom to interchange—words, goods, gifts—has been acknowledged.

As it happens, the rejection of the freedom to participate in the labor market is one of the ways of keeping people in bondage and captivity, and the battle against the unfreedom of bound labor is important in many third world countries today for some of the same reasons the American Civil War was momentous. The freedom to enter markets can itself be a significant contribution to development, quite aside from whatever the market mechanism may or may not do to promote economic growth or industrialization. In fact, the praise of capitalism by Karl Marx (not a great admirer of capitalism in general) and his characterization (in *Das Kapital*) of the American Civil War as “the one great event of contemporary history” related directly to the importance of the freedom of labor contract as opposed to slavery and the enforced exclusion from the labor market. As will be discussed, the crucial challenges of development in many developing countries today include the need for the freeing of labor from explicit or implicit bondage that denies access to the open labor market. Similarly, the denial of access to product markets is often among the deprivations from which many small cultivators and struggling producers suffer under traditional arrangements and restrictions. ***The freedom to participate in economic interchange has a basic role in social living.***

...

ORGANIZATIONS AND VALUES

Many other examples can be given to illustrate the pivotal difference that is made by pursuing a view of development as an integrated process of expansion of substantive freedoms that connect with one another. It is this view that is presented, scrutinized, and utilized in this book to investigate the development process in inclusive terms that integrate economic, social, and political considerations. ***A broad approach of this kind permits simultaneous appreciation of the vital roles, in the process of development, of many different institutions, including markets and market-related organizations, governments and local authorities, political parties and other civic institutions, educational arrangements and opportunities of open dialogue and debate (including the role of the media and other means of communication).***

Such an approach also allows us to acknowledge the role of social values and prevailing mores, which can influence the freedoms that people enjoy and have reason to treasure. Shared norms can influence

social features such as gender equity, the nature of childcare, family size and fertility patterns, the treatment of the environment and many other arrangements and outcomes. Prevailing values and social mores also affect the presence or absence of corruption, and the role of trust in economic or social or political relationships. The exercise of freedom is mediated by values, but the values in turn are influenced by public discussions and social interactions, which are themselves influenced by participatory freedoms. Each of these connections deserves scrutiny.

The fact that the freedom of economic transactions tends to be typically a great engine of economic growth has been widely acknowledged, even though forceful detractors remain. It is important not only to give the markets their due, but also to appreciate the role of other economic, social, and political freedoms in enhancing and enriching the lives that people are able to lead. This has a clear bearing even on such controversial matters as the so-called population problem. The role of freedom in moderating excessively high fertility rates is a subject on which contrary views have been held for a long time. While that great eighteenth-century French rationalist Condorcet expected that fertility rates would come down with “the progress of reason,” so that greater security, more education and more freedom of reflected decisions would restrain population growth, his contemporary Thomas Robert Malthus differed radically with this position. Indeed, Malthus argued that “there is no reason whatever to suppose that anything beside the difficulty of procuring in adequate plenty the necessaries of life should either indispose this greater number of persons to marry early or disable them from rearing in health the largest families.” The comparative merits of the two different positions—relying respectively on reasoned freedom and economic compulsion—will be investigated later in this study (the balance of evidence, I shall argue, is certainly more on Condorcet’s side). But it is especially important to recognize that this controversy is just one example of the debate between pro-freedom and anti-freedom approaches to development that has gone on for many centuries. That debate is still very active in many different forms.

INSTITUTIONS AND INSTRUMENTAL FREEDOMS

Five distinct types of freedom, seen in an “instrumental” perspective, are particularly investigated in the empirical studies that follow. These include (1) political freedoms, (2) economic facilities, (3) social opportunities, (4) transparency guarantees and (5) protective security. Each of these distinct types of rights and opportunities helps to advance the general capability of a person. They may also serve to complement each other. Public policy to foster human capabilities and substantive freedoms in general can work through the promotion of these distinct but interrelated instrumental freedoms. In the chapters that follow, each of these different types of freedom—and the institutions involved—will be explored, and their interconnections discussed. There will be an opportunity also to investigate their respective roles in the promotion of overall freedoms of people to lead the kind of lives they have reason to value. In the view of “development as freedom,” the instrumental freedoms link with each other and with the ends of enhancement of human freedom in general.

While development analysis must, on the one hand, be concerned with objectives and aims that make these instrumental freedoms consequentially important, it must also take note of the empirical linkages that tie the distinct types of freedom together, strengthening their joint importance. Indeed, these connections are central to a fuller understanding of the instrumental role of freedom.

A CONCLUDING REMARK

Freedoms are not only the primary ends of development, they are also among its principal means. In addition to acknowledging, foundationally, the evaluative importance of freedom, we also have to understand the remarkable empirical connection that links freedoms of different kinds with one another. Political freedoms (in the form of free speech and elections) help to promote economic security. Social opportunities (in the form of education and health facilities) facilitate economic participation. Economic facilities (in the form of opportunities for participation in trade and production) can help to generate personal abundance as well as public resources for social facilities. Freedoms of different kinds can strengthen one another.

These empirical connections reinforce the valuational priorities. In terms of the medieval distinction between “the patient” and “the agent,” this freedom-centered understanding of economics and of the process of development is very much an agent-oriented view. With adequate social opportunities, individuals can effectively shape their own destiny and help each other. They need not be seen primarily as passive recipients of the benefits of cunning development programs. There is indeed a strong rationale for recognizing the positive role of free and sustainable agency—and even of constructive impatience.

THE MASTER ALGORITHM

Domingos, Pedro. The Master Algorithm (pp. 235-237).

RELEVANCE:

Domingos The Master Algorithm is a readable approachable survey of machine learning. His goal to try to create a unified learner also leads him to some useful insights in the section below. You should notice echoes of Kauffman's pre-adaptations and Diamandis' Convergence in Domingo's concept of unifiers. They are all talking about the same elephant.

The Pieces of the Puzzle Fall into Place

Machine learning is both a science and a technology, and both characteristics give us hints on how to unify it. On the science side, unifying theories often begin with a deceptively simple observation.

Two seemingly unrelated phenomena turn out to be just two faces of the same coin, and like the first domino to fall, that realization sets off a cascade of others. An apple falling to the ground, the moon hanging in the sky: both are caused by gravity, and—apocryphal story or not—once Newton figured out how, gravity turned out to also account for the tides, the precession of the equinoxes, the trajectories of comets, and much else. In everyday experience, electricity and magnetism are never seen together: a lightning spark here, a rock that attracts iron objects there, both quite rare. But once Maxwell figured out how a changing electric field gives rise to magnetism and vice versa, it became clear that light itself is an intimate marriage of the two, and today we know that, far from rare, electromagnetism pervades all matter. Mendeleev's periodic table not only organized all the known elements into just two dimensions, it also predicted where new elements would be found. Darwin's observations aboard the Beagle suddenly began to make sense when Malthus's Essay on Population suggested natural selection as the organizing principle. When Crick and Watson [and Franklin] hit on the double helix structure as an explanation for the puzzling properties of DNA, they immediately saw how it might replicate itself, and biology's transition from stamp collecting (in Rutherford's pejorative words) to unified science had begun. In each of these cases, a bewildering variety of observations turned out to have a common cause, and once scientists identified it, they could in turn use it to predict many new phenomena. *Similarly, even though the learners we've met in this book seem quite disparate—some based on the brain, some on evolution, some on abstract mathematical principles—they in fact have much in common, and the resulting theory of learning yields many new insights¹⁶.*

Although it is less well known, many of the most important technologies in the world are the result of inventing a unifier, a single mechanism that does what previously required many. The Internet, as the name implies, is a network that interconnects networks.... The Internet's protocols are an Esperanto that gives each computer the illusion of talking directly to any other and that allows e-mail and the web to ignore the details of the physical infrastructure they flow over.... A microprocessor is an assembly of digital electronic components that can mimic any other assembly. Virtual machines allow the same

¹⁶ This is where the name Model Arsenal comes from. This simple idea that the right model is the essential tool for finding new solutions, and there is a special magic in combining models into processes creating virtual analytical factories. It is also an homage to the Venetian Arsenal, which was arguably the first factory line. The emergence of factories of ideas is what created the virtual world – the cloud and networks and software. Technological convergence is just process recombination across domains in new cycles of work -- exploring the adjacent possible with the new models in our arsenal.

computer to pose as a hundred different computers to a hundred different people at the same time and help make the cloud possible. Graphical user interfaces let us edit documents, spreadsheets, slide decks, and much else using a common language of windows, menus, and mouse clicks. The computer itself is a unifier: a single device capable of solving any logical or mathematical problem, provided we know how to program it. Even plain old electricity is a kind of unifier: you can generate it from many different sources—coal, gas, nuclear, hydro, wind, solar—and consume it in an infinite variety of ways. A power station doesn't know or care how the electricity it produces will be consumed, and your porch light, dishwasher, or brand-new Tesla are oblivious to where their electricity supply comes from. Electricity is the Esperanto of energy.

The Master Algorithm is the unifier of machine learning: it lets any application use any learner, by abstracting the learners into a common form that is all the applications need to know....

To recap: the unified learner we've arrived at uses MLNs (Markov Logic Networks) as the representation, posterior probability as the evaluation function, and genetic search coupled with gradient descent as the optimizer. If we want, we can easily replace the posterior by some other accuracy measure, or genetic search by hill climbing....

I wouldn't be so rash as to call this learner the Master Algorithm, however. For one, the proof of the pudding is in the eating, and although over the last decade this algorithm (or variations of it) has been successfully applied in many areas, there are many more to which it hasn't, and so it's not yet clear just how general purpose it is. Second, there are some important problems that it doesn't solve....

You can download the learner I've just described from <https://alchemy.cs.washington.edu>.

We christened it Alchemy to remind ourselves that, despite all its successes, machine learning is still in the alchemy stage of science. If you do download it, you'll see that it includes a lot more than the basic algorithm I've described but also that it is still missing a few things I said the universal learner ought to have, like crossover. Nevertheless, let's use the name Alchemy to refer to our candidate universal learner for simplicity.

Alchemy provides a unifier for machine learning in the same way that the Internet provides one for computer networks, the relational model for databases, or the graphical user interface for everyday applications. Of course, even if you use Alchemy with no initial formulas (and you can), that doesn't make it knowledge-free. The choice of formal language, score function, and optimizer implicitly encodes assumptions about the world. So, it's natural to ask whether we can have an even more general learner than Alchemy.

What did evolution assume when it began its long journey from the first bacteria to all the life-forms around today? I think there's a simple assumption from which all else follows: the learner is part of the world. This means that the learner as a physical system obeys the same laws as its environment, whatever they are, and therefore already "knows" them implicitly and is primed to discover them.

HOW TO WRITE A SENTENCE

Stanley Fish, How to Write a Sentence

RELEVANCE:

User interfaces (“UI”) collapse until they disappear. Or, the tools we use evolve such that we interact with them in ways which are natural to us, and which naturally work with our bodies and minds. This is widely recognized.

In the long-term, I believe language will be the UI in the digital world, just as it is the UI we use to communicate in the real world. Our spoken and written language and digital world languages will converge. The details of this argument aren’t that logically complicated.

Logically - breaking things into simple sentences functionally decimates a data set - it breaks it into its smallest pieces. A sentence on a relationship. A sentence on a value. etc. And then once you have the data in this form you can rebuild it in whatever way you want - or search for relationships you may have missed or all the relationships implied (formal concept analysis). The Resource Description Framework (“RDF”) is designed to be represent data sentences (subject, predicate, object) in a form called triples. It is modelled on simple sentences. So the data architecture already exists for the language interface – it’s just still in “code” or computer language terms instead of any vernacular. In that way it feels like computing is still waiting for its Gutenberg moment of code being translated into vernacular languages so that it isn’t a restricted few who can speak to the machines (or God in the case of the translation of the Bible).

Translating the bible was a big deal. But mobilizing language to do digital work will be a bigger deal. The Bible connected people with meaning in their spiritual lives – but by translating code into vernacular, people will end up with the ability to use words to do work on a scale unimaginable just 10 years ago.

RDF isn’t a secret - Bloomberg and others use it... but that’s because part of the architecture that Berners-Lee built in originally is that every triple and every query, every element of an RDF system comes with a URI or it’s own link. It’s designed as a complete linked data framework... in other words Bezos’ famous API everything memo was actually not brilliant but logically trivial - he could have just made the optimal architectural decision and had them build in RDF with those tools and all the links and APIs and other features are free... this took me a few years of working with Ivan and James to really get... a lot of what people are spending billions on (including capital one) is a bunch of band-aids on bad architecture. Now bad architecture is everywhere - visit any city - but good architecture is timeless.

And the basis of RDF and the semantic web is currently viewed as a failed dream. This was all actually designed in 2003 and hasn’t changed much. James added a fourth field in his RDF implementation - so each sentence also includes a time by default... and that means that in James RDF implementation you can also get versioning for free.... good architecture and clean logic is powerful.

The killer app is good sentences as defined by Stanley Fish... good sentences are written for an effect, define a set of relationships and do work... this was always the dream of the semantic web - but there weren’t other pieces of the puzzle... but now because of the “free linking / api” that comes with RDF and the requirement to have APIs, a lot of people are using RDF frameworks and tools... so the underlying stack of “open vocabularies” is there to be used in a tool for building good sentences...

From Stanley Fish (How to Write a Good Sentence and Read One):

.... That underlying form is the sentence itself, and even though it might seem to go without saying, let me say it again: If it is your goal is to write well-constructed sentences, it makes sense to begin by asking the basic and, one would have thought, obvious question “What is a sentence, anyway?” Well, my bottom line can be summarized in two statements: (1) a sentence is an organization of items in the world; and (2) a sentence is a structure of logical relationships. The first statement is insufficiently helpful because it is overbroad: lists, dictionaries, encyclopedias, library card catalogues, bedroom bureaus, and file cabinets also organize items in the world. What is distinctive about the organization a sentence performs? The answer is given in statement (2); it is a logical organization, an assertion that is also overbroad, but one that can be refined and narrowed...

... Let me say again that by “forms” I do not mean parts of speech or any other bit of abstract machinery. I mean structures of logic and rhetoric within which and by means of which meanings—lots of them—can be generated. The logical structures are the ones we have already met: the structure of relationships between actor, actions, and the objects acted upon. The rhetorical structures are structures of argument (that is what rhetoric is, the art of argument); they too are formal—abstract, contentless—but rather than being the forms that make random words into propositions (sentences), they are forms that link propositions together in more complex units. Relationships are also central to their operation, but they are relationships among statements, not the relationships that must be in place if there are to be statements at all.

... I want to pose an apparently simple question: What is it that we do when we make a sentence out of a random collection of words? What is it that we add to those words that causes them to form something we recognize as a sentence? The answer can be given in a single word, and that word is “relationships.”

... But if one understands that a sentence is a structure of logical relationships and that the number of relationships involved is finite, one understands too that there is only one error to worry about, the error of being illogical, and only one rule to follow: make sure that every component of your sentences is related to the other components in a way that is clear and unambiguous (unless ambiguity is what you are aiming at). And how do you do that? Not by learning rules, but by coming to know the limited number of relationships your words, phrases, and clauses can enter into, and becoming alert to those times when the relationships are not established or are unclear: when a phrase just dangles in space, when a connective has nothing to connect to, when a prepositional phrase is in search of a verb to complement, when a pronoun cannot be paired with a noun.

... When it comes to formulating a proposition, form comes first; forms are generative not of specific meanings, but of the very possibility of meaning. Despite the familiar proverb, it's not the thought that counts. Form, form, form, and only form is the road to what the classical theorists called “invention,” the art of coming up with something to say. It follows that familiarizing yourself with form independent of any content you might want to elaborate later is the way to learn how to write a sentence.

... You may be able to write a sentence; you still have to learn how to write a good sentence, and before you can do that, you have to know what a good sentence is. And this is where content, banished from the discussion so far, comes roaring back in. Content, the communication in a thrilling and effective way of ideas and passions, is finally what sentences are for.

... The important word in the previous sentence is “context” and it should be paired with another, “purpose.” People write or speak sentences in order to produce an effect, and the success of a sentence is measured by the degree to which the desired effect has been achieved. That is why the prescriptive advice you often get in books like Strunk and White's *The Elements of Style*—write short sentences, be direct, don't get lost in a maze of piled-up clauses, avoid the passive voice, place yourself in the background, employ figures of speech sparingly—is useful only in relation to some purposes, and unfortunate in relation to others. The first thing to ask when writing a sentence is “What am I trying to do?”

PART 3B: READINGS ON SOCIETY & COMPLEXITY

HUMAN SWARM

Moffett, Mark W.. The Human Swarm.

RELEVANCE:

Anthropology is useful for breaking us out of things we take for granted and rebuilding our assumptions. The selections below from the Human Swarm by Mark Moffett challenge our static or idealistic assumptions about societies and permanence and human progress,

Two key ideas to consider are that (1) societies have life cycles and all societies compete, fragment, evolve and eventually die. Most human societies on the planet today are inevitably declining and dying while a handful are winning and many more are just being born – fragmenting from the dying societies, and (2) humanity is fundamentally made up of many societies – it will never, can never by its natural and psychological characteristics, collapse into a single society. The dream of conquering the world – by an aggressive hegemon or a benevolent cosmopolitan is a dream planted in our psychology – but because we all share it, it become impossible. How can any one conquer the world by force or by persuasion, when everyone has that desire?

The Lifecycle of Societies

“We do not even know how to determine approximately the moment when a society is born and when it dies,” the eminent French scholar Émile Durkheim lamented well over a century ago. Despite the obvious practical and academic importance of the major questions regarding societies—how they are founded, how they develop, and how new societies replace them—no definitive answer has revealed itself since Durkheim’s assertion in 1895. Durkheim pointed out that even the biologists of his time had resolved little about the life and death of societies. Although those lifecycles have by now been well studied for certain groups of organisms, the topic has indeed been largely ignored by the natural sciences as a subject of general inquiry right up to the present day. As for sociologists and historians, their reflex tends to be to treat the birth or dissolution of any society—be it early Egypt or the former Czechoslovakia—as an affair exclusive to its time and place.

Of course, the devil is in the details. Nevertheless, the rise and fall of societies across the natural world indicates that our social groups have been engineered by evolution to come and go, much as the bodies of individual organisms are. This ebb and flow of societies is linked to how members perceive the identities of others. The behavior of animals and the ways in which human beings formulate our identities in changing social environments are closely tied to societal loss and rebuilding. Related to this deep question of identity is the critical topic of trauma and whether it is a sad necessity in the lifecycles of societies.

The dynamics of society genesis and transformation play out in a manner unique to each species, constituting the basic historical chronicle for its kind. That narrative depends on the rules by which the members of the species interact and identify each other, and the resources available at a certain time. Yet a theme emerges: Life requires the constant fulfilling of needs—food, sanctuary, mates. When those needs aren’t met, escalating physical and social stressors spur the decay of societies. More often than not, the troubles will be most severe when a society has overgrown what its environment can support. Even though a large society can overrun smaller neighbors, its swelling population aggravates competition among its own members, with the added burden on every individual to keep track of who is

who—should the societies of a species have opportunities to grow to a size where that is a problem. The members' proficiency at managing relationships and coordinating activities declines. This provokes a shift in allegiance to a subset of the society membership—splinter groups in which everyone fares better.

The separation of subgroups into independent societies is de rigueur in vertebrates. Several lionesses, for example, will leave a pride that's grown too big to feed all its members. Should an aggressive male join a pride, females rearing cubs fathered by another male may depart, splitting the pride to avoid infanticide by the newcomer. Lions in an oversized pride are forced to make a fresh start with those they know and get along with the best, as is typical of species that rely on individual recognition. Such a severing of relationships differs sharply from the phenomenon I've been calling fission: the temporary and casual separation of a society's members that regularly occurs in fission-fusion animals—species such as lions, chimps, and humans where individuals are free to separate and rejoin each other. ***When a society divides, there's little chance of piecing it back together.***

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BREAKING HUMAN SOCIETIES

Division, then, seems to be the usual road to birthing societies in humans and most other vertebrates. Dividing entails clear advantages across species—after all, both sides generally start off populous. The division of a human society is not likely to resemble the stress-free automated undertaking typical in honeybees, however. Groups of disgruntled mutineers don't rise up within insect societies, but people, after all, are typical quarrelsome vertebrates. Enough information exists that we can identify factors that would have caused societies made up of bands of hunter-gatherers to break down and evaluate how those factors may play into the dissolution of settled societies, including the nations of today.

Divisions wouldn't have been pushed along in nomadic hunter-gatherers by local social troubles, such as family disputes or social overstimulation for people who had little privacy. Given how fluidly people could move to bands located elsewhere in the territory, such conflicts could be resolved without the pain of a societal rupture. Tellingly, a dysfunctional band that split up left everyone's sense of who they were—their identity—unaffected. Life went on after people sorted out with whom in their society, they were more comfortable. A societal division would more typically be the outcome, instead, of rifts between broader groups of people, across many bands. It's here that the two-step course of action observed in our primate cousins and other mammals would have played out in our species: factions would have emerged to which members became increasingly attached, followed, often years later, by a severing of relationships.

What led to such factions is the question, since many of the factors that drive their occurrence in other vertebrates appear to matter little in human societies. Still, they are worth considering in turn. Shortages of food, water, mates, or safe havens, significant in divisions in other species, ultimately helped precipitate the fall of many human societies. Even so, resource shortfalls aren't necessary to the process. Because band societies were leaderless, their fate would not likely have hinged on the actions of specific individuals, in the sense described for the Gombe chimpanzees, which favored different contenders for dominant status. In any case, a division couldn't be coerced; people in bands would rise up against the idea. Difficulties in coordinating activities could have presented occasional problems, but there was usually little need for those living far apart in a band society to cooperate in any numbers anyway. Also, because band-dwelling people put little emphasis on blood kin outside their immediate families, weakening ties of biological kinship across expanding populations would not have mattered either. After

all, fictive kin, people one could call father, aunt, and so on, would still be everywhere in the society. However, staying connected with allies might have become more difficult as societies expanded beyond a thousand people, but probably not by much. And because of the use of shared markers large (rituals, language) and small (mannerisms, gestures), the presence of strangers—or leastways individuals who were ever more unfamiliar—would no longer have been an issue the way it could be for chimps or bonobos in a growing community.

Indeed, by sharing markers across their society, people add a twist to the mammalian pattern of separating into factions before dividing—and this shift from individual recognition to anonymous societies would have made a difference in the proximate means by which societies unraveled. The crucial role markers play in societal breakdowns might not be immediately obvious. After all, shared markers have the power to mitigate the tensions between individuals that drive other primates to sever their connections. A recurrent theme of history is that when humans identify with each other strongly, they not just endure but join together and flourish under all but the most brutal conditions. Starve or persecute them, pack them together or spread them apart, yet rest assured: the ties that bind people most tenaciously, beyond those to their immediate families, will be their identification with a society. Where relationships would have permanently eroded in a monkey troop or prairie dog coterie, markers give humans the resilience to stay faithful to other members. In point of fact, given the advantage of a big population in out-competing other societies, we might predict that once our ancestors used markers to distinguish insider from outsider, human societies could swell without bounds. After all, when the markers of a social insect are reliable, no more effort has to be expended to bond to an astronomically huge society than to a tiny one—Argentine ants continue to share an identity with the other ants in their supercolony even after it has spread across continents, obliterating all contenders from the territory.

However, there's a distinction to be made between the fixed molecular brew that defines an ant society and the indescribably diverse markers that bind a human one. Despite the social sturdiness markers give humans, over time the stability they confer cannot be trusted. Our markers aren't set in stone, but are subject to alterations, yielding social class distinctions, regional variations, and more. Even though population size isn't an issue for *Homo sapiens* in the same way it is for chimps, disruptive differences in markers are certain to materialize in our species when many individuals interact seldom—the usual situation when early humans spread out in bands. ***The more changes in a society's markers to accumulate without the members adjusting to them, the more factional that society is likely to be. Eventually—among bands of people, sooner rather than later, as I will illustrate next—every society reaches a breaking point.***

The Dynamic “Us”

The Walbiri people, living in the deserts north and west of Alice Springs, Australia, and famed for their tribal dances and art, were recorded by anthropologists in the 1950s as believing they had existed, with the same religious ties to the land. Yet for them as for all of us, societal stability is an illusion. We suffer from a cultural amnesia, a selective memory that lets us imagine our people's undergirding essence as bedrock, etched out by its valued labels of identity. In truth, markers are in flux. The number of stars on the US flag increased from 13 to 50 without undermining the citizens' connection to their nation—indeed, the multiplication of stars has been a point of pride. Even ways of doing things that seem essential to the life of a society, such as keeping slaves, mutate or die out. What matters in the long run isn't the specific markers we prize at the moment, but that whatever markers are in vogue ensure a society's separation from outsiders—and this means societies are open to transformation. Ever since the

ratcheting of human cultures accelerated in prehistory, societies have been subject to steady enhancement, reinterpretation, and outright reconstruction. The way things are properly done changes with the times without destabilizing a society or eroding the discontinuities between that society and others. When this social elasticity fails, however, we find ourselves on unstable ground.

IMPROVEMENTS AND INNOVATIONS

Over the long term, then, the boundaries between societies trump the markers that define them. Still, the members act to minimize changes affecting what they regard as their society's preeminent qualities. For preliterate peoples many ingredients of identity, from spiritual beliefs to dances, were preserved through the ages with surprising accuracy. Repetition and ritualization served to "comprise almost unbreakable 'codes' for the uninitiated," according to anthropologists at the University of Connecticut, all but guaranteeing that minutiae stuck. You don't have to turn to hunter-gatherer ceremonies and storytelling to be convinced of this point: early Greeks orally transmitted the Iliad and the Odyssey before inventing an alphabet. For those aspects of culture that require this kind of perseverance to learn, humans generally rise to the occasion—call it being mature, since the rites of passage central to most societies signify the taking on of the behaviors and responsibilities of adulthood. Despite this continuity of tradition, however, hunter-gatherers had no immutable standards about how to act, and certainly no means of enforcing their markers across the centuries. Early humans did not live in a vacuum, static and unreactive, even if the archaeological evidence indicates they underwent change at a near imperceptible pace.

Skills imperative for survival stayed most firmly in place, the clearest evidence of this being the consistency of stone tool types over long stretches of time. That didn't stop people from going so far as to reconfigure their livelihood when they had to, even though some stuck obstinately with their ways regardless of the dire consequences. Greenland's Vikings, for example, were apparently tethered to the homeland by sporadic trade and pressured by the church to keep their agricultural customs. A few of their communities may have starved after woeful attempts to raise imported livestock rather than take up the local Inuit practices of hunting whales and seals.

Nevertheless, a willingness to pursue opportunities is a trademark of being human. The Pumé are a superb exemplar of such adaptability. On the savannas of Venezuela, where farming would be hard, Pumé in hunter-gatherer bands feast on lizards, armadillos, and wild plants, while along the rivers, Pumé villages raise gardens of manioc and plantains. Among the Pumé these differences mean little. All of them carry out the same all-night Toja ritual, share the same language, and think of each other as Pumé.

The flexibility of human identities means that differences in subsistence aren't the cornerstone of differentiating our societies the way they are for animal species, with their niches or ecological roles. Yes, societies can opt for distinct approaches to nourish themselves, which can reduce competition. Coastal societies might reasonably rely on fishing, for example, while their neighbors hunt game—and the people could see these choices as part of what defined them. Yet societies occupying the same habitat can eat the same food and make the same tools, the only outward distinctions between them being matters of arbitrary variation in myths or dress.

Not all changes in identity come about on purpose. People enact the traditional ways as best they can, but faulty recall across the generations can alter their behavior inadvertently and at times detrimentally, such as when, as I instanced previously, the once ocean-savvy Tasmanians forgot how to fish. Written records slow but don't stop the loss, even for the things people treasure most. Poor recollection and new

frames of mind can affect perceptions of well-documented events when we envision the past in allegorical terms. Members of preliterate societies, dependent on each other's memories, lived out a game of telephone, in which sentences spoken from person to person grew garbled and unrecognizable—except for them the garbling could extend across everything they do.

Shifts in speech itself evidence this drift most vividly. Even in our age of globalized interchange, a rich diversity of languages and the dialects within them persist. The speech approximating that of the Midwestern states is often pointed to as standard for American English. After listening to this accent on television and radio for generations and adopting bits of it, however, English-speaking populations the world over have kept their own distinctive speech patterns and trajectories of language change. Midwesterners themselves continue to deviate from “their” standard: a shift in vowel sounds around the Great Lakes began in the 1960s, typified by the lengthening of the a sound in certain words so that “trap,” for instance, has come to sound like tryep. Linguists delight in such variations, recording them within all societies from !Kung Bushmen bands to British royalty.

Anything that can serve as a marker of identity, be it language or cuisine or gesture, is constantly being reshaped in this manner. Some shifts arguably are a consequence of the boredom people feel constantly doing things the same way. Novelties can be introduced bottom up, say with the influx of goods and ideas by trade or theft, or are an outcome of the diffusion of trends among the populace. The essayist Louis Menand summarized the evidence: “We get attracted to things that we see other people are attracted to, and we like things more the longer we like them.” Hunter-gatherers did not adopt fads with the regularity that we take for granted, from changing hemlines to the popularity of cell phone apps, nor did their cultures boast the shifting subcultures of modern life. Yet they painted their skin and played music in ways that varied, if only subtly. People no doubt tentatively espoused, and in time came to like more and more, novel social choices.

Complete novelties found their place, too. Imagine a hunter-gatherer coming up with something radical. If the innovation was invaluable, it would spread no matter its source. Otherwise, the reaction an individual had to it would depend on the innovator. People prefer to follow those who match their values, but they may cut a quirky close companion some slack. While hunter-gatherer bands had no clear leaders, novelties could be introduced top down by anyone with a modicum of influence. Such a role model could nudge everyone's choices in new directions, likely by arousing the subconscious impulse to mirror the behavior of admired others.⁹ For example, people could follow the advice of someone with spiritual skills, as this story told by an anthropologist who observed the Andaman Islanders lays bare:

Long-established customs may be altered overnight as the result of a “revelation” by some seer, only to have the new customs overthrown themselves in the course of time by the next “revelation.” I had an experience of this in the Onges when Enagaghe, a well-known seer, announced one day that he had received an order from the spirits about the way in which hunting trophies were to be displayed. No longer were pig jawbones to be impaled one behind the other on poles hung horizontally along the sloping roof of the hut a little above the hunter's bed.

Today much cultural variation emanates from teens, who lead the way in challenging “proper” behavior. Even though their choices can't extend outside a society's permissible wiggle room without a backlash, as time passes, everyone from hippie to skinhead affects the mainstream; counterbalancing the shift are older generations, who hold back the generational changes until their influence wanes. Such battles between young and old seem timeless, but whether they were waged in

hunter-gatherer societies isn't clear: most accounts of people living in bands focus on how children learned traditions rather than bucked or invented them. Kids are kids, though, and defiance appears integral to how a human child grows to be an independent being. As boys and girls are generally, the children of hunter-gatherers would have been open to new experiences, from messing about with hairstyles to exploring uncharted terrain. When chic ideas, methods, or products emerged in the distant past, it seems more likely than not a youth introduced them.

THE BIRTH OF AN OUTGROUP

That the details of how band societies split up are incomplete is to be expected, given that catching one of them at the moment of a division is almost impossible—in fact has never been done. The frequency with which languages are born might be taken as a rough guideline to how long societies endure. Languages drift apart over time much as genetic sequences drift between species in what's known as a molecular clock, and measurements of this linguistic drift suggest they divide on average about every 500 years. Not all societies develop what linguists judge to be their own language, however—some show only a difference in dialect. Hence a half millennium might overestimate the life span of societies. Still, the few estimates of the longevity of band societies suggest the figure isn't far off. Nor is this longevity unique: chimpanzee communities achieve a similar age.

Yet while the final sundering might occur once in approximately five centuries, the lead-up to that moment takes long enough to leave behind plenty of evidence as to what went on. By combining this fragmentary information with what we understand about how human groups split apart generally, I aim for a general picture of what the lifecycle of societies looked like throughout most of human existence.

Among bands the game-of-telephone effect was compounded by an inaccurate knowledge of what went on elsewhere. Variations crept in most prolifically when society members were seldom in contact. In an anonymous society, individuals living far apart might not need to know each other, yet if their markers were to remain the same, they did need to be aware of what distant members were doing. Certain factors could amplify the spatial variation between widely separated people in a band society. On a territorial border, obviously, exposure to outside ideas and goods was greater. Outlying bands with the most contact with foreign societies and the least exposure to other parts of their own society would come to differ from their compatriots elsewhere. Making things even more complicated, different stretches of the border would abut different neighbors. Hence bands faced sharply divergent problems and opportunities from place to place that compounded their differences in identity with others of their society. As an outcome, people at the territorial margins grew marginalized. Factions were born among these outliers.

A human attribute that helps keep a society together in the face of potentially disruptive diversity is that people can be blind to such differences even when confronted by them. Philosopher Ross Poole articulated this perfectly: “What is important is not so much that everyone imagines the same nation, but that they imagine that they imagine the same nation.” Even when discrepancies were noticed—in hunter-gatherers, perhaps during a reunion of bands for a feast—people would have sought to avoid openly expressing dissension around matters of identity if confrontation was likely, much as we do today. Nevertheless, at a certain point, even differences that once might have appeared random and inconsequential could come to be seen as important, and too uncomfortable to ignore. The reunions were times of animated gossip. Among the subjects must have been any oddball behavior, especially when people saw members with whom they had little or no acquaintance do something unexpected. Humans tend to project more negative motivations on strangers, and of course the larger a society is—even a society of hunter-gatherers—the more such poorly known others there are. With minds swayed one way

or another and no leader to compel anyone to conform, the stage would be set for ever more distinct and independent factions to emerge. Call them outgroups in the making.

In the same way that a chimpanzee's fate can be determined by which emerging faction it joins and thereby where it ends up living, and with whom, a person's fate could rest on his or her faction of choice—possibly more than on his or her choice of spouse. In making their decision, however, hunter-gatherers would often have had little to go on, especially since they were unlikely to have any idea what lay ahead. ***Early humans, like animals experiencing a society breakup, would seldom have gone through a division before. They could not fully comprehend their shifting circumstances or accurately envision an ideal outcome. To make matters worse, studies of decision making tell us that people can be remarkably uncertain about their best interests even when a lot is at stake—for example, often they go along with an idea they think is popular but which few believe. As regards to issues of identity, the cart can come before the horse. Whether a choice is bad or good may only be sorted out after people are forced to take a stand on the subject.***

Despite all that, it's more than likely most hunter-gatherers' choice of faction would be predictable. Humans find contentment in being with familiar others—for a band society, namely, those of their band and, probably, other bands nearby. People's connection to a certain general area within the territory—their "home turf"—might be a binder. Even the factions at Gombe formed around chimpanzees with a fondness for the same bit of earth. What made human factions very likely to correspond to where people spent most of their time was that newfangled markers would spread from their point of origin; because of this diffusion, hunter-gatherer factions, like many cultural distinctions today, would be regional.

Different factions didn't have to be adversarial, and certainly not right away. Like the Gombe chimpanzees that at first still socialized, the people remained identified with the original interconnected society. In recent times the Walbiri had four subgroups on good terms, each with its take on dreamings and rituals, while the Comanche fell into three factions with their own minor dialects, dances, and military associations. ***Given that human minds treat societies as species, I imagine we handle this variety within a society much as we do differences within an animal species, for example dog breeds that we (and, as it turns out, the dogs also) see as variations on a single theme; likewise we register members of other factions within our society as versions of our own "kind."***

Troubles mounted when the factions became a source of irritation. "The master defense against accurate social perception and change is always and in every society the tremendous conviction of rightness about any behavior form which exists," psychologist John Dollard averred. Once again it's important to recognize that it was the members themselves who decided what behavior was proper or offensive. Just about any difference could set off this "rightness" response and initiate the process of firming up the factions.

One imagines the division might be precipitated by either an accumulation of many peculiarities or that one especially bothersome difference; in his children's story *The Sneetches*, Dr. Seuss described a world where those who had stars on their bellies refused to associate with those who did not. Among the little changes that could turn into something of belly-star importance, language would be a chief contender, a truth made plain by the Tower of Babel story. Hunter-gatherer societies could develop several regional dialects as they grew. In the 1970s a linguist reported that the Dyirbalnan of Australia who live in the northern part of their territory not only have their own dialect but call themselves by a different name, suggesting a division wasn't far off.

Doubtless the black sheep effect could be a factor, where people grow hostile toward any member whose outrageous displays seem like an affront to conceptions of what their society is. Psychologists study black sheep as one or a few persons: rebellious teen turned criminal, maybe. But what if, as American sociologist Charles Cooley said, “the one who seems to be out of step with the procession is really keeping time to another music”? Such an outlier may not be treated as a black sheep among the company he or she keeps. People weed out some social variants and absorb others into their permissible diversity. What they allow may not match throughout the society, however. Among the likeminded, a person others may see as deviant can thrive, his or her choices emulated by those tuned in to the beat of a different drummer. The eating of the dead by the Ypety Ache, an act steeped in spiritual meaning yet misunderstood and feared by other groups of Ache, may have been a deal-breaker back when those groups parted ways.

Barriers in geography could bring about enough divergences that a society could cleave without any response to deviant behavior. Humans who became utterly isolated, such as the first Aboriginal arrivals to Australia, were cut loose to transform in any direction that suited the place. Elsewhere, topographical features led people to split up despite some contact with their former society-mates. Another group of Ache divided in the 1930s when a highway was constructed down the middle of their territory. Afraid of the foreigners who traveled that route, the hunter-gatherers gave the road a wide berth. With the social ties between them reduced almost to naught, the Yvytyruzu Ache drifted from the Northern Ache until both groups thought of themselves as independent people.

THE ULTIMATE SEVERING

Describing the fall of the Roman Empire, the nineteenth-century American senator Edward Everett wrote that the society had broken “into hostile atoms, whose only movement was that of mutual repulsion.” We can assume that for hunter-gatherers, too, the division would have been set in motion when each faction came to see the other as a hostile atom, its identity intolerable, its actions falling beyond the society’s boundaries of acceptable behavior, if only because of disagreements as to where those boundaries lay. *If a society gives the world meaning by conveying a story about how life should work, what had been one story would have split in two.*

No one has specifically modeled how a society might divide, but research by social psychologist Fabio Sani, much of it with colleague Steve Reicher, on schisms within different sorts of groups suggests factors that might come into play for entire societies, too. The Church of England split up after 1994 when the members who saw the ordination of women as contrary to the true nature of the church went their own way by devising other denominations. In a second example from the same time, the Italian communist party entered the mainstream and took a new name, causing a minority faction to splinter off into a new party that stuck to its principles and retained the party’s original symbols. In both situations the members who felt that the refinements enhanced their identity seized on the changes as mandatory. For them, those changes strengthened the group. The other members, though, interpreted the refinements as harmful deviations from the intangible essence that defined them, threatening their solidarity. *The belief that their identity would be subverted drove a wedge between them and supporters of the change.*

We might expect that most divisions have been prompted, now and in the past, by alterations in identity on both sides. Sani’s studies suggest there can be imbalances, however. The more conservative faction, the one espousing the least transformation—in hunter-gatherer times perhaps the people in the core of the territory who were buffered from outside influence and retained most of the society’s oldest attributes and original name—might have attracted those individuals we speak of today as nationalists,

who abhor change. In the view of such people, dangerous behavior, which might have originated with a single black sheep, would have spread. What had been one estranged person worthy of chastising, or worse, would have metastasized into a faction acting inappropriately, even maliciously, as a unit. Such a faction can elicit a strong commitment from its members because of how likeminded they are. Factions express a narrower set of behavior than present across the whole society, making their solidarity easy to come by. However, the more radical side—occupying the fringes of the territory—would have seen things the same way, and felt just as unified about their own viewpoint. Because they would have seen the changes they promoted as essential to strengthening the same society, for them the dissidents would have been those conservatives refusing the improvement.

When factions emerge around differences they find disagreeable, we can expect little effort by either side to see things from the other's perspective. Psychological studies show that this lessening of attention to the other side manifests even for factions that are just getting their start and so are still trivially different. As a result, communication breaks down and factions strengthen. To individuate the others becomes not merely hard, but destructive. Doing so might give us cause to question our beliefs when we are confident our motives are pure and they are in the wrong—evil, even. From the vantage point of both sides, then, the threat that sundered the society isn't a shortage of food or of places to live, even when the dire straits brought on by such misfortunes may factor into its demise. Rather it is defects in the collective identity that had once tied them together.

We have seen that people conceive of societies as if they were separate species. In fact, the metamorphosis of societies as they splinter off from each other has been labeled “pseudospeciation,” in effect turning what had been one species of human into two. In this splitting, markers serve a role similar to genes. As it does for biologists chronicling the origin of species, the pervasiveness of change makes figuring out who divided from whom in prehistory a challenge. The trials of anthropologists are made all the greater by the readiness with which societies trade, borrow, and steal. ***Darwin's final words about living things in Origin of Species apply equally to societies: “Endless forms most beautiful and most wonderful have been, and are being, evolved.”***

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“The past is a foreign country,” or so the writer L. P. Hartley claimed. The archetypal rules by which humans alter their identity thus ensure such a built-in obsolescence. Even the societies that prevail transmute with time to forms their original founders would reject as insufferably foreign. Given time, a society alters as inevitably as does a biological species that evolves until the living generations wouldn't recognize their predecessors if they met them, and paleontologists deem it worthy of a new species name.

So it was in countless repetition before the dawn of history, societies severing from each other, some wiped out yet even the winners in time changing beyond recognition to divide again. Every sundering would have been a source of heartbreak and agony, carried out for reasons that were momentous at that moment but ultimately forgotten. These divisions have been a part of the rhythm of life as primal as love or personal mortality, yet unfolding over a long crawl of generations beyond our ability to grasp. Win or lose, a society, like its individual members, is fleeting.

How many societies has the cycle birthed? If we can take estimates of the total number of languages to have ever existed as a rough indication, human societies have numbered into the hundreds of thousands. Since not every society has a distinct language, it's conservative to assume a million-plus

societies have come and gone, each made up of men and women confident of their society's significance, abiding existence, and success over its predecessors, of which yours is one.

Thus the sundering and death of societies is certain. In most vertebrate species, to review, the process has nothing to do with the markers that make people relaxed around each other. Chimpanzees have socially learned traditions, yet don't discriminate against those who act strangely. Furthermore there's no reason to think they come up with customs that some members take on and others spurn, bringing about a severing of a community. For humans, however, being part of a society comes with obligations of acting properly and abiding by its rules and expectations. Still, those rules and expectations are a moving target. Ever since humans first employed markers for societies, those societies have broken down around factions of people who take alternative views about their identity. Our psychology orchestrates the change by turning what once was familiar into the foreign.

In charting the birth of the societies of our hunter-gatherer forebears, I've said little that is specific to the rise and fall of nations, which will be one of the subjects of the section to follow. Before getting to this question I must address the social pathways that made such immense societies possible. As it turns out, what drove the success of civilizations was anything but dovishness. ***Societies that turned to conquering each other—in the course of time incorporating ethnicities and races—enter the picture now. The residual effects of their struggles, some ancient, some ongoing, linger in every corner of the earth's surface today.***

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DREAMS OF A UNIVERSAL SOCIETY

What we know about human cognition therefore doesn't bode well for those who think humanity may someday blanket the earth in one borderless population, leaving no person an outsider. Still, even though societies will never dissolve away, perhaps another scenario could play out that would effectively remove them from the picture. With the number of societies generally declining century after century, we might imagine that all the remaining nations of the world will someday drop their boundaries sufficiently to create a cosmopolitan community that is more significant to people than the societies themselves.

Some claim the internationalization of culture (think McDonald's, Mercedes-Benz, Star Wars) and connections (with Facebook bringing people together from Aa, Estonia, to Zu, Afghanistan) are a harbinger of a Berlin Wall-type collapse of state borders. This is false. ***Societies have never freely merged, and that won't change.*** Populations worldwide may devour KFC, Starbucks, and Coca-Cola while screening Hollywood blockbusters—and enjoy sushi, flamenco, French couture, Persian carpets, and Italian cars. They may adopt, and at times be swamped by, cosmopolitan trends. ***But no matter how many exotic influences nations absorb or how many foreign connections they make, such nations do not slip into disarray, and they retain their people's fierce devotion.*** After all, from time immemorial societies have taken what they wanted from the outside world to claim as their own, and have been all the stronger for doing so. Even the Statue of Liberty, the quintessential symbol of the United States, was designed and first erected on French soil by the same Mr. Eiffel who gave Paris its Tower.

Notwithstanding this rigid boundary maintenance, humankind can erect umbrella organizations comprised of many nations. Yet such a universal group would also fail to completely supersede our bonds to societies, as demonstrated by the most binding association of societies in the anthropological record. In northwest Amazonia reside 20 or so tribes, or language groups, known collectively as the

Tukanoans. Each has its own language or dialect, some similar, some mutually unintelligible. The tribes are so cross-connected as to be tied economically, each specializing in goods it exchanges with the others. Between them are what amount to obligatory trading relationships of an unusual sort: marriages within a tribe are improper. “Those who speak the same language with us are our brothers, and we do not marry our sisters,” the people say. Thus a bride marries into another tribe, where she learns the local tongue. In case you imagine the arrangement to be an aberration, similar across-the-board spousal interchanges are recorded for New Guinea.

One explanation for this arrangement is that it reduces inbreeding in very small societies. We see this in many nonhumans as well, such as in chimpanzees, where the females avoid mating with kin by similarly transferring between communities. When populations are minuscule, as those of the Tukanoans have been on occasion in the past, the only marriage options actually could be between brothers and sisters, an incestuous act to which our species has an innate abhorrence.¹⁴ This is a far more likely problem for the Tukanoans than for the countries of today. The psychological aversion to marrying siblings seems to overpower any trepidation the Tukanoans might have had about firmly linking their societies. Their compulsory spousal exchanges have created what to my mind are the tightest alliances ever recorded, currently totaling about 30,000 Amazonian souls. Yet for all that, the Tukanoan tribes remain clear and separate, each confined to specific areas.

Putting aside the unusual situation of the Tukanoans, the failure of alliances to supersede people’s affiliation to a society holds true universally, the United Nations and the European Union included.

These intergovernmental organizations don’t earn people’s emotional commitment because they lack ingredients that make them real for the members. The EU may be the most ambitious attempt at economic integration conceived; yet it will never supplant the nations within it. The members don’t see the EU as an entity worthy of their loyalty the way they do their countries, and for several reasons. First off, its borders are not fixed—in fact, are subject to revision as states enter or go. Additionally, its members have a history of conflict dating from the Middle Ages, and a split already exists from east to west between communist and capitalist cultures. To top all that off, the EU offers no grand foundation story, no venerable symbols or traditions, and there’s little sense anyone would fight and die for Europe as they might for their nation. That makes the EU a political coalition much like the Iroquois Confederacy was, though with less power. Each member country handles issues relating to its citizens’ identity and remains the focus of their self-worth, an outlook that makes membership in the EU secondary and disposable. An analysis of the 2016 Brexit vote shows that those most strongly thinking of themselves as English went against staying with the EU. Those voters saw an economic and peacekeeping tool as a threat to their national identity.

Financial and security issues hold the EU together. The same can be said for Switzerland, a country subject to perennial scrutiny because, as the four languages and complex territoriality of its people attest, its nationhood rests on a detailed social and political alliance between 26 local communities, or cantons. These self-governing settlements act in many respects as miniature nations nestled in a mountain landscape that enhances each one’s autonomy. “Each Canton has its own history, constitution, and flag, and some even have an anthem,” the political scientist Antoine Chollet reports, such that Swiss “citizenship refers to one who can vote and nothing more.” The formation of the Swiss confederation required rewriting accounts of the past to maintain a sense of equality between the cantons. This challenging step was necessary for the cantons to survive over the centuries when they were forced to negotiate their interests with far larger and more powerful neighboring countries.

The EU and Switzerland are regional entities kept together by the perceived need to counter hazards from outsiders, which gives both a reasonable chance of success. A global human union would have no such motivation, and would therefore be far more precarious. One possible means of attaining global unity might be to shift people's perception of who is an outsider—a point Ronald Reagan made often: "I occasionally think how quickly our differences worldwide would vanish," he remarked in an address to the United Nations, "if we were facing an alien threat from outside this world." Popular science fiction tales like *The War of the Worlds* depict all of humankind pulling together against a common enemy. Yet our societies will endure even this. Space aliens wouldn't make nations irrelevant any more than Europeans arriving in Australia caused Aborigines to dispense with their tribes. That would be so regardless of how much the aliens shattered the beliefs people held about their own societies, whose beloved differences would now look trivial by comparison. Moreover, when societies of our species turn to one another, whether for commercial advantage or to defend against aliens, that reliance doesn't diminish the weight they place on their differences. ***The notion of cosmopolitanism, the idea that the people of the world will come to feel a primary connection to the human race, is a pipe dream.***

THE WEIRDEST PEOPLE IN THE WORLD

Henrich, Joseph. The WEIRD People in the World

RELEVANCE:

A lot of things we take for granted are rooted in our cultural and cultural psychology. Jon Henrich's work lays out his hypothesis for where the unusual traits of Western societies may have come from and in the process offers a framework for thinking about cultures, psychology, group competition and cultural evolution.

In the context of our work investing in different cultures, Henrich's work highlights why many of the facts which seem obvious and true to us aren't true at all.

WEIRD Families

The families found in *WEIRD* societies are peculiar, even exotic, from a global and historical perspective. We don't have lineages or large kindreds that stretch out in all directions, entangling us in a web of familial responsibilities. Our identity, sense of self, legal existence, and personal security are not tied to membership in a house or clan, or to our position in a relational network. We limit ourselves to one spouse (at a time), and social norms usually exclude us from marrying relatives, including our cousins, nieces, stepchildren, and in-laws. Instead of arranged marriages, our "love marriages" are usually motivated by mutual affection and compatibility. Ideally, newly married couples set up residence independent of their parents, establishing what anthropologists call neolocal residence. Unlike patrilineal clans or segmentary lineages, relatedness among *WEIRD* people is reckoned bilaterally, by tracking descent equally through both fathers and mothers. Property is individually owned, and bequests are personal decisions. We don't, for example, have claims on the land owned by our brother, and we have no veto on his decision to sell it. Nuclear families form a distinct core in our societies but reside together only until the children marry to form new households. Beyond these small families, our kinship ties are fewer and weaker than those of most other societies. Though kinship does assert itself from time to time, such as when U.S. presidents appoint their children or in-laws to key White House posts, it usually remains subordinate to higher-level political, social, and economic institutions.

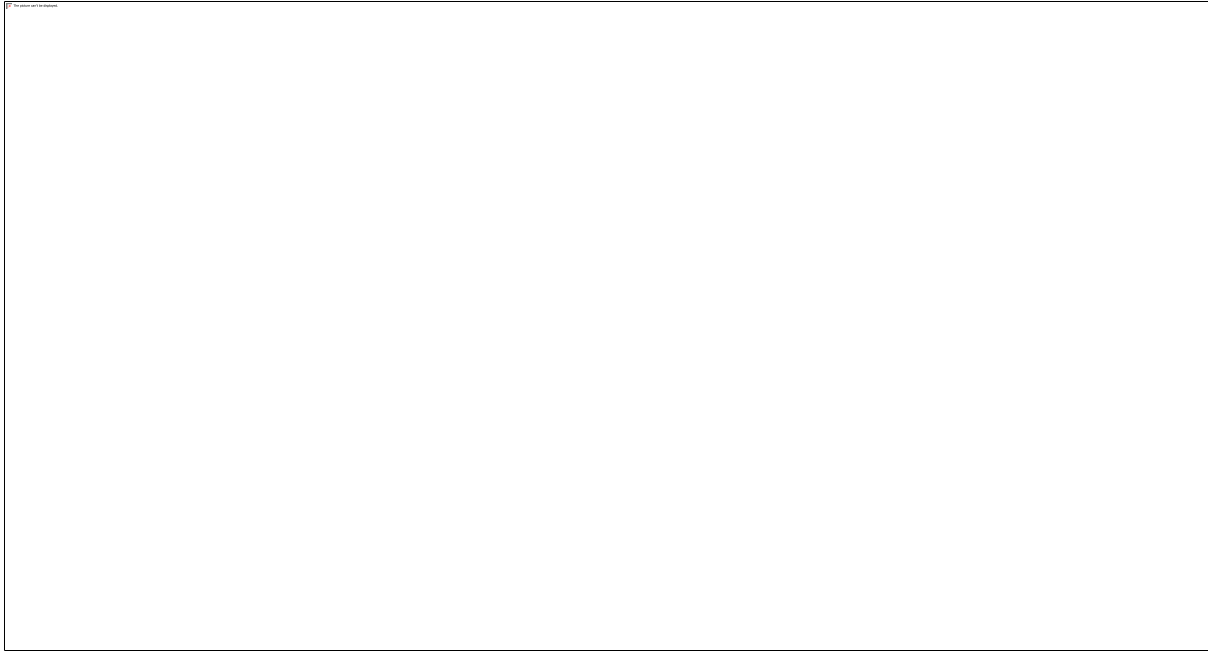
Let's begin by putting some numbers on the kinship patterns described above using the Ethnographic Atlas, an anthropological database of over 1,200 societies (ethnolinguistic groups) that captures life prior to industrialization. Table 5.1 shows five of the kinship traits that characterize *WEIRD* societies: (1) bilateral descent, (2) little or no cousin marriage, (3) monogamous marriage only, (4) nuclear family households, and (5) neolocal residence. The frequencies of these *WEIRD* kinship traits vary from a high of 28 percent, for bilateral descent, to a low of 5 percent, for neolocal residence. This suggests that most societies have long lived in extended family households, permitted polygamous marriage, encouraged cousin marriage, and tracked descent primarily through one parent. Taken separately, each trait is uncommon, but in combination, this package is extremely rare—*WEIRD*.



To see just how rare these patterns are, we can count how many of these kinship traits are possessed by each society in the Atlas. This gives us a score from zero to five that tells us how *WEIRD* a society is in terms of kinship. Figure 5.1 shows the results: over half of the societies in the Atlas (50.2 percent) possess zero of these *WEIRD* kinship traits, and 77 percent possess either zero or only one of these traits. At the other end, fewer than 3 percent of societies possess at least four of them, and only 0.7 percent possess all five traits. Notably, these tiny percentages include a small sampling of European societies, like the Irish and French Canadians of 1930. So, 99.3 percent of societies in this global anthropological database deviate from the *WEIRD* pattern.

The aspects of traditional kinship found in the Atlas open a window not only on the world prior to industrialization but also on the social norms that remain important even today. Consider this question: How many people do you personally know who married their cousins?

If you know none, that's *WEIRD*, since 1 in 10 marriages around the world today is to a cousin or other relative. Based on data from the latter half of the 20th century, Figure 5.2 maps the frequency with which people marry their first or second cousins or other close relatives (uncles, nieces). Remember that second cousins share a pair of great-grandparents. For the sake of simplicity, and because most marriages to relatives involve cousins, I'll refer to this as cousin marriage. At one end of the spectrum, we see that people in the Middle East and Africa marry relatives at least a quarter of the time, though in some places these numbers reach up above 50 percent—so over half of marriages are among relatives. In the middle, countries like India and China have moderate rates of cousin marriage, though it's worth knowing that in China, when the government began promoting “modern” (Western) marriage in the 1950s, it outlawed uncle-niece marriage and, later, first cousin marriage. By contrast, really *WEIRD* countries like the United States, Britain, and the Netherlands have rates of about 0.2 percent, or one-fifth of 1 percent.



So, how did WEIRD kinship become so unusual?

Many assume that the peculiar nature of WEIRD families is a product of the Industrial Revolution, economic prosperity, urbanization, and modern state-level institutions. This is sensible, and certainly appears to be what's happening in much of the world today, through globalization. As non-WEIRD societies have entered the global economy, urbanized, and adopted the formal secular institutions of WEIRD societies (e.g., Western civil codes, constitutions, etc.), their intensive kin-based institutions have often begun to slowly deteriorate, resulting in the spread of WEIRD kinship practices, particularly among educated urbanites. Nevertheless, against this onslaught of global economic and political forces, intensive kin-based institutions have proven themselves to be remarkably resilient.

In Europe, however, the historical order was reversed. First, between about 400 and 1200 CE, ***the intensive kin-based institutions of many European tribal populations were slowly degraded, dismantled, and eventually demolished by the branch of Christianity that evolved into the Roman Catholic Church***—hereinafter the Western Church or just the Church. Then, ***from the ruins of their traditional social structures, people began to form new voluntary associations based on shared interests or beliefs rather than on kinship or tribal affiliations***. In these European regions, societal evolution was blocked from the usual avenues—intensifying kinship—and then shunted down an unlikely side road.

The key point for now is that the dissolution of intensive kin-based institutions and the gradual creation of independent monogamous nuclear families represents the proverbial pebble that started the avalanche to the modern world in Europe. Now let's look at how this pebble was first inadvertently kicked by the Church.

Dissolving the Traditional Family

The roots of WEIRD families can be found in the slowly expanding package of doctrines, prohibitions, and prescriptions that the Church gradually adopted and energetically promoted, starting before the end of the Western Roman Empire. For centuries, during Late Antiquity and well into the Middle Ages, the Church's marriage and family policies were part of a larger cultural evolutionary process in which its

beliefs and practices were competing with many other gods, spirits, rituals, and institutional forms for the hearts, minds, and souls of Europeans. The Church vied against ancestor gods, traditional tribal deities such as Thor and Odin, the old Roman state religion (Jupiter, Mercury, etc.), and various Mediterranean salvation cults (Isis and Mithras, among others), as well as diverse variants of Christianity. These other Christian sects were serious competition and included the Nestorian, Coptic, Syrian, Arian, and Armenian Churches. The Goths, for example, who played a role in the fall of the Western Roman Empire, were not pagans but Arian Christians. Arians, major heretics in the Western Church, held the astonishing view that God the Son (Jesus) was created by God the Father at a particular point in time, making the Son subordinate to the Father.

Today, it's clear that the Western Church won this religious competition hands down. Christianity is the world's largest religion, having captured over 30 percent of the global population. However, 85–90 percent of modern Christians trace their cultural descent through the Roman Catholic Church, back to the Western Church in Rome, and not through the many other branches of Christianity such as the Orthodox or Oriental Churches. This outcome was far from clear when the Western half of the Roman Empire broke up. The Eastern Orthodox Church, as the state religion of the Byzantine Empire, was backed by powerful Roman state institutions and military might. The Nestorian Church, based in cosmopolitan Persia, had established missions in India by 300 CE and in China by 635, many centuries before the Roman Catholic Church would arrive in these places.

Why did the Western Church so dominate in the long run, not only exterminating or commandeering all of Europe's traditional gods and rituals but also outpacing other versions of Christianity?

There are many important elements to this story. For example, Rome's geographic location far from the main political action in Europe may have provided the pope—the bishop of Rome—with some freedom to maneuver. In contrast, other leading bishops, such as those in Constantinople, were under the thumb of the emperors of the Eastern Roman Empire. Similarly, much of northern Europe was relatively technologically backward and illiterate at this point, so the pope's missionaries might have had an easier job of making converts there, for the same reasons that North American missionaries were so successful at making converts in Amazonia during the 20th century. The locals were just more inclined to believe new religious teachings when missionaries showed up with fancy technologies and seemingly miraculous skills, like reading.

Complexities aside, the most important factor in explaining the Church's immense success lies in its extreme package of prohibitions, prescriptions, and preferences surrounding marriage and the family. Despite possessing only tenuous (at best) roots in Christianity's sacred writings, these policies were gradually wrapped in rituals and disseminated wherever possible through a combination of persuasion, ostracism, supernatural threats, and secular punishments. As these practices were slowly internalized by Christians and transmitted to later generations as commonsense social norms, people's lives and psychology were altered in crucial ways. These policies slowly transformed the experience of ordinary individuals by forcing them to adapt to, and reorganize their social habits around, a world without intensive kin-based institutions.

Throughout this process, the Church was competing not only with other religious complexes, but also with intensive kin-based institutions and tribal loyalties. By undermining intensive kinship, the Church's marriage and family policies gradually released individuals from the responsibilities, obligations, and benefits of their clans and houses, creating both more opportunities and greater incentives for people to devote themselves to the Church and, later, to other voluntary organizations. The accidental genius of Western Christianity was in “figuring out” how to dismantle kin-based institutions while at the same time catalyzing its own spread.

UP TO THE STARTING LINE

What did kinship look like among the tribes of Europe before the Church went to work? Unfortunately, we don't have the kind of detailed studies of kinship and marriage that anthropologists have provided for traditional societies in the 20th century. Instead, researchers have cobbled together insights from diverse sources, including (1) early law codes; (2) Church documents, including the many letters exchanged by popes, bishops, and kings; (3) travelers' reports; (4) saints' biographies; (5) Nordic and German sagas; (6) ancient DNA analysis (applied to burials); and (7) kinship terminologies preserved in ancient writings. Broadly speaking, these sources make it clear that prior to the Church's efforts to transform marriage and the family, European tribes had a range of intensive kin-based institutions that looked a lot like what we see elsewhere in the world.

Here are some broad patterns in the tribal populations of pre-Christian Europe:

1. People lived enmeshed in kin-based organizations within tribal groups or networks. Extended family households were part of larger kin-groups (clans, houses, lineages, etc.), some of which were called *sippen* (Germanic) or *septs* (Celtic).
2. Inheritance and post marital residence had patrilineal biases; people often lived in extended patrilineal households, and wives moved to live with their husbands' kinfolk.
3. Many kinship units collectively owned or controlled territory. Even where individual ownership existed, kinfolk often retained inheritance rights such that lands couldn't be sold or otherwise transferred without the consent of relatives.
4. Larger kin-based organizations provided individuals with both their legal and their social identities. Disputes within kin-groups were adjudicated internally, according to custom. Corporate responsibility meant that intentionality sometimes played little role in assigning punishments or levying fines for disputes between kin-groups.
5. Kin-based organizations provided members with protection, insurance, and security. These organizations cared for sick, injured, and poor members, as well as the elderly.
6. Arranged marriages with relatives were customary, as were marriage payments like dowry or bride price (where the groom or his family pays for the bride).
7. Polygynous marriages were common for high-status men. In many communities, men could pair with only one "primary" wife, typically someone of roughly equal social status, but could then add secondary wives, usually of lower social status.

Even at the core of the Roman Empire, intensive kin-based institutions remained central to social, political, and economic life. Roman families were organized around patriarchal patrilineages in which each man saw himself sandwiched in time between his great-grandfather and his great-grandsons. Even when they lived separately and had their own wives and children, adult men remained under the dominion of their fathers. Only male citizens without living fathers had full legal rights, control of family property, and access to tribunals; everyone else had to operate through the patriarch. It was within a father's power to kill his slaves or children. Inheritance rights, incest prohibitions, and exemption from giving legal testimony all extended out, along the patrilineal branches, to the descendants of one's father's father's father. Of course, the empire did develop legal mechanisms for inheritance by testament (wills), but during the pre-Christian period such testaments almost always followed custom and thus mostly came into play when matters were murky or disputes likely. Women remained under the control of either their father or their husband, although over time fathers increasingly retained control of their daughters even after marriage. Marriages were arranged (dowries paid) and adolescent brides went to reside in their husbands' homes (patrilocal residence). Marriage was monogamous by default, but Roman men had few sexual constraints on their behavior save for those that might conflict with other Roman men. Divorce became common in the empire when elites began ending their daughters' marriages in order to remarry them to ever more powerful families. Any

children born during the marriage stayed with their father's family, though the wife's dowry returned with her to her father. As for cousin marriage, the details are complex, and both law and custom changed over time; but in short, cousin marriage in some form was socially acceptable, and some elites did marry their cousins in Roman society (Brutus, St. Melania, and Emperor Constantine's four children). This continued until the Church started its relentless opposition.

THE MONKEY WRENCHES

Around 597 CE, Pope Gregory I—Gregory the Great—dispatched a mission to the Anglo-Saxon Kingdom of Kent in England, where King Æthelberht had married a Frankish Christian princess (eventually St. Bertha) some 17 years earlier. After only a few years, the missionary team had succeeded in converting Æthelberht, had begun to convert the rest of Kent, and had made plans to expand into nearby realms. These papal missionaries, unlike earlier Christian missionaries in places such as Ireland, had definite instructions regarding proper Christian marriage. Apparently, these policies did not go down well with the Anglo-Saxons, since the mission's leader, Augustine (later known as St. Augustine of Canterbury), soon wrote to the pope seeking clarification. Augustine's letter consisted of nine questions, four of which were focused on sex and marriage. Specifically, Augustine queried: (1) How distant must a relative be in order for a Christian marriage to be permissible (second cousins, third cousins, etc.)? (2) Can a man marry his stepmother or his brother's wife? (3) Can two brothers marry two sisters? (4) Can a man receive Communion after a sex dream?

Pope Gregory responded to each question in turn. To the first, after acknowledging its legality under Roman law, Gregory affirmed that first cousins, and certainly not anyone closer, were strictly prohibited from marrying. He then also confirmed that a man could not marry his stepmother or his dead brother's wife (no levirate marriage), even if they weren't related by blood. Although these responses meant that Augustine had his work cut out for him, the reply wasn't all bad news. The pope was fine with a pair of brothers marrying a pair of sisters, as long as the sets of siblings weren't related.

Almost two centuries later, in 786, a papal commission again arrived in England, this time to assess the progress on Christianizing the Anglo-Saxons. Their report indicates that, although many had been baptized, there were serious issues among the faithful surrounding (1) incest (i.e., cousin marriage) and (2) polygyny. To uproot these stubborn customs, the Church promulgated the notion of “illegitimate children,” which stripped the inheritance rights from all children except those born within legal—i.e., Christian—marriages. Prior to this, as in many societies, the children of secondary wives in polygynous unions had possessed some inheritance rights. For royalty, the sons of secondary wives could be “raised up” to succeed their father as king, especially if the king's primary wife was childless. Fighting this, by promoting the notion of “illegitimacy” and endowing itself with the power to determine who is legitimately married, the Church had seized a powerful lever of influence. These interventions made it substantially less appealing for cousins to marry or for women to become secondary wives.

Imposing these policies took centuries, in part because enforcement on the ground was so difficult. Throughout the ninth century, popes and other churchmen continued to complain to Anglo-Saxon kings about incest, polygyny, and illegitimacy, as well as the crime of having sex with nuns. In response, the Church could and sometimes did excommunicate elite men for marrying multiple women. By about 1000 CE, through its relentless efforts, the Church had largely prevailed in reshaping Anglo-Saxon (English) kinship.

The Anglo-Saxon mission is just one example of a much broader effort that reaches back before the fall of the Western Roman Empire (476 CE). Beginning in the fourth century, the Church and the newly Christian Empire began to lay down a series of new policies, again in fits and starts, that gradually corroded the pillars that supported intensive kinship. ***Keep in mind, however, that there is no single coherent program here, at least in the beginning. Things look scattershot and idiosyncratic for***

centuries; but slowly, the successful bits and pieces coalesced into what I'll call the Church's Marriage and Family Program—the MFP.

In undermining the intensive kin-based institutions in Europe, the MFP:

1. Prohibited marriage to blood relatives. These prohibitions were gradually extended to include quite distant relatives, up to sixth cousins. This essentially tabooed marriage or sex between those who shared one or more of their 128 great-great-great-great- grandparents.
2. Prohibited marriage to affinal kin within the circle of tabooed blood relatives. If your husband died, you couldn't then marry his brother, your brother-in-law. In the eyes of the Church, your husband's brother became like your real brother (incest!).
3. Prohibited polygynous marriage, including the taking of secondary wives, as well as the use of sex slaves and publicly supported brothels. Brothels were both legal and common in the Roman Empire, which may explain why Latin has 25 words for "prostitute."
4. Prohibited marriage to non-Christians (unless they have converted).
5. Created spiritual kinship, which established the institution of godparents. This institution provided a means to form new social bonds to care for children. Of course, you couldn't marry or have sex with spiritual kinfolk.
6. Discouraged the adoption of children. Mothers were to care for their own children; if they couldn't, the Church or godparents would provide.
7. Required both the bride and groom to publicly consent ("I do") to marriage. This suppressed arranged marriages and began to hitch marriage more firmly to romantic love.
8. Encouraged, and sometimes required, newly married couples to set up independent households—neolocal residence. The Church also encouraged the use of traditional marriage payments (e.g., dowry) to help fund this new residence.
9. Encouraged the individual ownership of property (land) and inheritance by personal testament. This meant that individuals could personally decide where their property went after their death.

To anyone other than an anthropologist, this might all sound boring or inconsequential, hardly the spark that ignited the blaze of Western civilization or the source of a major shift in people's psychology. However, by looking more closely, we can see how the Church's policies threw a barrage of monkey wrenches into the machinery of intensive kinship while simultaneously catalyzing its own spread. We'll first look at how the Church dismantled traditional marriage, then consider how it sapped the vigor of Europe's clans and kindreds, and finally see how it got rich on death, inheritance, and the afterlife.

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Downstream Transformations

As their intensive kin-based institutions dissolved, medieval Europeans became increasingly free to move, both relationally and residentially. Released from family obligations and inherited interdependence, individuals began to choose their own associates—their friends, spouses, business partners, and even patrons—and construct their own relational networks. Relational freedom spurred residential mobility, as individuals and nuclear families relocated to new lands and growing urban communities. This opened a door to the development and spread of voluntary associations, including new religious organizations as well as novel institutions such as charter towns, professional guilds, and universities. Such developments, underpinned by the psychological changes that I'll highlight over the next seven chapters, ushered in the Urban, Commercial, and Legal Revolutions of the High Middle Ages.

The impact of societal change on the Church itself is interesting, as it represents a kind of feedback between the social and psychological shifts wrought by the MFP and the subsequent evolution of

Catholic institutions. For example, the early monasteries in Anglo-Saxon England, before Pope Gregory's team arrived around 600 CE, tended to be family affairs. The offices of abbot and abbess passed among brothers or from mother to daughter. In Ireland, these practices continued for centuries, as monasteries were run by wealthy Irish clans and passed down as communal property. However, the destruction of kin-based institutions, combined with the eventual delegitimization of priests' children, gradually suppressed the strong intrusion of intensive kinship into the Church's organizations. Many monasteries required aspiring monks to cut their kin ties as a condition of membership, making them choose between the Church and their families. Beginning with Cluny Abbey (910 CE) and accelerating with the emergence of the Cistercian Order (1098 CE), monasteries became less like clan businesses and more like NGOs, with the democratic election of abbots, written charters, and a hierarchical franchise structure that began to balance local independence with centralized authority.

The Church's MFP reshaped the European family in a process that was largely complete 500 years ago. But does this really influence psychology today? Does growing up in less intensive kin-based institutions influence our motivations, perceptions, emotions, thinking styles, and self-concepts in significant ways? Is there a way to trace contemporary psychological variation back to the Church?

Psychological Differences

Families, and the Church Families represent the first institution we encounter upon arriving in the world, and, in most societies until recently, they have provided the central organizing framework for most people's lives. So, it makes sense that they might play a foundational role in shaping our minds and behavior. In Chapter 1, I showed global patterns of psychological variation in domains ranging from individualism, conformity, and guilt to impersonal trust, analytical thinking, and the use of intentionality in moral judgments. Here, I begin to lay down the evidence that this psychological variation arises, in part, as our minds adapt and calibrate to the culturally constructed environments we confront, especially while growing up. We'll examine how intensive kin-based institutions influence people's psychology and, more specifically, how the Church's dismantling of intensive kinship in medieval Europe inadvertently pushed Europeans, and later populations on other continents, toward a *WEIRDer* psychology.

To accomplish this, I'll first lay out two ways of measuring the intensity of kin-based institutions for different ethnolinguistic groups and countries around the world. Then, using a wide-angle lens, I'll show that kinship intensity can explain significant chunks of the cross-national psychological variation highlighted in Chapter 1. You'll see that the weaker a population's traditional kin-based institutions, the *WEIRDer* their psychology is today. Next, I'll use the historical spread of the Church to create a measure of the duration of the Marriage and Family Program for every country in the world. Think of this as a time-release dosage of the MFP, measured in centuries of exposure to the Church. Using this historical dosage measure, we'll first see that the stronger the MFP dosage ingested by a population, the weaker their kin-based institutions. Finally, we'll directly connect these MFP dosages to contemporary psychological differences. Strikingly, the stronger the historical MFP dosage for a population, the *WEIRDer* their psychology is today.

In the next chapter, I'll focus on the psychological variation within Europe, and even within European countries, as well as within China and India. These analyses not only confirm the expected relationships between psychological variation and both kinship intensity and the Church but should also fully immunize you against setting up the dichotomy of West vs. the Rest, or *WEIRD* vs. *non-WEIRD* in your mind. We are not observing fixed or essential differences among peoples but watching an ongoing cultural evolutionary process—influenced by multiple factors—playing out across geography and over centuries. Kinship Intensity and Psychology

Law, Science, and Religion

I think I can see the whole destiny of America contained in the first Puritan who landed on those shores, as that of the whole human race in the first man.

—Alexis de Tocqueville, *Democracy in America* (1835)

Western law, science, democratic government, and European religions have spread around the globe over the last few centuries. Even in countries without much real democracy or broad political representation, autocratic governments now often put on a big show that involves voting, elections, political parties, and campaigns. In places where the rule of law is weak, there are still written statutes and even inspiring constitutions that look like what you find in the United States, the United Kingdom, Germany, and France. Similarly, whenever I've traveled to remote communities, from Amazonia to the Pacific, I've found small Protestant congregations reading Bibles translated into the local languages. Where did these potent formal institutions and pervasive religions come from?

Many believe that these grand institutions, the bequests of Western civilization, represent the products of reason and the rise of rationality. These institutions—the rationalists argue—are what you get once you strip away Church dogma and apply “reason.” This is true even of Protestantism: many believed, and some continue to hold, that (some version of) Protestantism is what you get if you apply reason to the truths expressed in the Bible and toss away corrupt Church traditions. To the contrary, I would like to suggest a central role for the ongoing psychological changes wrought by cultural evolution during the Middle Ages—by the demolition of Europe's kin-based institutions (Chapters 5–8), the expansion of impersonal markets (Chapter 9), the rise of domesticated forms of intergroup competition (Chapter 10), and the growth of a broad, mobile division of labor in urban centers (Chapter 11). The WEIRD psychology that was emerging in fragmented communities across Europe, along with the accompanying changes in social norms, made people in these populations more likely to devise, endorse, and adopt particular kinds of ideas, laws, rules, policies, beliefs, practices, and arguments. Many modern ideas about law, government, science, philosophy, art, and religion that would have been “unthinkable,” aversive, or nonintuitive to people in most complex societies over most of human history began to “fit” the emerging proto-WEIRD psychology in medieval and Early Modern Europe. In many cases, these new ideas, laws, and policies were filtered and selected by relentless intergroup competition between voluntary associations, including among cities, guilds, universities, monasteries, scientific associations, and eventually territorial states.

Mapping the myriad connections and interactions between these societal developments and people's changing perceptions, motivations, worldviews, and decision biases could easily fill volumes. My goal here, however, is more modest. I want to illustrate how an increasingly WEIRD psychology likely midwived a few of the quintessentially Western formal institutions that came to dominate the legal, political, scientific, and religious domains of life in the latter half of the second millennium.

To warm up, let's consider four aspects of WEIRD psychology that likely had broad influences on the formal institutions built in Europe during the second millennium of the Common Era.

1. **Analytic thinking:** To better navigate a world of individuals without dense social interconnections, people increasingly thought about the world more analytically and less holistically/relationally. More analytically oriented thinkers prefer to explain things by assigning individuals, cases, situations, or objects to discrete categories, often associated with specific properties, rather than by focusing on the relationships between individuals, cases, etc. The behavior of individuals or objects can then be analytically explained by their properties or category memberships (e.g., “it's an electron”; “he's an extrovert”). Troubled by contradictions, the more analytically minded seek out higher- or lower-level categories or distinctions to “resolve” them. By contrast, holistically oriented thinkers either don't see contradictions or embrace them. In Europe, analytical approaches

gradually came to be thought of as superior to more holistic approaches. That is, they became normatively correct and highly valued.

2. **Internal attributions:** As the key substrates of social life shifted from relationships to individuals, thinkers increasingly highlighted the relevance of individuals' internal attributes. This included stable traits like dispositions, preferences, and personalities as well as mental states like beliefs and intentions. Soon lawyers and theologians even began to imagine that individuals had "rights."
3. **Independence and nonconformity:** Spurred by incentives to cultivate their own uniqueness, people's reverence for venerable traditions, ancient wisdom, and wise elders ebbed away. For good evolutionary reasons, humans everywhere tend to conform to peers, defer to their seniors, and follow enduring traditions; but, the incentives of a society with weak kin ties and impersonal markets pushed hard against this, favoring individualism, independence, and nonconformity, not to mention overconfidence and self-promotion.
4. **Impersonal prosociality:** As life was increasingly governed by impersonal norms for dealing with nonrelations or strangers, people came to prefer impartial rules and impersonal laws that applied to those in their groups or communities (their cities, guilds, monasteries, etc.) independent of social relationships, tribal identity, or social class. Of course, we shouldn't confuse these inchoate inklings with the full-blown liberal principles of rights, equality, or impartiality in the modern world.

These and related aspects of psychology were taking hold in small but influential populations scattered across western Europe by the High Middle Ages.

The Dark Matter of History

Humans are an intensely cultural species. For over a million years, the products of cumulative cultural evolution—our complex technologies, languages, and institutions—have driven our genetic evolution, shaping not only our digestive systems, teeth, feet, and shoulders but also our brains and psychology. The centrality of cultural change meant that, generation after generation, young learners had to adapt and calibrate their minds and bodies to an ever-shifting landscape of sharing norms, food taboos, gender roles, technical demands (e.g., projectile weapons and underwater foraging), and grammatical conventions. At the same time, cultural evolution has favored an arsenal of mind hacks, including rituals, socialization practices (e.g., bedtime stories), and games, that mold people's psychology in ways that allow them to navigate their culturally constructed worlds more effectively. ***The consequence of this culture-gene coevolutionary process is that to understand people's psychology we must consider not only our genetic inheritance but also how our minds have adapted ontogenetically and culturally to the local technologies and institutions—present or even a few generations past. Thus, we should expect a rich array of diverse cultural psychologies to go along with disparate societies. The cultural evolution of psychology is the dark matter that flows behind the scenes throughout history.***

To build up an understanding of the interaction between institutions and psychology, I began by drilling down to focus on our species' oldest and most fundamental institutions—those related to kinship and religion. Given their anchoring in our species' evolved psychology, it's not surprising that kin-based institutions culturally evolved as the primary way that mobile hunter-gatherers organize themselves and stretch out their cooperative networks. However, after the origins of sedentary agriculture, the need to control territory in the face of fierce intergroup competition drove the intensification of kin-based institutions, leading to the norm clusters that organize clans, cousin marriage, corporate ownership, patrilocal residence, segmentary lineages, and ancestor worship. As societies scaled up, the most successful political institutions remained heavily entwined with kinship. Even after the emergence of premodern states, with their military forces and tax-collecting bureaucracies, kin-based institutions still dominated life among both the elite and the lower strata. All of this means that some of the contemporary psychological variation we observe can be traced, through a variety of pathways, back to

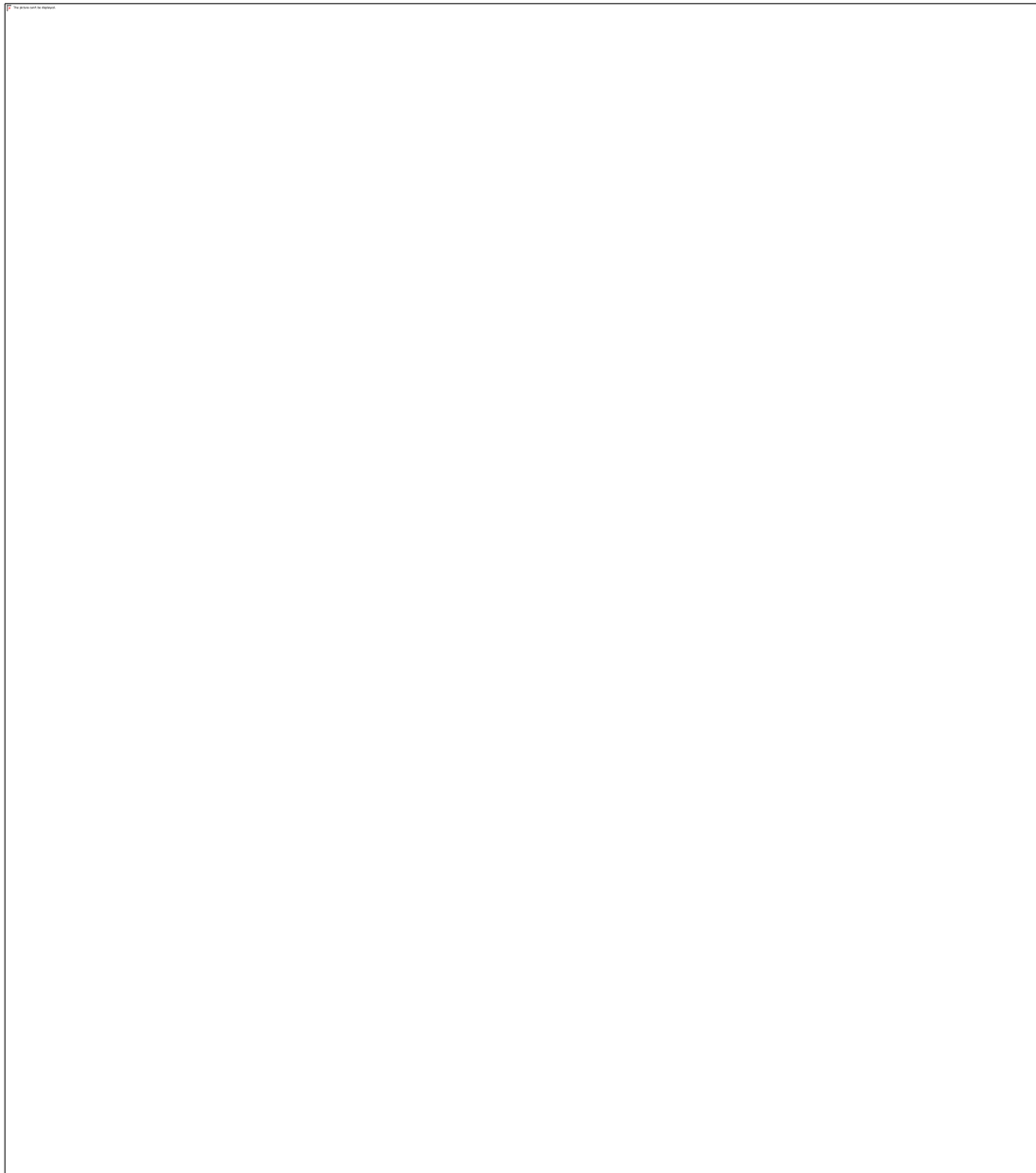
the ecological, climatic, and biogeographic factors that influenced the emergence of kin-based institutions and states.

Alongside kinship, both religions and rituals have also been culturally evolving over eons to harness aspects of our species' evolved psychology in ways that expand the social sphere and foster cooperation in larger groups. However, the emergence of diverse universalizing religions with the development of cosmopolitan empires provided new opportunities for cultural evolution to “experiment with” a variety of divine prescriptions and prohibitions related to marriage and the family. Some universalizing faiths endorsed marriage to close relatives (Zoroastrianism), while other faiths forbade previously common unions, such as those between cousins or affines. Similarly, some religions permitted men to marry as many wives as they wanted to (or could), while others limited them to only four and required equality among the wives (Islam). By the outset of the first millennia of the Common Era, the Roman Empire was a bubbling cauldron of religious competition that included the old Roman state religion, Judaism, Zoroastrianism, Mithraism, a potpourri of Christian faiths, and a panoply of local religions. From this cauldron, one strain of Christianity stumbled upon a peculiar set of taboos, prohibitions, and prescriptions regarding marriage and the family that eventually crystallized into the Church's Marriage and Family Program.

These prohibitions and prescriptions, which were infused into Christianizing populations under the Western Church over many centuries (arrow A in Figure 14.1), altered people's social lives (arrow B) and psychology (arrow C) by demolishing intensive kin-based organizations. These changes would have favored a psychology that was more individualistic, analytically oriented, guilt-ridden, and intention-focused (in judging others) but less bound by tradition, elder authority, and general conformity. The elimination of polygynous marriage and the tightening of constraints on male sexuality may also have inhibited male status-seeking and competition, which would have suppressed zero-sum thinking, impatience, and risk-seeking.

The social and psychological changes driven by the breakdown of intensive kinship opened the door to rising urbanization, expanding impersonal markets (arrows D and E), and competing voluntary associations like charter towns, guilds, and universities. By facilitating and enforcing impersonal interactions in various ways, urban centers and commercial markets further stimulated impersonal prosociality and impartial rule-following while incentivizing personal attributes like patience, positive-sum thinking, self-regulation, and time thrift (arrows F and G). By producing a growing division of labor, within which an expanding class of individuals could select their occupations and social niches, these new social environments may have fostered more differentiated personality profiles—expanding eventually into the WEIRD-5—and strengthened people's inclinations to think in dispositional ways about other individuals and groups.

The quiet fermentation of these psychological and social changes influenced the formation of governments, laws, religious faiths, and economic institutions in the latter half of the Middle Ages and beyond. Creating laws, for example, that focus on individuals and their properties (“rights”) just makes sense if one lives in communities with weak family ties, substantial relational mobility, and a developing individualistic psychology that parses the world in dispositional ways (“she's trustworthy”). By contrast, if one lives in a community where relational ties are central and people are primarily judged by their social and family connections, building law and government around individual rights doesn't seem like common sense. It doesn't “fit” people's psychological inclinations.

FIGURE 14.1. Outline of the main processes described in this book.

In the last chapter, I examined how cultural evolution—in the wake of the transformation of European kinship and rising urbanization—expanded Christendom’s collective brain and altered key aspects of people’s psychology in ways that would have catalyzed innovation, suppressed fertility, and propelled economic growth. We saw how these ongoing social and psychological changes gradually opened the flow of ideas, beliefs, practices, and techniques within a sprawling network of interconnected minds, each of which was motivated to produce new insights and challenge old assumptions. This occurred in myriad ways, including through the diffusion of literacy (due to Protestantism), the proliferation of scientific societies, and the accelerating flow of artisans, scholars, and merchants through far-flung European cities and towns. This expanding collective brain sparked the Enlightenment, drove the Industrial Revolution, and continues to propel economic growth around the globe.

Now, with this summary in mind, let's return to the three core questions that I posed back in Chapter 1:

1. *How can we explain global psychological variation, and specifically the variation highlighted throughout this book (Table 1.1)?*

Answer: To explain psychological variation broadly, we need to examine how history has unfolded in different ways in different places and consider the coevolution of people's minds with different institutions, technologies, and languages. Targeting the psychological patterns highlighted in Table 1.1, I focused on the evolution of institutions related to (1) intensive kinship, (2) impersonal markets and urbanization, (3) competition among voluntary associations, and (4) complex divisions of labor with substantial individual mobility.

2. *Why are WEIRD societies particularly unusual, so often occupying the extreme ends of global distributions of psychology and behavior?*

Answer: The sect of Christianity that evolved into the Roman Catholic Church stumbled onto a collection of marriage and family policies that demolished Europe's intensive kin-based institutions. This grassroots transformation of social life propelled these populations down a previously inaccessible pathway of societal evolution and opened the door to the rise of voluntary associations, impersonal markets, free cities, and so on.

3. *What role did these differences play in the Industrial Revolution and in the global expansion of Europe during the last few centuries?*

Answer: By the High Middle Ages, the social and psychological shifts induced by the Church had made some European communities susceptible to notions of individual rights, personal accountability, abstract principles, universal laws, and the centrality of mental states. This fertilized the psychological soil for the growth of representative governments, constitutional legitimacy, and individualistic religious faiths as well as the rise of Western law and science. These changes accelerated the ongoing social and psychological transformations that energized innovation and economic growth.

COMPLEXITY OF COOPERATION

Axelrod, Robert. The Complexity of Cooperation: Agent-Based Models of Competition and Collaboration: Agent-Based Models of Competition and Collaboration (Princeton Studies in Complexity)

RELEVANCE:

Axelrod is one of the first practitioners of agent-based modelling and his work has illustrated the stability of cooperation and collaboration in iterative games. The fact that order emerges from complex interactions is best captured in some of Kauffman's work, which Axelrod included in his class curriculum at Michigan. The excerpt below is useful as a discussion of agent-based modelling, and modelling in general.

When we think through future scenarios, we are implicitly running a simple agent-based model in our minds. My interest in agent-based modelling is in the potential for it to be a tool which we could use to test some of our simple assumptions and learn from interaction patterns that simulations could show us.

INTRODUCTION

The title of this book illustrates the dual purposes of the volume. One meaning of “The Complexity of Cooperation” refers to the addition of complexity to the most common framework for studying cooperation, namely the two-person iterated Prisoner’s Dilemma. Adding complexity to that framework allows the exploration of many interesting and important features of competition and collaboration that are beyond the reach of the Prisoner’s Dilemma paradigm.

The second meaning of “The Complexity of Cooperation” refers to the use of concepts and techniques that have come to be called complexity theory. Complexity theory involves the study of many actors and their interactions. The actors may be atoms, fish, people, organizations, or nations. Their interactions may consist of attraction, combat, mating, communication, trade, partnership, or rivalry. Because the study of large numbers of actors with changing patterns of interactions often gets too difficult for a mathematical solution, a primary research tool of complexity theory is computer simulation. The trick is to specify how the agents interact, and then observe properties that occur at the level of the whole society. For example, with given rules about actors and their interactions, do the actors tend to align into two competing groups? Do particular strategies dominate the population? Do clear patterns of behavior develop?

The simulation of agents and their interactions is known by several names, including agent-based modeling, bottom-up modeling, and artificial social systems. Whatever name is used, the purpose of agent-based modeling is to understand properties of complex social systems through the analysis of simulations. This method of doing science can be contrasted with the two standard methods of induction and deduction. Induction is the discovery of patterns in empirical data. For example, in the social sciences induction is widely used in the analysis of opinion surveys and macroeconomic data. Deduction, on the other hand, involves specifying a set of axioms and proving consequences that can be derived from those assumptions. The discovery of equilibrium results in game theory using rational-choice axioms is a good example of deduction.

Agent-based modeling is a third way of doing science. Like deduction, it starts with a set of explicit assumptions. But unlike deduction, it does not prove theorems. Instead, an agent-based model generates simulated data that can be analyzed inductively. Unlike typical induction, however, the simulated data come from a rigorously specified set of rules rather than direct measurement of the real world. Whereas the purpose of induction is to find patterns in data and that of deduction is to find consequences of assumptions, the purpose of agent-based modeling is to aid intuition. Agent-based modeling is a way of doing thought experiments. Although the assumptions may be simple, the consequences may not be at all obvious. Numerous examples appear throughout this volume of locally interacting agents producing large-scale effects. The large-scale effects of locally interacting agents are called “emergent properties” of the system. Emergent properties are often surprising because it can be hard to anticipate the full consequences of even simple forms of interaction.

There are some models, however, in which emergent properties can be formally deduced. Good examples include the neoclassical economic models in which rational agents operating under powerful assumptions about the availability of information and the capability of optimizing can achieve an efficient reallocation of resources among themselves through costless trading. But when the agents use adaptive rather than optimizing strategies, deducing the consequences is often impossible; simulation becomes necessary.

Throughout the social sciences today, the dominant form of modeling is based upon the rational-choice paradigm. Game theory, in particular, is typically based upon the assumption of rational choice. In my view, the reason for the dominance of the rational-choice approach is not that scholars think it is realistic. Nor is game theory used solely because it offers good advice to a decision maker, because its unrealistic assumptions undermine much of its value as a basis for advice. The real advantage of the rational-choice assumption is that it often allows deduction.

The main alternative to the assumption of rational choice is some form of adaptive behavior. The adaptation may be at the individual level through learning, or it may be at the population level through differential survival and reproduction of the more successful individuals. Either way, the consequences of adaptive processes are often very hard to deduce when there are many interacting agents following rules that have nonlinear effects. Thus the simulation of an agent-based model is often the only viable way to study populations of agents who are adaptive rather than fully rational.

Although agent-based modeling employs simulation, it does not aim to provide an accurate representation of a particular empirical application. ***Instead, the goal of agent-based modeling is to enrich our understanding of fundamental processes that may appear in a variety of applications. This requires adhering to the KISS principle, which stands for the army slogan “keep it simple, stupid.” The KISS principle is vital because of the character of the research community. Both the researcher and the audience have limited cognitive ability. When a surprising result occurs, it is very helpful to be confident that we can understand everything that went into the model.*** Although the topic being investigated may be complicated, the assumptions underlying the agent-based model should be simple. The complexity of agent-based modeling should be in the simulated results, not in the assumptions of the model.

Of course, there are many other uses of computer simulation in which the faithful reproduction of a particular setting is important. A simulation of the economy aimed at predicting interest rates three months into the future needs to be as accurate as possible. For this purpose, the assumptions that go into

the model may need to be quite complicated. Likewise, if a simulation is used to train the crew of a supertanker or to develop tactics for a new fighter aircraft, accuracy is important and simplicity of the model is not. ***But if the goal is to deepen our understanding of some fundamental process, then simplicity of the assumptions is important, and realistic representation of all the details of a particular setting is not.***

My earlier work on the Prisoner's Dilemma (Axelrod 1984) illustrates this theme. My main motivation for learning about effective strategies was to find out how cooperation could be promoted in international politics, especially between the East and the West during the Cold War. As it happened, my tournament approach and the evolutionary analysis that grew out of it suggested applications of which I had not even dreamed. For example, controlled experiments show that stickleback fish use the TIT FOR TAT strategy to achieve cooperation based upon reciprocity (Milinski 1987).

At a political science convention, a colleague came up to me and said she really appreciated my work and found it helpful for her divorce. She explained that my book showed her that she had been a sucker during her marriage, always giving in to her husband. I asked whether the book helped save her marriage. "No," she replied. "I didn't want to save my marriage. But it certainly helped with the divorce settlement. I started to play TIT FOR TAT, and once he learned that I couldn't be pushed around, I got a much better deal."

The fact that a single model, in this case the Prisoner's Dilemma, can be useful in understanding the dynamics between foraging fish and between divorcing people is not due to the accuracy of the model in representing the details of either situation. Instead, it is due to the fact that an extremely simple model captures a fundamental feature of many interactions. What the Prisoner's Dilemma captures so well is the tension between the advantages of selfishness in the short run versus the need to elicit cooperation from the other player to be successful in the longer run. The very simplicity of the Prisoner's Dilemma is highly valuable in helping us to discover and appreciate the deep consequences of the fundamental processes involved in dealing with this tension.

A moral of the story is that models that aim to explore fundamental processes should be judged by their fruitfulness, not by their accuracy. For this purpose, realistic representation of many details is unnecessary and even counterproductive. The models presented in the volume follow this same logic of simplicity. The intention is to explore fundamental social processes. Although a particular application may have motivated a given model, the primary aim is to undertake the exploration in a manner so general that many possible settings could be illuminated.

Taken as a whole, this book presents a set of studies that are unified in three ways. First, they all deal with problems and opportunities of cooperation in a more or less competitive environment. Second, they all employ models that use adaptive rather than rational agents. Although people may try to be rational, they can rarely meet the requirements of information or foresight that rational models impose (Simon 1955; March 1978). Third, they all use computer simulation to study the emergent properties of the interactions among the agents. Thus they are all agent-based models. The simulation is necessary because the interactions of adaptive agents typically lead to nonlinear effects that are not amenable to the deductive tools of formal mathematics.

....

For a long time, I had been eager to move beyond the two-person format of the basic Prisoner's Dilemma. I especially wanted to find out how cooperation could emerge when many people interacted with each other in groups rather than in pairs. It was well known that the straightforward extension of the Prisoner's Dilemma to an n-person version will not sustain cooperation very well because the players have no way of focusing their punishment on someone in the group who has failed to cooperate. Nevertheless, social norms do emerge and are often quite powerful means of sustaining cooperation. So I developed a "norms game" that allowed players to punish individuals who do not cooperate. It turned out that another twist was needed lest all the cooperators be tempted to let someone else be the one to bear the costs of disciplining the noncooperators. This led to a wide-ranging study of the mechanisms for promoting norms (Chapter 3).

Another form of cooperation occurs when people organize themselves into groups to compete with each other. This is clearly an example of collaboration in the interests of competition. It takes place in many forms, including alliances among nations, strategic partnerships among businesses, and coalitions among political parties in parliamentary democracies. Having worked on the problem of coalition formation in Italy as part of my dissertation in the late 1960's, I was struck by how political parties wanted to work with others who were similar to themselves (Axelrod 1970). Two decades later, I returned to this theme of choosing sides based upon affinity rather than strategic advantage. Working with a graduate student, Scott Bennett, I developed a model for how players choose sides. We found that the model actually did a good job of accounting for how European countries were aligned in World War II (Chapter 4).

The same model even worked well in accounting for how computer companies took sides in the competition to develop standards for the UNIX operating system (Chapter 5). This was work done with Scott Bennett and three collaborators from the Michigan Business School: Will Mitchell, Robert E. Thomas, and Erhard Bruderer.

An even deeper problem is how independent actors sometimes cooperate to such an extent that they give up most of their independence. The result is a new level of organization that behaves as an independent actor in its own right. Multicellular organisms evolved this way, and so have many large business organizations. My approach to analyzing how new levels of political actors can arise uses a model of war, threats, and commitments. The agent-based model and its results are provided in Chapter 6.

Whereas the model in Chapter 6 attributes the emergence of new actors to the dynamics of coping with conflict, I also wanted to study an even more fundamental question: how people become more alike so that they find it easier to work together in the first place. This led to a study of the process of social influence and the emergence of shared culture. Once again, the transformations of the post-Cold War environment helped emphasize the importance of returning to some very fundamental issues about the basis for cooperation within as well as between societies. The resulting model of social influence and cultural change is given in Chapter 7.

Two appendixes provide supporting material about agent-based modeling....

Associated with this volume is an Internet site:

<http://www-personal.umich.edu/~axe/ComplexCoop.html>

The site includes the source code and documentation for most of the models in this book. It also provides links to many topics related to complexity theory, agent-based modeling, and cooperation.

STUART A. KAUFFMAN

RELEVANCE:

You just need to read and understand this. Then re-read it. Kauffman takes a long time to understand, but he speaks very simply. I think I am just beginning to understand it. Many of our assumptions about unpredictability and uncertainty are explained by Kauffman. His vision of creativity is also what I believe is true, and the best way of seeing the world. Finally, his explanation that the world is not a machine is a needed refutation of all the Victorians and other determinists and reductivists in finance like the gentleman who runs the big hedge fund and write books on rules of the world and the economy as machine.

There are numerous useful analytical structures within Kauffman's descriptions – from the inevitable advance of forms or complexity through time to Darwinian pre-adaptations to the adjacent possible. Pay attention for useful concepts.

Kauffman, Stuart A. Reinventing the Sacred.

THE CYCLE OF WORK

An organized being is then not a mere machine, for that has merely moving power, but it possesses in itself formative power of a self-propagating kind, which it communicates to its materials though they have it not of themselves; it organizes them, in fact, and this cannot be explained by the mere mechanical faculty of motion. —Immanuel Kant

This quotation, from Kant's Critique of Judgment, points to the subject matter of this chapter: propagating organization of processes. As we will see, the phenomena are real and right before our eyes. Nevertheless, we appear to lack an adequate theory of what we are looking at. It is said that the Native Americans witnessing the arrival of Columbus did not "see" his ships, for they had no categories of thought in which to place ships. Such is our situation now. Sometimes science advances by seeing things anew. The evolution of life has yielded astonishing organization that links complex, spontaneous and nonspontaneous processes of enormous intricacy. Part of our natural astonishment at life stems from this outrageous, complex organization of process....

With the wide flowering of mathematics, physics, chemistry, biochemistry, biophysics, and biology, you would think that we would already have the scientific concepts in hand to understand the life that lies before us, but we do not. The manifold web of organized processes we call the biosphere largely reflects natural selection and the evolution of this specific biosphere and hence, while not contravening physical laws, is not reducible to physics. In this chapter I explore a web of concepts, hinted at by Kant, that concern propagating organization of process. From the larger perspective of this book that seeks to move beyond reductionism to a scientific worldview of emergence and ceaseless creativity, the evolutionary emergence of overwhelming organization of process is the most visible example of a phenomenon that is in no way accessible to reductionism.

For the nonscientific reader, the present chapter is an example of the struggle in science to formulate questions that are still unclear. The attempt to "find" the needed concepts for propagating organization is an example of the requirement for imagination and even wonder in science. There appears to be no

algorithm, or effective procedure, to find the concepts that we need. ***We do in fact live our lives forward, often without knowing, which requires all our evolved humanity, not just “knowledge.” ...***

The rise of molecular biology in the twentieth century brought to the study of living systems a focus on “information,” a concept that manages to be both restrictive and unclear. We have been enjoined to ignore energy, work, and other properties of cells and focus on the “information master molecules of life,” the genes, and the processes that genes control: transcription of RNA, translation to proteins, the differentiation of cells as they develop from the fertilized egg, and so forth. ***But “information,” particularly the one-way flow of information from DNA to RNA to proteins that remains molecular biology’s central dogma, is an inadequate framework for describing much of what we see in the cell.***

Cell biologists, for instance, have been struggling to understand the behavior of organelles called mitotic spindles, where molecular motors attach to one or more molecules called microtubules, themselves able to grow and shrink. The motors race back and forth holding onto the microtubules, and upon reaching the ends, gather the microtubules into a variety of complex mechanical structures that, among other things, form the mitotic spindles that move daughter chromosomes in sets to the two daughter dividing cells such that each gets a full complement of chromosomes. Here it is clear that mechanical work is done, for the chromosomes are moved bodily to the appropriate daughter cells. *More broadly, cells do some combination of mechanical, chemical, electrochemical, and other work, and work cycles in a web of propagating organization of processes that often link spontaneous and nonspontaneous processes. But the sorts of questions asked by molecular biology, with its emphasis on how one molecule “instructs” another, are of little help in understanding this web of propagating organization of processes and these work cycles....*

THE NONERGODIC UNIVERSE

I now wish to discuss what will turn out, among other things, to be the physical foundation for much of the evolution of diversity in the biosphere, of economic growth and even human history. ***The present chapter lays the foundations for the two that follow, where I will argue that the evolution of the biosphere and economy and history are all ceaselessly creative in ways that we typically cannot foretell.*** Thus, not only are we beyond reductionism into a scientific worldview that includes “emergence,” we will see that we are coming to a scientific worldview that takes us into subjects normally thought to lie outside the realm of science.

At levels of complexity above atoms, the universe is on a pathway, or trajectory, that will never repeat.

For example, in the evolution of the biosphere, from molecules to species, what arises is almost always unique in the history of the universe. *Using the physicist’s technical term, the evolution of molecules and species in the biosphere is vastly nonrepeating, or nonergodic.*

Now consider all possible proteins with a length of two hundred amino acids. Typical proteins inside you are about three hundred amino acids long, some several thousand amino acids long, so two hundred is conservative.

If there are twenty types of amino acids in biological proteins, how many proteins like this can there be? This is simple. At each position there are twenty choices, so the total number of possible proteins with a length of two hundred is twenty raised to the two hundredth power, or about 10^{260} . Could all these proteins arise at least once in the history of our universe since the big bang? The answer is an overwhelming no. Only a miniscule fraction of these proteins can have been constructed in the history of this universe. Thus, as we will see, the proteins that come into existence in the evolution of the biosphere are typically unique—to use Luigi Luisi’s phrase, they are never-before-born proteins.

Consider this simple calculation. There are about 10^{80} particles in the known universe. Of course, these are widely separated from one another, but let’s ignore that inconvenient fact. Suppose all these particles were busy colliding and building nothing but unique, never-before-born proteins of length two hundred

ever since the big bang. The universe is 10^{17} seconds old. Suppose it took 10^{-15} seconds, a femtosecond, a very fast chemical time scale, to make each protein.

It turns out that it would require 10^{67} repetitions of the entire history of the universe to create all possible proteins of length two hundred *just once*.

The fastest time scale in the universe is the Planck time scale, 10^{-43} seconds. If proteins of length two hundred were being built on the Planck time scale, it would not change the essential results: now it would take 10^{39} repetitions of the entire history of the universe to create all possible proteins length two hundred once.

This is quite profound. There is no way this universe could have created all possible proteins of length two hundred. The set that exists is a tiny subset of the possible. When a new protein is created, say via a mutation, it really is what Luisi happily calls a never-before-born protein. That is, as the biosphere advances into its chemical adjacent possible, it is persistently making unique molecules. ***Thus, as mutations occur so novel proteins are made, or new organic molecules are synthesized in evolution, the biosphere is persistently advancing into its adjacent possible. The adjacent possible is real. We are invading it much of the time.***

The same uniqueness occurs at the level of the evolution of species, the human economy, of human history, and human culture. To put the matter simply: we will never explore all the possibilities. History enters when the space of the possible is vastly larger than the space of the actual. At these levels of complexity, the evolution of the universe is vastly nonergodic...

It is interesting that these features may apply not only to the biosphere but to other features of the abiotic universe. For example, in chapter 7, I mentioned the giant, cold molecular clouds, the birthplaces of stars, that drift through many galaxies. Complex chemistry occurs in these clouds. For all we know, they are expanding forever into their chemically adjacent possible. Grains that form in the clouds can serve as surface catalysts for further chemistry, new molecules may bind to the grains' catalytic surfaces and modify the subsequent chemical reactions, hence the subsequent flow into the chemical adjacent possible. These grains aggregate up to the size of planetesimals. Almost certainly, they are unique on a grain level, so the giant clouds are almost certainly nonergodic.

Thus, at any level of complexity above atoms, from complex molecules upward, the universe cannot make all possible things in many repetitions of its own lifetime. While we are used to thinking of the second law as "the" arrow of time, this, too—the fact that the universe is nonergodic—is an arrow of time with respect to the time-reversibility of the fundamental laws of physics. I explore this issue further just below concerning "underpopulated vast chemical reaction graphs."

.... Each time a novel specific set of new never-before-born molecules appear, they create a new adjacent possible into which these new molecules may in turn flow. Fluctuations almost certainly do not damp out. Instead, the specific fluctuations that drive the matter on the reaction graph in particular directions thereby open up new adjacent possible pathways to be explored, possibly creating ever new "salients" of actual molecules on the vast reaction graph. Certainly, the total equilibrium distribution of atoms on all possible molecular species in the graph never remotely recurs. *History enters*, even in this simple chemical reaction system....

Almost certainly, the way the system keeps entering its ever adjacent possible on the underpopulated vast reaction graph creates a nonuniform (more technically, non-isotropic) flow with extra flux in certain directions, which in turn tends statistically to keep wandering off in some historically and contingently biased and unique directions. Fluctuations almost certainly do not damp out. While this kind of chemical reaction system requires detailed study, when the total mass on the graph is tiny compared to the number of molecular species, these guesses seem plausible. In the vast complexity of the giant, cold molecular

clouds in the galaxies, with their molecules and grains and planetesimals, something like this may be happening. If so, molecular diversity is exploding in our universe in ever unique ways.

The flow into the adjacent possible arises at levels of complexity above atoms, certainly for molecules, species, technologies, and human history. Here we must attend to the way the adjacent possible is entered. Salients are almost certainly created in specific “directions” in the space of possibilities, which in turn govern where the system can flow next into its new adjacent possible. These fluctuations almost certainly do not die out, but probably propagate in biased ways into the ever new adjacent possible. All this, as we shall see, applies to the human economy itself as one concrete case. As I shall discuss in detail in chapter 11, *there is an economic web of goods and services, where each good has neighbors that are its complements and substitutes. The structure of the web creates ever new economic niches in its adjacent possible for never-before-born, new goods and services, such that economic diversity has expanded. The structure of the economic web partially governs its own growth and transformation.*

Just possibly, as hinted in the last chapter, the biosphere and perhaps also the global economy enter the adjacent possible in such a way that the diversity of organized processes in the biosphere, or ways of making a living in both the biosphere and economy, is maximized. To state the matter both cautiously and boldly: Once we consider the nonergodic universe at these levels of complexity, and self-consistently self-constructing wholes like biospheres and a global economy, we are entitled to wonder whether there may be general laws that describe the overall features of such systems even if the details cannot be predicted.

We’re only beginning to see the implications of the nonergodicity of the universe. It is part of the persistent creativity in the universe, biosphere, economy, and history. Ceaseless creativity has become physically possible....

BREAKING THE GALILEAN SPELL

Now I want to make my outrageous claim: the evolution of the biosphere is radically nonpredictable and ceaselessly creative. If a scientific law is (as the physicist Murray Gell-Mann concisely put it) a compact statement, available beforehand, of the regularities of a process, then the evolution of the biosphere is partially beyond scientific law. Later we will see that other complex systems share this property. We cannot say beforehand what the regularities of the process are. I cannot overstate how radical what I am claiming is. Since Galileo rolled balls down incline planes and showed that the distance traveled varied with the square of the time elapsed, we scientists have believed that the universe and all in it are governed by natural laws, Newton’s, Einstein’s, Schrödinger’s. Let me call this belief the Galilean spell. Under this spell, we have believed reductionism for over 350 years. I am about to try to exorcize this wonderful Galilean spell that has underwritten so much outstanding science.

DARWINIAN PREADAPTATIONS

We have already discussed Darwinian adaptations as they relate to the heart. As I noted, the heart has several causal properties, including pumping blood and making heart sounds. In accounting for the pumping of blood as the “function” of the heart, we imagined Darwin telling us that it was by virtue of this causal property that hearts came into existence in the nonergodic universe. Now, armed with a few additional concepts, we can see that the evolution of the heart was a foray into the adjacent possible. More, the pumping of blood has causal consequences at the level of organisms and the future evolution of the biosphere. Hearts create salients in the adjacent possible that thereafter bias the way evolution occurs. Moreover, while the mechanisms of the heart betray no laws of physics, physics alone cannot predict the coming into existence in the universe of hearts with their structure and organization of processes, and with their specific causal powers. Thus, hearts are epistemologically and ontologically emergent with respect to physics. Without vitiating any law of physics, physical laws alone do not describe the causal

unfolding of the universe. This is already a radical claim, but nothing compared to challenging the Galilean spell.

One of Darwin's brilliant ideas is what is now called Darwinian preadaptation. Darwin noted that an organ, say the heart, could have causal features that were not the function of the organ and had no selective significance in its normal environment. But in a different environment, one of those causal features might come to have selective significance. By "preadapted" Darwin did not mean that some intelligence crafted the preadaptation. He simply meant that an incidental feature with no selective significance in one environment might turn out to have selective significance in another environment.

Preadaptations are abundant in biological evolution. When one occurs, typically, a novel functionality comes into existence in the biosphere—and thus the universe. The classic example concerns swim bladders in fish. These bladders, partially filled with air, partially with water, allow the fish to adjust their buoyancy in the water column. Paleontologists have traced the evolution of swim bladders from early fish with lungs. Some of these lived in oxygen-poor water. The lungs grow as outpouchings from the gut. The fish swallowed the oxygen-poor water, some of which entered the lungs, where air bubbles were absorbed, making it easier for the fish to survive. But now water and air were both in a single lung, and the lung was preadapted to evolve into a new function—a swim bladder that adjusted neutral buoyancy in the water column. With the evolution of the swim bladder a new function has entered the biosphere and universe—that of maintaining neutral buoyancy in a water column. This new function had causal consequences for the further evolution of the biosphere, the evolution of new species of fish and new, never-before-born proteins, and the coevolution of other species with these new species. ***So, the preadaptation changed the future evolution of the biosphere and physical content of the universe in the nonergodic universe above the level of atoms.***

Now I come to my radical question. Do you think you could say ahead of time, or finitely pre-state, all possible Darwinian pre-adaptations of all species alive today? Or could you pre-state all possible human pre-adaptations?

I have found no one who believes the answer is yes. We all appear to believe the answer is no.

Part of the problem with attempting to pre-state, or specify ahead of time, all possible pre-adaptations is that to do so we would have to pre-state all possible selective environments. Yet we have not the faintest idea of what all possible selective environments might be. More formally, we have no way to list all possible selective environments with respect to all causal features of organisms. How would we even get started on creating such a list? Thus, we cannot know beforehand the Darwinian preadaptations that will come to exist in the biosphere. After the fact, once the preadaptation has developed some new functionality, we may well be able to identify it. For swim bladders, the new functionality was the capacity to achieve neutral buoyancy in a water column. Only retrospectively can we sometimes account for the emergence of preadaptations. Paleontologists do this all the time.

But our lack of prior knowledge does nothing to slow down the biosphere's evolution. The profound implication of this is that virtually any feature or interconnected sets of features of an organism might, in the right selective environment, turn out to be a preadaptation and give rise to a novel functionality. Thus, ***the evolution of the biosphere is radically often un-pre-statable and unpredictable in its invasion of the adjacent possible on the unique trajectory that is its own biological evolution.***

This has very important implications. Consider how Newton taught us to do science: pre-state the configuration space (say the billiard table); identify the variables, forces among them, and initial and boundary conditions; and solve the dynamical equations for the forward evolution of the system. But we cannot follow Newton's mandate in the evolution of the biosphere, for the very deep reason that we do not know all the relevant variables beforehand. While we know the billiard balls on the table and the table with its boundaries, we do not know beforehand about swim bladders. Therefore, we cannot write down

the equations among the variables and solve for the forward evolution of the biosphere the way we can for the balls on a billiard table. We are precluded from following Newton. *The existence of Darwinian preadaptations means that Newton's way of doing science stops when it comes to the forward evolution of the biosphere.*

This, in turn, has implications for whether and to what extent the evolution of the biosphere is describable by “natural laws.” The status of laws in science has been much debated among scientists and philosophers of science. Are they in some metaphysical sense “real,” in the sense of prescriptively “governing” how things “must” unfold, or are they descriptions? *With Nobel laureate physicist Murray Gell-Mann, I adhere to the view that a law is a short, or compressed, description, available beforehand, of the regularities of the phenomenon it covers. But if we cannot pre-state, let alone predict, Darwinian preadaptations before they occur, and yet they occur, then we can have no law, in Gell-Mann's sense of a law, for the evolution of the biosphere by Darwinian preadaptations. **The same holds true, as we shall see, for the evolution of the economy or for the evolution of human culture.*** Then Darwinian preadaptations are literally partially lawless, though not in the sense that laws of physics are violated. An oxygen does not change to a nitrogen, but we cannot have a law-governed emergence of swim bladders. A more accurate way of saying this is that Darwinian preadaptations are not sufficiently covered by any natural law. Laws do still apply and are constraints. The oxygen really cannot change to nitrogen. But no natural law governs the emergence of swim bladders in the evolution of the biosphere.

I should stress that in saying that Darwinian preadaptations are not, apparently, law governed, I am not saying that the events that lead to the selection of a preadaptation in a novel environment may not be perfectly causal. That is, without invoking the probabilistic character of quantum mechanics, which may or may not play a role in the emergence of some preadaptations, there may well be perfectly good “classical physical” and biological reasons why and how swim bladders evolved from lungs. Indeed, multiple routes to the evolution of the swim bladder might all have been successful, even if evolution followed only one of these routes. This is another case of the “multiple platform” arguments. Not only can we not finitely pre-state all possible Darwinian preadaptations, also for any one, we cannot pre-state all possible pathways to its successful emergence in the biosphere. To say that Darwinian preadaptations may be classically caused but are not law governed is quite strange. Indeed, it is radical. Let us call such situations causally anomalous, using a phrase introduced by a philosopher of mind, Donald Davidson, to describe the relations of neural events to mental events. Causally anomalous phenomena are a huge departure from Laplace's reductionism, in which the entire future and past of the universe could be known from the current specification of the positions and velocities of all the particles in the universe. Clearly, we are moving far beyond reductionism.

If the evolution of the biosphere is partially lawless, as I believe is the evolution of the economy and human history, we are beyond Galileo's spell. ***It is not true, it begins to appear, that the unfoldings of the universe, biosphere, and human history are all fully describable by natural law.*** As we will see, this radical claim has, among its consequences, a radical and liberating creativity in the unfolding of the universe, biosphere, and human civilizations....

I come now to a difficult issue: can I prove that Darwinian preadaptations are partially lawless? I do not yet know how to do so. Moreover, it is not at all clear what could constitute such a proof. Let us look for a moment at Gödel's incompleteness theorem. Gödel showed that for any sufficiently rich mathematical system, such as arithmetic, there were statements that were true given the axioms of arithmetic, but that could not be deduced from those axioms. This theorem is stunning, of course. It shows, even for strictly mathematical systems, that they are persistently open in the precise sense that there are true, but not provable statements given the axioms of the systems. And if the true, but unprovable statements are added as new axioms, the enriched axiom set again has new true statements that cannot be deduced from the

enriched axiom set. Mathematics itself is not “complete,” it is persistently open, as it, by adding new axioms, invades a mathematical adjacent possible.

In a loose sense, Darwinian preadaptations, adding novel functionalities to the biosphere and allowing it, thereby, to invade the physical adjacent possible in new ways in its further evolution, are analogous to mathematics invading the mathematical adjacent possible by adding unprovable statements as new axioms.

In two later chapters, on the economy and mind, I will argue that the analogues of Darwinian preadaptations arise and are not algorithmic. *It is not at all clear how to prove that something is not algorithmic.*

My claim that we cannot finitely pre-state Darwinian preadaptations does not pertain to a mathematical world of symbols and equations. It pertains to the real world of atoms and organisms. What could constitute a proof that such Darwinian preadaptations are partially lawless? You might argue that some such natural law might, somehow, be found. We are entering new scientific and philosophic ground, as far as I can tell. I do not know what would constitute a proof of my claim.... How then, if at all, could one prove that I am correct, that Darwinian preadaptations cannot be finitely pre-stated, hence are partially lawless? I can think of no way to do so. Then my way forward is to proceed as I have above, to try to show us that there is no effective procedure to list all causal consequences of parts of organisms and their potential uses, alone or together with other parts of the organism, in all possible selective environments that might arise. This may be all the proof that we can ever have that the evolution of the biosphere of the biosphere is partially lawless. My claim may, itself, be beyond proof.

THE EVOLUTION OF THE ECONOMY

Let us turn to human action. In this chapter, I will move beyond the “hard” sciences and show how the ceaseless creativity of the universe is manifested in a particular social realm, the realm of economics. Like the biosphere, our human realm is endlessly creative in ways that typically cannot be foretold. Our common incapacity to know beforehand what Darwinian preadaptation will bring to biological evolution has an analogue in technological evolution—we can easily apply the idea of preadaptation there, too—and more broadly in the evolution of the economy. Later in this chapter, I will discuss economic webs among goods and services, and show how that web creates novel niches for ever newer goods and services. The global economy has exploded from perhaps a hundred to a thousand goods fifty thousand years ago to an estimated 10 billion today. The economic web invades its adjacent possible.

Like the biosphere, the “econosphere” is a self-consistently co-constructing whole, persistently evolving, with small and large extinctions of old ways of making a living, and the persistent small and large avalanches of the emergence of new ways of making a living. Because there are economic analogues of preadaptations, I hope to persuade you that how the economy evolves is often not foreseeable. Despite this, the evolution of the economy often remains whole as it evolves and adapts. On a broader scale, the same self-consistent co-construction and adaptive evolution to that which is unforeseeable is true of our civilizations, including the global civilization that may well be emerging and which we will, as humans, partially co-construct.

This discussion is not only scientifically important but critical in the practical world. The structure of the economic web that I will describe below influences its role in driving economic growth. I will show here that the growth of the web is self-amplifying, or autocatalytic, and that—as data confirm—economic growth is positively correlated with economic diversity. The more diverse the economic web, the easier is the creation of still further novelty. This has practical implications. Despite successes in achieving economic growth in Asia, including China, India, Korea, Taiwan, and Singapore, and in eastern Europe including Russia, much of the world today still lives in poverty. Understanding the economic web and its

modes of evolution may perhaps alleviate some of that poverty while achieving ecologically sustainable growth. I will argue here that there are severe limits to this ambition: our incapacity to predict technological evolution and other aspects of the evolving economy is an implicit condition of the creative universe. If we cannot extend science, Newtonian or otherwise, to the detailed evolution of the economic sphere, we must reexamine rationality in a broad framework. ***We must strive to understand how fully human agents manage to live their economic lives in the face of inescapable ignorance, thus where reason alone is an insufficient guide to our actions.*** We will have to re-examine how we use all the tools that we have evolved for 3.8 billion years, of cellular evolution, animal evolution, vertebrate evolution, mammalian evolution, and hominid evolution, to see our full human selves living our lives.

DARWINIAN PREADAPTATIONS IN TECHNOLOGICAL EVOLUTION¹⁷

The following story is said to be true: A group of engineers were trying to invent the tractor. They knew they would need a massive engine. They obtained a massive engine block and mounted it on a chassis, which promptly broke. They tried successively larger chassis, all of which broke. At last one of the engineers said, “You know, the engine block is so big and rigid, we can use the rigidity of the engine block itself as the chassis and hang the rest of the tractor off the engine block.” And indeed that is how tractors are made.

Now the rigidity of the engine block was an unused causal feature of the engine block that came to be used for a novel function—to serve as a chassis. It is a Darwinian preadaptation in the economic sphere: legally, it was an invention, presumably patentable.

Can we pre-state all technological preadaptations? In the extreme, obviously not: no one in Roman times foresaw cruise missiles. Even shortly after the invention of the computer, Thomas Johnson Watson Sr., the president of IBM, thought there would be a global market for perhaps nineteen IBM 701s, no one foresaw the Internet and World Wide Web.

More generally, the economist Brian Arthur points out that most inventions wind up being used for purposes unforeseen at the time of the invention. The use of the computer as a word processor is a familiar example. Such novel uses are precisely parallel to Darwinian preadaptations. With respect to both the frequency interpretation of probability and Laplace’s N door interpretation of probability, we appear to be precluded even from making sensible probability statements about the emergence of novel functions in the economy. As I described in the previous chapter, in the frequency interpretation of probabilities, we might toss a fair coin ten thousand times and predict that it would come up heads about five thousand times. But it is essential that we know, beforehand, the “sample space” of all the possible outcomes of tossing the coin ten thousand times. In Laplace’s example, we are confronted by N doors behind one of which is a treasure. We have no idea which door hides the treasure. Laplace tells us that the chance we pick the right door is 1/N. This is not a frequency interpretation of probability, but notice again that we know beforehand the sample space; it is the number of doors, N. In Darwinian preadaptations, biological or economic, we typically have no idea at all what the sample space is, what the possibilities are, before they are evolved or invented. By and large, we cannot make well-founded probability statements about the probability of the Internet in 1920, or 1840.

In short, typically we know little, even about the near future. This is, of course, an overstatement. The venture capital community would not exist if there were no investors with foresight and a capacity to

¹⁷ Preadaptations are also discussed as affordances, or thing features around which refactoring or redesigning take place. Facebook used the affordance of the printed facebook given to college students and re-imagined it. The affordance or preadaptation is the gateway to something new. Our thesis in power is that the power plant is an affordance that we be re-used, for instance. Our thesis in networking is that lighting is an affordance in the built environment which will be reused as network end points and other things. In materials, the affordance is the distribution system for heavy materials which enables scale and can just load in new advanced materials. One key imaginative or creative element of company analysis is how we look for preadaptations, or how we think of the existing affordances of a business.

assess the potential market for novel goods and services. But such judgments rely as much on intuition as on strict algorithms, that is, effective procedures to compute the probabilities of unknowable outcomes and inventions and optimize with respect to those probabilities. And even then, the best venture capitalists are more often wrong than right. Out of ten companies which receive venture capital funding, perhaps one will succeed. The others fail for a variety of reasons, including bad management and the fact that the business did not, in reality, have the promise that was supposed....

THE ECONOMIC WEB AND THE EVOLUTION OF FUTURE WEALTH

Economists employ the concepts of complementary and substitute goods. A hammer and a nail are complements since they are used together to create value. A nail and a screw are, largely, substitutes, since you can usually replace one with the other. ***Now imagine points in a large room for all 10 billion goods and services in the global economy. Draw green lines between points that are complements, and red lines between points that are substitutes. The resulting graph is a depiction of the global economic web as it stands today.*** Fifty thousand years ago the web would have contained a hundred to a thousand points. Thus, over time, the economic web has expanded into its adjacent possible. ***Our task is to understand how it expands, and what role the structure of the web itself plays in its own expansion. None of this is known. But it is virtually inconceivable that this evolution, which persistently creates new economic niches and destroys old ones, is not a central part of economic growth.***

Here is a first step. *Most novel goods and services enter the economy as either complements to or substitutes for existing goods and services.* There is no point in inventing the TV channel changer, for instance, until the television is invented and deployed reasonably widely, and there are multiple channels. So, the channel changer is a complement to the television. This simple example demonstrates that, *as the economic web evolves, it does persistently create new economic niches for new goods and services that fit functionally, hence sensibly, into the existing web. The web begets its own future in ways that we cannot foretell.* But, in addition, the television might become of use for other purposes, say long distance banking, as the computer has come to have new uses with respect to word processing, and the engine block that became the chase. Once that happens still new, unforeseeable, functional compliments may fit into the still new niches that the existing good, used for a new purpose, affords. Thus, it is not at all clear, even for existing goods and services, that we can actually prestate all the uses to which they might be put, hence which new economic niches they might create.

In considering the growth of the economic web, therefore, a central question is whether, on average, each new good or service affords less than one, exactly one, or more than one new complementary or substitute possibility in the adjacent possible. *If the answer is more than one, then the web (ignoring for the moment investment capital, adoption of new technologies, and other issues) can grow exponentially. That is, if each new good or service affords more than 1.0 new niches for yet further new goods and services, then new economic niches explode exponentially as the economic web grows in diversity. Under these circumstances the very diversity of the web “autocatalytically” drives its own growth into the adjacent possible, affording ever new economic niches—ever new ways of making a living.* This point is made even stronger if we bear in mind the new uses which existing goods might come to have, hence the unexpected new niches that might arise as the complements or substitutes for those new uses. In turn this raises the fascinating question of whether the emergence of new uses of existing goods itself depends upon the diversity of already existing goods in the economic web....

Both anecdotes and good economic data support the rough idea that the diversity of goods and services in an economy drives its growth. In her famous book *The Growth of Cities*, Jane Jacobs notes that in postwar Milan and its hinterland, and in Tokyo and its hinterland, a web of complementary technologies mutually spurred economic growth. Jose Scheinkman and colleagues studied a variety of American cities, normalized industries for total capitalization, and found a positive correlation between economic diversity

and growth. ***Thus, there are some demonstrated grounds to believe that the economic web autocatalytically drives its own growth into the economic adjacent possible, generating ever new economic niches and evolving future wealth.*** Another clue that economic diversity aids the innovation of new goods can be found in the story of the Wright brothers' airplane. Their first successful aircraft was a recombination between a modified boat's propeller, an airfoil, a light gas engine, and bicycle wheels. The more goods there are in an economy, the more potential recombinations there are among them. Put an umbrella down the smokestack of the Queen Mary and you get soot in the first-class cabins. Put the same umbrella behind a Cessna and you get an airbrake. ***Fifty thousand years ago, the recombination possibilities among the few goods and services were few. Now, if we just consider pairs of goods, N goods yields N^2 combinations, any of which has a chance of being useful.***

PROLOGUE

Kauffman, Stuart A.. *A World Beyond Physics*

Classical physics, our gift from Newton, is our world written in the passive voice: rivers flow, rocks fall, planets orbit, stars arc in the space-time deformed by their masses. There are no doings, only happenings: myriad, miraculous, but brute.

I broach 78 years as I sit to write, having angled to the kitchen to pick out a nectarine to eat. Yesterday, I clambered aboard the "Poised Realm," my 22-foot boat, to skiff across to the Crane Dock on Orcas Island to drive to Eastsound, Washington, to buy the nectarine I just retrieved as an afternoon snack. My heart thumps a bit, my own human heart. Most of my readers have a human heart as well.

Just where did my human heart, the nectarine, my kitchen, the boat, and Eastsound come from since the brute happening of the Big Bang 13.7 billion years ago?

Since Newton, we have turned to physics to assess reality: what is REAL. But physics will not tell us whence we come, how arrived, why the human heart exists, nor why I can buy nectarines in Eastsound, let alone what "buying" is.

We will talk of these things, for there is more to know than we know and more to say than we can say.

We are in a world beyond physics.

We are in a world of living creatures that construct themselves. Yet we lack the concepts to say it. A tree, from a seed, builds itself, launches itself upward toward the sun. We see it and do not yet know what to say. A forest builds itself, rooted, branched, quiet, as if longing. Our biosphere too grows in diversity into what it can become and has done so for some 3.7 billion years. A giraffe? Who knew three billion years ago? None could have known. And nectarines: Who could then have said?

We estimate that 50 to 90 percent of the 10 to the power of 22 (10^{22}) stars in the known universe have planets cycling them. If, as I believe and will say, life is abundant, the universe is rife with becoming, based on physics but beyond any physics we know.

The concept of perhaps 10^{22} biospheres staggers me. Yes, we thrill at Hubble's image of billions of galaxies, some 10^{11} of them. But are there 10^{22} biospheres, ebullient like ours? Not "a world beyond physics," but "worlds beyond physics," as vast as the vastness of the physics we know, almost unknowable.

We miss in our science the idea of a system that constructs itself. I will introduce the requisite concept due to Maël Montévil and Mateo Mossio (2015) called "Constraint Closure". These young scientists have found a, or maybe "the," missing concept of biological organization. We will grow to understand it clearly and build on it. The ideas are a tiny bit complex, but not very. We will get to them. ***But for now, we can think of constraint closure like this: it is a set of both constraints on the release of energy in***

non-equilibrium processes, and those processes, such that the system constructs its own constraints.

This is an amazing idea. Cells do this, automobiles do not.

Living systems achieve this constraint closure and do what are called “thermodynamic work cycles” by which they can reproduce themselves. Living systems also exhibit Darwin’s heritable variation, so can undergo his natural selection, hence evolve. I’ve written about that in some of my earlier books. But I was nagged by a feeling that there was something missing. With constraint closure a crucial puzzle piece is put into place.

But what evolves cannot be said ahead of time: what evolves emerges un-pre-statably—I know of no better word—and builds our biosphere of increasing complexity. We are its children: as are giraffes, nectarines, and sea cucumber. Some years ago, at his 70th birthday fest, a physicist friend of mine smiled at the way biologists see the world. Were biologists with Galileo on the tower of Pisa, they would have dropped red stones, orange stones, pink stones, blue stones, green stones, and so on. My physicist colleagues chuckled knowingly. Physicists seek to simplify to find laws, biologists to study how life became complex. So of course, the red stones were giraffes; the orange stones nectarines; the blue stones sea cucumbers; and the green stones, well, just us. The question is not whether the sea cucumber, giraffe, us, or the nectarine falls faster, but where did they come from in the very first place?

Physics won’t say. No one knows.

There is a world beyond physics.

Darwin taught that new species drive wedges into the crowded floor of nature to make room for their own existence: yes, but no. ***Creatures, by existing, create the very conditions for other creatures to come into existence. Species constitute the very cracks in the floor of nature that constitute the niches for yet new species to come into existence, creating yet more cracks for still more species to spring forth.***

The blossoming biosphere creates its own ever-new possibilities of becoming, yet more diverse and abundant. The same holds, almost unnoted, for the exploding global economy. New goods create niches for yet further new goods: the invention of the World Wide Web created niches for selling on the Web, hence eBay and Amazon; which in turn created content on the Web, hence niches for search engines like Google; and for businesses that try to game, the search algorithms to sell more stuff. Or think of all the iPhone apps: and apps upon apps, like the ad blockers that remove the sales pitches from what Safari shows.

We stumble into the world we make possible as we lumber forward, with no or little insight or foreknowledge. I can go to Eastsound to buy nectarines.

We think that in physics—Special and General Relativity, Quantum Mechanics and Quantum Field Theory with the Standard Model—we will find the foundations from which we can derive the world, the ultimate becoming. We cannot. The ultimate may rest on the foundations, but it is not derivable from them. This ultimate, an unknowable unfolding, slips its foundational moorings and floats free. As Heraclitus said the World Bubbles Forth.

THE WORLD IS NOT A MACHINE

Since the triumphs of Descartes, Newton, and Laplace and the birth of classical physics, we have come to regard physics as the answer to our questions about what reality “is.” In that search, we have come to think of the world as a vast machine. This Newtonian fundamental framework is wonderfully extended by Special and General Relativity. Quantum Mechanics, and Quantum Field Theory, alter some of the basic deterministic aspects of classical physics but not the view of reality as an enormous “machine.”

My thesis in this book is that, with respect to an evolving biosphere, ours and any in the universe, the “machine” thesis is wrong. Evolving life is not a machine. Elaborating how this is so will require some

patience on all our parts. The consequences of the change in world view here proposed cannot be anticipated, but I hope they will include *the realization that we are members of a living world of untellable creativity in its becoming*. Along with that, I hope, will come a profound joy—an expanded awareness, a heightened appreciation, and a deepened sense of responsibility for the living world. Time will tell.

C. P. Snow wrote his famous essay, “The Two Cultures,” decrying the split between the world of science and the world of art. Part of this split is the distinction between “mute” matter and the human imagination. But between these stands the evolving, living world, whether dumb of awareness or broadly conscious. I hope to show you that, unlike physics, where laws hold sway, no laws at all entail the becoming of the biosphere. *No one knows or can know what shall become as the biosphere evolves and shapes its own future in ways we cannot state in advance. They are “un-pre-statable.”* ***This lawless emergence, contingent yet not random, bespeaks a place between mute matter and Shakespeare. Life itself spans between physics and art.***

SO SIMPLE A BEGINNING

Parthasarathy, Raghuveer. So Simple a Beginning

RELEVANCE:

Many of the readings are from leading thinkers in Complexity. Complexity focuses on interactions of agents or element in systems and looks for principles to understand the systems. Traditionally a lot of analysts and scientists have focused on the agents or elements and missed the interactions, patterns and principles.

The passage below is Parthasarathy reflecting on principles in biophysics – echoing the rejection of the reductivism in Kauffman or Miller, and the potential in convergence and unification in Diamandis and Domingos. The readings are useful, in providing the basis for a shift in perspective, anchored in complexity and anthropology, which leads one to look at businesses in a different way.

UNDERSTANDING UNDERSTANDING

The most important theme of this book is that themes exist. The living world isn't just a collection of anatomical structures and biochemical reactions. There are principles, motifs, and mechanisms that unify all of its components and their activities. This statement shouldn't come as a surprise after the preceding 15 chapters. I've avoided until now, however, the question of whether this unity matters, either practically or aesthetically. This question underlies a tension present in much of contemporary science that I illustrate with a nonbiological example before we return to the living world.

There wasn't a singular experience that drove me to study physics, but one event I remember vividly occurred in my high school physics class, an activity in which we released a ball from a ramp onto a table, along which it rolled until plummeting off the table's edge to fall into a plastic cup on the floor. Our task: to predict beforehand where to place the cup to catch the ball. It may seem dull on paper, but to see that the ball did, in fact, land exactly where universal laws of motion say it should was amazing to me. We needn't memorize facts about balls, ramps, or cups, or even care about them; instead, there are general themes that guide us. Searching for and applying broad principles is central to physics and underpins the biophysical approach that I've championed throughout this book.

There is, however, another way we could have made our prediction. From prior experiments, we could have tabulated a vast list of outcomes. For a range of ramp heights, tilt angles, ball masses, ball materials, table heights, air temperatures, and whatever other parameters might possibly be relevant, we could have released balls and recorded where they landed. Then, for the conditions of our new roll, we could look up the outcome from our giant database. This, to use the current jargon, is the "big data" approach. There's nothing wrong with it, and it's often very successful. We've seen this already, in the correlations observed between genomic features and traits such as human height (chapter 14). These correlations enable predictions, but we have no understanding of why the correlations exist.

The contrast between understanding basic principles and cataloging information arises with increasing frequency throughout science, as our tools for generating and processing data become more powerful. Sometimes, as in the applications of Brownian motion (chapter 6), the basic principles are so profound and well understood that it would be absurd to ignore them. Sometimes, as in the genomics example above, looking for correlations is the only possible approach, at least for now. Sometimes it is unclear what to do. In chapter 4, we looked at the biological circuits that regulate genes, composed of repressor

and promoter proteins that orchestrate gene activity. To some researchers, identifying basic circuit motifs—feedback loops, oscillators, clocks, and so on—is the path to progress. To others, this is irrelevant; plugging in the details of all the interactions between all the components of a regulatory network, even if they number in the hundreds or thousands, is the more useful strategy. It isn't obvious whether the former, more biophysical approach or the latter, more phenomenological approach is better. They aren't mutually exclusive; pursuing both in parallel may be the best plan, especially as the detailed models might help uncover core concepts that are yet unknown. Still, the tension between these philosophies is real, and it colors debates about funding and research directions.

My own bias is toward illuminating basic, minimal sets of principles. After all, a giant database of falling ball trajectories would enable prediction of future experiments on falling balls, but it's the deeper understanding of Newton's laws of motion that gave us the trajectories that sent astronauts to the moon. Which approach is the more practical, therefore, may depend on the timescale: immediate application to some problem at hand, or to problems of the future that are still unimagined. Aside from the aims of practicality, there's also the deeply human appeal of understanding. Understanding is hard to define. Still, extracting from complexity simple yet powerful explanations has compelled us for as long we've been human, powering myths as well as science.

TWO ENDINGS TO OUR STORY

There is a way one may have expected this book would end. We began

A CRUDE LOOK AT THE WHOLE

Miller, John H.. A Crude Look at the Whole

RELEVANCE:

Best brief introduction to complexity that I have seen. A number of useful concepts in the following passage from the introduction to Miller book A Crude Look at the Whole. The extended metaphor around maps and mapmaking is worth focusing on and thinking about.

We too are mapmakers....

Introduction: True Places

It is not down in any map; true places never are. —Herman Melville, Moby Dick

Science is about mapmaking. It's about taking a complicated world and reducing it to some sparse set of markings on a map that provides new guidance across an otherwise incomprehensible, and potentially hostile, landscape. A good map eliminates as much spurious information as possible, so that what remains is just enough to guide our way. Moreover, when the map is well made we gain a deeper understanding of the world around us. We begin to recognize that rivers flow in certain directions, towns are not randomly placed, economic and political systems are tied to geography, and so on.

Maps—and science—are often more about what we leave out than what we put in. As Jorge Luis Borges catalogs in his one-paragraph-long short story “On Exactitude in Science,” “The Cartographers Guilds struck a Map of the Empire whose size was that of the Empire, and which coincided point for point with it. The following Generations, who were not so fond of the Study of Cartography as their Forebears had been, saw that that vast Map was Useless.”

Different maps—even of the same landscape—provide different insights into the world. A topographic map provides information on the various hills and dales in the world in just enough detail to be useful to a hiker. A road map, with its sparse set of major cities and the roads that connect them, provides just enough information for a cross-country drive. Divorcing a map from its purpose inevitably leads to frustration. Too little of the right kind of detail, or too much of the wrong kind, encumbers our ability to understand the world.

Science has proceeded by developing increasingly detailed maps of decreasingly small phenomena. At the heart of this reductionist strategy is a hope that once we have detailed maps of the smallest of parts, we can paste the mosaic together and have a useful map of Borges's Empire. That strategy fails, and while the result might please Borges's Cartographers Guild, the mosaic is as much a fool's errand as Borges envisioned.

The problem lies not in the incompleteness of our knowledge but in the dream—no, the fallacy—of reductionism. **Reductionism fails because even if you know everything possible about the individual pieces that compose a system, you know very little about how those pieces interact with one another**

when they form the system as a whole. Detailed knowledge of a piece of glass does not help you see, and appreciate, the image that emerges from a stained-glass window.

Over the past few decades a new science has been brewing. It is a science that recognizes that there are fundamental principles governing our world—such as emergence and organization—that appear in various guises across all of the nooks and crannies of science. For example, in physics, individual atoms organize into magnets, in biology, cells organize into organisms, and in economics, traders organize into markets. The universality of these principles was a surprise to scientists accustomed to thinking in terms of scientific disciplines, and by necessity, this new science transcends the traditional boundaries imposed by our current academic institutions. ***It is a science where simple things produce complexity and complex things produce simplicity. It is a science that embraces new investigative tools, such as computers serving as modeling substrates, in order to escape the bounds imposed by our usual collection of scientific tools,*** such as the various pieces of mathematics, largely derived in the late 1600s, that we so often rely on today. More fundamentally, it is a science that challenges our traditional notion that understanding comes from reducing things to their simplest components.

Alas, the new science we are after, the one that may hold sway over critical aspects of our life and destiny, is, as Herman Melville says, “not down in any map; true places never are.” Science as currently practiced—with psychology separate from economics, physics separate from biology, and on and on—has been remarkably productive. The creative destruction of scientific ideas, with its inherent quest to define the frontier by publicly disclosing, evaluating, and correcting ideas, has provided us with an engine of insight. The cost, however, is that individual fields have become increasingly separated from one another intellectually. ***Taking an exact look at a small piece of the world has become the academic norm and has almost fully displaced taking what my Santa Fe Institute colleague Murray Gell-Mann calls “a crude look at the whole.”***

That may seem a minor problem, but we see its importance when we look at the true places we wish to explore. Take any global-scale, societal challenge, such as financial collapse, climate change, terrorism, epidemics, revolution, or social change: not one neatly aligns with any particular academic field. Moreover, even if one did, the reductionist approach still may not let us understand the whole. The fundamental principles of complexity describe how even simple parts, once together, seemingly take on a life of their own. Having intimate knowledge of, say, each part of an engine, every bolt, piston, cam, and so on, tells us little about what happens when we put those pieces together and they begin to interact with one another. Moreover, such intimate knowledge gives us no insight into what would happen to the engine as a whole if, say, we increase the size of one of the cylinders.

Reduction gives us little insight into construction. And it is in construction that complexity abounds.

From agoras to amoebas, from bees to brains, from cities to collapse, and on up to zebra stripes, the world around us is an encyclopedia of complexity. Sometimes this complexity arises shaped by natural forces such as evolution, as in the consciousness that emerges from our brains. At other times we have a hand in its creation, as in the steady stream of prices that arises from the seemingly chaotic noise and gestures in a commodities trading pit. ***Without a science of complex systems, we have little chance to understand, let alone shape, the world around us.***

The initial academic discussions of complex systems can be traced back to at least 1776, when Adam Smith, in his *Wealth of Nations*, briefly discusses the “invisible hand” as a force that leads self-interested traders to unintentional, socially desirable outcomes. Of course, scientific propositions that are

based on an invisible hand are more akin to the invocation of a deity than to a scientific theory and are about as useful to an economist as one of Rudyard Kipling's just-so stories is to a biologist trying to explain how a leopard gets its spots.

The modern movement of complex-systems thinking can be tied to the beginnings of the atomic and information ages, when scientists such as Stanislaw Ulam and John von Neumann, using some of the world's first programmable electronic computers, began to blur the lines between traditional academic fields as they pursued questions such as whether a machine could be truly self-reproducing. Out of this effort arose a class of models that, starting with a collection of simple, well-defined pieces and interactions, results in a surprisingly rich set of global patterns.

The study of those patterns was an important step toward understanding not just the purpose of an animal's markings—say, camouflage—but also how they arise. Is it necessary that there be some master plan contained within the DNA of a leopard that specifies the color of each location on its skin, similar to how a digital image file directs the color of each pixel on a computer display, or is there a more universal explanation that can tell us how a leopard gets its spots?

The simple mathematical and computational models begun by Ulam and von Neumann have given us a lens through which to look at the origins of such complexity. ***We find that the combination of simple pieces, locally interacting with one another, is sufficient to lead to global behavior that is rather alien to its origins.*** Thus, the likely answer to how the leopard gets its spots—or how a lowly (but dangerous) sea snail gets its shell pattern, or even how the cacophony of a trading pit results in a well-organized set of trades and prices—is at once far simpler, far more universal, and far more fascinating than we might imagine.

Over the last few decades, the study of interacting systems has opened up a new frontier in our understanding of complex systems. Whether we consider abstract models running at the speed of light inside a computer or the carefully curated anthropological evidence of a century of rice farming, a small set of core principles governing complex systems has emerged. Interacting systems develop feedback loops among the agents, and these loops drive the system's behavior. Such feedback is moderated or exacerbated depending on the degree of heterogeneity among the agents. Interacting systems also tend to be inherently noisy, and such randomness can have surprising global consequences. Of course, who interacts with whom is a fundamental property of these systems, and such networks of interaction are an essential element of complex systems.

Core principles such as feedback, heterogeneity, noise, and networks can be used to understand new layers of complexity. For example, there are complex systems, such as your mind, that generate coherent and productive decisions in a completely decentralized manner, seemingly without control. Other systems, facing deeply embedded constraints such as getting oxygen to all of the cells in your body, lead to scaling laws that can take seemingly disconnected parts of the world and align them along a simple relationship. Yet other systems, such as the members of a social movement, self-organize into critical states that begin to exhibit a common characteristic behavior. Many interacting systems develop cooperation among the agents, a complex behavior that, once arisen, allows agents to shift into a new realm of opportunity, and we are now in a position to understand such a transition. Finally, by repurposing methods and ideas that were first developed at the dawn of the modern science of complex systems, we can generate a new theorem about the behavior of adaptive systems. These core principles

driving complex systems, and their application to understanding new layers of complexity, are the focus of the pages that follow.

One critical aspect of interactions is feedback. Sometimes feedback stabilizes the system, as happens when we install a not-too-touchy thermostat to control the furnace. Other times feedback causes a system to go out of control, as happens when we place a microphone too close to a loudspeaker, producing an ever-increasing screech. The recent growth of market interconnectivity has resulted in a system rife with feedbacks. This has come about through new communication links, the rise of derivative securities, and the use of high-speed, computer-automated trading. Indeed, these changes have easily outpaced our ability to truly understand their implications, and financial markets have become an unintentional Promethean experiment upon which we now base our economic livelihood.

An example is the “flash crash” of May 6, 2010, when a simple oversight in the programming of a trading computer in a suburb of Kansas City, Kansas, caused a temporary collapse of global markets. The havoc that ensued resulted in dramatic price changes on key stock indices and caused the shares of previously valuable mainstay companies to be sold for pennies (not pennies on the dollar, mind you, but pennies). A five-second trading pause invoked fifteen minutes into the crisis was, fortunately (and remarkably), sufficient to begin to restore the system, and the markets settled back into a more familiar pattern.

In 2008 a much larger event happened that resulted in a financial tsunami that rolled across the world’s economy, affecting the lives of billions and plaguing us to this day. Examining this crisis, we find a situation where any single agent on the economic landscape, from homeowners to mortgage brokers to rating agencies, was making sane decisions, yet the connections among these agents created a series of unfortunate feedback loops that destined the system to fail.

The economic collapse of 2008 represents a major failure for the profession of economics. Not only did economists fail to see the onslaught coming, but once the crisis arose, they had no idea how to deal with it. Part of this failure can be traced to the reductionist desire to break things down to simple parts. In the language of modern economic theory, this led to a reliance on “representative agents,” constructs that attempt to capture the behavior of, say, all consumers using a single mega-consumer. In part, such a choice arises from the fourteenth-century friar Father William of Ockham’s dictate to prefer simpler explanations to more complicated ones. Of course, Ockham still requires that the model, complicated or not, explain what we want to understand. ***In reality, the use of representative agents is also driven by the limitations of the modeling tools typically used by economists, as these tools can be deployed only if there is a high degree of homogeneity in the system.***

While homogeneity is a useful assumption—for both philosophical and practical reasons—the study of complex systems suggests that the behavior of heterogeneous systems may not be so easily averaged out. Whether we are looking at the temperature control of a honeybee hive or the likelihood of a riot, heterogeneous systems often function in ways that are different from homogeneous ones.

Recognizing heterogeneity not only changes our predictions about how a system will behave but also alters our policy prescriptions. Homogeneous systems tend to undergo rapid changes and oscillations, while heterogeneous ones tend to react more slowly. Thus, your ability to start, or quash, a social movement is tied to the degree of heterogeneity among the people involved. Similarly, markets may require some heterogeneity among the traders to remain stable.

Complex systems often have some inherent degree of randomness tied to the behavior of the agents or the structure of interactions. Perhaps surprisingly, such randomness can be useful. We often dread randomness in systems. Indeed, a key dictate in modern business management is to seek quality by removing all sources of randomness from any process. Given such imperatives, it is easy to think of randomness as a foe to be fought rather than as an opportunity to be embraced. The study of complexity suggests otherwise. ***Randomness is fundamental to Darwin’s theory of evolution, which relies on the notion that errors (variations) during reproduction will provide grist for the mill of selection and result in “endless forms most beautiful and most wonderful.”***

Darwin’s theory, and the role of randomness therein, is really about discovery on rugged landscapes. Our ability to discover new opportunities, whether new forms of animal life or novel technologies, is tied to both the ruggedness of the underlying landscape and our search skills. On simple landscapes, even simple searches can find good outcomes. On rugged landscapes, such searches founder.

Landscapes become more rugged as the elements that compose them interact more. Suppose we are seeking, say, a novel drug cocktail to fight some disease. If each drug we add to the mix has an effect that is independent of the others, then we can quickly find the best cocktail just by adding the drugs one at a time and keeping only the ones that improve the cocktail’s overall efficacy. However, if the drugs interact with one another, this simple search strategy breaks down, as the various interactions no longer provide a clear signal on how best to proceed.

It turns out that the introduction of randomness can greatly improve our ability to search on rugged landscapes. ***As James Joyce noted, “Errors . . . are the portals of discovery.”*** Just as evolution relies on variation to uncover most wonderful forms, introducing errors into a search can be a powerful strategy for discovery.

Accepting randomness in a system forces us to give up some control. Yet when we are facing hard problems, this may be the right thing to do if we want to improve the outcome. More generally, it may be the case that carefully controlled, centralized systems are more of a modern artifact, driven by reductionist thinking, than a universal norm. Indeed, there are plenty of examples where the principles of feedback, heterogeneity, and randomness conspire to create complex systems that are without centralized control, yet quite productive. Effective decentralized decision making may be one of the best new old ideas to emerge from complex systems.

When we think about decision making, our natural tendency is to focus on our own decisions. Over the last few decades entire academic fields have been devoted to understanding how humans make decisions. While unraveling the mysteries of our deciding brain is a worthy enterprise, it is far too easy to overlook the vast number of decisions that take place elsewhere in the biological world. To take just one example, bacteria exist in environments that contain both useful and harmful chemicals, and thus they constantly must make life-and-death decisions about where to move, given the trade-offs among various opportunities. How is this possible without a brain? Even more intriguing, humans (presumably using a brain) and bacteria (presumably not using one) demonstrate similar patterns of choice errors in simple experiments.

The notion that one doesn’t need a brain to make good decisions is startling. From the lone bacterium on up to large-scale social systems such as honeybee hives and financial markets, we are surrounded by

decision making. How can a swarm of honeybees make good decisions? The queen is not the leader. She leads a rather insular life, serving as a well-tended egg-laying machine, able to emit only signals about her health and existence, rather than operating instructions to the rest of the hive.

Karl von Frisch's discoveries about honeybee communication in the late 1940s inspired generations of scientist to undertake the careful observation and analysis of honeybee behavior. Through this work, we are beginning to understand how a colony can sort out its various options and make good decisions without any central leadership. One particularly important decision for a colony—the difference between its perpetuation and demise—is finding a new location when the old one becomes too crowded.

A swarm of bees solves the problem of finding a new location through the use of a few simple rules and feedback mechanisms. Scout bees, after identifying a potential new site, advertise it to other scouts. The better the site, the more vigorously the scout promotes it. This decentralized process allows the sites to be sorted out and suitably investigated, and ultimately it results in the swarm tending to choose the best site relatively quickly without any central direction.

Understanding such decentralized processes has numerous benefits. It solves an interesting, life-or-death case of honeybee natural history. It also shows how decentralized mechanisms can be used to solve hard problems. This suggests an approach that we might be able to hijack for our own use in, say, coordinating computer networks or large-scale human organizations. Finally, and perhaps most profoundly, such decentralized mechanisms give us new insights into related phenomena. For example, perhaps bees are to neurons as hives are to brains. Are swarm decisions akin to human consciousness?

Complexity arises in systems of interacting agents. Take some agents with simple behavior, connect them together in a particular way, and some global behavior will result. Alter the connections and, often, new global behavior arises. Given this, knowing how patterns of interactions—that is, networks—influence behavior is fundamental to understanding complex systems.

Even in simple models such as lakeside neighbors competing to keep up with one another, interesting patterns begin to emerge. Starting from such a simple system, we can alter the connections slightly and find radically different behavior taking over. Indeed, by introducing only a few long-range connections, we find that it may be a small world after all, where anyone can connect to anyone else using only a few intermediaries. If neighbors can connect to one another, they can influence one another. Thus, the networks that define neighborhoods drive system-wide behavior. This behavior is often surprising. For example, a well-mixed world where neighbors are tolerant of others easily segregates into neighborhoods of homogeneous types.

One of the more surprising principles coming out of the complexity that abounds is the existence of scaling laws. Starting in the late 1800s, biologists began to notice that, when appropriately scaled, various physical and physiological features of a variety of organisms aligned in a simple way. A simple rule links the metabolism of a single cell to that of a blue whale. Knowing the heart rate and weight of, say, a mouse allows us to predict the heart rate of, say, a thousand-pound cow. The ability to make such predictions is tied to the fundamental constraints that govern such complex systems. In this case, limits on how densely we can pack the pathways needed to provide resources to the organism drive the scaling.

Scaling laws arise in other complex systems as well. The size of cities or firms tends to follow well-defined laws, with the largest having twice the size of the second-largest, three times that of the third-largest, and so on. Similarly, in a book, the word that is most commonly used is twice as likely to occur

as the next most commonly used word. Even the number and death tolls of wars are governed by a scaling law.

Knowing the scaling laws that govern our lives provides a portal into our future. For example, over the last century we have seen a trend toward urbanization. More than half of the world's population now lives in urban areas. Is such a trend good or bad for humanity? The answer to this question is tied to the coefficients of various scaling laws of cities. These will tell us whether more urbanization will allow us to use fewer resources, be more inventive, and so on. Similarly, the scaling laws of wars may hint at how many conflicts with how many deaths we are likely to see in the future.

In complex social systems we often see the emergence of cooperation. Agents in systems can either compete or cooperate with one another. Competition makes you slightly better off, while cooperation makes you much better off. Unfortunately, most social systems have incentives that favor, at least individually, competition over cooperation. Such systems can easily end up with the inferior, competitive outcome.

Notwithstanding incentives to compete rather than cooperate, complex social systems may find ways to achieve the cooperative and socially superior outcome. On the island of Bali, farmers have been farming the picturesque rice terraces sustainably for more than a thousand years. This cooperation persists despite what would appear to be overwhelming economic incentives to compete with one another for the scarce water. However, by carefully unraveling the complex dynamics that govern this ecosystem and applying the principles of feedback and networks discussed above, we can resolve this apparent anomaly. Oddly, the neighborhood feedbacks from the presence of damaging crop pests and diseases realign each farmer's incentive to share water, and with such sharing, society is better off. Moreover, the newfound need for coordinated cropping opens up a niche for an elaborate religious institution with various shrines and temples tied to the irrigation systems.

We can also formulate an abstract model from which we can observe and understand the emergence and persistence of cooperation. We find that in a world red in tooth and claw, where competition can easily overwhelm the system, slight variations in competitive strategies provide a means by which cooperation can emerge. Cooperative agents develop a way to communicate so as to recognize one another. By doing so, they get the benefits of cooperation while minimizing losses when they encounter competitive agents. Through such a mechanism, cooperation can emerge and be sustained.

The final principle we will explore is that of self-organized criticality. Consider grains of sand being slowly piled on a table. As we drop each grain, it might land on a stable spot and increase the pile, but in doing so it makes that spot less stable than it was before. Alternatively, it might land on an unstable part of the pile, triggering an avalanche.

Over time, this interplay of stability and instability self-organizes the sand pile into a critical state. Once this happens, we find that avalanches of all sizes are possible (the distribution of which is described by a scaling law), with smaller ones far more likely than larger ones.

One implication of the sand pile is that once we enter the critical regime, the dropping of a single grain of sand can cause, on rare occasions, an avalanche encompassing the entire pile. Various social systems may evolve toward similar critical states. ***We might find ourselves in a world governed by the mathematics of the sand pile.*** Stock markets may be subject to numerous routine adjustments as typical world events transpire. Yet these same types of events will, on rare occasions, lead to a massive

readjustment. Civilizations may be governed by political systems that tend to push people toward critical states, where small events occasionally result in the collapse of an ancient civilization or, as we saw in the Arab Spring, modern governments.

We will conclude our exploration of complexity by following an arc that begins with our desire to understand atomic interactions at the start of the atomic and information ages and ends with a new fundamental theorem about complex adaptive systems. In the early 1950s Nicholas Metropolis and others developed an algorithm to explore interacting molecular systems. At the core of this algorithm is a set of simple manipulations that ultimately allows one to recover a critical piece of information that is impossible to generate directly; as if by magic, this algorithm produces the unknowable. Related algorithms have become a critical component in our emerging analytic age, as they solve a key problem in applying the eighteenth-century statistical ideas of Presbyterian minister Thomas Bayes to real-world problems ranging from targeted web advertisements to driverless cars.

At the heart of complex adaptive systems are agents searching for better outcomes. With a few simplifications, the key aspects of this search behavior can be linked to elements of the algorithm above. Thus, agents in such systems are, unknowingly, performing a dance governed by a cosmic algorithm. Given this connection, we derive a new theorem of complex adaptive systems that embraces the magic inherent in the algorithm. This new theorem implies that as agents adapt in these complex systems, their adaptations are governed by probabilities tied to their underlying fitness. While agents are more likely to be found concentrating on the better solutions, there is always a (lower) chance that they will find themselves in suboptimal circumstances. This is a result that is at once both gratifying and humbling, as it suggests that while agents will often find the best outcomes, they will inevitably fail on occasion.

Complexity abounds. Exploring its core principles will take us on a journey across the scientific landscapes outlined above. It is a journey marked by awe, inspiration, and ultimately insights that are critical to our scientific understanding of the world around us and to our ability to survive when confronted by our most challenging problems. It is a journey about true places, where the maps are not always well formed, but they are suggestive enough to be of use given our innate desire and need to explore this frontier.

GREAT MENTAL MODELS VOLUME 3

Beaubien, Rhiannon ; Leizrowice, Rosie. The Great Mental Models Volume 3: Systems and Mathematics Latticework Publishing Inc.

via Shane Parrish & Farnam Street.

RELEVANCE:

Enough has been said about “mental models” that I’ve come to cringe at the term. Our “model arsenal” comes from the idea that models aren’t meant to be kept in mind but used as tools for competitive advantage. Two sample “models” of bottlenecks and critical mass are useful to keep in mind and are worth explicitly adding to the readings.

Regarding Bottlenecks, rate limiting factors are essential to watch for. These are often either where economic rents are extracted or points of control. A strong argument can be made that the key to understanding any process or value chain begins with understanding the rate limiting steps -- the bottleneck or constraints.

In economic and business terms, the classic “bottleneck” is the customer or owning the customer relationship. Note how I am framing what a bottleneck is somewhat differently than the reading below and think about why. Note also how bottlenecks relate to previous complexity readings – particularly Kauffman’s closing idea about sets of auto-catalytic constraints being the key to the way order emerges in physics and biology.

Understanding the idea of critical mass, and the simple 10% rule or “tipping points” is essential for analyzing the scaling of innovation and change in industries. One can debate whether zero to one is more important than the 10% tipping point – but from an investment perspective it isn’t worth getting involved in going from zero to one if it isn’t going make it to ten... So analytically, one way to use the idea of critical mass is in this context. This links up to other logical and biological structures – like allostery and logically and strategically dominant approaches. But critical mass is a key check point. It makes sense to put it with bottlenecks, because critical mass is functionally the rate limiter or bottleneck on scaling innovation – can it get to 10%?

Note, when we talk about non-linearity, we are talking about critical mass – or the points of inflection such that systems shift. So, we talk about critical mass a lot, we have just named it by its effect, which is non-linearity around the point of critical mass or tipping point.

Since future economic growth will be driven by innovation as populations stabilize or even decline – understanding and deliberately thinking through the roles of bottlenecks in encouraging innovation and critical mass in limiting it – well it will matter. Patterns of innovation will take center stage.

BOTTLENECKS

All systems have parts that are slower than others. The slowest part of a system is called the “bottleneck” because, as the neck of a bottle limits the amount of liquid that can flow through, bottlenecks in systems limit the amount of outputs they can produce. Using bottlenecks as a model gives us insight into how a limiting factor can hurt or help us.

No one wants to be a bottleneck, which is easily conceptualized as that person who makes everyone else wait. We see this behavior in people who can’t delegate. If you have to make every decision yourself, there’s likely a long line of people twiddling their thumbs while waiting for you to move their projects forward.

Bottlenecks tend to create waste as resources pile up behind them. In manufacturing, they limit how much you can produce and sell. If you work in an industry that depends on timely information, then you risk inputs becoming irrelevant before they make it through the bottleneck.

A bottleneck is also the point that is most under strain. It can be the part that is most likely to break down or has the most impact if it does. In trying to improve the flow of your system, focusing on anything besides the bottleneck is a waste of time. You will just create more pressure on the bottleneck, further increasing how much it holds you back by generating more buildup.

Every system has a bottleneck. You cannot completely eliminate them because once you do, another part of the system will become the new limiting factor. You can, however, anticipate bottlenecks and plan accordingly. Or you can leverage the need to overcome them as an impetus for finding new ways of making a system work. Sometimes you can overcome bottlenecks by adding more of the same, such as dedicating more resources to ease the pressure on a bottleneck. But sometimes the sole solution is to rethink that part of the system.

What you want to avoid is opening up one bottleneck only to create additional, worse ones for yourself later on. If bottlenecks are unavoidable, we at least want them to be in a less disruptive place.

Although the terms are sometimes used interchangeably, a bottleneck is different from a constraint. A bottleneck is something we can alleviate; a constraint is a fundamental limitation of the system. So a machine that keeps breaking down is a bottleneck, but the fact there are twenty-four hours in the day is a constraint.

You also need to be aware of false dependencies disguised as bottlenecks. We often hear explanations in the form of “I won’t do X before Y.” Most statements of this type are only in place to make you feel good about procrastinating when you are the bottleneck. For example, you might say you will start writing every day once you move house and have a dedicated desk for it. If the bottleneck is a lack of a suitable workspace, then moving will alleviate it. But if the bottleneck is something else, like time or ideas, you’re setting up a false dependency. The bottleneck will still be there once you move. Even if the problem is your workspace, you could still find ways to make progress, such as by going to the library or a coffee shop. Anything you do now will make it easier to get into the habit of daily writing.

If you think you’ve identified a bottleneck, it’s a good idea to do what you can to validate that this factor is indeed the limiting factor. Otherwise you might end up solving the wrong problem.

Liebig's law of the minimum refers to the idea that a plant's growth will be limited by the nutrient that is least available. Yield is thus constrained by resource limitation.

The Trans-Siberian Railway

How you deal with your bottleneck can have huge impacts on the overall quality of your system. Often we tend to just deal with our bottleneck whatever the cost. But since there is always one in every system, anticipating the consequences of how we deal with our bottleneck is important. Some bottlenecks are better to have than others because they are easier to organize the rest of our system around.

The building of the Trans-Siberian Railway (TSR) was a complicated project with many moving parts that borrowed from future resources to address its bottleneck. It is both an inspiring and cautionary tale, as sometimes dealing with bottlenecks in the most expedient way possible can cause significant issues later on.

The TSR was a massive undertaking. Not only is it the longest railway in the world, but the challenges faced in building over that distance were unmatched at the time. The railway spans the entirety of Russia from St. Petersburg in the west, close to Finland, to Vladivostok on the Pacific Ocean just east of North Korea.

As W. Bruce Lincoln describes in *The Conquest of a Continent*, there were multiple sources of challenge to building the TSR:

Construction crews would have to work thousands of miles away from their bases of supply. Rails and bridge iron would have to be brought to Siberia from foundries as far away as St. Petersburg and Warsaw, ties would have to be cut in European Russia and shipped across the Urals because almost no hardwoods grew in the steppe or the taiga, and stone for bridge piers and abutments would have to be transported from quarries on the western frontier of Mongolia. Then, as the tracklayers moved deeper into Siberia, terrain and climate would magnify the obstacles. Then endless forest, the gorges cut from solid rock, the mountains of the Transbaikals, the treacherous permafrost, the short winter days, and the deep, deep Siberian cold all presented obstacles on a scale that the world's builders had yet to face.

Given the scope of the undertaking—9,458 kilometers, two to three times longer than the transcontinental railways that had been built in North America at the time—it's not surprising that addressing a bottleneck could have far-reaching consequences.

One critical point about bottlenecks is they can move around systems. You fix one only to introduce another. In *The Chip*, T. R. Reid gives an example of shifting a bottleneck that threatened an entire system. He describes a textile factory that started falling far behind its normal production rate. To figure out what was going on, the factory manager followed the output process on the production floor. They found the employees constantly had to rethread their sewing machines because a cheaper thread they'd bought kept breaking. To save 15¢ per spool, they were losing \$150 per hour in production output.

The TSR was a complex undertaking, so attempts to alleviate a bottleneck could easily cause unanticipated consequences. Problems beset the project. There was a continual shortage of local supplies. There were limits to construction schedules caused by the seasonal weather. The fact that all decisions about the railway had to go through a central committee in St. Petersburg with a weeks-long communication delay created uncoordinated short-term solutions to problems.

In addition, the deepest lake in the world sits in the middle of the route. Originally the main line stopped on one side of the lake and goods were ferried across to the rail continuation on the other shore. This situation created a huge physical bottleneck in terms of the movement of goods and people until a track was completed around its southern shore decades after the main line was built.

Finally, and perhaps most critically for the actual construction, was the bottleneck created by the extreme shortage of labor, which had significant impacts on the functioning of the other parts of the system.

The railway was built as five separate projects that were worked on simultaneously. One of the consequences of deciding to work this way was that the railroad wasn't treated as one project when it came to labor. Each of the five projects competed with the others for part of the same pool of resources.

The desire to shorten the total building time resulted in a trade-off that augmented the labor bottleneck. As Christian Wolmar explains in *To the Edge of the World*, this construction approach created a competition for resources that, combined with a low local population density, meant that there weren't enough locally available workers. They had to be imported. In order to ease the pressure on the labor bottleneck, skilled workers were imported from all over Europe to work along the length of the TSR. For the eastern section, thousands of workers were brought in from China, Japan, and Korea. And on all parts, but in particular the middle section that ran through the Siberian prison camp area, convicts were used in the construction of the track.

The enormous time pressures placed on the men charged with building the railway meant that the labor bottlenecks were often addressed with excessive sums paid to contractors. Thus labor absorbed most of the money available for the project. As Wolmar describes, local peasant contractors were unsupervised, and there was no competitive tendering process. Contractors "often asked for extra payment, once work had begun, as they knew that there was no alternative supplier because the imperative was to get the job done quickly."

The problem with easing the pressure on the labor bottleneck by subcontracting the work to those without sufficient experience manifested on the train track itself. The incentive for the contractors was to pocket money in the short term. The labor shortage could be solved with money, yet there was only so much funding available. Something had to give. What got sacrificed was safety. "With very little supervision of the work, contractors boosted their profits by skimping on material or building to below the required standard, resulting in embankments that were too narrow, insufficient ballast, inadequate drainage, and a host of other failings." To save money on building materials because the labor was so expensive (and the cut directly pocketed by the subcontractor so high), the inclines in many places were too steep and the curves too tight. It was a dangerous railway.

For the TSR, solving the labor shortage created a materials bottleneck because the money used to solve the labor problem meant there wasn't enough left over to purchase quality materials. Accordingly, the central committee thousands of miles away could not keep up with the demands to solve workmanship and safety issues. They were unable to react fast enough, so the integrity of the track was compromised.

Despite the remarkable achievement of building the TSR, the cost-cutting on materials and the shortening of the route through unsuitable terrain with steep grades and tight corners meant it had problems from the start. Wolmar explains that "almost as soon as each section of the line was completed, improvements had to be made to ensure it was functioning properly." Even with the sustained effort, the

locomotives wore out quickly, goods were shipped painfully slowly, and accidents and deaths occurred all the time.

Spending money without quality assurance only moves problems into the future. Russia effectively had to build the same railroad multiple times because the first track was almost unusable.

On a project with the scope of the TSR, bottlenecks are inevitable. Identifying them and planning how to manage them is part of the process of construction. The lesson here is to be careful how we address bottlenecks so that we don't create huge problems for ourselves later on.

Often when we encounter a bottleneck, we keep patching over it so it bounces back to being the bottleneck again. On the TSR, the money used to solve the labor shortage also created incentives to keep that shortage going. Throwing money at the problem without understanding the system is unlikely to yield the intended benefits.

Instead of addressing bottlenecks as they appear, your time might be better spent on a root-cause fix that makes a foundational improvement that leads many bottlenecks to disappear indefinitely. One way to achieve foundational improvement is to simulate conditions you are likely to face to try to find bottlenecks ahead of time. Instead of merely fixing the problem, we can solve a bottleneck by asking how the system could be designed to not have that problem exist in the first place. Addressing bottlenecks is a never-ending job and must always be factored into your planning.

Bottlenecks and innovation

Bottlenecks inspire innovation. When a limit emerges, we're often forced to try something new to alleviate it. Many inventions come about as the result of shortages of resources that prompt people to find alternatives. Innovating as a response to bottlenecks is common during wars when necessary materials may be unobtainable. Looking at the past century, many things we now use regularly were invented at times of conflict to alleviate bottlenecks in supply.

Nylon was the first synthetic fiber, and today we use it in everything from swimwear and fishing nets to seat belts and tents. Being light, strong, and waterproof, it is versatile and practical. Nylon was invented in the early 1930s as an alternative to silk and began commercial production toward the end of the decade. The United States obtained most of its silk from Japan at the time but found itself risking losing access due to rising tensions between the nations. Nylon eliminated that bottleneck by providing an alternative material manufactured in the United States.

While it was invented in response to a shortage, it proved to have advantages over silk in common products as well as new uses. In particular, nylon stockings were popular during the early 1940s, before nylon's production was diverted for military purposes. It served an essential wartime role as parachute and tent material. DuPont, the inventors of nylon, decided not to trademark it so it would seem like a material in itself, not a brand. By being available for experimentation and development, nylon continues to find many uses.

Similarly, the United States had difficulties obtaining rubber during World War II due to conflict with Japan. Ameripol, a synthetic rubber that didn't rely on access to Asian natural resources, was invented by chemist Waldo Semon* as an alternative. Not having any rubber would have been disastrous for the war, as it was integral for practically every item and device used in the fighting, in particular as a

material for tires. Without rubber, as innocuous as it seems, vehicles like planes and tanks wouldn't have been able to operate. It's very possible that without the rapid effort to invent a viable form of synthetic rubber and develop the capacity to produce almost a million tons of it, the Allies would have lost the war.¹⁰ Now most rubber is synthetic.

Medical science tends to advance the fastest during wars. Facing new demands and shortages of essential supplies, people find creative ways to deal with injuries and diseases. During the American Civil War, dozens of new types of prosthetic limbs were invented, and surgeons became more adept at using ligatures. At the start of the war, the mortality rate from infections was 60%. By the end, it was 3%. During World War II, production capabilities for penicillin skyrocketed.

During World War I, many people became malnourished or undernourished due to food rationing. Nutrients became a bottleneck. Lack of adequate food was problematic for children, many of whom developed rickets (soft bones due to vitamin D deficiency). Many soldiers suffered serious bone breakages. Kurt Huldshinsky, a doctor working in Berlin, discovered he could cure rickets by seating children in front of an ultraviolet lamp. Research after the war identified why this worked: a sun lamp simulates sunlight and prompts the body to produce vitamin D, thereby helping to alleviate the bottleneck in access to nutritious food caused by the war. Today sun lamps are a common medical tool for everything from skin conditions to seasonal affective disorder.

The need to overcome the effects of a lack of nutritious food led to the invention of an alternative way of meeting people's nutritional needs. Wartime medical innovations developed as a response to bottlenecks have, in many cases, ended up benefiting everyone.

Conclusion

Although we tend to view bottlenecks as something negative, they can nudge us toward new ideas and better ways of solving the same problems. Using bottlenecks as a mental model not only helps us conceptualize them in the literal sense when looking at a system but also teaches us that they are not something to fear. When there are no severe bottlenecks, we may not think to improve things. After they emerge, we're forced to be more creative and watch that we don't transfer the pain somewhere worse. In the long run, they can make us better off.

CRITICAL MASS

A system becomes "critical" when it is on the verge of changing from one state to another. The final unit of input before the change has a disproportionate impact. It is the proverbial straw to break the camel's back. Before a critical mass is reached, the camel can support the amount of weight it's required to carry. Then the weight passes a threshold where any additional amount is disastrous, and the final straw tips the camel into another state. Once a system passes a certain threshold and enters a critical state, it only takes a tiny nudge to change it.

When a system changes from one state to another, we say it has achieved critical mass, also known as reaching the tipping point. In social systems, critical mass tends to mean when enough people have adopted something, such as a belief or product, that its growth can sustain itself. In his 1978 book

Micromotives and Macrobehavior, game theorist Thomas Schelling wrote, “The generic name for behaviors of this sort is critical mass. Social scientists have adopted the term from nuclear engineering, where it is common currency in connection with atomic bombs.”

The amount of energy required for a system to achieve critical mass is variable. Different systems have different properties and thus require varying amounts of inputs to tip from one state to another.

Heated water is at critical mass when it is hot enough to change from liquid to gas. There is a massive difference between 211 and 212 degrees Fahrenheit. One is the boiling point; the other is not. In business, critical mass is the point where a business makes enough money to no longer need outside investment or the point where the financial growth of a company becomes self-perpetuating. In epidemiology, critical mass can refer to the point where enough people are vaccinated in a population to prevent an infectious disease from spreading to vulnerable people who cannot be immunized.

Using critical mass as a model helps us understand the effort required to achieve sustained change. Systems have certain inflection points where they change from one state to another. It doesn't help us to focus solely on the tipping point and ignore the work required to bring a system there. Because the inputs that tip a system into a new state tend not to have a linear effect—the final unit of input that leads to the change in state has an outsized marginal impact—we are disproportionately impressed by them. We are also disproportionately affected by them. But the straw only breaks the camel's back when there is already a lot of weight on it. Putting one piece of straw on a camel isn't always going to have the same effect.

The critical mass lens also helps us identify the parts of a system we can target to advance change. In social systems, for example, we don't need to spend equal effort changing everyone's mind. We can instead focus our efforts on changing the minds of opinion leaders to more quickly progress change.

Systems in a critical state tend to be precarious, but they don't stay that way for long because they're so easily tipped. Getting insight into what could be the straw is valuable, as is recognizing when a system is poised on the edge of instability. A pencil balanced on its end may appear at equilibrium as it remains upright. But it could topple at the slightest disruption, so it is not stable.

Sidebar: The Overton Window

Day by day, people's minds don't change that much. It's unusual for someone to wake up one day and decide to completely change their tack on a pertinent issue. But in the long run, over decades, the ideas in the mainstream alter drastically. Fringe ideas become mainstream, and mainstream ideas become fringe. One way to understand this phenomenon is by considering the Overton Window, a concept developed by Joseph P. Overton in the 1990s as part of his work for the Mackinac Center for Public Policy.

The Overton Window refers to the range of ideas considered acceptable for politicians to propose as policy. Ideas outside of that, no matter how good, cannot gain widespread support. They're too extreme for the current climate and are best avoided lest they harm one's chances of re-election. Over time, the Overton Window shifts. Some politicians may advance far-out ideas in a deliberate attempt to move the Window further from the norm and make more moderate ideas more palatable.

Ideas move in a progression: unthinkable ≥ radical ≥ acceptable ≥ sensible ≥ popular ≥ policy.

For instance, the suffrage movement shifted the Overton Window to make the idea of women being able to vote move from unthinkable to policy. Now to suggest otherwise would be unthinkable.

Politicians must prioritize the Overton Window over their personal beliefs. It's important to recognize that the Overton Window isn't universal. The conservative political positions of one country may be considered liberal in another.

The value of the Overton Window as a concept is that it encourages us to recognize that attitudes and opinions are not static. What we consider acceptable today may one day be unacceptable. Ideas that are fringe and wacky now may one day be mainstream.

The work required for change

We like to tell stories about the tipping points. We look at the landmark cases or individual actions that sparked a cascade of change in the past and wonder how we can re-create them to push our current system into a new state. Using the mental model of critical mass, however, reminds us that it is equally important to pay attention to the effort involved in the buildup.

In September 1893, New Zealand became the first self-governing country to give most adult women the right to vote in parliamentary elections. American women would not earn this right for another 27 years and British women for 25. An important thing to understand about women's suffrage in New Zealand is that it was far from a sudden change, even if it may have appeared as such from afar. Through the efforts of many people over many years, there was a slow shifting of the Overton Window to the point where women being able to vote became reasonable in the minds of enough people to shift the voting system into a new state.

Certain unusual aspects of New Zealand's society and events in its history helped lay the groundwork for changes in the voting system even before the official suffrage movement began. Many of the people living in New Zealand had settled there in recent decades and desired to create a fairer society than the European one they had left behind. Seeing as the population was small (under 750,000 in 1893, including 40,000 Maori), fewer minds needed changing to create a critical mass. The movement received support from prominent male politicians early on, which aided in getting a foothold in Parliament.

Women receiving equal access to education in New Zealand was another key factor in the buildup of opinion change to critical mass. Due to the campaigning of educationalist Learmonth Dalrymple, girls received the same secondary education as boys, with the first school for girls opening in 1871.⁷ Dalrymple also successfully ensured women were able to attend university, where they made up half the student body by 1893. Greater education led to improved employment prospects outside of the home, beyond the customary option of domestic labor. More and more women entered the workforce once they had better education, gaining social influence in areas such as teaching, journalism, medicine, and the arts. When they faced worse working conditions than men, New Zealander women began to unionize.

In many ways, the New Zealand suffrage movement was entwined with the temperance movement, which sought to restrict or prohibit the consumption of alcohol. Throughout the 19th century, alcohol became a growing problem in many countries, leading to poverty, violence, crime, and harm to family life. For New Zealanders, it was particularly harmful among men working in the agricultural, maritime, and industrial industries. As in many other countries, it was reported that many men drank away their wages before even making it home on the weekend, leaving their wives and children bereft. Seeing as women tended to suffer the most as a result of widespread heavy drinking, they were influential in the movement.

Although the temperance movement never achieved the aim of total prohibition in New Zealand, it constructed a framework for women to politically organize. Alongside unionization, it gave women the confidence that they could have influence if they worked together in sufficient numbers with clear goals and a sense of focus. As Patricia Grimshaw writes in *Women's Suffrage in New Zealand*, "Women, on the practical side, learned the arts of organization, administration, and leadership which could be turned to use in later years in their own cause. Women, on the ideological side, entered a sphere in which a new outlook on their basic rights developed rapidly, spurring them to aim at the realization of their full rights as women."

All of this work culminated in the suffrage movement, led by Kate Sheppard, which built upon the social change that had been growing since the country's founding. In the early 1890s, Sheppard organized several petitions in favor of women being able to vote, which she presented to Parliament. Despite initial failures, the movement kept trying, gaining more and more support each year. By 1893, her petition amassed 32,000 signatures, a number all the more impressive considering the tiny population of the country at the time. After numerous attempts, the bill passed by a whisker. The changes in opinion had reached critical mass.

In turn, women earning the right to vote in New Zealand helped motivate and inspire similar movements elsewhere because it showed wider suffrage was possible. After World War II, women's political emancipation spread around the globe, a visible symbol of wider improvements.¹³ Once you pass a tipping point, the whole nature of a system changes. It develops new properties, and new things are possible. New Zealand was that tipping point for women's suffrage in many other countries.

When we look back at significant social changes, it's important to recognize the work involved in building a critical mass. Women getting the vote in New Zealand was the result of years of effort on many different fronts to build the capabilities needed to change opinions. As social norms regarding women voting started to change, the movement gained the critical mass necessary for petitions in Parliament to be the final straw that resulted in a new state.

We can learn from the mental model of critical mass that changing a system doesn't require changing everything about it. Changing a small percentage of its parts can shift the whole thing into a new state. Getting people to alter their beliefs doesn't mean convincing everyone; once you pass a threshold, the change perpetuates itself.

Sidebar: Minority Opinions

Sometimes, people change their minds a lot in a short time. Although it can seem as though this shift occurs overnight, what really happens is that things change slowly until

a critical mass of people hold a viewpoint. Interestingly, a majority is not required for things to tip and result in almost everyone changing their minds. Once opinion leaders hold a viewpoint, it spreads easier because people who don't hold this same viewpoint face negative consequences. Targeting opinion leaders can accelerate getting to the tipping point.

Researchers at Rensselaer Polytechnic Institute identified the percentage of a population necessary for social change as 10%. They stated that this holds true regardless of the type of network. However, other research suggests the number is much higher, with around 25% of a population being the tipping point. Past this point, a minority view can replace the prevalent status quo. The researchers attribute this to most people not being as committed to their opinions as they imagine, meaning they're liable to change their mind as those around them do. The 25% figure is likely to vary depending on the extent of the stake people have in their viewpoints and the social clout of the minority seeking to change things.

Sidebar: Madeline Pollard

Unprecedented legal decisions are often the most visible sign of significant social change. They can appear sudden and dramatic, but they are usually the result of a slow build of changing opinions that becomes impossible to ignore at a certain point. One such case that stands out is that of Madeline Pollard and her legal battle with a congressman William C. P. Breckinridge.

In 1894, the attention of Americans was ensnared by Pollard's suing Breckinridge for breach of promise, seeking compensation of \$50,000. The story went far beyond the two of them and was symptomatic of the wider social changes that would soon earn women the right to vote.

The pair met when Pollard was a struggling seventeen-year-old student with no parents and he was a successful politician three decades older. They began a lengthy affair, during which he financially supported her and she gave birth to, then gave up, two of his children. Pollard believed Breckinridge intended to marry her only to discover he had lied. She decided to sue, even though women had been laughed out of court for similar cases only a few years earlier. The trial was vicious, and the media lapped it up.³ But Pollard ended up winning, a victory that set an important precedent. It helped highlight the double standards of the time. While we could look at the case as a turning point, it was really the result of changing opinions building up until a critical mass was reached.

Organic cities

In nuclear physics, critical mass refers to the minimum amount of fissile material needed to start a self-sustaining reaction. You can pile up the uranium, and nothing will happen until a high enough density is reached. To focus on what prompted the change from inert to active in a nuclear reaction isn't all that interesting. It's just one more bit of uranium. The more interesting question is, how much is needed to

kickstart the reaction to continue without further inputs? The lens of critical mass is thus a useful one to apply to other situations where we'd like to produce self-sustaining reactions, such as cities.

Cities are complex systems where planners have often misidentified the elements required to create enough density to produce self-sustaining interactions. In cities, it's not the amount of infrastructure that produces interactions, it's how that infrastructure is laid out. A certain number of interactions are required for a city to function well and adapt to meet the needs of those living in it. What makes a city safe, interesting, prosperous, and creative isn't the buildings or streets. It's how the infrastructure fosters interactions and relationships between people.

Jane Jacobs wrote extensively in *The Death and Life of Great American Cities* about how to achieve self-sustaining interactions in cities and why they are important. She argued that when we isolate parts of cities, we miss the many interconnected functions that they perform. For example, "A city sidewalk by itself is nothing. It is an abstraction. It means something only in conjunction with the buildings and other uses that border it, or border other sidewalks very near it."

The system formed by a sidewalk and its users is what makes an area both safe and interesting. When the area around a sidewalk is subject to active mixed uses—homes, cafés, shops, and so on—there are always eyes upon it and people passing through. These people do not need to know each other or even talk to each other. It is enough that they see each other, are aware they are watching and being watched, and observe each other's behavior. Jacobs writes, "The basic requisite for such surveillance is a substantial quantity of stores and other public places sprinkled along the sidewalks of a district; enterprises and public places that are used by evening and night must be among them especially."

It is the interplay of people that ensures a sidewalk is safe and places limits on antisocial behavior. People moderate how they act knowing someone is or might be watching. Any antisocial behavior that does break out is likely to be swiftly halted by the interventions of bystanders. An organic, unorganized system of control enforced by social norms is more immediate and effective than the use of police, although the threat of them being called plays a role.

To understand why neighborhood safety breaks down in certain situations, we need to consider a sidewalk as needing a minimum number of interactions in order to function as part of a city system instead of just a piece of concrete. You'd feel safer at night walking along a street lined with bars open late than one with stores that close at 5:00 p.m. You'd feel safer walking along a main street passing the fronts of houses than an alleyway only visible from a couple of windows. You would feel safer on a crowded street than an empty one with a police officer present. You feel safer on a sidewalk that is part of a whole system of self-sustaining interactions. The more people that are using a sidewalk, at different times and for varied purposes, the better it functions as a safe space. This is true outside of cities, but other factors are likely to be relevant for safety in towns or rural areas.

It is this "intricacy of sidewalk use, bringing with it a constant succession of eyes" that also makes an area lively and interesting, and therefore a desirable destination. ***Activity attracts more activity.*** Many people using an area brings economic benefits, which further attracts more businesses, especially more unusual and specialized ones, which in turn attract more people.

Visual activity is appealing to the eye. We like to watch things happening and other people going about their days, so crowds attract more crowds. People who watch attract more people who do by making an

area safer, and people who do attract more people who watch by making an area interesting. ***It is a feedback loop that is dependent on a myriad of uses and interactions.***

Our experience of a city also has to do with the relationships it fosters with other people. Again, interactions are paramount. Both social isolation and a lack of privacy are risks in cities. An individual may contend with both throughout the course of a day in a poorly designed city. Ideally we gravitate toward spaces where we can have controllable levels of interaction with others. In *Happy City*, Charles Montgomery says, “The richest social environments are those in which we feel free to edge closer together or move apart as we wish. They scale not abruptly but gradually, from private realm to semi-private, to public; from boardroom to living room to porch to neighborhood to city.”

One instance of an environment that allows individuals to regulate their level of interaction but also accommodates a wide variety of users is a plaza or square. Present at varying scales as a standard feature of the city, squares are mixed-use areas in which one could, among other things, meet a friend or date, enjoy a coffee at a sidewalk café, watch a street performer, exercise a dog or child, attend a protest, or strike up a conversation with a stranger. The surrounding bustle allows for both safety and interest. You feel safe meeting a date or talking to a stranger because there are enough people around to take note if something goes wrong. But you also feel comfortable having a private conversation with a friend or writing in a notebook because no one is likely to pay much attention to you in particular with so much going on.

Space alone does not create this environment. An overly large square can actively repel people as it feels overexposed and empty. A popular one must have enough activity and reasons to visit in a small enough space to create the density needed for spontaneous and ongoing interactions to occur.

Strøget is a network of pedestrianized streets in Copenhagen created in the 1960s as part of an effort to switch the city’s focus from cars to pedestrians and cyclists. Closing off the area to cars was a controversial move. People didn’t believe Danes would simply want to mingle in a public space, considering the country’s typically cold weather and lack of a preexisting café culture.

But today, Strøget is a lively area with an annual peak of 120,000 visitors braving the icy Danish winter on the last Sunday before Christmas.

Strøget works because it manages to combine many possible uses in one area, making it a busy and engaging place to visit. It facilitates the interactions needed for a city to function in a way that promotes needs like psychological safety and controllable social engagement. Pedestrians and cyclists pass through on their way to somewhere else. Shoppers visit both luxury boutiques and chain stores. Others visit the theater or church. Street performers attract audiences, as do peaceful demonstrations. People sit at pavement cafés or wander along eating inexpensive street food. Tourists visit museums and art galleries. People are willing to brave the cold because the area has so much to offer.

Architect Jan Gehl has since applied the principles of Strøget to other parts of Copenhagen and cities around the world. The city was able to promote a new sort of street culture by recognizing that a bustling public space is the product of a number of factors coming together at a certain density.²³ Street culture is not specific to certain cultures; it is about having the right kind of spaces for it. These spaces recognize that a minimum of interactions must be maintained in order for them to function in a self-sustained way. The actual architecture should be invisible because the focus is on the people and on bringing out their best qualities.

The idea that infrastructure needs to promote and facilitate a certain number of interactions and not just look good explains why some attempts to design and build cities from scratch have been riddled with problems. If you focus on the infrastructure first and just build a list of requirements—like houses, stores, and streets—you increase the chance that your city won't function well. Rather, the infrastructure needs to be designed to facilitate a critical mass of interactions—something that planned cities often miss. Planned cities may try to design out those interactions because designers see them as a waste of time, or they may even want to discourage any type of organized action.

Planned cities often segregate different functions, like workplaces and homes, ignoring the benefits of mixed-use areas, which are the standard in natural cities. Visually this looks ordered and pleasing, but it doesn't promote the interactions cities require. It's useful for people to be able to access resources close to where they live. It cuts down on commuting and increases time people can spend on relationships.

Mixed-use areas combining residential and commercial elements create more interactions than those segregated for one function. Planned cities may segregate roads and sidewalks for pedestrians, viewing driving and walking as distinct activities. But this prevents people from being able to hail a taxi and combine the two functions.

Brasília, the federal capital of Brazil, was designed in the 1950s and '60s to replace Rio de Janeiro as the capital. Architect Oscar Niemeyer and urban planner Lucio Costa crafted a vision of a utopian city from scratch. ***In the visual sense, Brasília is a stunning World Heritage site. From the air, it has a beautiful birdlike form. As a place to live, it functions less well.***

Areas of Brasília have specific uses: people live in one part, work in another, and shop in another. Without mixed-use areas or much catering to pedestrians, the city cannot form a street culture. Communities cannot cohere due to a lack of areas where people can mingle. Visual order—Brasília is laid out in a grid—does not translate into good function.

Everything in Brasília was built new and modern to similar specifications. Yet cities need buildings of varying age and quality to allow for people at different income levels. The architects and designers never planned for low-income housing, despite Brasília needing inexpensive labor as much as any other city.²⁵ As a result, unofficial areas have sprung up around the city that house its poorer residents. Only by deviating from the plan can it function at all.

So although Brasília contains the same parts as a typical city, those parts do not facilitate much interaction. It seems as if its designers believed that the arrangement of the infrastructure is irrelevant to city functioning. But how the infrastructure is laid out is critical because it facilitates the critical mass of interactions cities need to be able to adapt and grow to meet the needs of those living in them.

Conclusion

The mental model of critical mass gives us insight into the amount of material needed for a system to change states. We can categorize material as interactions, connections, or efforts. When enough material builds up, systems reach their tipping point. When we keep going, we get sustainable change. Using critical mass as a lens on situations where you want different outcomes helps you identify both the design elements you need to change and the work you need to put in.

FALSTAFF: GIVE ME LIFE

Bloom, Harold. Falstaff: Give Me Life (Shakespeare's Personalities Book 1)

RELEVANCE:

Falstaff is life, and in Falstaff one can see all the lessons of life – but they are hidden in the perspective of an old man who has not been accomplished in any way except in his zest for life and collection of friends. For professional purposes, the relevance of Falstaff is in his critiques of authority. In recruiting men for battle: “spare me the great men and grant me the spare men” as any will do as well to die on the field. This is before he plays dead on the battlefield himself, explaining when Hal catches him that, dear boy, “discretion is the better part of valor.” Falstaff sees life, drinks it in and shares it with us. We learn from that. We can learn to see life too – not just the illusions of we are taught to believe in. Falstaff challenges us to think for ourselves.

The heroic vitalists in literature include Rabelais’ Panurge, Chaucer’s Wife of Bath, and Cervantes’ Sancho Panza. Falstaff reigns over them. John Ruskin taught that the only wealth is life. Sir John Falstaff, the Socrates of Eastcheap, embodies that truth.

What is the essence of Falstaffianism? My late friend and drinking companion Anthony Burgess told me it was freedom from the state. Anthony and I never quite agreed on that though indeed no societal standards ever could abide Falstaff. I recall telling Burgess that for me the essence of Falstaffianism was: do not moralize. Computing Falstaff’s flaws is trivial: he bulges with them. Hal, like his father Bolingbroke, is the essence of hypocrisy. They are Machiavels. Bolingbroke, who becomes Henry IV, is a usurper and a regicide. His nonsensical obsession is that he will expiate the murder of Richard II by leading yet another crusade to capture Jerusalem. He dies indeed in the chamber of his palace called Jerusalem. Hal, when he becomes Henry V, leads a land grab to capture France. A crusade is what one might expect of Prince Hal, who hungers like Hotspur for what both call honor. Falstaff destroys the validity of that appetite in a reply to Hal in act 5, scene 1 in the first part of Henry IV:

Hal: Why, thou owest God a death. [Exit.]

Falstaff: ’Tis not due yet; I would be loath to pay him before his day.
 What need I be so forward with him that calls not on me?
 Well, ’tis no matter; honour pricks me on. Y
 ea, but how if honour prick me off when I come on? how then?
 Can honour set to a leg? No: or an arm? no: or take away the grief of a wound? no.
 Honour hath no skill in surgery, then? no. What is honour? a word.
 What is in that word honour? what is that honour? air.
 A trim reckoning! Who hath it? he that died o’ Wednesday. Doth he feel it? no.
 Doth he hear it? no. ’Tis insensible, then. Yea, to the dead.
 But will it not live with the living? no. Why? detraction will not suffer it.
 Therefore I’ll none of it. Honour is a mere scutcheon: and so ends my catechism.
act 5, scene 1, lines 126–40

If there could be a religion of vitalism this would do very well for its catechism. Falstaff mocks faith, killing the insane notion that we owe God our death. Knowingly he also mocks both Hal and himself. Disreputable and joyous, he speaks to a world that goes from violence to violence.

Falstaff immediately became the most popular of all Shakespearean personalities, and remains so. The audiences at the Globe and the readers who purchased quartos found little to moralize against in Sir John. His being overflows and that excess brings new meanings to our minds. Exuberance in itself is a shadowy virtue and can be dangerous to the self and to others, but in Falstaff it generates more life.

...

In that sense the sublime Falstaff is bad news. Against myself I cite **Orson Welles, whose *Chimes at Midnight remains a neglected masterpiece***. Welles made the film, an adaptation of the Henriad, and played it as tragedy. The film had a supporting cast of brilliant stars including Keith Baxter as Hal, John Gielgud as Henry IV, Jeanne Moreau as Doll Tearsheet, Margaret Rutherford as Mistress Quickly, and Ralph Richardson as the Narrator. ***Welles called Falstaff a “gloriously life-affirming good man . . . defending a force—the old England—which is going down. What is difficult about Falstaff . . . is that he is the greatest conception of a good man, the most completely good man, in all drama. His faults are so small and he makes tremendous jokes out of little faults. But his goodness is like bread, like wine.”***

I may be unique in my total agreement with Orson Welles. Is there anyone else in Henry IV whose goodness is like bread, like wine? They are scurvy politicians like the King and the brilliant Prince Hal and most of the rebels. They are smug thugs like Prince John and high-spirited killing machines like the captivating Hotspur and Douglas. Falstaff's followers—Bardolph, Nym, the outrageous Pistol—are entertaining rogues, and Mistress Quickly and Doll Tearsheet are better company than the Lord Chief Justice. Justice Shallow is charmingly absurd and his crony Silence augments the irreality.

Falstaff is as bewildering as Hamlet, as infinitely varied as Cleopatra. He can be apprehended but never fully comprehended. There is no end to Falstaff. His matrix is freedom but he dies for love.

Oliver Goldsmith in his “A Reverie at the Boar's Head Tavern, Eastcheap” is a beacon:

The character of old Falstaff, even with all his faults, gives me more consolation than the most studied efforts of wisdom. I here behold an agreeable old fellow, forgetting age, and showing me the way to be young at sixty-five. Sure I am well able to be as merry, though not so comical as he. Is it not in my power to have, though not so much wit, at least as much vivacity? Age, care, wisdom, reflection, begone! I give you to the winds. Let's have t'other bottle; here's to the memory of Shakespeare, Falstaff, and all the merry men of Eastcheap!

....

Why did Shakespeare invent Falstaff?

Literary character is always an invention and indebted to prior inventions. Shakespeare invented literary character as we know it. He reformed our expectations for the verbal imitation of personality and the reformation appears to be permanent and uncannily inevitable. The Bible and Homer powerfully render personages yet their characters are mostly unchanging. They age and die within their stories but their modes of being do not develop.

Shakespeare's personalities do. The representation of character in his plays now seems normative and indeed became the accepted mode almost immediately. Shakespeare's personalities have little in common with those of Ben Jonson or Christopher Marlowe. Shakespeare's originality in portraying women and men finds itself upon The Canterbury Tales of Geoffrey Chaucer.

Throughout Shakespeare, vitality transmutes into doom-eagerness. Richard II, the protagonist of the history that begins the Henriad, is a moral masochist whose luxurious self-indulgence in despair augments his overthrow by the usurper Bolingbroke, who thus becomes Henry IV. In the personality of Richard II, Shakespeare prefigures that element in all of us by which we render bad situations even worse through our own hyperbolic language.

Falstaff is different. His zest for life pervades his torrent of language and laughter. Hotspur is the incarnation of doom-eagerness. His mode though is opposite to that of Richard II. His vaunting language assaults the frontiers of what is possible. Hal, his father's son, distrusts his own vitalism, and yet goes to Falstaff to be confirmed in it. The royal pupil proves unforgiving toward his teacher. Kings have no friends, only followers, and Sir John Falstaff is no man's follower.

Directors, actors, playgoers, readers need to understand that Falstaff, most magnificent of wits, is tragicomic. Unlike Hotspur and Hal, he is not one of the fools of time. Dr. Johnson said that love was the wisdom of fools, and the folly of the wise. I cannot think of a better description of my hero Sir John Falstaff.

HOW TO READ AND WHY

Harold Bloom

RELEVANCE:

To read fully is an amazingly imaginative act. It expands the world. And if the sentence does work, then the sentences on the page aren't just ink turned by a reader, but a "program" that reaches through time and into your imagination from the page. Reading is a superpower.

PROLOGUE: WHY READ?

Going on seventy, one doesn't want to read badly any more than live badly, since time will not relent. I don't know that we owe God or nature a death, but nature will collect anyway, **and we certainly owe mediocrity nothing, whatever collectivity it purports to advance or at least represent.**

Because my ideal reader, for half a century, has been Dr. Samuel Johnson, I turn next to my favorite passage in his *Preface to Shakespeare*:

This, therefore, is the praise of Shakespeare, that his drama is the mirror of life; that he who has mazed his imagination in following the phantoms which other writers raise up before him may here be cured of his delirious ecstasies by reading human sentiments in human language, by scenes from which a hermit may estimate the transactions of the world and a confessor predict the progress of the passions.

To read human sentiments in human language you must be able to read humanly, with all of you. You are more than an ideology, whatever your convictions, and Shakespeare speaks to as much of you as you can bring to him. That is to say: Shakespeare reads you more fully than you can read him, even after you have cleared your mind of cant. No writer before or since Shakespeare has had anything like his control of perspectivism, which outleaps any contextualization we impose upon the plays. Johnson, admirably perceiving this, urges us to allow Shakespeare to cure us of our "delirious ecstasies." Let me extend Johnson by also urging us to recognize the phantoms that the deep reading of Shakespeare will exorcise. One such phantom is the Death of the Author; another is the assertion that the self is a fiction; yet another is the opinion that literary and dramatic characters are so many marks upon a page. A fourth phantom, and the most pernicious, is that language does the thinking for us.

Still, my love for Johnson, and for reading, turns me at last away from polemic, and towards a celebration of the many solitary readers I keep encountering, whether in the classroom or in messages I receive. We read Shakespeare, Dante, Chaucer, Cervantes, Dickens, Proust, and all their peers because they more than enlarge life. Pragmatically, they have become the Blessing, in its true Yahwistic sense of more life into a time without boundaries. **We read deeply for varied reasons, most of them familiar: that we cannot know enough people profoundly enough; that we need to know ourselves better; that we require knowledge, not just of self and others, but of the way things are.** Yet the strongest, most authentic motive for deep reading of the now much-abused traditional canon is the search for a difficult pleasure... and a pleasurable difficulty seems to me a plausible definition of the Sublime, but a higher pleasure remains the reader's quest. There is a reader's Sublime, and it seems the only secular transcendence we can ever attain... I urge you to find what truly comes near to you, that can be used for weighing and for considering. **Read deeply, not to believe, not to accept, not to contradict, but to learn to share in that one nature that writes and reads.**

THE PROPER STUDY OF MANKIND

An Anthology of Essays by Isiah Berlin

Hardy, Henry. The Proper Study Of Mankind (pp. 436-438). Random House. Kindle Edition.

RELEVANCE:

Jim Collins and Phil Tetlock both use the Hedgehog and the Fox metaphor in their work. Collins notably cites Berlin's essay but misses the meaning completely – that Tolstoy's genius was in being a fox who wanted to be a hedgehog. Collins' own Hedgehog Principle could have benefitted from the insight that a key to excellence involved being a fox (seeing the details) but wanting to be a hedgehog (finding simplicity).

Complexity also lurks within the tension between the hedgehog and the fox. Remember from Miller, Axelrod, Kauffman, Paransarathy, ... the whole study of complexity is about how a simple set of rules or constraints explodes through interactions in a system into unexpected and often unpredictable patterns. The simplicity of many hedgehogs explodes into the world in a way that would bring a fox glee.

In the context of scientific inquiry and what we do as analysts – one needs to be a fox who wants to be a hedgehog. The scientist looks for simple forces and constraints that drive and limit the chaos of a set of reactions; We look at a set of complex transactions and economic interactions for a similar underlying simplicity. One must be a fox who wants to be a hedgehog to be able to look at all the complexity of a business try to capture the essence of it in a few sentences in an investment thesis.

The Hedgehog and the Fox: An Essay on Tolstoy's View of History

*A queer combination of the
brain of an English chemist with
the soul of an Indian Buddhist.*

- E. M. de Vogüé

THERE IS a line among the fragments of the Greek poet Archilochus which says:

'The fox knows many things, but the hedgehog knows one big thing.'

Scholars have differed about the correct interpretation of these dark words, which may mean no more than that the fox, for all his cunning, is defeated by the hedgehog's one defence. But, taken figuratively, the words can be made to yield a sense in which they mark one of the deepest differences which divide writers and thinkers, and, it may be, human beings in general. For there exists a great chasm between those, on one side, who relate everything to a single central vision, one system, less or more coherent or articulate, in terms of which they understand, think and feel – a single, universal, organising principle in terms of which alone all that they are and say has significance – and, on the other side, those who pursue many ends, often unrelated and even contradictory, connected, if at all, only in some de facto way, for some psychological or physiological cause, related to no moral or aesthetic principle.

These last lead lives, perform acts and entertain ideas that are centrifugal rather than centripetal; their thought is scattered or diffused, moving on many levels, seizing upon the essence of a vast variety of experiences and objects for what they are in themselves, without, consciously or unconsciously, seeking to fit them into, or exclude them from, any one unchanging, all-embracing, sometimes self-contradictory

and incomplete, at times fanatical, unitary inner vision. The first kind of intellectual and artistic personality belongs to the hedgehogs, the second to the foxes; and without insisting on a rigid classification, we may, without too much fear of contradiction, say that, in this sense, Dante belongs to the first category, Shakespeare to the second; Plato, Lucretius, Pascal, Hegel, Dostoevsky, Nietzsche, Ibsen, Proust are, in varying degrees, hedgehogs; Herodotus, Aristotle, Montaigne, Erasmus, Molière, Goethe, Pushkin, Balzac, Joyce are foxes.

Of course, like all over-simple classifications of this type, the dichotomy becomes, if pressed, artificial, scholastic and ultimately absurd. But if it is not an aid to serious criticism, neither should it be rejected as being merely superficial or frivolous: *like all distinctions which embody any degree of truth, it offers a point of view from which to look and compare, a starting-point for genuine investigation.*

Thus we have no doubt about the violence of the contrast between Pushkin and Dostoevsky; and *Dostoevsky's celebrated speech about Pushkin has, for all its eloquence and depth of feeling, seldom been considered by any perceptive reader to cast light on the genius of Pushkin, but rather on that of Dostoevsky himself, precisely because it perversely represents Pushkin – an arch-fox, the greatest in the nineteenth century – as being similar to Dostoevsky, who is nothing if not a hedgehog; and thereby transforms, indeed distorts, Pushkin into a dedicated prophet, a bearer of a single, universal message which was indeed the centre of Dostoevsky's own universe, but exceedingly remote from the many varied provinces of Pushkin's protean genius.* Indeed, it would not be absurd to say that Russian literature is spanned by these gigantic figures – at one pole Pushkin, at the other Dostoevsky; and that the characteristics of other Russian writers can, by those who find it useful or enjoyable to ask that kind of question, to some degree be determined in relation to these great opposites. To ask of Gogol, Turgenev, Chekhov, Blok how they stand in relation to Pushkin and to Dostoevsky leads – or, at any rate, has led – to fruitful and illuminating criticism.

But when we come to Count Lev Nikolaevich Tolstoy, and ask this of him – ask whether he belongs to the first category or the second, whether he is a monist or a pluralist, whether his vision is of one or of many, whether he is of a single substance or compounded of heterogeneous elements – there is no clear or immediate answer. The question does not, somehow, seem wholly appropriate; it seems to breed more darkness than it dispels. Yet it is not lack of information that makes us pause: Tolstoy has told us more about himself and his views and attitudes than any other Russian, more, almost, than any other European, writer. Nor can his art be called obscure in any normal sense: his universe has no dark corners, his stories are luminous with the light of day; he has explained them and himself, and argued about them and the methods by which they are constructed, more articulately and with greater force and sanity and lucidity than any other writer. Is he a fox or a hedgehog? What are we to say? Why is the answer so curiously difficult to find? Does he resemble Shakespeare or Pushkin more than Dante or Dostoevsky? Or is he wholly unlike either, and is the question therefore unanswerable because it is absurd? What is the mysterious obstacle with which our enquiry seems faced?

I do not propose in this essay to formulate a reply to this question, since this would involve nothing less than a critical examination of the art and thought of Tolstoy as a whole. I shall confine myself to suggesting that the difficulty may be, at least in part, due to the fact that Tolstoy was himself not unaware of the problem, and did his best to falsify the answer. ***The hypothesis I wish to offer is that Tolstoy was by nature a fox, but believed in being a hedgehog; that his gifts and achievement are one thing, and his beliefs, and consequently his interpretation of his own achievement, another; and that consequently his ideals have led him, and those whom his genius for persuasion has taken in, into a systematic misinterpretation of what he and others were doing or should be doing.....***

THE PLEASURE OF FINDING THINGS OUT

Feynman, Richard P.. The Pleasure of Finding Things Out (p. 2). Basic Books. Kindle Edition.

RELEVANCE:

It's Feynman. Curiosity and imagination start with being able to see. Science offers a key tool to seeing what's not on the surface.

These reading are all about different tools and ways of seeing. We use different analytical tools to see, and we also shift our attention and mode from creative imagination to analytical realism to structured inquiry to hypothesis testing to reflection and calibration and back again.

The Beauty of a Flower

I have a friend who's an artist and he's sometimes taken a view which I don't agree with very well. He'll hold up a flower and say, "Look how beautiful it is," and I'll agree, I think.

And he says, "you see, I as an artist can see how beautiful this is, but you as a scientist, oh, take this all apart and it becomes a dull thing." And I think that he's kind of nutty.

First of all, the beauty that he sees is available to other people and to me, too, I believe, although I might not be quite as refined aesthetically as he is; but I can appreciate the beauty of a flower.

At the same time, I see much more about the flower than he sees. I can imagine the cells in there, the complicated actions inside which also have a beauty. I mean it's not just beauty at this dimension of one centimeter, there is also beauty at a smaller dimension, the inner structure. Also, the processes, the fact that the colors in the flower evolved in order to attract insects to pollinate it is interesting—it means that insects can see the color. It adds a question: Does this aesthetic sense also exist in the lower forms? Why is it aesthetic? All kinds of interesting questions which shows that a science knowledge only adds to the excitement and mystery and the awe of a flower. It only adds; I don't understand how it subtracts.

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